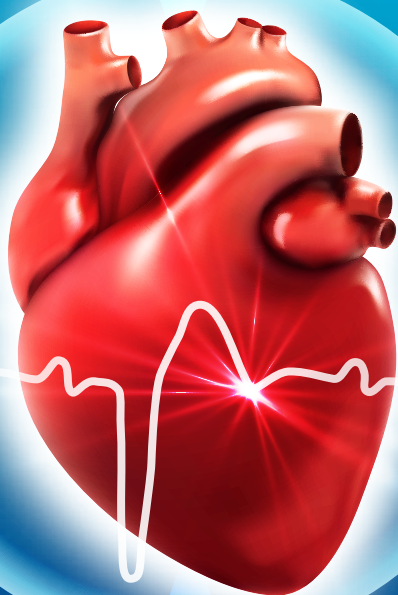


CLINICAL PRACTICE GUIDELINES  
**MANAGEMENT OF  
HEART FAILURE 2023**  
5<sup>th</sup> Edition



Ministry of Health Malaysia



Academy of Medicine Malaysia



National Heart Association of Malaysia

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**STATEMENT OF INTENT**

This clinical practice guideline (CPG) is meant to be a guide for clinical practice, based on the best available evidence at the time of development. Adherence to these guidelines may not necessarily guarantee the best outcome in every case. Every healthcare provider is responsible for the management of his/her patient based on the clinical picture presented by the patient and the management options available locally.

**PERIOD OF VALIDITY**

This CPG is issued in 2023 and will be reviewed in 5 years or sooner if new evidence becomes available.

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Electronic version available on the following website:  
<http://www.moh.gov.my>  
<http://www.acadamed.org.my>

This is the 5<sup>th</sup> update to the Clinical Practice Guidelines on Heart Failure (published 2000, 2007, 2014 and 2019). It supersedes the previous CPGs on Heart Failure.

**MESSAGE FROM THE DIRECTOR GENERAL OF HEALTH**

Heart failure is a condition associated with significant morbidity and mortality. In recent years, there have been advancements in both treatment and diagnostics, to enable all stakeholders in healthcare to improve the management of this condition.

Over 23 years have passed since the publication of the first Clinical Practice Guidelines in Malaysia for Heart failure. Subsequent editions in 2007, 2014, and 2019 reflected the growing body of evidence generated both internationally and locally. This latest edition incorporates the latest science in this field.

I am pleased to see that the goal of management has evolved, from reducing symptoms, improving functional capacity, quality of life and patient survival, to reducing heart failure-related hospitalisation, and reducing the socio-economic impact. The paradigm shift is timely to reduce the burden of hospital admissions due to heart failure.

In addition, I can see new approaches to managing heart failure, including the introduction of early implementation and optimization of foundational heart failure medications, and a structured, multidisciplinary strategy is essential to embark seamless care between hospitals and primary care.

With 25 key messages and 22 key recommendations, I am sure these guidelines will provide all healthcare providers the necessary information that will lead to optimal care of patients with heart failure in Malaysia.

I congratulate the Writing Committee members who have worked hard to produce these guidelines, and I thank all stakeholders who have contributed to its publication. I look forward to the active dissemination of the new information, and its translation into clinical practice.

**Dr Muhammad Radzi Abu Hassan**  
*Director-General of Health Malaysia*

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**RATIONALE AND PROCESS OF GUIDELINES DEVELOPMENT**

Cardiovascular disease is an important cause of morbidity and mortality in Malaysia. Heart Failure (HF), the end stage of most diseases of the heart, is a common medical problem encountered in primary care and is an important cause of hospital admissions and readmissions with a significant impact on hospital expenditure. As the population ages, the prevalence of HF is expected to increase.

The 1st Clinical Practice Guidelines (CPG) in HF was published in 2000 with revisions in 2007, 2014 and 2019. Since then, there have been many new developments in this field. Thus, the publication of this 5th edition is timely. This CPG stresses on the early implementation and optimization of the “Foundational HF” medications. It also proposes a structured multidisciplinary strategy for the seamless care of patients with HF between hospital and primary care.

This CPG was drawn up by a committee appointed by the National Heart Association of Malaysia and Ministry of Health. It comprises cardiologists, nephrologists, family medicine and general physicians and pharmacists from the government, private sectors, and the public Universities.

**Objectives:**

The objectives of this CPG are to:

- Critically review the latest scientific evidence, since the last CPG, in the diagnosis and management of HF.
- Incorporate this new evidence in the daily management of our patients and adapt it to our local needs and available resources.
- Develop clinical pathways for HF.
- Establish HF Clinics and provide guidance on step down care of these patients within primary healthcare.

**Process**

The last CPG published in 2019 was used as a basis. Apart from addressing previous clinical questions that needed to be updated, the Expert Panel also formulated new questions that needed to be addressed. These clinical questions have been divided into sections and each member was assigned one or more topics.

A review of current medical literature on HF from 1<sup>st</sup> Sept 2018 (the date of the last CPG) till 31<sup>st</sup> August 2022 was performed. Literature search was carried out using the following electronic databases - PubMed and Cochrane Database of Systematic Reviews. The following Medical Subject Headings ( MESH) terms or free text terms were used either singly or in combination:

“Heart Failure”, “Congestive Cardiac Failure”, “Acute Heart Failure”, “Chronic Heart Failure” “Right Heart Failure”, “Left Heart Failure” [MeSH], “Heart Failure Reduced Left Ventricular Function”, “Heart Failure Preserved Left Ventricular Function” [MeSH], Acute decompensated heart failure, tachycardia-induced cardiomyopathy, heart failure mid-range, refractory heart failure, terminal heart failure, end stage heart failure, cardio-oncology



The search was filtered to clinical trials and reviews, involving humans, and published in the English language. The relevant articles were carefully selected from this huge list. In addition, the reference lists of all relevant articles retrieved were searched to identify further studies. Experts in the field were also contacted to obtain further information. International guidelines on HF - the American Heart Association / American College of Cardiology and European Society of Cardiology - were also studied.

All literature retrieved was appraised by members of the Expert Panel and all statements and recommendations made were collectively agreed by the group. The grading of evidence and the level of recommendation used in this CPG was adapted from the American College of Cardiology / American Heart Association and the European Society of Cardiology (Page10).

After much discussion, the draft was then drawn up and submitted to the Technical Advisory Committee for Clinical Practice Guidelines, Ministry of Health Malaysia and key health personnel in the major hospitals of the Ministry of Health and the private sector for review and feedback.

#### **Clinical Questions Addressed:**

There were several topics and subtopics that were formulated addressing the diagnosis and therapy of HF.

For **diagnosis**: In a person presenting with shortness of breath:

- What features in the history and clinical examination would make one suspect this patient is having HF?
- What diagnostic tests help confirm the clinical suspicion of HF with reasonable sensitivity and specificity?
  - ◆ ECG
  - ◆ Chest X-ray
  - ◆ Natriuretic peptides
  - ◆ Echocardiogram
  - ◆ Lung ultrasound

For **therapy**, the topics and subtopics were formulated using the Population Intervention Comparison and Outcome (PICO) method as follows:

**P: Population** - Persons with confirmed HF (including both gender and the elderly) and who had:

- Reduced left ventricular (LV) function (LVEF < 40%) - HF<sub>r</sub>EF
- Preserved LV function (LVEF > 50%) - HF<sub>p</sub>EF
- Mildly reduced LV function (LVEF: 40-50%) - HF<sub>mr</sub>EF
- HF with improved LV function - HF<sub>imp</sub>EF

These patients could be:

- Congested (Volume overload),
- Hypotensive (Cold) or,
- Combination of congestion and hypotension.

The etiology could be: (either singly or in combination)

- Coronary artery disease
- Atrial Fibrillation (AF)
- Diabetes



- Chronic Kidney disease
  - ◆ Not on renal replacement therapy
  - ◆ On renal replacement therapy

**I: Intervention:**

- Non-pharmacological therapy
- Pharmacological therapy:
  - ◆ Diuretics
  - ◆ Angiotensin Converting Enzyme Inhibitors (ACE-I),
  - ◆ Angiotensin Receptor Blockers (ARB)
  - ◆ Angiotensin Receptor and Neprilysin Blockers ( ARNI)
  - ◆  $\beta$ -blockers
  - ◆ Mineralocorticoid Antagonists (MRA)-both steroidal and non steroidal
  - ◆ Sodium Glucose cotransporter-2 inhibitors (SGLT2-i)
  - ◆ Statins
  - ◆ etc
- Surgery :
  - ◆ Valve surgery/percutaneous intervention
  - ◆ Coronary artery bypass surgery
- Device therapy
  - ◆ Cardiac resynchronisation therapy
  - ◆ Catheter ablation
  - ◆ Pacemaker therapy

**C: Comparison:**

- Non-pharmacological therapy vs no non-pharmacological therapy
- Diuretics vs no diuretics
- ACE-I vs no ACE-I
- etc

**O: Outcome:**

- Improvement in symptoms.
- Reduction in:
  - ◆ Hospital readmissions for HF
  - ◆ Major Adverse Cardiovascular (CV) Events - MACE - (Myocardial Infarction (MI), stroke, CV death).
  - ◆ All-cause mortality

**Type of Question - Involves:**

- Therapy - pharmacotherapy, surgery, device therapy
- Harm -
  - ◆ Worsening of symptoms and readmission rate.
  - ◆ Increase in MACE.
  - ◆ Increase in bleeding risk and stroke rate.
  - ◆ Adverse effects due to pharmacotherapy.
- Prognosis - reduction in MI, heart failure, CV death and improvement in all-cause mortality.

**Type of Study**

- Systematic review and meta-analysis.
- Randomised controlled studies.
- Cohort studies.



Thus, there were numerous clinical questions formulated.

Examples of some of these Clinical Questions:

- For a person with HF<sub>r</sub>EF and congested (volume overload) will the use of diuretics lead to an improvement in symptoms, hospital readmission, cardiac event rate and / or all-cause mortality?
- For a person with HF<sub>r</sub>EF and not congested (volume overload) will the use of diuretics lead to an improvement in symptoms, hospital readmission, cardiac event rate and / or all-cause mortality?
- For a person with HF<sub>r</sub>EF and congested (volume overload) will the use of ACE-I lead to an improvement in symptoms, hospital readmission, cardiac event rate and / or all-cause mortality?
- For a person with HF<sub>r</sub>EF and CAD, will coronary artery bypass surgery lead to an improvement in symptoms, hospital readmission, cardiac event rate and / or all-cause mortality?
- For a person with HF<sub>p</sub>EF and congested (volume overload) will the use of ACE-I lead to an improvement in symptoms, hospital readmission, cardiac event rate and / or all-cause mortality?

**Target Group and Target Population:**

This guideline is developed for all healthcare providers involved in the management of HF in individuals (> 18 years) with and at risk of HF.

**Facilitators and Barriers:**

The main barrier for the successful implementation of this CPG is:

1. The lack of knowledge of healthcare providers in the:
  - Diagnosis of HF.
  - Management of HF - initial treatment and long-term follow-up.
  - Optimization of therapy and when to refer to tertiary centers.
2. Availability of Natriuretic Peptide testing - BNP, NTProBNP - for early diagnosis of HF in Primary Health Care Clinics.
3. Availability of Foundational HF Medications in Primary Health Care Clinics (ARNI - Not available, SGLT2- available in small quota and only for use in diabetic patients.)

**Applicability of the Guidelines and Resource Implications:**

These guidelines were developed considering our local health resources. Blood investigations, chest radiographs, ECGs and echocardiograms are common in almost all public health facilities. Most of the drugs used to treat HF - diuretics, ACE-I, ARB,  $\beta$ -blockers have been approved for use in Malaysia and available in public hospitals as generics.

This guideline aims to educate health care professionals on strategies to optimize existing resources in the timely management of patients with HF.

**Implementation of the Guidelines:**

The implementation of the recommendations of a CPG is part of good clinical governance. To ensure successful implementation of this CPG we suggest:

- Increasing public awareness of CVD and HF in general and educating them on the importance of seeking early medical attention.
- Continuous medical education and training of healthcare providers on the implementation and optimization of the Foundational HF medications. This can be done by road shows, electronic media, and in-house training sessions.
- Clinical audit by individual hospitals and units to ensure compliance using the suggested performance measures in Section 19, pages 160.

**Period of Validity of the Guidelines:**

These guidelines need to be revised at least every 5 years to keep abreast with recent developments and knowledge that is being learnt.

Dr. Jeyamalar Rajadurai  
*Chairperson*

**Table 1: GRADES OF RECOMMENDATIONS AND LEVELS OF EVIDENCE**

| GRADES OF RECOMMENDATION |                                                                                                                                                      |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>I</b>                 | Conditions for which there is evidence and/or general agreement.                                                                                     |
| <b>II</b>                | Conditions for which there is conflicting evidence and/or divergence of opinion about the usefulness/efficacy of a procedure/therapy.                |
| <b>II-a</b>              | Weight of evidence/opinion is in favor of its usefulness/efficacy.                                                                                   |
| <b>II-b</b>              | Usefulness/efficacy is less well established by evidence/opinion.                                                                                    |
| <b>III</b>               | Conditions for which there is evidence and/or general agreement that a procedure/therapy is not useful/effective and, in some cases, may be harmful. |

| LEVELS OF EVIDENCE |                                                                                       |
|--------------------|---------------------------------------------------------------------------------------|
| <b>A</b>           | Data derived from multiple randomized clinical trials or meta-analyses.               |
| <b>B</b>           | Data derived from a single randomized clinical trial or large non-randomized studies. |
| <b>C</b>           | Only consensus of opinions of experts, case studies or standard of care.              |

*Adapted from the American College of Cardiology Foundation/ American Heart Association and the European Society of Cardiology*

*(Available at: [http://assets.cardiosource.com/Methodology\\_Manual\\_for\\_ACC\\_HA\\_Writing\\_Committees](http://assets.cardiosource.com/Methodology_Manual_for_ACC_HA_Writing_Committees) and at <http://www.escardio.org/guidelines-surveys/escguidelines/about/Pages/rules-writing.aspx>).*

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**WHAT'S NEW IN THE GUIDELINES**

What is new is:

1. The concept of “Foundational Heart Failure” medications and advising on the strategy of:
  - Initiating all about the same time.
  - Titrating them up relatively quickly (preferably within 3 months) to maximally tolerated or target doses.
2. The class of SGLT2-i as a HF medication based on recent clinical evidence.

|                      | <b>4th Ed CPG Heart Failure 2019 (Old)</b>                                                         | <b>5th Ed CPG Heart Failure 2023 (New)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------------|----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Goal of management   | Focused on reducing symptoms, improving functional capacity, quality of life and patient survival. | <ul style="list-style-type: none"><li>• In addition to the previously mentioned goals, the emphasis is on making a reduction in HF related hospitalizations as an important goal of management to reduce the socio-economic impact of the disease.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                     |
| Definition of HF     | Was defined solely based on clinical symptoms and signs.                                           | <ul style="list-style-type: none"><li>• Is defined based on clinical symptoms and signs that should be supported objectively by either an elevation of natriuretic peptides and/or evidence of pulmonary or systemic congestion.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                       |
| Classification of HF | Was classified by clinical presentation only.                                                      | <ul style="list-style-type: none"><li>• Is classified by clinical presentation as well as LVEF categories and Stages of HF.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Diagnosis of HF      | -                                                                                                  | <ul style="list-style-type: none"><li>• Places more emphasis on using objective testing with Natriuretic Peptides for the diagnosis of HF in addition to clinical symptoms &amp; signs.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Acute HF             | -                                                                                                  | <ul style="list-style-type: none"><li>• Introduces the concept of 3 phases:<ul style="list-style-type: none"><li>➢ <b>Phase 1</b> - Urgent treatment and stabilization usually in the emergency department.</li><li>➢ <b>Phase 2</b> - In-hospital management.</li><li>➢ <b>Phase 3</b> - Discharge and Post discharge.</li></ul></li><li>• Recognizes that in the period immediately following discharge, the patient is “vulnerable” for decompensation and advises closer follow up, preferably in the HF clinic.</li><li>• Provides a discharge care plan &amp; summary to facilitate follow up. (Appendix III, IV, pages 164-165)</li></ul> |



|                       |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Precipitating Factors | - | <ul style="list-style-type: none"> <li>Introduces the acronym “<b>CHAMPION</b>” as a quick reminder of the precipitating factors for acute HF. (page 73)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Pharmacotherapy HFrEF | - | <ul style="list-style-type: none"> <li>Introduces the concept of “<b>Foundational HF Medications</b>”: <ul style="list-style-type: none"> <li>➢ RAS blockers (ACE-I, ARB, ARNI)</li> <li>➢ <math>\beta</math>-blockers</li> <li>➢ MRA</li> <li>➢ SGLT2-i</li> </ul> </li> <li>Recommends that these be initiated while the patient is in hospital and up titrated to maximally tolerated or target doses as soon as possible, preferably within 12 weeks post- discharge.</li> <li>Recommending that the dose of diuretic be reduced when the patient is no longer congested.</li> <li>Provides a section on patient profiling (section 10.2) and how to initiate and titrate “<b>Foundational HF Medications</b>”. This is complemented with Flowcharts III, IV and V, pages 37-39</li> </ul> |
| SGLT2-i               | - | <ul style="list-style-type: none"> <li>Introduces this class of medications in the management of all categories of LVEF to reduce HF hospitalizations and improve CV outcomes.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Diuretics             |   | <ul style="list-style-type: none"> <li>Recommends that the dose of diuretics be reduced when the patient is no longer volume overloaded.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Atrial Fibrillation   | - | <ul style="list-style-type: none"> <li>Recommends rhythm control instead of rate control for recent onset AF of &lt; 1 year duration, wherever possible.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| HFpEF                 |   | <ul style="list-style-type: none"> <li>Provides more recommendations on the diagnosis and management of HFpEF.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| New Sections          |   | 11. Chronic HF due to HFmrEF<br>12. Chronic HF due to HFimpEF<br>14.2 Valvular Heart Disease and HF<br>14.3 Cardiomyopathy and HF<br>14.8 Coronavirus 2019 (Covid-19) +/- Vaccine and HF<br>15.3 Palliative and End of Life Care<br>16. Heart Failure Rehabilitation<br>17.4 Telemedicine/Telehealth                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

**GLOSSARY**

| Abbreviation | Description                                                |
|--------------|------------------------------------------------------------|
| ACC          | American College of Cardiology                             |
| ACE-I        | Angiotensin Converting Enzyme Inhibitor                    |
| ACHD         | Adult Congenital Heart Disease                             |
| ACS          | Acute Coronary Syndrome                                    |
| ADHF         | Acute Decompensated Heart Failure                          |
| AF           | Atrial Fibrillation                                        |
| Acute HF     | Acute Heart Failure                                        |
| AHA          | American Heart Association                                 |
| AMI          | Acute Myocardial Infarction                                |
| ALCAPA       | Anomalous left coronary artery from pulmonary artery       |
| ARB          | Angiotensin Receptor Blocker                               |
| ARNI         | Angiotensin Receptor-Nepilysin Inhibitor                   |
| ASD          | Atrial Septal Defect                                       |
| ASV          | Adaptive Servo-Ventilation                                 |
| AV Node      | Atrioventricular Node                                      |
| AVM          | Arteriovenous Malformation                                 |
| AVSD         | Atrio-Ventricular Septal Defect                            |
| Bd           | Bis Die (twice daily)                                      |
| BiPaP        | Bi-level Positive Airway Pressure                          |
| BNP          | B-Type Natriuretic Peptides                                |
| BP           | Blood Pressure                                             |
| BT Shunt     | Blalock-Taussig shunt                                      |
| CABG         | Coronary Artery Bypass Graft                               |
| CAD          | Coronary Artery Disease                                    |
| cCTGA        | Congenitally Corrected Transposition of the Great Arteries |
| CCU          | Cardiac Care Unit                                          |
| CHD          | Coronary Heart Disease                                     |
| CIN          | Contrast Induced Nephropathy                               |
| CKD          | Chronic Kidney Disease                                     |
| cMR          | Cardiac Magnetic Resonance                                 |
| CPG          | Clinical Practice Guidelines                               |
| CR           | Cardiac Rehabilitation                                     |
| CrCl         | Creatinine Clearance                                       |
| CRT          | Cardiac Resynchronisation Therapy                          |
| CSA          | Central Sleep Apnea                                        |
| cTn          | Cardiac troponins                                          |
| CV           | Cardiovascular                                             |



| Abbreviation | Description                                         |
|--------------|-----------------------------------------------------|
| CVD          | Cardiovascular Disease                              |
| CPAP         | Continuous Positive Airway Pressure                 |
| DBP          | Diastolic Blood Pressure                            |
| DM           | Diabetes Mellitus                                   |
| DOAC         | Direct Oral Anticoagulants                          |
| DVT          | Deep Venous Thrombosis                              |
| ECG          | Electrocardiogram                                   |
| ECMO         | Extracorporeal Membrane Oxygenator                  |
| ED           | Erectile Dysfunction                                |
| EF           | Ejection Fraction                                   |
| EOD          | Every Other Day                                     |
| EP           | Electrophysiology                                   |
| ER           | Emergency Room                                      |
| eGFR         | Estimated Glomerular Filtration Rate                |
| ERA          | Endothelin Receptor Antagonists                     |
| ESC          | European Society of Cardiology                      |
| FiO2         | Fraction of Inspired Oxygen                         |
| GFR          | Glomerular Filtration Rate                          |
| GLP-1        | Glucagon Like Peptide 1                             |
| HCM          | Hypertrophic Cardiomyopathy                         |
| HF           | Heart Failure                                       |
| HFimpEF      | Heart Failure with Improved Ejection Fraction       |
| HFmrEF       | Heart Failure with Mildly Reduced Ejection Fraction |
| HFpEF        | Heart Failure with Reduced Ejection Fraction        |
| HFpEF        | Heart Failure with Preserved Ejection Fraction      |
| HIIT         | High Intensity Interval Training                    |
| HR           | Heart Rate                                          |
| HRQoL        | Health Related Quality of Life                      |
| ICD          | Implantable Cardioverter-Defibrillator              |
| IHD          | Ischemic Heart Disease                              |
| IMT          | Inspiratory Muscle Training                         |
| IV           | Intravenous                                         |
| KDIGO        | Kidney Disease Improving Global Outcomes            |
| LBBB         | Left Bundle Branch Block                            |
| LV           | Left Ventricle                                      |
| LVAD         | Left Ventricular Assist Device                      |
| LVEF         | Left Ventricular Ejection Fraction                  |
| LVH          | Left Ventricular Hypertrophy                        |
| LVT          | Left Ventricular Thrombus                           |



| Abbreviation     | Description                                             |
|------------------|---------------------------------------------------------|
| MACE             | Major Adverse Cardiovascular Events                     |
| MI               | Myocardial Infarction                                   |
| mPCWP            | Mean Pulmonary Capillary Wedge Pressure                 |
| MOH              | Ministry of Health Malaysia                             |
| MRI              | Magnetic Resonance Imaging                              |
| MRA              | Mineralocorticoid Receptor Antagonist                   |
| NP               | Natriuretic Peptides                                    |
| NTProBNP         | N-terminal (NT)-pro hormone B-Type Natriuretic Peptides |
| NSAIDs           | Non-Steroidal Anti-inflammatory Agents                  |
| NYHA             | New York Heart Association                              |
| Od               | Once daily                                              |
| OMT              | Optimal Medical Therapy                                 |
| OSA              | Obstructive Sleep A                                     |
| PA               | Pulmonary Artery                                        |
| PAH              | Pulmonary Arterial Hypertension                         |
| PCI              | Percutaneous Coronary Interventions                     |
| PCWP             | Pulmonary Capillary Wedge Pressure                      |
| PDA              | Patent Ductus Arteriosus                                |
| PLE              | Protein Losing Enteropathy                              |
| PPCM             | Peripartum Cardiomyopathy                               |
| PDE-5 inhibitors | Phosphodiesterase-5 Inhibitors                          |
| PSG              | Polysomnography                                         |
| PUFA             | Polyunsaturated Fatty Acids                             |
| RAS              | Renin Angiotensin System                                |
| RHD              | Rheumatic Heart Disease                                 |
| RV               | Right Ventricle                                         |
| SBP              | Systolic Blood Pressure                                 |
| SC               | Subcutaneous                                            |
| SCD              | Sudden Cardiac Death                                    |
| SCr              | Serum Creatinine                                        |
| SDB              | Sleep Disordered Breathing                              |
| SGLT2-i          | Sodium-Glucose Cotransporter-2 Inhibitor                |
| SpO2             | Pulse Oximeter Oxygen Saturation                        |
| STEMI            | ST Segment Elevation Myocardial Infarction              |
| Tds              | Ter die sumendus (three times per day)                  |
| TOF              | Tetralogy of Fallot                                     |
| VAD              | Ventricular Assist Device                               |
| VHD              | Valvular Heart Disease                                  |
| VPC              | Ventricular Premature Contraction                       |
| VSD              | Ventricular Septal Defect                               |
| VT               | Ventricular Tachycardia                                 |

**KEY MESSAGES****# 1: Epidemiology and Goals of Management**

- In Malaysia, patients with HF are almost 10-15 years younger than those in western countries. Comorbidities such as hypertension, diabetes and chronic kidney disease are common in our patients.
- HF is associated with a huge socio-economic and humanistic burden.
- Most of the costs related to HF are related to re-hospitalizations and inpatient hospital care.
- Important goals of management include:
  - Preventing readmissions thus reducing both the socio-economic and humanistic burden of the disease.
  - Improving symptoms, functional capacity, and quality of life of these patients.
  - Improving patient survival.

**#2: Definition**

- HF is a clinical syndrome due to any structural or physiological abnormality of the heart resulting in its inability to meet the metabolic demands of the body or its ability to do so only at higher-than-normal filling pressures.
- The diagnosis is made by the presence of characteristic symptoms and/or signs and supported by the findings of either an elevation of natriuretic peptides and/or evidence of pulmonary or systemic congestion.

**#3: Classification & Stages of HF**

- HF can be **classified** according to:
  - **Clinical Presentation:**
    - ◆ Acute heart failure (Acute HF)
    - ◆ Chronic heart failure (Chronic HF)
  - **Left ventricular ejection fraction (LVEF):**
    - ◆ **Reduced (LVEF  $\leq$  40%)** - Heart failure with reduced ejection function (HFrEF).
    - ◆ **Mildly reduced (LVEF 41% - 49%)** - Heart Failure with the LVEF being in the mildly reduced range (HFmrEF).
    - ◆ **Preserved (LVEF  $>$  50%)** - Heart failure with preserved ejection fraction (HFpEF).
    - ◆ **Improved Ejection Fraction** - HF with an initial baseline LVEF of  $\leq$  40%, a  $\geq$  10-point increase from baseline LVEF following treatment, and a second measurement of LVEF of  $>$ 40%. (HFimpEF).
  - **Stages of HF are:**
    - ◆ A - "At Risk"
    - ◆ B - "Pre HF"
    - ◆ C - "Symptomatic HF with previous or current symptoms"
    - ◆ D - "End Stage HF"
- The severity of symptomatic HF and the exercise capacity of the patient can be assessed using the New York Heart Association (NYHA) functional Class. (Table 9, page 50)

**#4: Pathophysiology**

- The main pathophysiology of HF is the inability of the heart to provide sufficient cardiac output to meet the perfusion and oxygenation requirements of the body while maintaining normal filling pressures.
- This results in compensatory mechanisms which can also aggravate HF by increasing ventricular afterload and preload to the point where pulmonary and/or systemic congestion and edema occur.
- HF may be due to either impaired cardiac contractility or impaired relaxation and compliance. Both mechanisms may co-exist in the same patient and depending on the predominant pathophysiology, the patient may present as **HFrEF** or **HFpEF**.
- In **HFrEF**, cardiac output is reduced due to depressed myocardial contractility, irrespective of the etiology.
- In **HFpEF**, there is impaired left ventricular (LV) filling due to decreased relaxation (during early diastole) and/or reduced compliance (early to late diastole) leading to elevated LV filling pressures at rest or during exercise resulting in dyspnea.
- **HFmrEF**, is a clinical entity between **HFrEF** and **HFpEF**. With regards to etiology, it is more like **HFrEF** (high prevalence of CAD) but in terms of prognosis, it is like **HFpEF**.
- In **HFimpEF**, patients have resolution of symptoms and signs of HF either spontaneously or with treatment, but HF is known to frequently relapse especially after withdrawal of pharmacological treatment. These patients have a lower mortality and risk of hospitalization compared to **HFrEF** and **HFpEF**.

**# 5: Etiology**

- The common underlying causes of HF in adults are:
  - Coronary artery disease (CAD) - accounting for almost 60-66% of HF in Malaysia.
  - Hypertension.
  - Dilated cardiomyopathy - idiopathic, familial.
  - Valvular heart disease - an important cause is rheumatic heart disease (RHD) among the rural and urban poor.
  - Diabetic cardiomyopathy.
- Comorbidities are common and include hypertension, diabetes, and dyslipidemia.

**#6: Prevention of HF**

- Prevention and early intervention, wherever appropriate, should be the primary objective of management.
- There is robust clinical data that by appropriate and timely interventions, HF can be prevented, and cardiac function improved.
- It should focus on those who are in:
  - **Stage A - "At Risk"** - Individuals who are at high risk of developing cardiac disease but who do not have symptoms or signs of HF and still have structurally normal hearts.
  - **Stage B - "Pre HF"** - Individuals with structural cardiac disease and who have either normal or impaired cardiac function but do not, yet, have signs and symptoms of HF.

**# 7: Acute HF**

- Acute HF may present as:
  - De Novo HF - first occurrence of HF or,
  - Acute decompensated HF (ADHF) - this is a more common presentation occurring in a previously stable patient with HF who has now deteriorated.
- Three important phases should be considered in the management of these patients.
  - **Phase 1** - Urgent treatment and stabilization usually in the emergency department.
  - **Phase 2** - In-hospital management.
  - **Phase 3** - Discharge and Post discharge.

**# 8: Chronic HF due to LV reduced Function (HFrEF)**

- **Optimal HF medications are:**
  - Diuretics- to be titrated according to the presence of congestion.
  - **Foundational HF medications:**
    - ◆ Renin-angiotensin system (RAS) blockers:
      - Angiotensin converting enzyme inhibitor (ACE-I) **or**
      - Angiotensin receptor blocker (ARB) **or**
      - Angiotensin receptor neprilysin inhibitors (ARNI).
    - ◆  $\beta$ -blockers.
    - ◆ Mineralocorticoid Antagonists (MRA).
    - ◆ Sodium glucose cotransporter 2 inhibitors (SGLT2-i).
  - Other drugs (when necessary) include:
    - ◆ Ivabradine.
    - ◆ Nitrates.
- For the doses and the Grades of Recommendation and Levels of Evidence of these medications for HFrEF, see Table 3 & 4, page 40-42).

**# 9: Arrhythmias**

- Arrhythmias are common in HF. These include:
  - Atrial Fibrillation.
  - Ventricular arrhythmias.
  - Bradyarrhythmias.

**# 10: Surgery for HF**

- Patients with HF should undergo surgery if the pathology causing the HF (e.g., CAD, valve lesions) is amenable to surgical treatment.

**#11: HFmrEF**

- HFmrEF includes symptomatic HF patients with LVEF between 41-49% and is a clinical entity between HFrEF and HFpEF.
- There are a limited number of randomized control trials focusing on management of this category of patients - most of the evidence is derived from post-hoc or sub-group analysis of previous HF trials.

**#12: Chronic HF due to HFimpEF**

- HFimpEF includes patients with a:
  - Baseline LVEF of <40% and a
  - >10-point increase from baseline LVEF following treatment to
  - A second measure of LVEF > 40%.



- LV function and structural abnormalities do not fully normalize despite improvement in symptoms, functional capacity, and near normalization of biomarkers.
- Symptoms and signs may however relapse and cardiac biomarkers may increase if HF treatment is withdrawn.
- Foundational HF medications should be continued to prevent relapse of symptoms and subsequent deterioration of LV function.

**#13: Chronic HF due to HFpEF**

- HFpEF is a heterogeneous disease that is highly prevalent, accounting for up to 50% of all patients with HF and is becoming the dominant form of HF in aging populations worldwide.
- In this guideline HFpEF refers to LVEF  $\geq$  50%. LVEF is a continuous variable and the EF cut-offs used in definitions are therefore arbitrary.
- HFpEF differs from HFrEF in that HFpEF patients are older, more often female with AF, CKD and have more non-CV comorbidities.
- The main hemodynamic finding in HFpEF is an elevation in LV filling pressures i.e. end diastolic LV pressure (LVEDP), initially only on exertion and later even at rest resulting in dyspnea.

**# 14: HF and Diabetes**

- HF and diabetes mellitus (DM) often co-exist, each increasing the likelihood of developing the other. Whether this indicates a causal effect or just a comorbidity is still unclear.
- HF per se is associated with a high morbidity and mortality and concomitant DM compounds this risk. This is particularly so in patients with HFpEF.

**# 15: HF and Valvular Heart Disease (VHD)**

- VHD is an important cause of HF.
- In the young, RHD is an important cause. The incidence of RHD in rural and urban poor is high.
- In the older population, degenerative valve disease is more common.

**# 16: HF and Cardiomyopathies**

- Cardiomyopathies are a heterogeneous group of myocardial disorders which frequently present as HF.
- They can be inherited (familial/genetic) or acquired.
- They are usually classified according to anatomic and physiologic features into:
  - Dilated cardiomyopathy (DCM)
  - Hypertrophic cardiomyopathy (HCM)
  - Restrictive cardiomyopathy (RCM)
  - Arrhythmogenic cardiomyopathy
  - Unclassified cardiomyopathy
- In general, treatment of patients with cardiomyopathy encompasses the Foundational HF Medications as well as etiology-specific therapy (if available).

**# 17: HF Due to Arrhythmias and Conduction Abnormalities**

- HF due to arrhythmias and conduction abnormalities are potentially reversible.
- Successful treatment of the arrhythmia by drug therapy or catheter ablation can result in normalization of LV function.

**#18: Cardio-Oncology**

- Heart disease and cancer are often linked due to common etiologic factors and chemotherapeutic treatment strategies.
- Chemotherapy-induced cardiotoxicity is not common:
  - Clinical HF occurs in 1-5%.
  - An asymptomatic decrease in LV function occurs in the range of 5% to 20%.
- It can develop in a subacute, acute (within 2 weeks of termination of drug administration) or in a chronic manner.
- One must consider both drug efficacy and toxicity in choosing chemotherapeutic agents.

**#19: HF and Chronic Kidney Disease**

- Cardiac and chronic kidney disease often occur together, and this increases the complexity and costs of care, and may interact to worsen prognosis.
- During treatment of Acute HF, a significant proportion of patients will develop varying degrees of worsening renal function (WRF) usually in the first three to five days of hospitalization.
- WRF may not always indicate a poor outcome-especially if it is due to overdiuresis and hypotension or due to drug therapy.

**#20: HF and Pregnancy**

- About 0.5 - 4% of pregnant women have cardiac disease. HF remains the most common complication among these women regardless of the cause.
- Women with cardiac disease should be assessed:
  - Before conception to assess their risk and to be advised accordingly **and**
  - Early in the pregnancy to optimize the outcome of the pregnancy.
- Maternal cardiovascular risk can be assessed using the modified World Health Organization (WHO) or NYHA classification. (Appendix XI & XII, page 171-173)
- Level of Care will depend on the maternal CV risk.
  - **Low risk:** can be managed at their local center after review by a family medicine specialist/physician or cardiologist.
  - **Moderate risk:** should be managed at a tertiary center by a multidisciplinary team with cardiac expertise.
  - **High risk:** should be referred early to a tertiary center for assessment.
- Patients with LVEF < 30% and those in NYHA Class III and IV should be strongly advised not to get pregnant. If pregnant, termination should be considered.

**# 21: HF and Covid - 19 Infection and Vaccine**

- In hospitalized patients with COVID-19, the prevalence of HF varied between 4% and 21%. About 8% - 33% of these patients required critical care.
- Patients with HF who develop COVID-19 had an overall mortality rate between 20% and 40%.
- Myocarditis is a rare but serious complication of SARS-CoV-2 infection as well as COVID-19 mRNA vaccination.
- The rate of myocarditis or pericarditis across different age groups:
  - After SARS-CoV-2 infection - 12.6-114 per 100,000 for males and 5.4-61.7 per 100,000 for females.
  - After mRNA COVID-19 vaccination - 0-35.9 per 100,000 for males and 0-10.9 per 100,000 for females.

**# 22: HF and Acquired Congenital Heart disease (ACHD)**

- HF is the most common cause of mortality in ACHD patients, accounting for 17- 42% of all deaths.
- Patients with isolated simple defects generally do well with mortality rates like those in the general population.
- Patients with complex heart defects such as systemic RV, single ventricle palliated with Fontan circulation, and unrepaired cyanotic CHD with Eisenmenger physiology comprise the majority of HF- related deaths.

**#23: Advanced HF**

- All patients with severe symptomatic HF despite OMT and no other alternative therapeutic options have a poor prognosis. There should be a discussion with the patient and family on the choice of further management.
- Further options include:
  - Heart transplant if the patient is eligible.
  - If they are not eligible or a donor heart is not available, a Left Ventricular Assist Device (LVAD) as a destination therapy or a bridge to heart transplant.
- Patients who either choose not to or are ineligible for the available options, should be referred for palliative care.

**# 24: HF Rehabilitation**

- Fatigue & breathlessness leads to individuals restricting their physical activity and this in turn, leads to deconditioning.
- Exercise training in patients with HF is safe and leads to an improvement in functional capacity, exercise duration, and health related quality of Life.
- HF is now endorsed as an indication for Cardiac Rehabilitation, and it spans through out the continuum of HF care.

**# 25: Level of Care, Shared Care, Monitoring and Follow - Up & Telemedicine /Telehealth**

- The care of patients with HF should ideally take place in a multidisciplinary system, allowing for shared care between the hospital (secondary or tertiary settings) and community (primary setting).
- HF clinics will serve as an intermediary between in-patient hospital care and community primary care.
- When stable with optimized **Foundational HF medications**/treatment plans, patients can be discharged to the community with appropriate care plans to primary care.
- Telemedicine services may improve healthcare accessibility and geographical limitations. It can complement existing HF clinic services and provide venues for early escalation of HF therapy before decompensation.



## KEY RECOMMENDATIONS

| Recommendations                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Grade of Recommendation/<br>Level of Evidence |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| <p><b>#1: Diagnosis &amp; Essential Investigations</b></p> <ul style="list-style-type: none"> <li>• In making a diagnosis of HF, a detailed history and a thorough physical examination are important.</li> <li>• The clinical suspicion of HF should be supported by either raised natriuretic peptides and/or evidence of pulmonary or systemic congestion.</li> <li>• Important basic investigations: <ul style="list-style-type: none"> <li>➢ Echocardiogram</li> <li>➢ ECG</li> <li>➢ Chest Radiograph</li> <li>➢ Blood - Full Blood Count, urea, creatinine, serum electrolytes (sodium and potassium) liver function, serum glucose, lipid profile.</li> <li>➢ Natriuretic Peptides: NT - Pro BNP or BNP <ul style="list-style-type: none"> <li>♦ They are a useful “rule out” test in the diagnosis of HF in patients presenting with acute dyspnea.</li> <li>♦ In chronic HF, NPs may be persistently elevated and show substantial biological variation. In this setting, NPs should be interpreted with careful clinical assessment and with all other clinical information.</li> <li>♦ A value of NT - Pro BNP of <math>\leq 1,000</math> pg/ml during treatment was associated with better CV outcomes.</li> </ul> </li> </ul> </li> </ul> | <p>I,C</p> <p>I,C</p> <p>I,C</p> <p>I,A</p>   |
| <p><b>#2: Prevention of HF</b></p> <ul style="list-style-type: none"> <li>• In the prevention of HF in patients in <b>Stage A</b>: <ul style="list-style-type: none"> <li>➢ All CV risk factors should be treated to target - BP, Lipids, and glucose.</li> <li>➢ Optimal profiles of the 7 CV risk factors - smoking, body mass index, physical activity, diet, cholesterol, blood pressure, and glucose have been shown to be associated with a lower lifetime risk of HF.</li> </ul> </li> <li>• In the prevention of HF in patients in <b>Stage B</b>: <ul style="list-style-type: none"> <li>➢ Patients with Acute Coronary Syndrome, Hypertension, Stable CAD and arrhythmias should be treated timely and according to the guidelines.</li> <li>➢ In patients who have impaired LV function (LVEF&lt;40%): <ul style="list-style-type: none"> <li>♦ Treat the underlying cause and avoid/treat precipitating factors early.</li> <li>♦ The following medications should be instituted early to prevent progression of the HF and improve CV outcomes.</li> </ul> </li> </ul> </li> </ul> <p>□ In the <b>asymptomatic patient in NYHA Class I</b>:</p> <ul style="list-style-type: none"> <li>- ACE-I</li> </ul>                                  | <p>I,A</p> <p>I,A</p> <p>I,A</p>              |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>- <math>\beta</math>-blockers</li> <li>- SGLT2-i</li> </ul> <p><b>□ In symptomatic patients in NYHA Class II-IV:</b></p> <ul style="list-style-type: none"> <li>- All the Foundational HF medications (RAS blockers, MRA, <math>\beta</math>-blockers and SGLT2-i)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <p>I,A</p> <p>I,A</p> <p>I,A</p>                                                            |
| <p><b>#3: Non-pharmacological Measures</b></p> <ul style="list-style-type: none"> <li>● In managing HF, non-pharmacological measures play a vital role. It is important to:             <ul style="list-style-type: none"> <li>➤ Educate the patient and family about the disease, treatment options and prognosis. There should be frank end-of-life discussions and advanced care planning with the patient and family. This is important as there is a potential risk of sudden death, and they should put their affairs in order.</li> <li>➤ Encourage lifestyle measures:                 <ul style="list-style-type: none"> <li>◆ Proper balanced diet to prevent malnutrition and cardiac cachexia.</li> <li>◆ Weight management -avoid obesity.</li> <li>◆ Regular physical activity and exercise training.</li> <li>◆ Smoking cessation.</li> </ul> </li> <li>➤ Individualize fluid intake - A general recommendation is 1-1.5 Liters per day in patients with normal renal function. During periods of acute decompensation and in patients with more advanced stages of HF, more fluid restriction may be necessary.</li> <li>➤ Provide advice regarding sexual activities and pregnancy.</li> <li>➤ Address psychosocial problems encountered by the patient and the family.</li> </ul> </li> </ul> | <p>I,B</p> <p>Ila,B</p> <p>I,B</p> <p>I,B</p> <p>I,A</p> <p>I,C</p> <p>Ila,B</p> <p>I,C</p> |
| <p><b>#4: Acute HF</b></p> <ul style="list-style-type: none"> <li>● Phase 1 - Urgent treatment and stabilization             <ul style="list-style-type: none"> <li>➤ Assessment and management must be prompt and done concurrently.</li> <li>➤ Rapid recognition and making the diagnosis of HF based on symptoms and signs and quickly assessing its severity. Precipitants are as in the acronym:<br/> <b>"C H A M P I O N"</b> <ul style="list-style-type: none"> <li>◆ <b>C</b> - Coronary Artery Disease - Acute Coronary Syndrome (Myocardial infarction/Ischemia).</li> <li>◆ <b>H</b> - Severe and uncontrolled hypertension.</li> <li>◆ <b>A</b> - Arrhythmias.</li> <li>◆ <b>M</b> - Mechanical e.g., Acute valvular dysfunction (e.g., acute mitral regurgitation from chordal rupture); cardiac tamponade.</li> <li>◆ <b>P</b> - Pulmonary embolism, pulmonary infections.</li> <li>◆ <b>I</b> - Infections e.g., urinary tract, Covid.</li> <li>◆ <b>O</b> - Other medications (medications that increase fluid retention and/or have negative inotropic effects).</li> <li>◆ <b>N</b> - Non-compliance with treatment especially oral diuretics and or dietary/fluid restriction.</li> </ul> </li> </ul> </li> </ul>                                                                            | <p>I,C</p>                                                                                  |



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| <p>➤ Look for non-cardiovascular comorbidities- Diabetes, chronic kidney disease, thyroid disease (both hyper and hypothyroidism), chronic lung disease, anemia.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                              |
| <p><b>#5: Acute HF</b></p> <ul style="list-style-type: none"> <li>● <b>Phase 2 - In-hospital management</b> <ul style="list-style-type: none"> <li>➤ After initial clinical assessment, management should be instituted as in Flow Chart II, Page 35.</li> <li>➤ For grading of recommendations and levels of evidence, see Table 2, Page 36.</li> </ul> </li> <li>● <b>Phase 3 - Discharge and Post discharge</b> <ul style="list-style-type: none"> <li>➤ Discharge planning is a very important process after an episode of hospitalization for HF to reduce HF related readmissions.</li> <li>➤ The patient should be given a discharge summary and a discharge care plan. (Appendix III &amp; IV, pages 164-165)</li> <li>➤ <b>Prior to discharge</b>, the patient should be: <ul style="list-style-type: none"> <li>◆ Carefully assessed to ensure that there are no longer signs of congestion <b>and</b></li> <li>◆ If the LVEF &lt; 40%, already initiated on Foundational HF medications i.e., RAS blockers, MRA, β-blockers and SGLT2-i <b>and</b></li> <li>◆ Given a discharge care plan and summary (Appendix III &amp; IV, pages 164-165) <b>and</b></li> <li>◆ Informed of the need and, if indicated, given appointments for further cardiac work up e.g., coronary angiography and cardiac rehabilitation.</li> </ul> </li> <li>➤ <b>At the Follow up visit:</b> <ul style="list-style-type: none"> <li>◆ The first FU visit should be within 2 weeks of discharge.</li> <li>◆ Review that instructions in the discharge care plan (Appendix III &amp; IV, pages 164-165) were followed and identify/determine reasons if otherwise.</li> <li>◆ The Foundational HF medications should be up titrated to maximally tolerated or target doses as soon as possible, preferably within 12 weeks post-discharge.</li> </ul> </li> </ul> </li> </ul> | <p>I,C</p> <p>I,C</p> <p>I,A</p> <p>I,C</p> <p>I,C</p> <p>I,C</p> <p>I,A</p> |
| <p><b>#6: Chronic HF due to HFrEF: Pharmacotherapy</b></p> <ul style="list-style-type: none"> <li>● <b>Foundational HF Medications - For doses of the medications, Grades of recommendation and Levels of Evidence, see Table 3 &amp; 4, pages 40-42)</b> <ul style="list-style-type: none"> <li>➤ The Foundational HF Medications should be initiated at about the same time and up titrated to their target or maximally tolerated doses.</li> <li>➤ Once the patient is stable and no longer volume overloaded, the dose of diuretics may be down titrated to the lowest maintenance dose and in selected patients, may be discontinued especially if LVEF has improved to &gt; 40-45% following treatment.</li> <li>➤ In general, starting low doses of the 4 different classes would be preferred over up titration of each of the individual drugs</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <p>I,A</p>                                                                   |



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| <p>to the maximally tolerated dose before initiating the next drug.</p> <ul style="list-style-type: none"> <li>➤ These drugs are preferably all initiated, albeit at low doses, when the patient is admitted with HF so that at the time of discharge, the patient is on all of them.</li> <li>➤ In elderly patients who often have multiple comorbidities, it is important to consider pill burden, compliance, and the potential of adverse events due to drug-drug interaction. Education of the patient, family and caregivers is important.</li> <li>➤ For the initiation and up titration of these HF drugs see section 10.2 and Flowcharts III, IV &amp; V, pages 37-39.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                       |
| <p><b>#7: Arrhythmias</b></p> <ul style="list-style-type: none"> <li>● In HF patients who present <b>with palpitations, near faints and syncope</b>, arrhythmias are sometimes difficult to detect. If the clinical suspicion is high, the following may be considered: <ul style="list-style-type: none"> <li>➤ Prolonged ECG rhythm monitoring using 3-day, 7-day or even 1-month rhythm monitors <b>or</b></li> <li>➤ An implantable loop recorder <b>or</b></li> <li>➤ Using a smart watch that can monitor ECG heart rhythm.</li> </ul> </li> <li>● <b>Atrial Fibrillation</b> <ul style="list-style-type: none"> <li>➤ To be treated by: <ul style="list-style-type: none"> <li>◆ Rate <b>or</b></li> <li>◆ Rhythm control (if onset &lt; 1 year)</li> <li>◆ To anticoagulate with DOAC or Vitamin K antagonists</li> </ul> </li> </ul> </li> <li>● <b>Ventricular arrhythmias</b> <ul style="list-style-type: none"> <li>➤ Identify contributing factors such as electrolyte imbalances, ischemia and drugs.</li> <li>➤ Consider an Implantable cardioverter defibrillator (ICD) for secondary prevention in: <ul style="list-style-type: none"> <li>◆ Patients resuscitated from SCD due to ventricular fibrillation or hemodynamically unstable sustained ventricular tachycardia.</li> <li>◆ Prior MI (&gt; 40 days) and LVEF ≤ 40% with non-sustained VT <b>AND</b> inducible sustained VT or VF during an EP study.</li> <li>◆ Patients with chronic HF and LVEF ≤ 35% who experience syncope of unclear origin.</li> <li>◆ Prior MI (&gt; 40 days) and 3 months after revascularization, LVEF ≤ 35% and NYHA class II-III.</li> <li>◆ No prior MI, LVEF ≤ 35%, on optimal medical treatment, and in NYHA II or III</li> </ul> </li> </ul> </li> <li>● <b>Bradyarrhythmias</b> <ul style="list-style-type: none"> <li>➤ The following patients should be considered for pacemakers: <ul style="list-style-type: none"> <li>◆ Significant symptomatic bradyarrhythmias.</li> <li>◆ Trifascicular bundle branch blocks.</li> <li>◆ Permanent or paroxysmal third-or high-degree atrioventricular (AV) blocks.</li> </ul> </li> <li>➤ Prior to implanting a conventional pacemaker, the need for an ICD or Cardiac Resynchronisation Therapy (CRT) device should be considered.</li> </ul> </li> </ul> | <p><b>IIa,B</b><br/><b>IIa,A</b><br/><b>I,A</b></p> <p><b>I,C</b></p> <p><b>I,A</b></p> <p><b>I,A</b></p> <p><b>IIa,B</b></p> <p><b>I,A</b></p> <p><b>I,B</b></p> <p><b>I,A</b></p> <p><b>I,A</b></p> |



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| <ul style="list-style-type: none"> <li>➤ Conduction system pacing (which includes His bundle and left bundle branch area pacing) is a new pacing modality. The early results appear promising but evidence on its indications, safety and efficacy is still lacking.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                               |
| <p><b>#8: Chronic HF due to HF<sub>r</sub>EF</b></p> <ul style="list-style-type: none"> <li>• Surgery For HF <ul style="list-style-type: none"> <li>➤ <b>CAD</b> <ul style="list-style-type: none"> <li>♦ Patients with HF should undergo surgery if they have: <ul style="list-style-type: none"> <li>❑ Ischemia demonstrated as angina or on non-invasive testing <b>and</b></li> <li>❑ An anatomy that is suitable for revascularization (left main stem or triple vessel disease).</li> </ul> </li> </ul> </li> <li>➤ <b>Valve Heart Disease (VHD)</b> <ul style="list-style-type: none"> <li>♦ All patients with VHD should be assessed periodically on the need for early intervention before they begin to develop symptoms of reduced effort tolerance and decompensate.</li> <li>♦ Patients who are assessed to require intervention should be seen by a heart team to help decide the timing and type of intervention.</li> <li>♦ The indications for valve intervention are as in the Appendix V, page 166.</li> </ul> </li> </ul> </li> </ul> | <p>I,C</p> <p>I,C</p> <p>I,C</p>              |
| <p><b>#9: Chronic HF due to HF<sub>mr</sub>EF</b></p> <ul style="list-style-type: none"> <li>• The management of these patients include: <ul style="list-style-type: none"> <li>➤ Optimal treatment of CV risk factors.</li> <li>➤ <b>SGLT2- i.</b></li> <li>➤ <b>Diuretics</b> for patients who remain symptomatic and show signs of congestion.</li> <li>➤ <b>β-blockers, RAS blockers</b> (ARNI, ACEI or ARB) and <b>MRAs</b> can be considered to reduce the risk of HF hospitalization and CV death.</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <p>I,C</p> <p>I,A</p> <p>I,C</p> <p>Ila,B</p> |
| <p><b>#10: Chronic HF due to HF<sub>imp</sub>EF</b></p> <ul style="list-style-type: none"> <li>• Symptoms and signs may relapse, and cardiac biomarkers may increase if HF treatment is withdrawn.</li> <li>• Foundational HF medications should be continued to prevent relapse of symptoms and subsequent deterioration of LV function.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <p>Ila,B</p>                                  |
| <p><b>#11: Chronic HF due to HF<sub>p</sub>EF</b></p> <ul style="list-style-type: none"> <li>• Timely and early diagnosis of HF<sub>p</sub>EF leads to a better outcome.</li> <li>• The management of patients with HF<sub>p</sub>EF includes: <ul style="list-style-type: none"> <li>➤ Encouraging lifestyle measures <ul style="list-style-type: none"> <li>♦ Weight reduction</li> <li>♦ Exercise training</li> </ul> </li> <li>➤ Managing comorbidities such as hypertension, CAD, CKD, atrial fibrillation, obesity according to guidelines.</li> <li>➤ Pharmacotherapy: <ul style="list-style-type: none"> <li>♦ <b>Diuretics</b> - for volume overload.</li> </ul> </li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                                                         | <p>I,B</p> <p>I,A</p> <p>I,C</p> <p>I,C</p>   |

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| <ul style="list-style-type: none"> <li>♦ <b>SGLT2-i</b> - decreases HF hospitalizations and CV mortality.</li> <li>♦ <b>RAS blockers</b> (ARB, ACE-I, ARNI) - As a group, RAS blockers have not been shown to reduce total or CV mortality. There was a suggestion of benefit with ARNI in women with LVEF &lt; 60%, while for men the benefit was restricted to LVEF &lt; 45%.</li> <li>♦ <b>MRA</b> - it may be considered to decrease combined CV mortality and HF hospitalizations, particularly among patients with LVEF on the lower end of this spectrum.</li> <li>♦ <b>β-blockers</b> - No good data that they are beneficial in the treatment of HFpEF although they are often prescribed for the management of comorbidities such as CAD or AF.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <p>I,A<br/>IIb,B</p> <p>IIb,B</p>                                                                                                           |
| <p><b>#12: HF and Diabetes</b></p> <ul style="list-style-type: none"> <li>● <b>Prevention of HF:</b> <ul style="list-style-type: none"> <li>➢ Lifestyle measures are important. They have been shown to lower the risk for HF, particularly HFpEF. These measures include: <ul style="list-style-type: none"> <li>♦ Attainment of appropriate weight, <i>and</i></li> <li>♦ Increased physical activity <i>and</i></li> <li>♦ Smoking cessation.</li> </ul> </li> <li>➢ The BP and LDL-C should be treated to target: <ul style="list-style-type: none"> <li>♦ BP 130-139/70-79 mmHg</li> <li>♦ LDL-C according to CV risk category.</li> </ul> </li> </ul> </li> <li>● <b>Treatment of Diabetes - Glucose lowering Agents</b> <ul style="list-style-type: none"> <li>➢ Glycemic targets for patients with DM and HF should be individualized.</li> <li>➢ Current recommendations suggest a target range of HbA1c 7%-8%, while minimizing adverse effects of treatments particularly hypoglycemia.</li> <li>➢ Glucose lowering drugs: <ul style="list-style-type: none"> <li>♦ With proven CV benefits: <ul style="list-style-type: none"> <li>□ SGLT2-i.</li> <li>□ GLP-1 receptor agonists.</li> <li>□ Metformin.</li> </ul> </li> <li>♦ With no proven CV benefits but which will help with glucose control: <ul style="list-style-type: none"> <li>□ DPP4 inhibitors.</li> </ul> </li> <li>♦ With no proven CV benefits and need to be used with caution: <ul style="list-style-type: none"> <li>□ Insulin -This has been associated with increased all-cause mortality and hospitalization for HF especially in patients with low HbA1c &lt; 7%.</li> <li>□ Sulfonylureas.</li> </ul> </li> <li>♦ That need to be avoided: <ul style="list-style-type: none"> <li>□ Thiazolidinediones</li> </ul> </li> </ul> </li> </ul> </li> </ul> | <p>I,A<br/>I,B<br/>I,B</p> <p>I,A<br/>I,A</p> <p>I,A</p> <p>I,A<br/>IIa,A<br/>IIa,B</p> <p>IIa,B</p> <p>IIa,B</p> <p>IIa,B</p> <p>III,A</p> |
| <p><b>#13: HF Due to Arrhythmias and Conduction Abnormalities</b></p> <ul style="list-style-type: none"> <li>● In managing cardiomyopathy due to arrhythmias (SVT, AF or frequent PVCs or VT), radiofrequency ablation is the preferred</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <p>I,A</p>                                                                                                                                  |

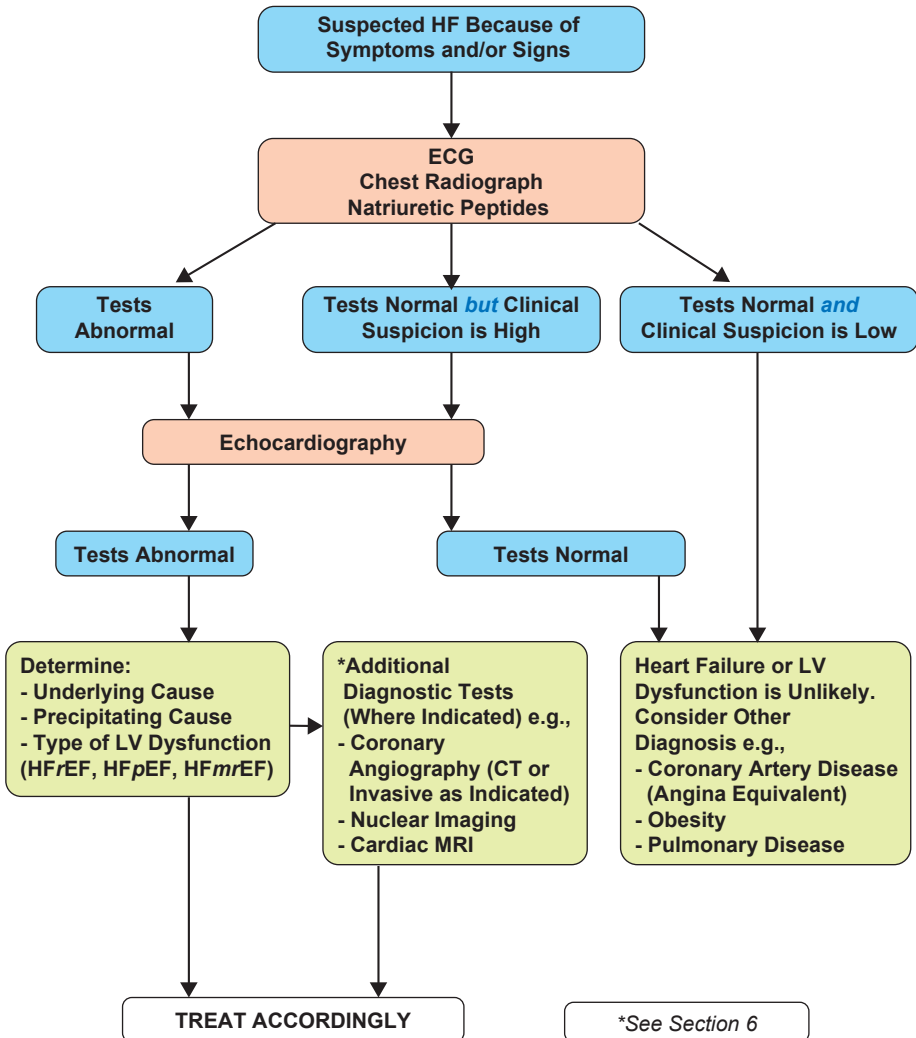
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| therapy, since most antiarrhythmic drugs are contraindicated in the presence of HF.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                            |
| <ul style="list-style-type: none"> <li>In patients with HF who have bradyarrhythmias and where pacing is indicated, biventricular pacing (Cardiac Resynchronisation Therapy) is the pacing mode of choice.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | I,A                        |
| <b>#14: Cardio-Oncology</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                            |
| <ul style="list-style-type: none"> <li>Close collaboration between the oncologist and the cardiologist is important.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | I,C                        |
| <ul style="list-style-type: none"> <li>Patients undergoing chemotherapy should have careful clinical evaluation and assessment. Specifically:             <ul style="list-style-type: none"> <li>All CV risk factors should be treated adequately.</li> <li>High risk patients should be identified and in these patients:                 <ul style="list-style-type: none"> <li>A pre-treatment cardiac echocardiogram is advisable. If the LVEF &lt; 50%, they should be referred to a cardiologist.</li> <li>Reassessing and repeating (if necessary) imaging studies during and after treatment.</li> <li>Assessing cardiac biomarkers when indicated - troponins and/or Natriuretic Peptides.</li> </ul> </li> <li>Considering cardio-protection prior to/during treatment using β-blockers, MRA and/or ACE-I/ARB if:                 <ul style="list-style-type: none"> <li>EF &lt; 50%,</li> <li>EF drops by &gt; 10%</li> <li>Abnormal global longitudinal strain (GLS) (&gt; 15% drop).</li> </ul> </li> </ul> </li> </ul> | I,C<br><br>Ila,B           |
| <b>#15: HF and CKD</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                            |
| <ul style="list-style-type: none"> <li>A multi-disciplinary approach with early referral to a nephrologist is recommended.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | I,C                        |
| <ul style="list-style-type: none"> <li>Almost all <b>Foundational HF Medications</b> can be used in patients with eGFR ≥ 30mls/min/1.73m<sup>2</sup></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | I,A                        |
| <ul style="list-style-type: none"> <li>In patients with eGFR &lt; 30mls/min/1.73m<sup>2</sup>, the following drugs can be used:             <ul style="list-style-type: none"> <li>Diuretics - usually higher maintenance doses.</li> <li>Careful use of RAS blockers.</li> <li>β-blockers.</li> <li>SGLT2-i (eGFR &gt;20mls/min/1.73m<sup>2</sup>).</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | I,B<br>I,B<br>I,B<br>Ila,B |
| <ul style="list-style-type: none"> <li>Closely monitor electrolytes and kidney function. The baseline renal function will determine how frequently this should be done.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                            |
| <ul style="list-style-type: none"> <li>RAS blockers and SGLT2-i can lead to an initial drop in eGFR but this should not be a reason to automatically stop or down titrate these agents. See table 18, page135 for the management of RAS blockers in response to changes in renal function.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | I,C                        |
| <ul style="list-style-type: none"> <li>Occasionally ultrafiltration and renal replacement therapy (hemodialysis) may be necessary.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | I,C                        |
| <b>#16: HF and Pregnancy</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                            |
| <ul style="list-style-type: none"> <li>HF in pregnancy should be managed by a multidisciplinary team consisting of physicians, obstetricians, and pediatricians.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | I,C                        |

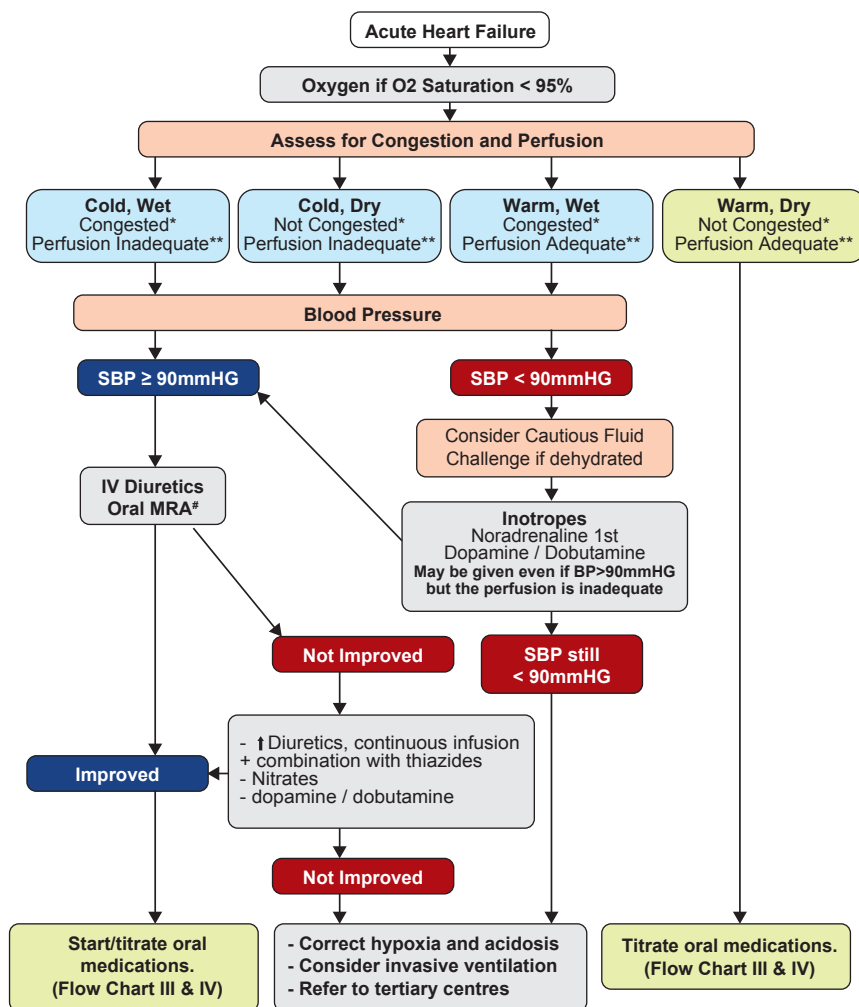


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| <ul style="list-style-type: none"> <li>● HF that develops during pregnancy can be managed with the judicious use of diuretics, digoxin, nitrates, <math>\beta</math>-blockers (most commonly metoprolol) and/or hydralazine.</li> <li>● For postpartum women with severe acute HF caused by Peripartum cardiomyopathy and LVEF &lt;35%, Foundational HF medications to improve LVEF recovery and prophylactic anticoagulation are recommended.</li> </ul>                                                                                                                                           | <p>I,C</p> <p>I,A</p>                         |
| <p><b>#17: HF and ACHD</b> (Appendix XIII-XV, pages 174-177)</p> <ul style="list-style-type: none"> <li>● The principles of managing HF in ACHD HF are: <ul style="list-style-type: none"> <li>➢ First to access and address all reversible causes.</li> <li>➢ If HF control is still not optimal, initiate pharmacotherapy.</li> <li>➢ Arrhythmias to be treated appropriately by pacing or ICD as indicated.</li> <li>➢ To consider cardiac resynchronization and more advanced therapies if these patients continue to have worsening HF despite optimal medical therapy.</li> </ul> </li> </ul> | <p>I,C</p> <p>I,C</p> <p>I,B</p> <p>Ila,C</p> |
| <p><b>#18: Advanced HF</b></p> <ul style="list-style-type: none"> <li>● Heart transplantation is well-established for refractory end stage HF.</li> <li>● Patients with severe symptomatic HF despite OMT and no other alternative therapeutic options should be considered for palliative care.</li> <li>● Older people with multiple comorbidities and generally poor prognosis, should be considered for palliative care even at an early stage of the disease.</li> </ul>                                                                                                                       | <p>I,A</p> <p>I,C</p> <p>I,C</p>              |
| <p><b># 19: HF Rehabilitation</b></p> <ul style="list-style-type: none"> <li>● Cardiac Rehabilitation should be recommended to all stable HF patients, in NYHA II-III.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                   | <p>Ila,B</p>                                  |
| <p><b>#22: Performance Measures</b></p> <ul style="list-style-type: none"> <li>● Performance measures are used with the goal of improving quality of care for HF.</li> <li>● This includes: <ul style="list-style-type: none"> <li>➢ Process performance which measures the aspects of care that are delivered to a patient and</li> <li>➢ Outcome measures which focus on hard endpoints such as mortality and hospitalization.</li> </ul> </li> <li>● For the quality indicators in HF, see section 19, page 160 &amp; Appendix XVI, page 178)</li> </ul>                                         |                                               |



Flow Chart I: Algorithm for the Diagnosis of Heart Failure\*



**Flow Chart II: Management of Acute Heart Failure**

\***Congestion:** Peripheral oedema, orthopnoea, paroxysmal nocturnal dyspnoea, lung crepitations, jugular venous dilatation, hepatojugular reflux, congested hepatomegaly, gut congestion, ascites.

\*\***Hypoperfusion:** Cold peripheries, capillary refill time more than 2 seconds, diaphoresis, oliguria, dizziness, Confusion, narrow pulse pressure, hypotension.

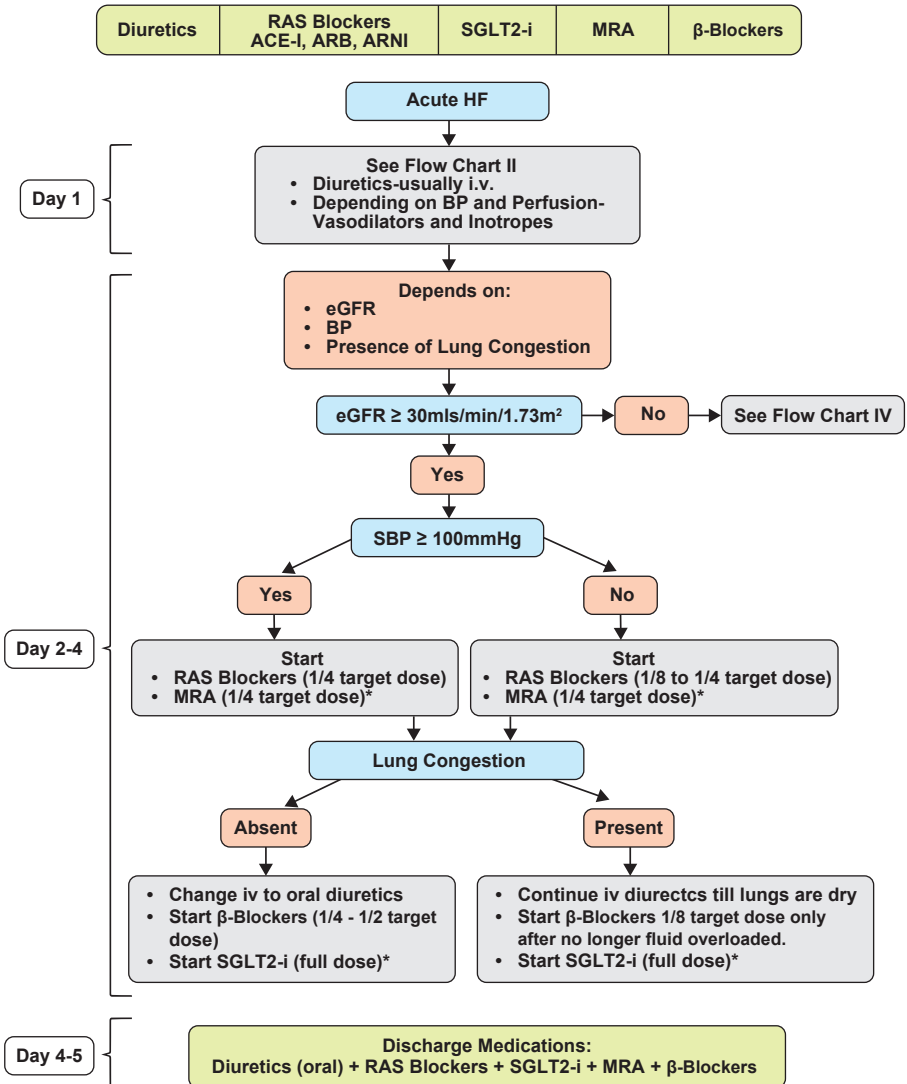
#Administration of MRA will depend on eGFR ≥ 30 mls/min/1.73m<sup>2</sup> and serum potassium levels < 5.0 mmol/l

**From onset, evaluate to identify correctable/reversible lesions - CHAMPION**



**Table 2: Grading of Recommendations in the Management of Acute HF**

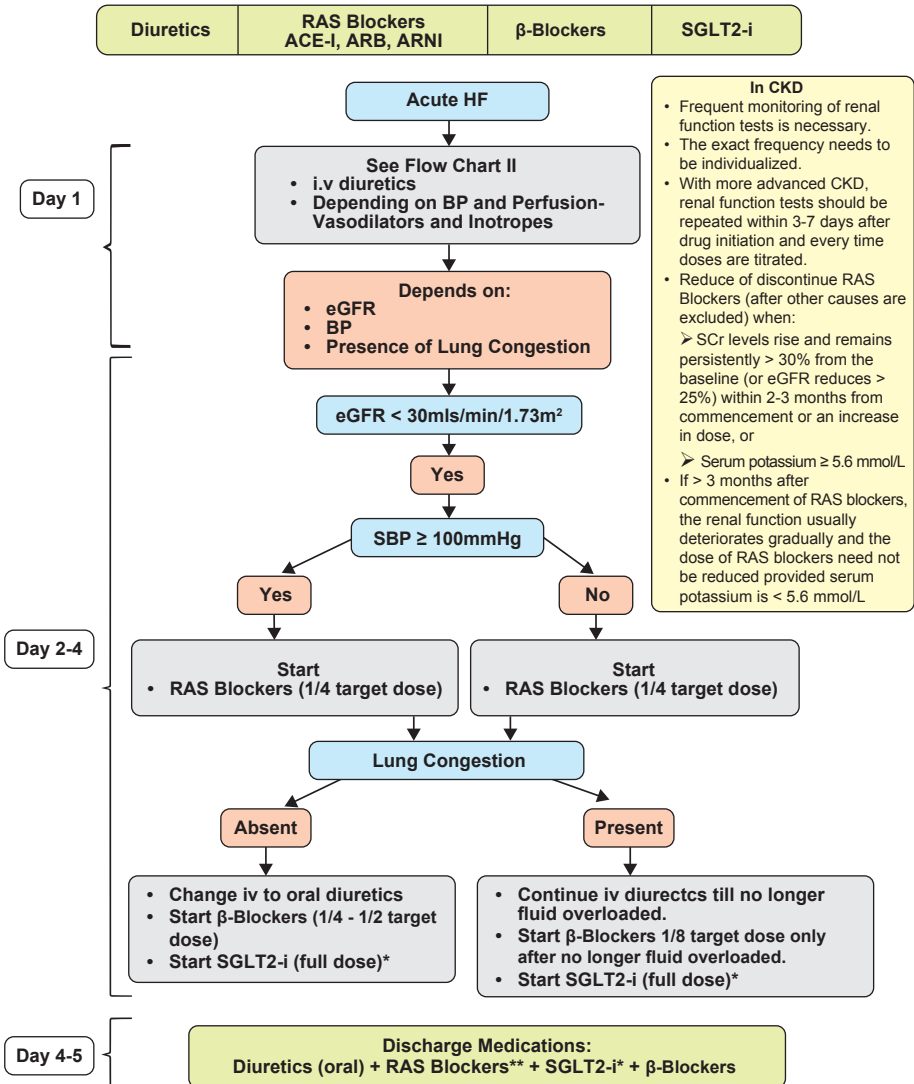
| Intervention                                                                                                                                     | Grades of Recommendation | Levels of Evidence | Comments                                                                                                                                                                             |
|--------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>INITIAL MANAGEMENT CONSISTS OF:</b>                                                                                                           |                          |                    |                                                                                                                                                                                      |
| Oxygen                                                                                                                                           | I                        | C                  | Maintain the oxygen saturation $\geq 95\%$ .                                                                                                                                         |
| Diuretics                                                                                                                                        | I                        | B                  | To be given preferably when SBP $\geq 90\text{mmHg}$ .                                                                                                                               |
| MRA                                                                                                                                              | I                        | A                  | To be given if eGFR $\geq 30 \text{ mLs/min/1.73m}^2$ and serum potassium levels $< 5.0\text{mmol/l}$ . May be initiated within 24-48 hours of admission.                            |
| <b>NOT RESPONSIVE TO INITIAL TREATMENT AND SBP <math>\geq 90\text{mmHg}</math></b>                                                               |                          |                    |                                                                                                                                                                                      |
| Diuretics                                                                                                                                        | IIa                      | B                  | Continuous infusion +/- combination with thiazides.                                                                                                                                  |
| Nitrates                                                                                                                                         | I                        | B                  | Most useful if there is concomitant myocardial ischemia, severe hypertension or aortic or mitral regurgitation.                                                                      |
| Noradrenaline                                                                                                                                    | IIa                      | B                  | Indicated for peripheral hypoperfusion despite an adequate filling status.                                                                                                           |
| Dopamine                                                                                                                                         | IIb                      | B                  | Indicated for peripheral hypoperfusion despite an adequate filling status.                                                                                                           |
| <b>NOT RESPONSIVE TO INITIAL TREATMENT AND SBP <math>&lt; 90\text{mmHg}</math></b>                                                               |                          |                    |                                                                                                                                                                                      |
| Noradrenaline                                                                                                                                    | IIa                      | B                  | Indicated to increase the BP.                                                                                                                                                        |
| Dopamine                                                                                                                                         | IIb                      | B                  | Indicated to increase the BP.                                                                                                                                                        |
| IABP                                                                                                                                             | IIa                      | B                  | Indicated as a bridge till myocardial recovery or heart transplant.                                                                                                                  |
| Ventricular Assist (VAD)                                                                                                                         | IIa                      | B                  | Indicated as a bridge till myocardial recovery or heart transplant. Occasionally as destination (definitive) therapy.                                                                |
| <b>ORAL MEDICATIONS IN ACUTE HF</b><br>(Initiated when patient is hemodynamically stable, off all i.v. inotropes and SBP $\geq 100\text{mmHg}$ ) |                          |                    |                                                                                                                                                                                      |
| RAS blockers (ACE-I or ARB in ACE-I intolerant patients)                                                                                         | I                        | A                  |                                                                                                                                                                                      |
| ARNI                                                                                                                                             | IIa                      | B                  |                                                                                                                                                                                      |
| $\beta$ -Blockers                                                                                                                                | I                        | A                  |                                                                                                                                                                                      |
| SGLT2-i                                                                                                                                          | I                        | A                  | No increase in i.v. diuretic dose and off all iv vasodilators, including nitrates within the last 6 hours.<br>The dose of diuretics may need to be reduced on initiation of SGLT2-i. |

**Flow Chart III: Initiating Foundational HF Medication**

\*The dose of diuretics may need to be down titrated upon initiation of SGLT2-i and MRA. The initiating drug doses may occasionally need to be modified depending on the patient's clinical condition. SGLT2-i should be initiated when patients have not been on inotropes for at least 24 hours, SBP > 100mmHg and there is no increase in the diuretic dose in the last 6 hours. SGLT2-i may also be initiated post discharge .



**Flow Chart IV: For Patients with eGFR < 30mls/min/1.73m<sup>2</sup>  
(Initiating Foundational HF Medication)**

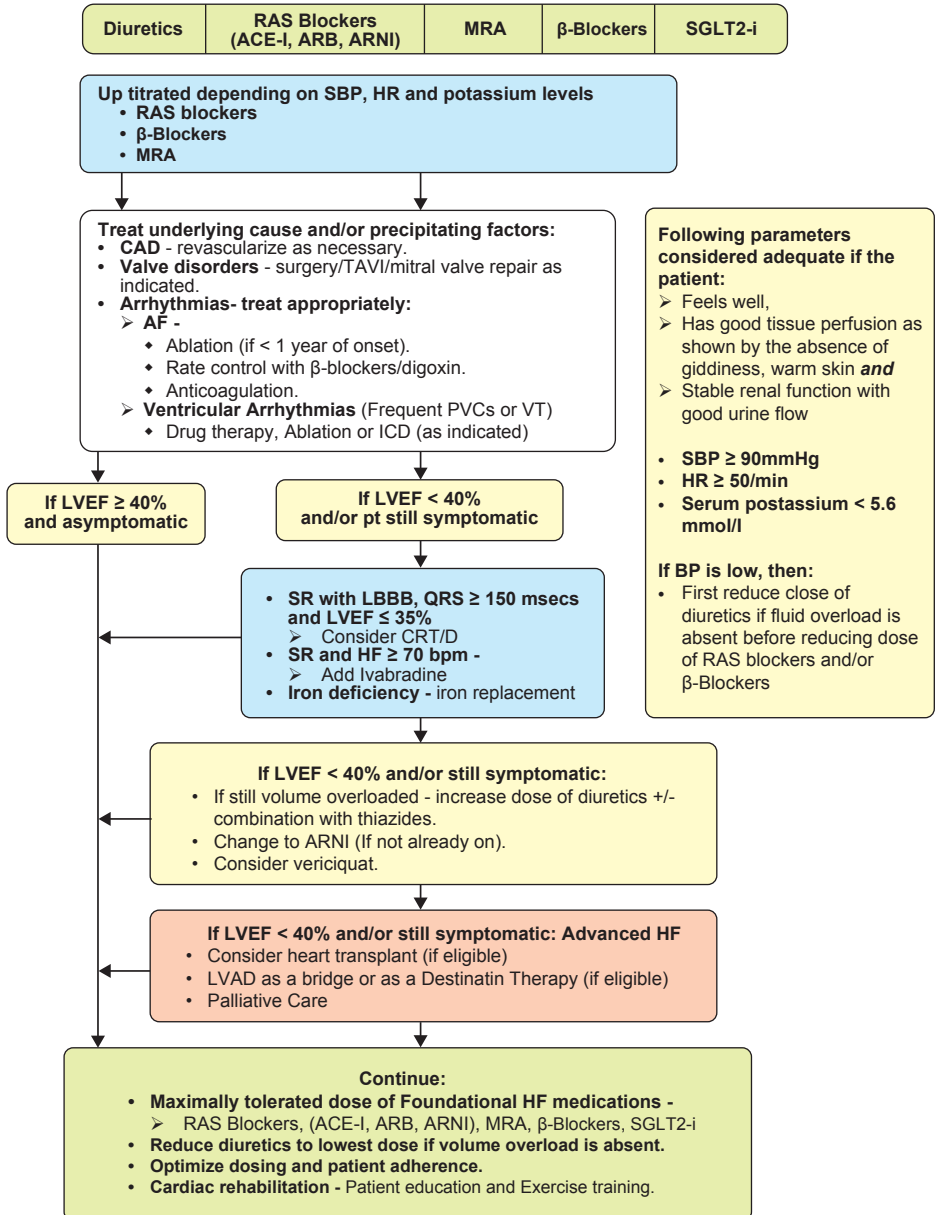


\*SGLT2-i should only be initiated if eGFR ≥ 20mls/min/1.73m<sup>2</sup>. Caution should be exercised when it is started concurrently with the RAS-blocker. The dose of diuretics may need to be down titrated upon initiation of SGLT2-i.

\*\*ARNI should be used with caution if eGFR < 30mls/min/1.73m<sup>2</sup>. It is not recommended in patients with end-stage renal disease.



**Flow Chart V: Optimizing Management of HF<sub>rEF</sub>  
Foundational HF Medications**




**Table 3: Doses of Foundational HF Medications**

| Drug                 | Initial Daily Dose(s) | Target Dose(s)      | 1/8 Target Dose | 1/4 Target Dose | 1/2 Target Dose | Tablet Strength Available | Renal Dose For CrCl < 30ml/min                                  |
|----------------------|-----------------------|---------------------|-----------------|-----------------|-----------------|---------------------------|-----------------------------------------------------------------|
| <b>ACE-I</b>         |                       |                     |                 |                 |                 |                           |                                                                 |
| Captopril            | 6.25mg BD             | 50mg TDS            | 6.25mg          | 12.5mg          | 25mg            | 12.5mg, 25mg              | Yes 6.25mg-12.5mg bd/tds                                        |
| Enalapril            | 2.5mg OD              | 10-20mg BD          |                 | 2.5-5mg         | 5-10mg          | 5mg, 10mg, 20mg           | Yes 2.5mg OD                                                    |
| Lisinopril           | 2.5–5mg OD            | 20-40mg OD          | 2.5mg           | 5-10mg          | 10-20mg         | 5mg, 10mg, 20mg           | Yes 2.5mg OD<br>CrCl 10-30: 5mg OD<br>CrCl <10: 2.5mg OD        |
| Perindopril          | 2mg OD                | 8-16mg OD           |                 | 2-4mg           | 4-8mg           | 2mg, 4mg, 8mg             | Yes CrCl 15-30: 2mg eod<br>CrCl <15 & HD: 2mg on dialysis day** |
| Ramipril             | 2.5mg OD              | 10mg OD             | 1.25mg          | 2.5mg           | 5mg             | 2.5mg, 5mg, 10mg          | Yes 1.25mg OD<br>CrCl <40: 25% normal dose                      |
| <b>ARBs</b>          |                       |                     |                 |                 |                 |                           |                                                                 |
| Candesartan          | 4-8mg OD              | 32mg                | 4mg OD          | 8mg             | 16mg            | 8mg, 16mg                 | Yes 4mg OD                                                      |
| Losartan             | 25-50mg OD            | 50-150mg OD         |                 | 12.5-37.5mg     | 25-75mg         | 25mg, 50mg, 100mg         | Yes No adjustment                                               |
| Valsartan            | 40mg OD               | 160mg BD            |                 | 40mg            | 80mg            | 40mg, 80mg, 160mg         | Yes Use with caution                                            |
| <b>ARNI</b>          |                       |                     |                 |                 |                 |                           |                                                                 |
| Sacubitril/valsartan | 100mg (49/51mg) BD*   | 200mg (97/103mg) BD | 25mg            | 50mg            | 100mg           | 50mg, 100mg, 200mg        | Yes 50mg BD                                                     |



| Drug                 | Starting Dose | Target Dose | 1/8 Target Dose | 1/4 Target Dose | 1/2 Target Dose | Tablet Strength Available | Renal Dose For CrCl < 30ml/min |                                                  |
|----------------------|---------------|-------------|-----------------|-----------------|-----------------|---------------------------|--------------------------------|--------------------------------------------------|
| β-Blocker            |               |             |                 |                 |                 |                           |                                |                                                  |
| Bisoprolol           | 1.25mg OD     | 10 mg OD    | 1.25mg          | 2.5mg           | 5mg             | 2.5mg, 5mg, 10mg          | Yes                            | No adjustment                                    |
| Carvedilol           | 3.125mg BD    | 25 mg BD    | 3.125mg         | 6.25mg          | 12.5mg          | 6.25mg, 12.5mg, 25mg      | Yes                            | No adjustment                                    |
| Metoprolol Tartrate* | 12.5-25mg BD  | 100 mg BD   | 12.5mg          | 25mg            | 50mg            | 50mg, 100mg               | Yes                            | No adjustment                                    |
| Nebivolol+           | 1.25mg OD     | 10 mg OD    | 1.25mg          | 2.5mg           | 5mg             | 5mg                       | Yes                            | No adjustment                                    |
| MRA                  |               |             |                 |                 |                 |                           |                                |                                                  |
| Spironolactone       | 12.5-25mg OD  | 50 mg OD    |                 | 12.5mg          | 25mg            | 25mg                      | Yes                            | Contraindicated                                  |
| Eplerenone           | 25 mg OD      | 50 mg BD    |                 | 12.5mg          | 25mg            | 25mg, 50mg                | Yes                            | Contraindicated                                  |
| Finerenone           | 20mg OD       | 20mg OD     |                 |                 |                 | 10mg, 20mg                | Yes                            | eGFR 25-60: 10mg OD<br>eGFR <25: contraindicated |
| SGLT2-i              |               |             |                 |                 |                 |                           |                                |                                                  |
| Dapagliflozin        | 10mg OD       | 10mg OD     |                 |                 |                 |                           |                                | CrCl <25: limited experience                     |
| Empagliflozin        | 10mg OD       | 10mg OD     |                 |                 |                 |                           |                                | CrCl <20: not recommended                        |

\* The evidence-based form of metoprolol in HF is metoprolol succinate (CR/XL), which is not available in Malaysia. The only form of metoprolol available in Malaysia is metoprolol tartrate

+ Shown to reduce the composite endpoint of all-cause mortality and CV hospitalization, but not all-cause mortality.

IBM Micromedex Drug Ref (2023) (cited 14<sup>th</sup> January 2023) & product leaflet available from Quest3 (accessed date: 13<sup>th</sup> January 2023)

\*\* At the discretion of the attending physician

CrCl unit: ml/min; eGFR unit: ml/min/1.73m<sup>2</sup>



**Table 4: Grading of Recommendations and Levels of Evidence in the Management of HFrEF (LVEF<40%)**

| Intervention                                                                      | Grades of Recommendation | Levels of Evidence | Comments                                                                                                                                                                                              |
|-----------------------------------------------------------------------------------|--------------------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>INDICATED FOR FLUID RETENTION IN NYHA II - IV</b>                              |                          |                    |                                                                                                                                                                                                       |
| <b>Diuretics</b>                                                                  | I                        | B                  | No randomized trial to show improvement in survival.                                                                                                                                                  |
| <b>INDICATED IN ALL PATIENTS</b>                                                  |                          |                    |                                                                                                                                                                                                       |
| <b>ACE-I</b>                                                                      | I                        | A                  | Improves survival and delays progression in all classes of HF.                                                                                                                                        |
| <b>ARB</b>                                                                        | I                        | A                  | In ACE-I intolerant patients.                                                                                                                                                                         |
| <b>ARNI<br/>(Instead of ACE-I)</b>                                                | I                        | B                  | Improves survival and delays progression in all classes of HF when compared to ACE-I.                                                                                                                 |
| <b>β-Blockers</b>                                                                 | I                        | A                  | Improves survival and delays progression in all classes of HF.                                                                                                                                        |
| <b>SGLT2-i</b>                                                                    | I                        | A                  | Improves survival and delays progression in all classes of HF.                                                                                                                                        |
| <b>Mineralocorticoid<br/>Receptor<br/>Antagonists</b>                             | I                        | A                  | Improves survival and reduces hospitalizations in moderate to severe HF and in post MI patients with mild HF.                                                                                         |
| <b>IN ADDITION TO THE ABOVE, THE FOLLOWING ARE INDICATED IN SELECTED PATIENTS</b> |                          |                    |                                                                                                                                                                                                       |
| <b>ARB<br/>(instead of ACE-I)</b>                                                 | I                        | B                  | In patients post MI and LVEF < 40%, Valsartan was shown to be comparable to captopril.                                                                                                                |
| <b>Digoxin</b>                                                                    | I                        | B                  | In patients with HF and AF                                                                                                                                                                            |
|                                                                                   | IIa                      | B                  | No effect on survival. Reduces hospitalizations when added to optimal medical therapy.                                                                                                                |
| <b>Ivabradine</b>                                                                 | IIa                      | B                  | Reduces hospitalizations when added to optimal medical therapy in patients in sinus rhythm and heart rate ≥ 70bpm                                                                                     |
| <b>ICD<br/>(Implantable<br/>Cardioverter<br/>Defibrillator)</b>                   | I                        | A                  | Improves survival in patients with resuscitated cardiac arrest, VF, or sustained VT                                                                                                                   |
|                                                                                   | I                        | A                  | Improves survival in patients > 40 days post MI, LVEF ≤ 30%, with non-sustained VT <b>AND</b> inducible sustained VT or VF during an EP study and on optimal medical treatment, and in NYHA II or III |
|                                                                                   | I                        | A                  | Improves survival in patients with prior MI and > 40 days post MI and 3 months after revascularization, LVEF ≤ 35% and NYHA class II-III                                                              |
|                                                                                   | I                        | B                  | Improves survival in patients (no prior MI), LVEF ≤ 35%, on optimal medical treatment, and in NYHA II or III                                                                                          |
| <b>CRT<br/>(Cardiac<br/>Resynchronisation<br/>Therapy)</b>                        |                          |                    | Improves survival in patients having <b>all of the following</b> :<br>sinus rhythm, LVEF ≤ 35%, LBBB <b>and</b> QRS duration on resting 12-lead ECG: ≥ 150ms                                          |
|                                                                                   | I                        | A                  |                                                                                                                                                                                                       |
|                                                                                   | IIa                      | B                  | ≥ 120-149ms                                                                                                                                                                                           |
| <b>Pacemaker</b>                                                                  | I                        | A                  | For significant symptomatic bradyarrhythmias, trifascicular BBB, third-or-high-degree AV blocks.                                                                                                      |



**Table 5: Patient Profiling and Titration of “Foundational HF Medications”**

|                                  |                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------------------------|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Blood Pressure</b>            | <b>Systolic BP &lt; 90 mmHG</b>  | <ul style="list-style-type: none"> <li>• Detect and treat causes of hypotension, such as hypovolemia from over diuresis, bleeding, infection etc.</li> <li>• If the patient is euvolemic, reduce the dose of diuretics to the lowest maintenance. Occasionally it may be possible to remove the diuretic completely especially if the LVEF &gt; 40-45%.</li> <li>• Remove all non “Foundational HF” Medications that may also cause hypotension e.g., - nitrates, calcium channel blockers, alpha blockers.</li> <li>• The dose of “Foundational HF” Medications only needs to be reduced or temporarily stopped if there is symptomatic hypotension.</li> <li>• SGLT2-i and low-dose MRA have minimal effects on BP and may be continued.</li> </ul> |
|                                  | <b>Systolic BP &gt; 110 mmHG</b> | <ul style="list-style-type: none"> <li>• The dose of RAS blockers and <math>\beta</math>-blockers should be up titrated in turn till the target or maximally tolerated dose.</li> <li>• Whether the RAS blocker or the <math>\beta</math>-blocker should be up titrated first will depend on the patient’s heart rate, renal function, and potassium levels.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Heart rate (Sinus Rhythm)</b> | <b>HR &gt; 70/bpm</b>            | <ul style="list-style-type: none"> <li>• The dose of <math>\beta</math>-blocker should first be up titrated to the target or maximally tolerated dose.</li> <li>• Ivabradine may be added to the <math>\beta</math>-blocker to achieve the target HR.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                                  | <b>HR &lt; 50/bpm</b>            | <ul style="list-style-type: none"> <li>• Discontinue non-dihydropyridine calcium channel blockers e.g., diltiazem and verapamil, digoxin, or other antiarrhythmic drugs such as amiodarone.</li> <li>• If the HR is still &lt; 50/bpm, the dose of ivabradine should first be reduced.</li> <li>• If the HR is still &lt; 50/bpm or the patient has symptomatic bradycardia, then the dose of <math>\beta</math>-blocker should be reduced or temporarily discontinued.</li> </ul>                                                                                                                                                                                                                                                                    |
| <b>Atrial Fibrillation</b>       | <b>Heart Rate (HR)</b>           | <ul style="list-style-type: none"> <li>• The optimal resting ventricular rate in patients with AF and HF is unknown. It is not unreasonable to aim for a HR &lt; 110bpm.</li> <li>• If the patient is troubled by palpitations or there is deterioration of LV function, then a more strict rate control of 80bpm may be targeted.</li> <li>• The ventricular rate should however, be maintained &gt; 70 bpm. Lower HR have been associated with worse outcomes.</li> <li>• Excessive rate control, which may be associated with an increase in pauses, carries a risk.</li> <li>• The optimal ventricular rate during exercise is also uncertain, but may be &lt; 110bpm during light exercise.</li> </ul>                                           |
|                                  | <b>Systolic BP &lt; 90mmHg</b>   | <ul style="list-style-type: none"> <li>• <math>\beta</math>-blockers may be stopped and replaced with digoxin for rate control.</li> <li>• This action may allow for the up titration of RAS blockers as the SBP improves to <math>\geq</math> 90mmHG.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Renal Function</b>            | <b>Potassium</b>                 | <ul style="list-style-type: none"> <li>• RAS blockers and MRA to be sequentially initiated if serum potassium is &lt; 5.5 mmol/l.</li> <li>• Consider reducing or discontinuing the dose of RAS Blockers and MRA if serum potassium is persistently &gt; 5.5 mmol/l despite other measures to reduce hyperkalemia.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                         |
|                                  | <b>Serm Creatinine</b>           | <ul style="list-style-type: none"> <li>• Consider reducing or discontinuing RAS Blockers (after excluding other precipitating factors) when SCr levels remain <math>\geq</math> 30% from the baseline (or eGFR reduces <math>\geq</math> 25%) and if these occur within two from commencement or dose increase.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                            |

months



**Table 6 Grading of Recommendations and Levels of Evidence in the Management of HFpEF (LVEF > 50%)**

| Intervention                                                               | Grades of Recommendation | Levels of Evidence | Comments                                                                                                                                                                                                                                                                                                                                                            |
|----------------------------------------------------------------------------|--------------------------|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>LIFESTYLE MEASURES</b>                                                  |                          |                    |                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Overweight / Obesity</b>                                                | I                        | B                  | <ul style="list-style-type: none"> <li>A caloric restriction diet is feasible and safe and should ideally be combined with exercise.</li> <li>Bariatric Surgery in patients with HFpEF and obesity was associated with improved symptoms and reduction in HF hospitalizations.</li> </ul>                                                                           |
| <b>Exercise Training</b>                                                   | I                        | A                  | <ul style="list-style-type: none"> <li>This is safe and improves exercise capacity and quality of life.</li> </ul>                                                                                                                                                                                                                                                  |
| <b>IDENTIFYING AND TREATING THE UNDERLYING CAUSE(S) AND CO-MORBIDITIES</b> |                          |                    |                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Hypertension</b>                                                        | I                        | A                  | <ul style="list-style-type: none"> <li>Improved BP control has been shown to reduce morbidity and hospitalizations for HF.</li> </ul>                                                                                                                                                                                                                               |
| <b>Tachyarrhythmias (Persistent or Paroxysmal AF)</b>                      | IIa                      | B                  | <ul style="list-style-type: none"> <li><b>Rate Control</b> with <math>\beta</math>-blockers or non-dihydropyridine calcium channel blockers (verapamil, diltiazem) alone or in combination.</li> </ul>                                                                                                                                                              |
|                                                                            | IIa                      | A                  | <ul style="list-style-type: none"> <li><b>Rhythm Control</b> in patients with recent onset AF &lt; 1 year duration or paroxysmal AF.</li> </ul>                                                                                                                                                                                                                     |
| <b>Anti Coagulation</b>                                                    | I                        | A                  | <ul style="list-style-type: none"> <li>To reduce the risk of thromboembolic events.</li> </ul>                                                                                                                                                                                                                                                                      |
| <b>Others</b>                                                              | I                        | C                  | <ul style="list-style-type: none"> <li>Treat CAD, Diabetes, CKD appropriately according to guidelines.</li> </ul>                                                                                                                                                                                                                                                   |
| <b>PHARMACOTHERAPY</b>                                                     |                          |                    |                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Diuretics</b>                                                           | I                        | C                  | <ul style="list-style-type: none"> <li>To relieve congestion.</li> </ul>                                                                                                                                                                                                                                                                                            |
| <b>RAS Blockers</b>                                                        | IIb                      | B                  | <ul style="list-style-type: none"> <li>Trial data show a reduction in HF hospitalizations, but no reduction in all-cause or CV mortality in HFpEF.</li> <li>With ARNI, there was a suggestion of benefit in patients with LVEF &lt; 57% (in women benefits of ARNI were sustained up to LVEF 60%, while for men the benefit was restricted to LVEF 45%).</li> </ul> |
| <b>MRA</b>                                                                 | IIb                      | B                  | <ul style="list-style-type: none"> <li>It may be considered to decrease HF hospitalizations, particularly among patients with LVEF on the lower end of this spectrum.</li> </ul>                                                                                                                                                                                    |
| <b>SGLT2-i</b>                                                             | IIa                      | A                  | <ul style="list-style-type: none"> <li>These have been shown to decrease HF hospitalizations and CV mortality.</li> <li>As more trial data becomes evident, the grading <b>may</b> more upwards to I, A</li> </ul>                                                                                                                                                  |
| <b><math>\beta</math>-Blockers</b>                                         |                          |                    | <ul style="list-style-type: none"> <li>No good data to show that <math>\beta</math>-blockers are beneficial in the treatment of HFpEF.</li> <li>Often prescribed for treatment of co-morbidities.</li> </ul>                                                                                                                                                        |

**1. INTRODUCTION****1.1 Epidemiology of Heart Failure**

**Heart failure (HF) is a clinical syndrome and is the end stage of most heart diseases.** Globally, it affects more than 60 million individuals and in Asia, the prevalence is estimated to be between 1.3% and 6.7%.<sup>1</sup> In general, the incidence of HF has stabilized but the prevalence appears to be increasing due to the ageing of the population and increasing number of survivors post myocardial infarction.<sup>1-4</sup>

**In Asia, patients with HF are younger with a mean age of 60 years** and two-thirds are < 65 years.<sup>5</sup> This is 10 to 15 years younger than patients in Europe (71-79 years) and the USA (mean 72 years).<sup>6-10</sup> Most of the Asian patients were males and about two-thirds (64%) had 2 or more comorbid conditions such as hypertension (51.9%), coronary artery disease (CAD, 50.2%), or diabetes (40.4%).<sup>5</sup> There were regional variations but Southeast Asians (Malaysia, Indonesia, Singapore, Thailand, and Philippines) had the highest burden of comorbidities, particularly CAD, hypertension, diabetes mellitus, obesity and chronic kidney disease.<sup>11</sup>

In the ASIAN-HF registry, the 1-year all-cause mortality for the whole population was 9.6%, varying from 4.4% in Japan to 21.4% in Indonesia.<sup>11</sup> Southeast Asians had the highest mortality (13.6%) compared with South Asians (8.3%) and Northeast Asians (8.9%).<sup>11</sup>

**In Malaysia for Acute HF**, (see Table 7, page 48), the:<sup>11-17</sup>

- **Median age** of patients was 59-64 years. In Thailand, the mean age of patients with HF was 65.3 years.<sup>34</sup>
- **Median length of hospital stay (LOS)** was 3-5 days - comparable to that in the US.<sup>18</sup>
- **In-hospital mortality** varied from <3% to as high as 7.5%. In the European Society of Cardiology (ESC) HF-Long term (LT) registry, the in-hospital mortality was 4.9% and in the Get With The Guidelines-HF (GWTG-HF) registry, the in-hospital mortality was <3%.<sup>18,19</sup>
- **30-day mortality** was high at 11-15%.
- **1-year mortality** was also similarly high at 33-49.5%.
- **30-day and 1-year all cause readmission rates** were 4-18% and 24-76% respectively compared to the 19% and 53% seen in a meta-analysis of 27 HF registries.<sup>20</sup>

In general, in addition to hospitalizations due to HF, all-cause hospitalizations were also increased and observed in up to 60% of patients.<sup>21</sup> In Malaysia, cardiovascular causes accounted for half of all 30-day readmissions (50.1%) with HF specifically accounting for 27.8%.<sup>17</sup> Readmission rates were higher in the older age and those with CKD.<sup>17</sup> Length of stay increased with each hospitalization and the time in between hospitalizations decreased with each subsequent hospitalization.<sup>17</sup>

The prognosis of HF has improved over time, but mortality remains high.<sup>1,6-9,22,23</sup> In the European Society of Cardiology (ESC) HF-Long term (LT) registry, 1-year mortality was 23.6% for acute HF and 6.4% for chronic HF between 2011 and 2013.<sup>19</sup> In ambulatory patients with HF, pooled analysis of cohorts from three North-western European countries, showed that outcomes have significantly improved during the last two decades. (Period 1:1995-2005 vs period 2: 2006-2015).<sup>24</sup> Improvement in all-cause mortality can be explained partly by demographic differences and also by better utilization of Optimal Medical Therapy.<sup>25</sup>



## 1.2 Socio-Economic Consequences of Heart Failure

The socio-economic burden of HF is huge. In high-income countries, 1-2% of the total healthcare expenditure is spent on HF.<sup>21</sup> In the US, total cost for HF was estimated to be \$30.7 billion in 2012, with 68% attributable to direct medical costs.<sup>26</sup> A more contemporary review estimated annual healthcare costs for HF patients in the western world, to amount to 25,000 Euro.<sup>1</sup> Most of the costs was linked to inpatient care and re-hospitalizations. In Malaysia, in 2014, it was estimated that 3.6% of the GDP is being spent on total healthcare with only 1.8% spent on HF.<sup>27</sup> This cost is expected to rise with an ageing and rapidly expanding population.

The overall global economic cost of HF is highly variable from country to country.<sup>21,28</sup> In general, the costs of treatment of a HF patient is higher:<sup>28-30</sup>

- than other diseases such as asthma, coronary artery disease, chronic obstructive pulmonary diseases, diabetes, hyperlipidemia and hypertension.
- in the presence of comorbidities such as diabetes, obesity and chronic kidney disease.

Healthcare costs for HF include:<sup>27</sup>

- Direct costs - this accounts for 60% of total costs.
- Indirect costs - account for 40%. This includes premature mortality, 'presenteeism' (the lost productivity that occurs when employees are not fully functioning in the workplace because of an illness, injury, or other condition), disability, sick absenteeism (short- and medium-term absence from work) and costs of caregiver's absenteeism.

In general, in most low and medium economies like Malaysia, the indirect costs of HF in terms of premature mortality, morbidity, lost earning potential and unpaid care costs outweigh the direct costs.<sup>27</sup>

In Asia, there is substantial variations in healthcare spending. In a non-interventional, retrospective study conducted through medical chart audit for the year 2014, the average cost of HF hospitalization varied from USD 4,513 in Taiwan to USD 1,443 in public hospitals in Malaysia (adjusted to 2015 USD).<sup>31</sup> This wide variation in costs could be partly explained by the fact that in Malaysia, healthcare is heavily subsidized by the government and the wide use of generic medications. There was also less use of more expensive medications such as Angiotensin Receptor Neprilysin Inhibitors and device therapy.

In a costing study on HF conducted in Hospital Queen Elizabeth II, Kota Kinabalu, the mean and median annual cost of HF per person were USD 5,428 (MYR 22,649) and USD 591 (MYR 2,466) respectively in 2017.<sup>32</sup> This is much higher than that seen in the earlier study because it includes interventional procedures. Inpatient cost accounted for 90.6% of the total cost and was mainly attributable to percutaneous coronary intervention (PCI) procedures and hospitalization.<sup>32</sup>

In another cost analysis done in 3 public hospitals in Sabah, Kelantan and Pulau Penang looking only at patients with chronic HF from 2016-2018, the mean total cost per HF patient per-year (PPPY) was USD 1,971 (MYR 8,224)  $\pm$  USD 1,255 (MYR 5,236), of which inpatient cost accounted for 74.7% of the total cost.<sup>33</sup> Medication costs (42.0%) and procedure cost (40.8%) contributed to the largest proportion of outpatient and inpatient costs. HF patients with preserved LVEF had the highest mean total cost of PPPY, at USD 2,410 (MYR 10,056)  $\pm$  USD 1,226 (MYR 5,115).<sup>33</sup>



**Thus, an important goal in management of HF is to prevent readmissions, reducing both direct and indirect costs.**

### **1.3 Humanistic Burden of Heart Failure**

A diagnosis of HF also has a huge humanistic burden.<sup>21,27</sup> This refers to the impact of an illness on the patient's health-related quality of life (HRQoL), activities of daily living, caregiver(s) 's health, caregiver(s)'s quality of life, patients' treatment satisfaction and compliance with their specific treatment regimens.<sup>21,28</sup> In general, most patients prefer improved HRQoL to length of survival.

The goals of management of HF include:

- Preventing hospitalizations and unplanned hospital visits - this will reduce both the socio-economic and humanistic burden of HF.
- Reducing symptoms, improving functional capacity and quality of life, and thus improving the humanistic burden of HF.
- Improving patient survival.

This guideline provides evidence-based recommendations to help health care providers in the management of their patients with HF. Beyond the Clinical Practice Guidelines (CPG), clinical management needs to be individualized considering the patient's overall health goals, values, perspective, and preferences.

Sound clinical judgment plays an important role in formulating appropriate patient-centered care plans.

#### **Key Message # 1: Epidemiology and Goals of Management**

- In Malaysia, patients with HF are almost 10-15 years younger than those in Western Countries. Comorbidities such as hypertension, diabetes and chronic kidney disease are common.
- HF is associated with a huge socio-economic and humanistic burden.
- Most of the costs related to HF are related to inpatient hospital care and re-hospitalizations.
- Important goals of management include:
  - Preventing readmissions thus reducing both the socio-economic and humanistic burden of the disease.
  - Improving symptoms, functional capacity, and quality of life.
  - Improving patient survival.


**Table 7 : Epidemiology of HF in Asia & Malaysia**

|                         | Asian HF registry <sup>11</sup>                                                                                                                                                               | Lim YMF et al <sup>12</sup>                                                                                        | MYHF <sup>13,14</sup>                                                             | Sharif RER et al <sup>14</sup>                       | Ling HS et al <sup>15</sup>                                                             | Azmee et al <sup>16</sup>                                                                             |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------|-----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Source of Data          | Prospective longitudinal study of outpatient (at least 1 episode of decompensated HF in the previous 6 months) and hospitalized patients 1 <sup>st</sup> Oct 2012 - 31 <sup>st</sup> Oct 2016 | 10 years retrospective data from 1 <sup>st</sup> Jan 2007 to 31 <sup>st</sup> Dec 2016 of pts hospitalized with HF | Prospective observational study on hospitalized patients from Aug 2019 - Dec 2020 | Single center Retrospective data Jan 2012 - Dec 2016 | Single center Prospective observational study on hospitalized pts Sept 2017 to Oct 2018 | Single center Retrospective on hospitalized pts 1 <sup>st</sup> Jan 2009 to 31 <sup>st</sup> Dec 2018 |
| Population              | 6480 patients aged > 18 years With symptomatic HF                                                                                                                                             | 105,399 > 20 years incident HF hospitalizations in a MOH hospital                                                  | 2673 patients > 18 years                                                          | 1307 patients > 18 years Acute HF patients           | 117 patients >18 years Acute HF patients                                                | 3923 patients Adult HF pts                                                                            |
| Mean Age                | 61.6 years                                                                                                                                                                                    | 64.1 years                                                                                                         | 60.17 years                                                                       | 63.4 years                                           | 59 years                                                                                | 62 years                                                                                              |
| % Women                 | 27%                                                                                                                                                                                           | 44%                                                                                                                | 33.2%                                                                             | 53.6%                                                | 41.1%                                                                                   | 37.7%                                                                                                 |
| % HF/EF                 | 81%                                                                                                                                                                                           | NA                                                                                                                 | 64.6%                                                                             | 40.8%                                                | 48.6%                                                                                   | 62.9%                                                                                                 |
| Length of Hospital Stay | NA                                                                                                                                                                                            | 3 days (median)                                                                                                    | 7.3 days (mean)<br>3 - 4 (median)                                                 | 5.3 ± 3.4 days (mean).                               | 5 days (median)                                                                         | 9.2 days (mean)                                                                                       |
| All Cause Readmission   |                                                                                                                                                                                               |                                                                                                                    |                                                                                   |                                                      |                                                                                         |                                                                                                       |
| 30 days                 | NA                                                                                                                                                                                            | 18.1%                                                                                                              | 12.9%                                                                             | 4.1%                                                 | 11.2%                                                                                   | 6.8%                                                                                                  |
| 1 year                  | NA                                                                                                                                                                                            | NA                                                                                                                 | NA                                                                                | 76.1%                                                | NA                                                                                      | 24.7%                                                                                                 |
| In Hospital Mortality   |                                                                                                                                                                                               | 5.3%                                                                                                               | 2.8%                                                                              | 1.7%                                                 | 7.5%                                                                                    | 7.2%                                                                                                  |
| 30 days Mortality       |                                                                                                                                                                                               | 11.2%                                                                                                              | 7.8%                                                                              | 15.7%                                                | 13.1%                                                                                   | NA                                                                                                    |
| 1 year Mortality        | 9.6% HF/EF:10.6% HFpEF: 5.4%,                                                                                                                                                                 | 33.1%                                                                                                              | NA                                                                                | 49.7%                                                | NA                                                                                      | NA                                                                                                    |

**2. DEFINITION**

**HF is a clinical syndrome** due to any structural or physiological abnormality of the heart resulting in its inability to meet the metabolic demands of the body or its ability to do so only at higher-than-normal filling pressures. This is supported by either an elevation of natriuretic peptides and/or evidence of pulmonary or systemic congestion.<sup>35</sup>

This may be accompanied by signs and symptoms of systemic hypoperfusion and/or volume overload. Patients may have typical symptoms (e.g., breathlessness, ankle swelling and fatigue) and signs (e.g., elevated jugular venous pressure, ankle edema, pulmonary crackles and displaced apex beat). Occasionally, some patients may present without signs or symptoms of volume overload. Older patients, often, present with atypical symptoms such as delirium, reduced appetite, immobility, incontinence, and falls.

Most commonly, HF is due to myocardial dysfunction- either systolic, diastolic, or both. However, pathology of the valves, pericardium, and endocardium, and abnormalities of heart rhythm and conduction can also cause HF.

Occasionally, non-cardiac disease, e.g., anemia, pulmonary, renal, thyroid, or hepatic disease may have symptoms and signs like those of HF, but in the absence of cardiac dysfunction, these do not fulfil the criteria for HF. However, these pathologies can coexist with HF and exacerbate the condition.

**Key Message #2: Definition**

- HF is a clinical syndrome due to any structural or physiological abnormality of the heart resulting in its inability to meet the metabolic demands of the body or its ability to do so only at higher-than-normal filling pressures.
- This is supported objectively by either an elevation of natriuretic peptides and/or evidence of pulmonary or systemic congestion.

**3. CLASSIFICATION**

**HF can be classified by phenotypes.** These descriptions may reflect the:

- Temporal characteristics (e.g., acuteness, chronicity)
- Affected circulatory systems (e.g., left vs right)
- Clinical context and trajectory (e.g., decompensated, improved, advanced, end-stage).

**Another commonly used classification is by the severity of LV systolic function as assessed by LV ejection Fraction (LVEF).**<sup>35</sup> (Table 8, page 50)

**Table 8: Classification Of Heart Failure According To LVEF**

| Ejection Fraction Terminology                           | LVEF                                                                                                                                                        |
|---------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Heart Failure with Reduced Ejection Fraction (HFrEF)    | $\leq 40\%$                                                                                                                                                 |
| Heart Failure with mildly reduced LVEF (HFmrEF)         | 41 - 49%                                                                                                                                                    |
| Heart Failure with Preserved Ejection Fraction (HFpEF)  | $\geq 50\%$                                                                                                                                                 |
| Heart Failure with Improved Ejection Fraction (HFimpEF) | HF with a baseline LVEF of $\leq 40\%$ , a $\geq 10$ -point increase from baseline LVEF following treatment, and a second measurement of LVEF of $> 40\%$ . |

**Stages of HF include<sup>35</sup>:**

- **A** - "At Risk"
  - Asymptomatic without structural cardiac disease but 'at risk' of developing HF.
- **B** - "Pre HF"
  - Asymptomatic but with structural and functional cardiac abnormalities that can lead to HF. They may have normal or mildly reduced LV function.
- **C** - "HF"
  - Symptomatic HF, either previous or current symptoms.
- **D** - "Advanced HF"
  - Marked symptoms interfering with daily activities of living and with recurrent hospitalizations.

The severity of symptomatic HF can be assessed by the New York Heart Association (NYHA) Functional Class. (Table 9, page 50)

**Table 9: New York Heart Association Functional Classification for Patients with Heart Disease**

| CLASS            | FUNCTIONAL CAPACITY                                                                                                                                                                  | 1 YEAR MORTALITY |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| <b>CLASS I</b>   | No limitation. Ordinary physical activity does not cause undue fatigue, dyspnea or palpitation.                                                                                      | 5 - 10%          |
| <b>CLASS II</b>  | Slight limitation of physical activity. Such patients are comfortable at rest. Ordinary physical activity results in fatigue, palpitation, dyspnea or angina.                        | 10 - 15%         |
| <b>CLASS III</b> | Marked limitation of physical activity. Although patients are comfortable at rest, less than ordinary activity will lead to symptoms.                                                | 15 - 20%         |
| <b>CLASS IV</b>  | Inability to carry on any physical activity without discomfort. Symptoms of congestive failure are present at rest. With any physical activity, increased discomfort is experienced. | 20 - 50%         |

**Key Message #3: Classification & Stages of HF**

- HF can be classified according to:
  - **Clinical Presentation:**
    - ◆ Acute heart failure (Acute HF)
    - ◆ Chronic heart failure (Chronic HF)
  - **Left ventricular ejection fraction (LVEF):**
    - ◆ **Reduced (LVEF ≤ 40%)** - Heart failure with reduced ejection function (HFrEF).
    - ◆ **Mildly reduced (LVEF 41%-49%)** - Heart Failure with the LVEF being in the mildly reduced range (HFmrEF)
    - ◆ **Preserved (LVEF > 50%)** - Heart failure with preserved ejection fraction (HFpEF)
    - ◆ **Improved Ejection Fraction** - HF with a baseline LVEF of < 40%, a >10-point increase from baseline LVEF following treatment, and a second measurement of LVEF of >40%. (HFimpEF)
  - **Stages of HF are:**
    - ◆ **A** - "At Risk"
    - ◆ **B** - "Pre HF"
    - ◆ **C** - "Symptomatic HF with previous or current symptoms"
    - ◆ **D** - "Advanced HF"
- The **severity of symptomatic HF** and the **exercise capacity** of the patient can be assessed using the New York Heart Association (NYHA) functional Class (Table 9, page 50).

**4. PATHOPHYSIOLOGY**

The **main pathophysiology** of HF is the **inability of the heart to provide sufficient cardiac output to meet the perfusion and oxygenation requirements of the body while maintaining normal filling pressures**. This may be due to either **impaired cardiac contractility or impaired relaxation and compliance**. Both mechanisms may co-exist in the same patient and depending on the predominant pathophysiology, the patient may present as HFrEF or HFpEF.

This will result in the following compensatory mechanisms:

- A higher ventricular end diastolic pressure - This is a compensatory mechanism to increase stroke volume by the Frank Starling mechanism.
- Increasing ventricular volume and wall thickness through ventricular remodelling.
- Neuro-hormonal activation of the:
  - Sympathetic nervous system
  - Renin-angiotensin-aldosterone system
  - Vasopressin

This neuro-hormonal activation is aimed at increasing stroke volume and cardiac output by:

- An increase in heart rate and ventricular contraction.
- Vasoconstriction of arterial resistance vessels to maintain blood pressure.
- Venous constriction to increase venous preload.
- Salt and water retention to increase preload.

In general, these neuro-hormonal responses are compensatory mechanisms. However they can also aggravate HF by increasing ventricular afterload and increasing preload to the point where pulmonary and/or systemic congestion and edema occur.



#### 4.1 HFrEF

In HFrEF, cardiac output is reduced due to depressed myocardial contractility, irrespective of the etiology. This leads to a cascade of pathophysiological changes as outlined above. There are effective medical and device therapies that have been shown to have a survival benefit in HFrEF. In this group of patients, their LVEF can remain as < 40% or improve to  $\geq 40\%$  when they are reclassified as Heart Failure with Improved Ejection Fraction (HFimpEF).

#### 4.2 HFpEF

In HFpEF there is impaired left ventricular (LV) filling due to decreased relaxation (during early diastole) and/or reduced compliance (early to late diastole) leading to elevated LV filling pressures at rest or during exercise. These hemodynamic changes are accompanied predominantly by signs of pulmonary and/or venous congestion and occasionally systemic hypoperfusion as well.

Clinical studies seem to indicate that HFrEF and HFpEF are mechanistically distinct pathophysiological entities.<sup>36</sup> The transition from HFpEF to HFrEF is rare.<sup>37</sup>

HFpEF appears to be a multi-organ, systemic syndrome with heterogeneous clinical manifestations and high comorbidity burden.<sup>36</sup>

There is limited data available on therapies that improve survival in HFpEF unlike those with HFrEF. In this group of patients, their LVEF can deteriorate to between 41 to 49% (HFmrEF) or rarely < 40% (HFrEF).

#### 4.3 HFmrEF

The prevalence of HFmrEF is between 10-25%.<sup>38</sup> (Table 10, page 53). It is a clinical entity between HFrEF and HFpEF. With regards to etiology, it is more similar to HFrEF (high prevalence of CAD) but in terms of prognosis, they are more like HFpEF in terms of survival rates, low risk of CV events but a higher risk of non-CV adverse events than those with HFrEF.<sup>36,38,39</sup>

**HFmrEF represents a transitional phase** and the LVEF can:

- Remain in the 41 to 49% range (HFmrEF) or
- Worsen to  $\leq 40\%$  (HFrEF) or
- Improve to  $\geq 50\%$  (HFimpEF).

Patients with HFmrEF had the best outcomes, compared to the high rates of mortality seen in patients with HFrEF and the high rates of HF readmissions seen in patients with HFpEF.<sup>6</sup> Only 1/3 of patients with HFmrEF transitioned during follow up, with the lowest mortality seen in patients transitioning to HFpEF.<sup>40</sup>

Clinical trials on therapies for HFpEF that enrolled patients with an LVEF of >40% or  $\geq 45\%$  did not demonstrate a clear treatment effect. Post hoc analysis however suggests that some therapies for HFrEF might also be effective in HFmrEF.

#### 4.4 HFimpEF

In HF with improved EF (HFimpEF), the baseline LVEF is  $\leq 40\%$  but either spontaneously or following treatment, there is a  $\geq 10\%$  increase from baseline LVEF, and a second measurement of LVEF is >40%.<sup>35</sup> These patients have resolution of symptoms and signs of



HF. "HF in remission" is a term that is sometimes used but **the HF is known to frequently relapse especially after withdrawal of pharmacological treatment.**<sup>41</sup>

In a large meta-analysis, the prevalence of **HFimpEF** was 22.64%.<sup>42</sup> They had a lower mortality and risk of hospitalization compared **HFrEF** and **HFpEF**.<sup>42</sup>

#### Key Message #4: Pathophysiology

- The main pathophysiology of HF is the inability of the heart to provide sufficient cardiac output to meet the perfusion and oxygenation requirements of the body while maintaining normal filling pressures.
- In **HFrEF**, cardiac output is reduced due to depressed myocardial contractility, irrespective of the etiology.
- In **HFpEF** there is impaired left ventricular (LV) filling due to decreased relaxation (during early diastole) and/or reduced compliance (early to late diastole) leading to elevated LV filling pressures at rest or during exercise and dyspnea.
- **HFmrEF** is a clinical entity between **HFrEF** and **HFpEF**. With regards to etiology, it is more like **HFrEF** (high prevalence of CAD) but in terms of prognosis, it is more like **HFpEF**.
- In **HFimpEF**, patients have resolution of symptoms and signs of HF but HF is known to frequently relapse especially after withdrawal of pharmacological treatment. They have a lower mortality and risk of hospitalization compared **HFrEF** and **HFpEF**.

**Table 10: Prevalence of HFrEF, HFpEF and HFmrEF in Registries**

|                | ESCHF-LT <sup>43</sup><br>HF <sup>44</sup>                                                      | GWG-<br>HF <sup>45</sup>                                                       | Swedish<br>Registry <sup>5</sup>                                                             | ASIAN HF                                                                                                                                                                                     | MYHF <sup>13</sup>                                                                |
|----------------|-------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Source of Data | Prospective Observational study of Outpatients and hospitalized patients April 2011 to Jan 2015 | Prospective Observational study of hospitalized between Jan 2005 and Sept 2013 | Prospective Observational study of Outpatients and hospitalized patients between 2000 - 2018 | Prospective longitudinal study of outpatient (at least 1 episode of decompensated HF in the previous 6 months) and hospitalized patients 1 <sup>st</sup> Oct 2012- 31 <sup>st</sup> Oct 2016 | Prospective observational study on hospitalized patients from Aug 2019 - Dec 2020 |
| No             | 16,354                                                                                          | 99,825                                                                         | 75,518                                                                                       | 6,480                                                                                                                                                                                        | 2,673                                                                             |
| HFrEF          | 59.8%                                                                                           | 49.0%                                                                          | 52.5%                                                                                        | 81.0%                                                                                                                                                                                        | 64.6%                                                                             |
| HFmrEF         | 24.2%                                                                                           | 12.8%                                                                          | 23.5%                                                                                        | NA                                                                                                                                                                                           | 11.3%                                                                             |
| HFpEF          | 16.0%                                                                                           | 38.1%                                                                          | 24.1%                                                                                        | 19.0%                                                                                                                                                                                        | 21.6%                                                                             |

**5. ETIOLOGY**

HF is not a complete diagnosis. **It is important to identify the underlying disease and the precipitating cause(s), if present**, so that disease-specific treatment can be initiated early.

The common underlying causes of HF in adults are:

- Coronary artery disease (CAD)
- Hypertension
- Dilated cardiomyopathy-idiopathic, familial
- Valvular heart disease
- Diabetic cardiomyopathy

The commonest cause in Malaysia is CAD accounting for almost 60-66% of HF.<sup>14,15,17</sup> Comorbidities are common and include hypertension, diabetes, and dyslipidemia.<sup>14,15,17</sup> Valvular Heart Disease, especially rheumatic heart disease, is an important cause among the rural and urban poor.

Other causes of HF include:

- Congenital heart disease
- Cor pulmonale
- Pericardial disease: constrictive pericarditis, cardiac tamponade
- Hypertrophic cardiomyopathy
- Viral myocarditis
- Acute rheumatic fever
- Toxic:
  - Alcohol
  - Cardiotoxic chemotherapy e.g., doxorubicin, cyclophosphamide, trastuzumab (Herceptin), immune checkpoint inhibitors,
  - Radiotherapy
  - Medications such as clozapine
  - Substance abuse: methamphetamine and cocaine
- Endocrine and metabolic disorders: thyroid disease, acromegaly, pheochromocytoma.
- Collagen vascular disease: systemic lupus erythematosus, polymyositis, polyarteritis nodosa.
- Tachycardia induced cardiomyopathy e.g., uncontrolled atrial fibrillation.
- Infiltrative cardiac disease e.g., amyloid, hyper-eosinophilic syndrome, Sarcoid.
- Neuromuscular Disorders: Friedreich's Ataxia, Muscular Dystrophy.
- Storage Disorders: Haemochromatosis, Glycogen Storage Disease, Fabry disease.
- Endomyocardial disease: Carcinoid, Endomyocardial Fibrosis.
- Miscellaneous
  - High output HF e.g., severe anemia, large A-V shunts/malformations.
  - Peripartum cardiomyopathy.
  - Stress (Takotsubo) cardiomyopathy.



**Patients with Chronic HF may occasionally develop acute decompensation.** Factors that can contribute to this acute decompensation are listed in Table 11, Page 55. The more important causes that need to be recognized and treated appropriately are as in the acronym:

**C H A M P I O N**

- C** - Coronary artery disease - Acute myocardial infarction/myocardial ischemia
- H** - Hypertensive emergencies
- A** - Arrhythmias (e.g., atrial fibrillation)
- M** - Mechanical causes such as acute valve regurgitation
- P** - Pulmonary - e.g., pulmonary embolism, pneumonia
- I** - Infections - e.g., urinary tract infection, Covid
- O** - Other medications such as NSAIDs and COX-2 inhibitors,
- N** - Non-compliance to medications

**Key message #5: Etiology**

- The common underlying causes of HF in adults are:
  - Coronary artery disease (CAD) - accounting for almost 60-66% of HF in Malaysia
  - Hypertension
  - Dilated cardiomyopathy-idiopathic, familial.
  - Valvular heart disease
  - Diabetic cardiomyopathy
- Comorbidities are common and include hypertension, diabetes, and dyslipidemia.

**Table 11: Factors Contributing to Decompensation in a Patient with Stable HF**

**Patient factors**

- Non-compliance to medications
- Dietary indiscretion especially salt and fluid intake
- Inappropriate medications e.g., NSAIDs and COX-2 inhibitors
- Alcohol consumption

**Cardiac causes**

- Superimposed myocardial ischemia or infarction (often asymptomatic)
- Hypertensive emergencies
- Arrhythmias
- Pulmonary embolism
- Secondary mitral or tricuspid regurgitation

**Systemic conditions**

- Superimposed infections
- Anemia
- Thyroid disease
- Electrolyte disturbances
- Worsening renal disease

**Others**

- Urinary retention
- Severe emotional or physical stress
- Medications with negative inotropic effect, e.g., Verapamil

**6. DIAGNOSIS**

I, C

**The diagnosis of HF depends on a detailed history focusing on symptoms, risk factors and a thorough physical examination.** Patients with HF may present differently but tend to be consistent in their presentation whenever they develop a recurrent episode.

**6.1. Symptoms and Signs**

I, C

**The clinical suspicion of HF should be supported by objective evidence of cardiac dysfunction.** Breathlessness with orthopnea, paroxysmal nocturnal dyspnea (PND), reduced exercise tolerance and ankle swelling are the characteristic symptoms of HF. Orthopnea and paroxysmal nocturnal dyspnea are markers of high filling pressures and contribute to sleep-disordered breathing.

Signs which are more specific for HF are an elevated jugular venous pulse (JVP), and a third heart sound. These signs are associated with adverse outcomes in patients with HF and asymptomatic LV dysfunction.<sup>46,47</sup> The presence of a raised JVP, a positive jugulo-venous reflux and hepatomegaly generally indicate a raised right atrial pressure of > 8 mmHg.<sup>48</sup> A raised JVP has a good sensitivity (70%) and specificity (79%) of left sided congestion.<sup>48</sup>

A fourth heart sound is due to atrial contraction and is more frequent in patients with HFpEF. It is absent in patients with atrial fibrillation (AF).

These signs may be accompanied by a laterally displaced apical impulse and a cardiac murmur. Other supportive signs include peripheral edema, tachycardia, narrow pulse pressure, pulmonary crepitations, hepatomegaly and ascites. These clinical findings may be transient and resolve completely following initial therapy.

However, these signs are difficult to detect and are not always easily reproducible in the elderly, the obese and in patients with chronic lung disease. In a small study on patients with chronic HF, signs of congestion (rales, edema and JVP elevation) were absent in 42% of patients with a PCWP  $\geq$  22 mmHg.<sup>49</sup> Occasionally symptoms and signs of volume overload may be absent, and the patient may present with fatigue only.

In the elderly, as exertional dyspnea increases, they begin to lead more sedentary lives. Fatigue is common and this is likely due to low cardiac output, peripheral hypo-perfusion and skeletal muscle deconditioning and should not be considered due to ageing only.<sup>50</sup> Atypical symptoms, such as confusion, memory deficit, sleepiness, episodes of delirium, irritability, syncopal states, fatigue, anorexia, and reduced level of activity are common manifestations of HF especially after age 80.<sup>50</sup> Other co-existing conditions such as infections, anemia, delirium etc may complicate the clinical picture making the diagnosis of HF difficult.<sup>51</sup> Bilateral leg edema is common in the elderly and may be due to other causes such as venous insufficiency, chronic kidney disease or lymphedema besides HF.<sup>51</sup>

In patients presenting with dyspnea, acute LV failure can sometimes mimic an acute exacerbation of bronchial asthma or chronic obstructive pulmonary disease. Thus, a proper history and clinical examination is essential. Similarly, in patients with CKD presenting with dyspnea, a concomitant diagnosis of HF may be missed.

Exercise capacity in a patient with heart disease is assessed by the New York Heart Association (NYHA) functional classification. (Table 9, page 50)



I, A

The clinical suspicion of HF should be supported objectively with either raised natriuretic peptides and/or evidence of pulmonary or systemic congestion.<sup>35</sup> A value of NT-Pro BNP > 125pg/ml or BNP >35pg/ml is used as a cut off value for the diagnosis of HF in the non-acute setting.<sup>35</sup> (Table 12, page 60)

Objective evidence of cardiac dysfunction by diagnostic tests will further support the diagnosis of HF. It will help establish the etiology and the precipitating factors of the syndrome.

**Key message # 5:**

- In making a diagnosis of HF, a detailed history and a thorough physical examination is important.
- The clinical suspicion of HF should be supported objectively with either raised natriuretic peptides and/or evidence of pulmonary or systemic congestion.



## 6.2 Investigations (Flowchart 1, page 32)

| BASIC INVESTIGATIONS                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>12 Lead Electrocardiogram</b><br><b>Chest Radiograph</b>                                                                                                            | <ul style="list-style-type: none"> <li>- To assess heart rate, rhythm, QRS morphology, QRS duration, QRS voltage, evidence of ischemia, LV hypertrophy and arrhythmias.</li> <li>- To look for pulmonary congestion, cardiomegaly, and presence of underlying lung pathology.</li> <li>- Patients with HFpEF may have a normal cardiac size.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Blood Tests</b>                                                                                                                                                     | Full Blood Count, urea, creatinine, serum electrolytes (sodium and potassium) liver function, serum glucose, lipid profile.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Urinalysis</b>                                                                                                                                                      | To look for proteinuria.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| OTHER IMPORTANT INVESTIGATIONS                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Echocardiography</b>                                                                                                                                                | <p>This will allow assessment of:</p> <ul style="list-style-type: none"> <li>● LV chamber size, volume, and systolic function.</li> <li>● LV wall thickness, evidence of scarring and wall motion abnormalities.</li> <li>● Diastolic function of the heart.</li> <li>● Valvular structure and function.</li> <li>● Congenital cardiac abnormalities.</li> <li>● LV mechanical dyssynchrony.</li> <li>● Pulmonary hypertension.</li> </ul> <p>It is the most useful and widely available test to establish the diagnosis in patients suspected of HF.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Natriuretic peptides in (NP):</b> <ul style="list-style-type: none"> <li>● Brain Natriuretic Peptide (BNP) or</li> <li>● N-Terminal Pro BNP (NT-Pro BNP)</li> </ul> | <p>BNP and NT-Pro BNP are a family of hormones secreted by the ventricles response to wall stress.</p> <p>They are useful in the following situations:</p> <p><b>DIAGNOSIS OF HF</b></p> <p><b>A) <u>De Novo HF-1st episode of HF</u></b></p> <p><b>A1. Emergency Setting</b></p> <ul style="list-style-type: none"> <li>● NPs are useful as a 'rule out' test for patients presenting with acute dyspnea. A level of &lt; 100pg/ml for BNP and &lt; 300pg/ml for NT-Pro BNP makes the diagnosis of acute HF unlikely.<sup>52-56</sup></li> <li>● There is no statistical difference between the diagnostic accuracy of plasma B type natriuretic peptide (BNP) and NT-Pro BNP.<sup>55</sup></li> <li>● These levels are affected by renal function and gender.<sup>55-57</sup> (See Table 12, Page 60 for the optimal cut off values of NP to exclude or diagnose HF in patients with dyspnea.)</li> <li>● A high level supports the diagnosis of acute HF and very high levels correlate with the severity of HF and adverse outcomes.<sup>52-57</sup></li> <li>● The values of both BNP and NT-Pro BNP are predictive of both in-hospital and 1-year mortality, irrespective of LVEF.<sup>55-60</sup></li> <li>● The discharge value as well as percent change from admission to discharge were more predictive of subsequent mortality and/or rehospitalization than admission values.<sup>61</sup> A decrease by &gt; 30% at discharge indicates a better prognosis than when values fail to decrease or actually increase.<sup>62</sup></li> </ul> |

**A2. In the community setting (Primary Care):**

- They are a useful “rule out” test in the diagnosis of HF in patients presenting with dyspnea.<sup>57,63,64</sup> A point-of-care test allows early diagnosis.
- Changes in the levels of BNP and NT-Pro BNP predict risk of hospital admissions for HF.<sup>57,65</sup>
- A decline in NT-ProBNP over time suggests favourable LV remodeand improved clinical outcomes.<sup>66,67</sup>

**B) Acute Decompensation in patients with Chronic HF**

- In patients with chronic HF, NPs may be persistently elevated and show substantial biological variation.<sup>68</sup>
- When these patients present with acute symptoms, a change of 100% or more from the stable concentration suggests decompensation.<sup>57</sup>
- A combination of symptoms, weight gain and NP concentration may, however, be the best way to diagnose early decompensation in these patients.

NP levels are affected by:

- Atrial fibrillation (AF)<sup>69,70</sup> - levels are increased even in the absence of HF.
- Age<sup>71,72</sup> - Levels of NP increase with age.
- Renal function.<sup>57,73,74</sup>
- Obesity<sup>75,76</sup> - Levels are reduced in obesity.
- Certain medications such as Angiotensin Receptor Neprilysin Inhibitor (ARNI) may interfere with the interpretation of BNP levels.

In cases of suspected HF, a point-of-care NP test allows early confirmation of the diagnosis. However:

- A raised NP level may be due to other causes besides HF. (Appendix I, Page 161)
- For this reason, NPs should only be used in conjunction with careful clinical assessment and investigation and interpreted with all other clinical information.

**GUIDING THERAPY:**

- The results of studies on the use of NP to guide therapy in HF are conflicting.<sup>57,77-80</sup>
- A value of NT-Pro BNP of  $\leq 1,000$  pg/ml during treatment was associated with better CV outcomes.<sup>81</sup>



| ADDITIONAL INVESTIGATIONS WHEN INDICATED:             |                                                                                                                                                                                                                                                                                                                      |
|-------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Blood Tests</b>                                    | <ul style="list-style-type: none"><li>• Serum cardiac biomarkers: to look for myocardial necrosis - troponins, creatine kinase - muscle/brain band (CKMB).</li><li>• Iron studies (ferritin, serum iron, transferrin saturation).</li><li>• Serum calcium and magnesium.</li><li>• Thyroid function tests.</li></ul> |
|                                                       | Other less common tests that may be considered include: <ul style="list-style-type: none"><li>• Gamma-glutamyl transferase (GGT)</li><li>• Viral studies</li></ul>                                                                                                                                                   |
| <b>Tests for Myocardial Ischemia and/or Viability</b> | <ul style="list-style-type: none"><li>• Treadmill exercise test</li><li>• Stress echocardiography (exercise or pharmacological)</li><li>• Radionuclide studies</li><li>• Cardiac magnetic resonance imaging (Cardiac MRI)</li></ul>                                                                                  |
| <b>Invasive Tests</b>                                 | <ul style="list-style-type: none"><li>• Coronary angiography</li><li>• Cardiac catheterisation</li><li>• Endomyocardial biopsy</li></ul>                                                                                                                                                                             |
| <b>Others</b>                                         | <ul style="list-style-type: none"><li>• Holter electrocardiography, loop recorders</li><li>• Pulmonary function test</li></ul>                                                                                                                                                                                       |

**Table 12: Optimal Natriuretic Peptide Cut Points for Diagnosis or Exclusion of Heart Failure among Patients with Dyspnea.** <sup>56,57,64</sup>

|                                         | BNP (ng/L) | NT-Pro BNP (ng/L)  |
|-----------------------------------------|------------|--------------------|
| <b>Emergency Setting</b>                |            |                    |
| Heart Failure Rule Out                  | < 100      | < 300              |
| Heart Failure Possible                  | > 400      | Age < 50 y: > 450  |
|                                         |            | Age 50 - 75: > 900 |
|                                         |            | Age > 75: > 1800   |
| <b>Community Setting (Primary Care)</b> |            |                    |
| Heart Failure Rule Out                  | < 35       | < 125              |

**Key Recommendations # 1: Diagnosis & Essential Investigations**

- In making a diagnosis of HF, a detailed history and a thorough physical examination are important.
- The clinical suspicion of HF should be supported objectively by either raised natriuretic peptides and/or evidence of pulmonary or systemic congestion.
- Important basic investigations:
  - Echocardiogram.
  - ECG.
  - Chest Radiograph.
  - Blood - Full Blood Count, urea, creatinine, serum electrolytes (sodium and potassium) liver function, serum glucose, lipid profile.
  - Natriuretic Peptides: NT-Pro BNP or BNP
    - ◆ They are a useful “rule out” test in the diagnosis of HF in patients presenting with acute dyspnea.
    - ◆ In chronic HF, NPs may be persistently elevated and show substantial biological variation. In this setting, NPs should be interpreted with careful clinical assessment and investigation and with all other clinical information.
    - ◆ A value of NT-Pro BNP of  $\leq 1,000$  pg/ml during treatment was associated with better CV outcomes.

**7. PREVENTION****FOUNDATION OF MANAGEMENT OF HEART FAILURE**

|                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Stage A<br/>“At Risk”</b> | <b>Individuals who do not have structural cardiac disease but are at high risk of developing HF/CAD.</b><br>Hypertension<br>Diabetes<br>Obesity and the Metabolic Syndrome<br>Cigarette smoking<br>Familial hypercholesterolemia<br>Multiple CV risk factors<br>Family history of Cardiomyopathy<br>Thyroid Disease<br>Renal Disease<br>Cardiotoxins<br><i>Others:</i><br>Sleep Disordered breathing,<br>Connective Tissue Disease,<br>Chronic Pulmonary disease and Pulmonary Hypertension |
| <b>Stage B<br/>“Pre HF”</b>  | <b>Individuals with structural cardiac disease and who either have normal or impaired myocardial function.</b><br>Established CAD<br>Hypertension with LVH<br>Cardiac Valve disease<br>Congenital Heart disease<br>Arrhythmias                                                                                                                                                                                                                                                              |



**Prevention of HF should always be the primary objective and the foundation of management.** It should focus on those who are in:

- **Stage A - “At Risk”** - Individuals who are at high risk of developing cardiac disease but who do not have symptoms or signs of HF and have structurally normal hearts.
- **Stage B - “Pre HF”** - Individuals with structural cardiac disease and who have either normal or impaired cardiac function but do not have signs and symptoms of HF.

There is robust clinical data that by appropriate and timely interventions, HF can be prevented in these individuals and cardiac function improved.

### 7.1 STAGE A - “AT RISK”

Individuals who are at high risk of developing HF/CAD but who do not yet have structural heart disease. These include individuals with:

- **Hypertension -**
  - This is an important risk factor for the development of HF especially in the elderly.
  - In the Framingham Heart Study, 91% of the participants with HF had a previous diagnosis of hypertension.<sup>82</sup> In multivariable analysis, the population-attributable risk of HF in the Framingham population was 39% and 59% for male and females respectively.<sup>82</sup>
  - In a meta-analysis of 23 hypertension trials, the risk of development of HF was comparable to that of stroke: 8.5 and 9.1 events per 1000 patients respectively.<sup>83</sup>
- **Diabetes -**
  - This is a risk factor for the development of HF independent of the presence of coexisting hypertension or CAD and after adjusting for other CV risk factors such as age, hyperlipidemia, and smoking.<sup>84</sup>
  - In the Framingham population cohort, the presence of diabetes increased the risk of HF by 2x in men and 5x in women.<sup>85</sup>
  - The prevalence of HF in individuals with diabetes is 4x higher than that of the general population.<sup>86</sup>
- **Obesity and Metabolic Syndrome -**
  - In the Framingham Heart Study, obese individuals had a doubling of the risk of HF when compared to those with a normal body-mass index.<sup>85</sup> There was a continuous gradient of the risk of developing HF with increasing body-mass index in both sexes which persisted even after adjusting for other CV risk factors, presence of Left Ventricular Hypertrophy and CAD.<sup>87</sup>
  - A meta-analysis showed that there was a 41% increase in HF per 5-unit increment in BMI and a threshold of risk at 23-24 kg/m<sup>2</sup>.<sup>88</sup>
  - HFpEF is more common than HFrEF in obese individuals. Those who are overweight and with class 1 obesity have 38% and 56% higher risk of HFpEF, respectively, independent of other CV risk factors.<sup>89,90</sup>
  - However, there exists an obesity paradox in HF - Patients with overweight or class 1 obesity have better clinical outcomes than patients with normal weight for similar degrees of HF.<sup>90</sup>
  - Metabolic Syndrome - After adjusting for other risk factors for HF, the presence of metabolic syndrome at baseline was also a strong predictor of the development of subsequent HF.<sup>91,92</sup>



- **Cigarette Smoking -**

- This is a risk factor for HF independent of traditional CV risk factors.
- Current smoking, smoking  $\geq 20$  cigarettes/day and smoking burden ( $\geq 15$  pack-years) were significantly associated with incident HF hospitalization in comparison with never smoking.<sup>93</sup>
- In older adults, both current and past cigarette smoking was shown to increase HF risk after adjusting for other HF risk factors.<sup>94</sup>
- Smokers had 2x the risk of developing HF<sub>rEF</sub> and HF<sub>pEF</sub> compared to never / nonsmokers after a median of 13 years of follow up.<sup>95</sup>

- **Familial Hypercholesterolemia -**

- This leads to premature CAD and subsequent HF.<sup>96</sup>

- **Multiple CV Risk Factors** or who already have evidence of atherosclerotic disease in other vascular beds (e.g., cerebral, peripheral vascular disease).

- **Family History of Cardiomyopathy -**

- These usually manifest in younger individuals as HF, arrhythmias, or syncope.
- A family history of premature cardiomyopathy death was associated with an increase in the risk of developing cardiomyopathy ranging from 6 - to 400-fold, depending on age, kinship, gender, and number of affected family members.<sup>97</sup>
- In the Framingham Offspring Study, parental HF was connected to asymptomatic left ventricular dysfunction and increased the likelihood of overt HF in the offspring.<sup>98</sup>

- **Thyroid Disorders -**

- Both hyper and hypothyroidism increase the risk of HF.<sup>99</sup>
- A meta-analysis showed that the risk of HF is increased with a TSH  $< 0.10$  mIU/L and  $> 10.0$  mIU/L among older adults.<sup>99-101</sup> The risk of a low-output HF in hyperthyroid patients has been reported to be between 6 and 15%.<sup>99</sup>

- **Kidney Disease -**

- Microalbuminuria is a strong and independent indicator of increased risk of CV events and death among individuals with and without diabetes, with stable CAD and those with HF.<sup>102-104</sup> It is a CV risk marker - reflecting subclinical vascular damage in the kidneys and other vascular beds.<sup>105</sup>
- There is a close relationship between the heart and the kidneys, each condition causes or exacerbates the other, compounded by the accompanying anemia.<sup>106</sup>
- The incidence of de novo HF in known CKD is in the range of 17% to 21% and for post-kidney transplant it is 18% at 3 years.<sup>107</sup>
- The prevalence of HF increases greatly as the patient's renal function deteriorates, and, at end-stage kidney disease, can reach 65-70%.<sup>108</sup>

- **Cardiotoxins -**

- Alcohol - Heavy drinking has been shown to increase the risk of HF, whereas light-to-moderate drinking (up to 1 drink per day for women and up to 2 drinks per day for men) has been associated with a lower risk of HF.<sup>108</sup>
- Chemotherapeutic agents.
- Substance abuse such as cocaine, amphetamine, antidepressants.

- **Healthy Sleep Pattern -**

- In a prospective study, adherence to a healthy sleep pattern was associated with a lower risk of HF, independent of the conventional risk factors.<sup>109</sup> Healthy sleep factors were defined as early chronotype ("morning" person or "morning" than "evening" person); sleeping 7 to 8 hours/day; never/rarely or sometimes insomnia symptoms; no snoring; and no excessive daytime sleepiness.



- The National Sleep Foundation recommends 7 to 9 hours for young adults and adults, and 7 to 8 hours of sleep for older adults.<sup>110</sup>
- Sleep-disordered breathing includes Obstructive Sleep Apnea (OSA) and Central Sleep Apnea (CSA) with Cheyne-Stokes respiration.
- The relationship between sleep disordered breathing and HF is complex. It may just be an association rather than a cause-and-effect relationship.<sup>111,112</sup>
- It is a frequent comorbidity in HF, the prevalence ranging from 11% to as high as 40-50% in symptomatic HF.<sup>111,112</sup>
- Connective tissue diseases such as rheumatoid arthritis, SLE -
  - There is a high prevalence of atherosclerotic coronary disease in these systemic inflammatory conditions.<sup>113</sup> In addition, HF can occur secondary to chronic myocardial and pericardial inflammation, valvular disease and/or pulmonary hypertension.
  - Chronic pulmonary disease with pulmonary hypertension- These lead to right HF which in turn also affects left heart function and causes a complex clinical syndrome affecting multiple organ systems.<sup>114</sup>

### 7.1.1 Preventive Strategies in Stage A

#### ● Treating Hypertension to Target -

**I,A**

- Reducing systolic BP to 140-145mmHg, has been shown in several trials in older patients with systolic hypertension to reduce new onset HF from 36-64%.<sup>115-118</sup>

**IIa,B**

- Further lowering of the target SBP from  $\leq 140$  to  $\leq 120$  in non-diabetic hypertensive individuals who were age  $\geq 50$  years and at high CV risk was also shown to reduce the development of HF by as much as 37%.<sup>119</sup>

#### ● Diabetes -

**I,A**

- Treat A1c to target - Poor glycaemic control increases the risk of HF, each 1% increase in A1c increases the risk by 8-32%.<sup>120,121</sup>

**I,A**

- In patients at high CV risk and/or with established CVD, sodium-glucose cotransport-2 inhibitors (SGLT2-i) have been shown to reduce hospitalization for HF and improve CV outcomes.<sup>122-127</sup>

#### ● Smoking Cessation -

**I,B**

- Quitting smoking appears to have a substantial and early effect (within two years) on decreasing morbidity and mortality in patients with left ventricular dysfunction mainly due to a reduction in CV events.<sup>128</sup>
- After > 15 years of smoking cessation, the risk of HF and death for most former smokers becomes like that of never-smokers.<sup>129</sup>

#### ● Familial Hypercholesterolemia -

**I,A**

- Treating lipid levels to target reduces risk of MI and thus prevents the development of HF.<sup>130,131</sup>

#### ● Multiple CV Risk Factors -

**I,A**

- Optimal profiles in the 7 CV risk factors -smoking, body mass index, physical activity, diet, cholesterol, blood pressure, and glucose- are associated with a lower lifetime risk of HF.<sup>132,133</sup>

#### ● Family History of Cardiomyopathy -

- Screening and early treatment of first-degree relatives of patients with known heritable cardiomyopathy.



I,C

**• Thyroid Disease -**

- Detecting and treating thyroid disease early especially in older adults to prevent thyroid heart disease.

I,C

**• Kidney Disease -**

- Early detection and appropriate management of kidney disease and the coexisting anemia. In adult CKD patients, however, KDIGO has advised that the Hb be maintained < 11.5 gm/100ml.<sup>134</sup> These levels are associated with lower mortality and less frequent hospitalization rates.<sup>134</sup>
- Preservation of kidney function by avoiding nephrotoxic agents such as non-steroidal anti-inflammatory agents and the use of proven kidney-protective agents.

**• Alcohol -**

- In the absence of large, randomized trials, light-to-moderate drinking cannot be recommended to lower the risk of HF. Advise abstinence.

I,C

**• Chemotherapy -**

- Identifying and monitoring at risk individuals prior to administration of cardiotoxic chemotherapy.

III,B

**• Sleep Disordered Breathing -**

- To date, the use of servo-ventilation and/or Continuous Positive Airway Pressure (CPAP) for central and /or obstructive sleep apnea (OSA) has not been shown to prevent HF or to improve CV outcomes.<sup>135-140</sup>

**• Others:**

I,A

- Regular exercise - A minimum physical activity of at least 150 minutes per week of moderate intensity activity has been recommended to reduce CV risk and prevent ischemic heart disease.<sup>89,141</sup>

I,A

- Maintain ideal body weight - Maintenance of normal weight and physical activity during adult life have been associated with lower incidence of HF.<sup>89,141,142</sup>
- n-3 Polyunsaturated Fatty Acids (PUFA) supplements -

- ♦ To date, there have been no primary prevention trials on the use of these supplements to prevent cardiac disease.<sup>143</sup>

IIb,B

- ♦ The cumulative findings from RCTs on the prevention of HF and or CV events in patients at high CV risk and or established CVD have been mixed.<sup>143</sup>

**7.2. STAGE B - “Pre HF”**

Individuals with structural cardiac disease but who do not have signs and symptoms of HF. Cardiac function may be normal or impaired. The following strategies have been shown to prevent the development of HF.

**7.2.1. In the presence of normal cardiac function**

I,A

- Timely triage and appropriate treatment of patients with acute coronary syndromes.<sup>144,145</sup>

I,A

- Patients with CAD should be treated appropriately with guideline directed medical therapy (antiplatelet agents, statins,  $\beta$ -blockers, and Renin Angiotensin System blockers) and intervention as necessary.<sup>146</sup>

I,A

- Patients with hypertension and left ventricular hypertrophy (LVH) should have their blood pressure control optimized. Regression of LVH has been shown to be associated with a lower incidence of new onset HF.<sup>147,148</sup>

I,C

- Patients with significant valve disease (moderate and above) should be assessed for



progression and undergo timely intervention as indicated.

- Patients with arrhythmias, when indicated, should be referred for evaluation and treatment.
- Patients with congenital heart disease should have their cardiac lesions corrected with appropriate follow-up looking for progression and sequelae.

I,C

### 7.2.2. In the presence of impaired LV function (LVEF < 40%) irrespective of symptoms:

I,A

- Treat the underlying cause wherever possible.
- The evidence for the benefits of pharmacotherapy in patients with **asymptomatic LV dysfunction in NYHA Class I** has been mainly derived from studies conducted in patients with LVEF < 40% with  $\beta$ -blockers and ACE-I.<sup>149-152</sup>
- SGLT2-i reduce hospitalization for HF in patients with and without established CVD, irrespective of diabetes status.<sup>127,128,153,154</sup>
- In **symptomatic patients in NYHA Class II-IV**- All the 4 Foundational HF medications (RAS blockers, MRA,  $\beta$ -blockers and SGLT2-i) improve CV outcomes.<sup>155,156</sup>

I,A

I,A

#### Key message #6: Prevention of HF

- Prevention and early intervention, wherever appropriate, should be the primary objective of management of patients at risk for HF.
- There is robust clinical data that by appropriate and timely interventions, HF can be prevented in these individuals and cardiac function improved.
- It should focus on those who are in:
  - **Stage A - "At Risk"** - Individuals who are at high risk of developing cardiac disease but who do not have symptoms or signs of HF and have structurally normal hearts.
  - **Stage B - "Pre HF"** - Individuals with structural cardiac disease and who have either normal or impaired cardiac function but do not have signs and symptoms of HF.

#### Key Recommendations #2: Prevention of HF

- In the prevention of HF for patients in **Stage A**:
  - All CV risk factors should be treated to target.
  - Optimal profiles of the 7 CV risk factors - blood pressure, glucose, cholesterol, absence of smoking, body mass index and physical activity has been shown to be associated with a lower lifetime risk of HF.
- In the prevention of HF for patients in **Stage B**:
  - Patients with Acute Coronary Syndrome, Hypertension, CAD, arrhythmias, valve, and congenital heart disease should be treated timely and according to the guidelines.
  - In patients who have impaired LV function (LVEF < 40%):
    - ◆ Treat the underlying cause and avoid/treat early precipitating factors.
    - ◆ In **asymptomatic patients in NYHA Class I**: ACE-I,  $\beta$ -blockers, and SGLT2-i.
    - ◆ **Symptomatic patients in NYHA Class II-IV**: All the 4 Foundational HF medications (RAS blockers, MRA,  $\beta$ -blockers and SGLT2-i) should be instituted early to improve CV outcomes.

**8. NON-PHARMACOLOGICAL MEASURES****8.1 Education**

I,B

**HF patients and their family members should be educated on the definition, causes, signs, symptoms, and the progressive and relapsing nature of the disease, emphasizing self-care wherever possible.** Increased awareness helps patients adhere to therapeutic instructions whereas knowledge leads to reduced anxiety and better sense of control.<sup>157</sup> Self-care is important and improves long term adherence to management strategies.<sup>157</sup>

Patients and their family should<sup>158</sup>:

- Be educated on self-care which includes:
  - Maintenance (e.g., taking medication, exercising, and adhering to a healthy diet)
  - Monitoring (e.g., regular weighing) **and**
  - Management (e.g., changing diuretic dose in response to symptoms)
- Recognize the changes in their signs and symptoms - a sudden weight gain - more than 2kg in 3 days is a sign of worsening HF.
- Know when to contact their healthcare provider.
- Understand the indication, dosing, side effects and drug interaction of each medication they are prescribed.
- Be warned about self-medication and potential drug interactions.
- Adhere to treatment and be informed of the potential complications resulting from non-adherence to prescribed medications.
- Be provided with prognostic information to enable them to make realistic long-term decisions and plans. There should be frank end-of-life discussions and advanced care planning with the patient and family. This is important as there is a potential risk of sudden death, and they must put their affairs in order.

In advanced HF, treatment options must be discussed tactfully and realistically with the patient and family.

**Structured telephone support and non-invasive home telemonitoring have been shown to reduce all-cause and cardiac hospitalizations, length of hospital stays and all-cause and cardiac mortality.**<sup>159-161</sup> Patients involved actively in educational process through telemedicine are more likely to accept positive lifestyle changes.<sup>157</sup>

Telemedicine (See also section 17.4)

- Allows more frequent monitoring of the patient's status using smart phones and other mobile devices.
- Has been found to be effective in improving patient's self-care abilities.<sup>159</sup>
- Enables HF patients to be cared for while staying at home, thus reducing hospitalizations.

They are however barriers for the widespread implementation of telemedicine. These include:<sup>162</sup>

- Issues related to patient privacy and data safety although recently, many platforms have been developed to address these issues.



- Costs- it would be more cost effective to identify:
  - The most relevant biological parameters to monitor.
  - The HF sub-populations who may gain from telehealth interventions (e.g., older patients with more comorbidities).
  - In which specific healthcare subsets these interventions should be implemented e.g., immediate post discharge.

### 8.2. Exercise training-see also Section 16 - HF Rehabilitation

**I,B**

Several systematic reviews and meta-analyses support **exercise training as an integral part of the non-pharmacological treatment of HF**.<sup>163-169</sup> Exercise also leads to an improvement in the patient's Health Related Quality of Life (HRQoL) - symptoms, a greater sense of wellbeing, and better functional capacity.<sup>163</sup>

Exercise training:

- Is safe in patients with stable HF.<sup>163-170</sup>
- In patients with HF<sub>r</sub>EF, has been shown to reduce overall and HF-specific hospitalizations.<sup>163-169</sup>
- In patients with HF<sub>p</sub>EF has limited clinical data. Small trials show that exercise training leads to an improvement in exercise capacity and quality of life.<sup>169,170</sup>

Exercise-based rehabilitation:

- Reduced the risk of hospital admissions.<sup>163-169</sup>
- Improved health related quality of life. (HRQoL).<sup>163-170</sup>
- Enhanced exercise capacity.<sup>163-170</sup>
- Did not result in an obvious improvement in cardiac function although in some small studies, it has been shown to reverse left ventricular remodelling.<sup>171</sup>

Regular aerobic exercises are encouraged in NYHA I - III patients. These include:<sup>172</sup>

- Aerobic and endurance training - Moderate intensity continuous exercises such as walking, treadmill, stationary bicycle as well as swimming with a target goal of 5 days per week, 30 minutes per session.
- Interval training and high intensity interval training (HIIT) - This has been shown to be safe and results in greater improvements in exercise tolerance in selected patients with HF. In a trial on patients with HF<sub>p</sub>EF, however, HIIT did not confer any advantage compared to guideline based physical exercise.<sup>173</sup>
- Inspiratory muscle muscle training (IMT) - the addition of IMT has been shown to reduce dyspnea, improve exercise time and HRQoL.

### 8.3 Diet and nutrition

It has been widely accepted that sodium intake should be restricted in patients with HF especially in those with symptoms. **However, there is little clinical evidence to support this.**<sup>174-176</sup> Sodium restriction to < 100gm/day did not reduce clinical events.<sup>175</sup>

**I,B****IIa,C**

**The current recommendation is to avoid adding salt and flavouring sauces such as soya sauce, tomato ketchup and chilli sauce while cooking or at the table.** Refer to Appendix II, Page 162-163 on salt content of common Malaysian food.

**IIa,B**

A good balanced diet plays an important role in preventing energy depletion which can lead to cardiac cachexia and malnutrition. Cachexia is associated with increased mortality.<sup>177</sup>



A review found that the combination of personalized nutrition intervention with conventional treatment led to a decrease in all-cause mortality and hospital readmission.<sup>178</sup>

There are at present, however, few existing dietary strategies proven to improve clinical outcomes in HF.<sup>179,180</sup> The role of routine supplementation of micronutrients as a component of HF management is unclear.<sup>179,180</sup>

#### 8.4 Fluid restriction

**Ila,C** **The current evidence on fluid restriction is mixed.**<sup>181,182</sup> As with salt, excessive fluid restriction can also lead to worse outcomes.<sup>181,182</sup> This may also be due to reverse causality - sicker patients tend to take less salt and water.

**I,C** Fluid intake should be individualized. **A general recommendation is 1-1.5 liters per day in patients with normal renal function.**

In older persons, particularly those living with dementia and at the end of life, fluid intake may be more liberal.

Temporary fluid restriction can be considered in decompensated heart failure and/or patients with hyponatremia.<sup>182</sup>

##### 8.2.1.5 Lifestyle measures

These include:

- I,B** • **Weight Monitoring** - Patients should be encouraged to monitor their own weight. In obese patients, weight loss should be encouraged.<sup>183</sup>
- Ila,B** • **Alcohol** - Heavy binge drinking should be avoided in patients with HF as it can predispose to supraventricular arrhythmias especially atrial fibrillation and lead to acute cardiac decompensation.<sup>184,185</sup> Patients with alcoholic cardiomyopathy should abstain from alcohol.<sup>185</sup>
- I,B** • **Smoking** should be stopped.<sup>128,186</sup>

##### 8.2.1.6 Sexual activity, pregnancy, and contraception

As many as 60% to 87% of HF patients have sexual problems.<sup>187</sup> Erectile dysfunction (ED) is common and many have misconceptions that their symptoms (20%) or medications (10%) is the cause.<sup>185</sup> Proper counselling is necessary to address these concerns, to allay anxiety, avoid skipping medications and prevent the use of traditional potency enhancing drugs or herbs.<sup>187</sup>

**Ila,B** **It is imperative that enquiries on sexual activities or dysfunction be addressed to provide a holistic approach to patient care.**<sup>188</sup> The physician must take over the initial approach since patients are often embarrassed to initiate the topic. Some helpful tools to initiate the conversation include:

- PLISSIT (permission, limited information, specific suggestion, and intensive therapy).<sup>189</sup>
- Needs of Sexual Counselling Scale for Chronic Heart Failure (NSCS-CHF).<sup>190</sup>
- Sexual Adjustment Scale (SAS).<sup>190</sup>



Patients should be taught:

- To pay attention to their symptoms of HF.
- The potential dangers (worsening chest pain, dyspnea, giddiness and/or palpitations) and how to manage them when they occur during sexual activities.
- To defer sexual activities if in NYHA III-IV.
- Not to resume until his/her heart condition stabilizes.
- To modify sexual practices to accommodate impaired effort tolerance.

HF patients need to be told that certain cardiac medications have important side effects and drug interactions:

- Nitrates may dangerously interact with drugs for erectile dysfunction - phosphodiesterase - 5-inhibitors (Viagra, Cialis, Levitra).
- $\beta$ -blockers may contribute towards worsening erectile dysfunction, but it is important that HF patients remain compliant to them.

I,C

**Patients with LVEF < 30% and those with NYHA III and IV should be advised against pregnancy because of high maternal mortality.**<sup>191,192</sup> If pregnant, termination of pregnancy should be considered.<sup>192</sup>

I,C

When pregnancy is contraindicated, a multi-disciplinary approach with pre-conception counselling and proper contraceptive advice becomes paramount to the safety of the patient by preventing an unwanted pregnancy. Contraceptive counselling should begin early. In the absence of randomized controlled studies, the choice of contraceptive method is almost always based on expert opinion of the attending cardiologists, obstetrician, and the patient's choice.

Generally, Tier I methods which includes permanent sterilization and long-acting reversible contraceptives such as intrauterine devices [IUDs] and implants, have typical-use 1-year failure rates of < 1%. Given their excellent safety and efficacy profile, they should be recommended for women with underlying CVD. The risk of infective endocarditis with IUD is low.<sup>193,194</sup>

### 8.2.1.7 Sleep disorders

Sleep disorders are common in HF with prevalence rates of 50-75%, the prevalence increasing with age, obesity, male sex, Atrial Fibrillation, and poorer LV systolic function.<sup>195</sup> These sleep disorders include short sleep time, low sleep quality, and sleep-disordered breathing.<sup>196</sup> Sleep disordered breathing (SDB) includes either:

- Central Sleep Apnea (CSA) or
- Obstructive Sleep Apnea (OSA) or
- Coexisting OSA and CSA.

Sleep disturbances could also be due to pulmonary congestion leading to breathlessness and cough, and nocturnal diuresis due to diuretics and anxiety.

OSA may occur in the normal population or in HF patients, while CSA, which may present as Cheyne-Stokes respiration, is more associated with HF.<sup>197</sup> Independent predictors of SDB include older age, male gender, obesity, low ejection fraction and the presence of AF.<sup>196</sup> Attended in-hospital polysomnography (PSG) is the gold standard in diagnosing OSA and CSA. However, a screening using overnight pulse oximetry is useful to preselect a patient for PSG.



SDB affects HRQoL and leads to harmful effects on cardiac function, arrhythmias (both atrial fibrillation and malignant ventricular arrhythmias) and poorer prognosis due to the repetitive hypoxemia, hypercapnia and swings in blood pressure and intrathoracic pressure.<sup>196,197</sup>

**I,C**

**OSA patients are encouraged to lose weight and to exercise moderately. Alcohol and sedatives before sleep are best avoided.**<sup>196</sup> These patients are best managed in a multi-disciplinary manner.

**IIa,B**

**CPAP improves daytime sleepiness. However, none of the evidence so far indicates an improvement in terms of cardiovascular or all-cause mortality or hospital admissions.**<sup>196,198-201</sup>

In patients with HF and CSA, the use of Adaptive Servo Ventilation (ASV) was associated with an increase in all cause and CV mortality mainly driven by an increase in sudden cardiac death.<sup>137</sup>

**III,B**

**ASV is contraindicated in patients with HF and CSA.**<sup>135-140</sup>

**I,C**

**As CSA tends to worsen when HF worsens, optimizing medical therapy remains the main strategy in CSA.**<sup>197</sup>

### 8.2.1.8 Psychosocial support

About 35% patients with HF have clinical depression and about 40% have severe anxiety.<sup>202,203</sup> There is an association between HRQoL scores and survival.<sup>204</sup> Most patients prefer improved HRQoL to longevity.<sup>205,206</sup>

Patients with HF who have psychological problems have been shown to have increased morbidity and hospital readmission rates, to be less adherent to their medical regimen, and to have an overall increase in cost of care. Effective, homebased HF management, has been shown to improve HRQoL, reduce expenditure and re-hospitalization rate.<sup>207,208</sup>

**I,C**

**Thus, it is important that family members and carers are included during counselling sessions.** Depressive symptoms may affect adherence and should prompt referral to the appropriate specialists for psychological support. Inpatient treatment and hospitalizations are the main drivers for a decrease of HRQoL and efforts should be made to avoid this.<sup>21</sup>

### Key Recommendation # 3: Non-Pharmacological Measures

- In managing HF, non-pharmacological measures play a vital role. It is important to:
  - Educate patient and family about the disease, treatment options and prognosis. There should be frank end-of-life discussions and advanced care planning with the patient and family. This is important as there is a potential risk of sudden death, and they should put their affairs in order.
  - Encourage lifestyle measures:
    - ◆ Proper balanced diet to prevent malnutrition and cardiac cachexia.
    - ◆ Weight management - avoid obesity.
    - ◆ Regular physical activity and exercise training.
    - ◆ Smoking cessation.
  - Individualize fluid intake - A general recommendation is 1-1.5 liters per day in patients with normal renal function.
  - Provide advice regarding sexual activities and pregnancy.
  - Address psychosocial problems encountered by the patient and the family.

**9. ACUTE HEART FAILURE**

Acute heart failure is a heterogeneous clinical syndrome of new or worsening symptoms and signs of HF. It may occur as:

- **De Novo HF** - first occurrence of HF **or**,
- **Acute decompensated HF (ADHF)** - this is a more common presentation occurring in a previously stable patient with HF who has now deteriorated.

**In general, patients presenting with De Novo HF tend to do better than previously stable HF patients admitted with worsening symptoms ie ADHF.**<sup>209,210</sup> Patients admitted with ADHF tend to have a greater comorbidity burden and are symptomatic despite being on proven medical therapy.<sup>209</sup>

When previously stable patients present with ADHF, their risk of death is increased.<sup>211</sup> Only one quarter to one third of these patients survive 5 years after a HF related hospitalization. Patients who have decompensated because of under or suboptimal treatment tend to do better when compared to those who have decompensated on OMT.<sup>212,214</sup>

The onset can be:<sup>215</sup>

- **Sudden** with the patient presenting in acute respiratory distress and/or cardiogenic shock - This is a more common presentation of De Novo HF following an acute cardiac injury e.g., Acute MI **or**
- As a **slow progressive deterioration over days (up to weeks)** prior to hospital admission - this is a more common presentation of ADHF.

Three important phases should be considered in the management of these patients.<sup>216</sup>

- **Phase 1** - Urgent treatment and stabilization usually in the Emergency Room. (ER).
- **Phase 2** - In-hospital management.
- **Phase 3** - Discharge and Post discharge “vulnerable” period.

**9.1 PHASE 1- Urgent treatment and stabilization (See Flowchart II, page 35)**

Assessment and management must be prompt and done concurrently.

- Rapid recognition and making the diagnosis of HF based on symptoms and signs and quickly assessing its severity.
- Maintaining adequate oxygenation and perfusion of vital organs.
- Identification and stabilization of life-threatening hemodynamics (heart rate and rhythm, blood pressure and organ/tissue perfusion).
- Relieving clinical symptoms, signs and preventing end organ damage.

Once the diagnosis of Acute HF has been made, the **initial management** should focus on: (see also section 8.2 & Flowchart II, page 35)

- Adequate oxygenation-SpO<sub>2</sub> ≥ 95%.
- Intravenous (i.v.) diuretics to relieve congestion if present.
- Stabilization of hemodynamics and perfusion with vasodilators +/- inotropes.
- Identifying and treating the precipitating factors. An aide memoire to common precipitants is: (see Table 11, Page 55)

**C H A M P I O N**

- **C** - Coronary artery disease - Acute Coronary Syndrome (Myocardial infarction/Ischemia).
- **H** - Severe and uncontrolled hypertension.
- **A** - Arrhythmias.
- **M** - Mechanical e.g., Acute valvular dysfunction (e.g., acute mitral regurgitation from chordal rupture); cardiac tamponade
- **P** - Pulmonary embolism, pulmonary infections.
- **I** - Infections e.g., urinary tract, Covid.
- **O** - Other medications (Medications that increase fluid retention and/or have negative inotropic effects).
- **N** - Non-compliance to treatment especially oral diuretics and or dietary/fluid restriction.

In 40% of patients admitted with Acute HF however, a clear underlying precipitant may not be identified.<sup>217</sup>

- **Identify non-cardiovascular comorbidities** - These can complicate management affecting the choice and doses of medications used. The presence of these non-cardiovascular comorbidities tends to worsen the prognosis. The more common comorbidities include:
  - Diabetes (Section 14.1)
  - Chronic kidney disease (Section 14.6)
  - Thyroid disease (Both hyper and hypothyroidism)
  - Chronic lung disease
  - Anemia
- It is important that **precipitating factors be identified as soon as possible after presentation and treated immediately** according to guidelines.
- The patient **should be seen by a specialist as soon as possible after admission** to co-ordinate the initial and further management. In the United Kingdom National Heart Failure Audit 2022, in-hospital mortality was lower in patients who had been seen by a specialist especially a cardiologist.<sup>218</sup>

**9.1.1. Classification of Acute HF**

Based on the initial clinical assessment, the patient may be classified as:<sup>215,219</sup>

- Wet or dry depending on their fluid status.
- Cold or warm depending on their perfusion status.

**Table 13: Classification of Acute HF According to Clinical Presentation and a Guide to Management**

| Warm / Wet                                                                                                                                                                                                                                                                                                                       | Warm / Dry                                                                                                                                                                                                                                                                                                                       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>SBP</b> : Adequate ( $\geq 90$ mmHg)<br><b>Perfusion</b> : Adequate<br><b>Fluid Status</b> : Congested<br><b>Management:</b> <ul style="list-style-type: none"> <li>• Diuretics - Yes</li> <li>• Vasodilators - Yes</li> <li>• Inotropes - No</li> </ul>                                                                      | <b>SBP</b> : Adequate ( $\geq 90$ mmHg)<br><b>Perfusion</b> : Adequate<br><b>Fluid Status</b> : Not congested.<br><b>Management:</b> <ul style="list-style-type: none"> <li>• Diuretics - No</li> <li>• Vasodilators - No</li> <li>• Inotropes - No</li> <li>• Adjust Oral Medications</li> </ul>                                |
| Cold / Wet                                                                                                                                                                                                                                                                                                                       | Cold / Dry                                                                                                                                                                                                                                                                                                                       |
| <b>SBP</b> : Low ( $< 90$ mmHg) or Adequate ( $\geq 90$ mmHg)<br><b>Perfusion</b> : Poor<br><b>Fluid Status</b> : Congested<br><b>Management:</b> <ul style="list-style-type: none"> <li>• Diuretics - Yes</li> <li>• Vasodilators - If BP <math>\geq 90</math>mmHg</li> <li>• Inotropes - Yes irrespective of the BP</li> </ul> | <b>SBP</b> : Low ( $< 90$ mmHg)<br><b>Perfusion</b> : Poor<br><b>Fluid Status</b> : Not congested<br><b>Management:</b> <ul style="list-style-type: none"> <li>• Diuretics - No</li> <li>• Vasodilators - If BP <math>\geq 90</math>mmHg</li> <li>• Inotropes - Yes</li> </ul> <p><b>Consider fluid challenge cautiously</b></p> |

The classification in the table above will help guide the initial management.

Thus according to the clinical presentation, the patient may be: (Table 13, Page 74)

- **Warm and Wet** - Adequate perfusion but congested\*\* (lungs and/or periphery).
- **Cold and Dry** - Hypoperfusion\* and dehydrated/not congested\*\*.
- **Cold and Wet** - Hypoperfusion\* and congested\*\* (lungs and/or periphery).
- **Warm and Dry** - Adequate perfusion and dehydrated/not congested.\*\* These patients have either mild HF or are in the compensated stage of HF.

**\*Hypoperfusion:** cold peripheries, capillary refill time more than 2 seconds, diaphoresis, oliguria, dizziness, confusion, narrow pulse pressure, hypotension.

**\*\*Congestion:** peripheral edema, orthopnoea, paroxysmal nocturnal dyspnea, lung crepitations, jugular venous dilatation, positive hepatojugular reflux, congested hepatomegaly, gut congestion, ascites.

**Most patients admitted with Acute HF are in the ‘wet-warm’ category.<sup>220</sup>**

**The goal of therapy is to make them ‘dry-warm’- optimal perfusion and fluid status.**



### 9.1.2. Investigations

**I,C**

**Essential Investigations** in Acute HF include: (See Section 6.2)

- **Electrocardiogram** - ECG.
- **Chest Radiograph** - Chest X-Ray - About 20% of patients with congestion, however, may exhibit a normal Chest X-Ray.<sup>221</sup>
- **Blood Investigations** - FBC, serum electrolytes, urea, creatinine, cardiac biomarkers (troponins, CKMB, BNP or NT-Pro BNP), liver function tests.
- **Echocardiography** - To assess LV function, Pulmonary Artery Pressure and to estimate right and left sided filling pressures. An IVC diameter less than 21 mm and which collapses > 50% suggests normal right atrial pressures.<sup>222</sup>

**Other Investigations** which may be performed if indicated and/or available:

- **Blood Investigations** - Thyroid function tests, serum lactate and pH.
- **Blood Gases** - If oxygen saturation is still < 90% despite initial treatment.
- **Lung Ultrasound** - To look for extravascular fluid in the lungs (B-lines). These are often observed in patients with HF, but can also occur in other conditions, such as non-cardiogenic pulmonary edema and interstitial lung disease.<sup>223</sup> Lung ultrasound is, in general, more sensitive in ruling out interstitial edema and pleural effusions than Chest X-Ray.<sup>224</sup>

### 9.1.4. Decision for hospitalization and care-setting

**I,C**

Initial care in the **critical care unit (ICU/CCU)** should be considered for **high-risk patients** with features such as:

- Hemodynamic instability.
- Arrhythmias.
- Hypoperfused state-cold peripheries, capillary refill time more than 2 seconds, diaphoresis, oliguria, dizziness, confusion, narrow pulse pressure, hypotension.
- Need for invasive ventilatory support.
- Oxygen saturation (SpO<sub>2</sub>) < 90% despite supplemental oxygen.

**I,C**

**The remaining patients with Acute HF can be managed in a high-dependency unit or normal ward** depending on the clinical circumstances. However clinical deterioration may occur and hence, frequent re-assessments are necessary.

**I,C**

Step-down care from the ICU/CCU is dictated by clinical improvement. Similarly, should the patient not improve, he should be considered to be transferred to a tertiary hospital with a Cardiology Unit.

### 9.1.5. Response to Therapy

**Response should be assessed continuously** using the following parameters:

- **Symptoms and signs of HF** - this requires repeated thorough clinical examinations looking for resolution or worsening of clinical signs.
- **Vital signs**
  - Oxygen saturation - SpO<sub>2</sub>.
  - Heart rate.
  - Blood pressure - including looking for a postural drop whenever possible.
  - Respiratory rate.



- Urine output.
- Body weight.

**● Investigations**

- Urea, creatinine, serum potassium and sodium.
- Invasive hemodynamic monitoring may be considered in patients if there are uncertainties in diagnosis, or, for the more severe cases, that despite pharmacological treatment, present refractory symptoms (particularly with hypotension and hypoperfusion). Invasive monitoring includes:
  - ◆ Arterial pressure line.
  - ◆ Central venous pressure line and pulmonary artery catheter (PAC). This would allow a more accurate assessment of the fluid status of the patient and allow better titration of medications. It is only recommended in cardiogenic shock and in this setting, PAC is associated with lower mortality and in-hospital cardiac arrest.<sup>225-227</sup>

**I**b**,B**

**An adequate response** would be reflected by **all** the following:

An improvement in the patient's clinical condition and symptoms,

- Warm peripheries,
- Decrease in heart rate,
- Decrease in respiratory rate,
- An improvement in oxygen saturation **and**
- An improvement in the urine output.

Generally, **a SBP  $\geq$  90mmHg would be considered adequate** if the patient has all the following:

- Feels well,
- Has good tissue perfusion as shown by the absence of giddiness, warm skin **and**
- Stable renal function with good urine flow.

## **9.2. PHASE 2: In - Hospital Management (Flow Chart II, page 35, Table 2, page 36 & Table 14, page 80)**

The management of patients with Acute HF is largely based on clinical judgement and experience rather than on randomized controlled trials. Most clinical trials have been small and of low quality.

### **9.2.1. Pharmacotherapy**

#### **9.2.1.1. Oxygen**

**I,C**

- Measurement of oxygenation by pulse oximetry ( $SpO_2$ ) is recommended.

**I,B**

- **Supplemental oxygen therapy is recommended when the  $SpO_2 < 94\%$  and should be titrated to achieve  $SpO_2 \geq 95\%$ .**

**III,B**

- Routine use in non-hypoxic patients is not recommended as it can cause deleterious effects such as vasoconstriction and a reduction in cardiac output.<sup>228-235</sup>

**IIa,B**

- **Supplemental oxygen therapy** can be delivered as:<sup>236</sup>
  - **Nasal Prongs** - at a flow rate of 2-6 l/min gives approximately 24-50%  $FiO_2$ .
  - **Simple Face Mask** - at a flow rate 5-10 l/min gives 35-60%  $FiO_2$ . Low flow rates below 5 l/min may cause carbon dioxide rebreathing and increased resistance to inspiration and thus should not be used.



➤ **Venturi Masks** - Gives concentrations of 24-60%  $\text{FiO}_2$ . A 60% venturi mask gives approximately 50%  $\text{FiO}_2$ .

➤ **Non-Rebreathing Reservoir Mask** - gives concentration of between 60-80%  $\text{FiO}_2$ .

**IIa,B**

- **Non-Invasive ventilation (NIV)** reduces respiratory distress and may decrease the need for intubation although data regarding mortality are less conclusive.<sup>237,238</sup> It includes:<sup>238,239</sup>

➤ **High Flow Nasal Cannula (HFNC)** - This seems more effective and better tolerated than conventional oxygen therapy using nasal prongs or mask.<sup>240-242</sup>

➤ **Continuous Positive Airway Pressure (CPAP)** - can be used without a ventilator and does not require specialized training.

➤ **Non-Invasive Pressure Support Ventilation (NIPSV - Also Called Bilevel or BiPAP)**. This requires a ventilator and specialized training.

- CPAP and NIPSV should be considered early in patients with respiratory distress (respiratory rate > 25 breaths/min,  $\text{SpO}_2 < 90\%$ ) despite high-flow oxygen administration. There are no significant differences in clinical outcomes when comparing CPAP with BiPAP and the choice will depend on the equipment and expertise that is available.<sup>237,243,244</sup>

**I, C**

- Intubation may be considered in patients with respiratory failure, who cannot be managed with NIV techniques and who show signs of exhaustion and respiratory muscle fatigue.

- **Some helpful indicators of respiratory failure include:**

- Hypoxemia ( $\text{PaO}_2 < 60\text{mmHg}$ ),
- Hypercapnia ( $\text{PaCO}_2 > 50\text{mmHg}$ ), and
- Acidosis ( $\text{pH} < 7.35$ )

### 9.2.1.2. Diuretics

- **Diuretics is the cornerstone of therapy in patients who are fluid overloaded (wet).**

- i.v. diuretics should be administered as early as possible to relieve congestion and provide symptomatic relief. Studies on the impact of early administration of i.v. diuretics on in-hospital mortality, however, have shown mixed results.<sup>245,246</sup>

**I, B**

- **i.v. furosemide 40-100mg is the diuretic of choice.**<sup>48,247,248</sup> This initial dose should be individualized depending on the severity of the clinical condition.

- **In individuals who are diuretic naive, a reasonable initial dose is 20-40 mg.**

**IIa,B**

- **Patients who have already been on diuretics or have chronic kidney disease**, may require a higher dose. A reasonable initial strategy in these patients is **a daily dose of 2.5 times the previous oral dose on a mg-to-mg basis**, administered as twice-daily boluses.<sup>48,247</sup>

- Further doses can be adjusted according to clinical response, blood pressure, urine output and renal function.

- In general, with a sufficient dose of a loop diuretic agent, urine output should measurably increase within 2 hours. If there is not an adequate response to the initial dose, there is no need to wait until the next scheduled dose to increase dosing.<sup>48</sup>

- An adequate response is a urine output >100ml/ hour during the first 6 hours.<sup>48</sup>

- If the urine output is:

➤ **Excessive (a negative balance of > 3L/day)** and associated with a drop in systolic BP, then consider:

- ◆ Temporarily stopping the diuretic **or**
- ◆ Reducing the dose by 50% **or**
- ◆ Increasing the dosing interval **or**
- ◆ Changing to oral administration.



- ♦ If worsening renal function occurs after the 1st i.v. dose, then:
  - Stop the diuretic **and**
  - Rehydrate the patient with i.v. fluids **and**
  - Restart later after the renal function improves, at a much lower preferably oral dose.

➤ **Adequate** and patient is:

- ♦ **Still congested** - continue the same i.v. dose of diuretic at 12 hourly intervals.
- ♦ **No longer congested but patient is still dyspneic** - consider other causes of dyspnea e.g., pulmonary embolism, chest infection etc.

➤ **Inadequate (< 600ml in the first 6 hours) (rule out urinary retention as a cause)** - As HF progresses, diuretic resistance may develop and responsiveness to loop diuretics diminish. Diuretic resistance is defined as an impaired sensitivity to diuretics resulting in reduced natriuresis and diuresis limiting the possibility to achieve euvolemia.<sup>48</sup>

**This can be overcome by:**

- ♦ **Uptitration of the dose of loop diuretics** - this is the preferred strategy. This can be done by:
  - Increasing the dose of i.v. furosemide and/or dosing intervals.
  - Continuous infusion - To date, there has been no difference between continuous infusion or bolus dosing of furosemide for all-cause mortality, length of hospital stay and electrolyte disturbances, but continuous infusion was superior to bolus administration with regard to diuretic effect, safety profile and reduction in brain natriuretic peptides.<sup>249-252</sup>

I,C

- ♦ **Combination of low dose dopamine and low dose furosemide** was as effective as high-dose furosemide with a suggestion of less worsening of renal function.<sup>252,253</sup> The addition of dopamine however, did not result in any difference in CV outcomes.<sup>252</sup>

IIa,B

- ♦ **Using bumetanide**, a second generation loop diuretic, because of its more predictable absorption.<sup>254</sup>

IIa,B

- ♦ **Combination of thiazides and loop diuretics for “total nephron block.”**<sup>255-257</sup> It may also be used in patients with refractory edema and advanced renal impairment.<sup>258,259</sup> However, this combination has been associated with hypokalemia, hyponatremia, worsening renal function and increased mortality.<sup>255</sup>

IIa,B

- ♦ **Metolazone** is a once-daily oral thiazide diuretic. It is given in combination with a loop diuretic in patients with severe HF and refractory edema.<sup>260-262</sup> At present, there is inadequate data to show that it is superior to the other thiazides in this setting.<sup>262</sup> It can be used in advanced renal impairment.<sup>263</sup>

IIb,B

- ♦ **Other agents, when used in combination with i.v. loop diuretics**, that have been shown to produce rapid and persistent diuresis and weight loss but without, however, an impact on CV outcomes. These include :

IIa,B

- **Tolvaptan**<sup>264-266</sup>
- **Nesiritide**<sup>267,268</sup>
- **Acetazolamide**<sup>269</sup>

- ♦ When combination therapy is used, there can be a marked diuresis. Careful monitoring of fluid and electrolyte balance and BP, including orthostatic hypotension, is essential.

- ♦ **Target 0.5 - 1kg decrease in body weight/day when the patient is volume overloaded.**<sup>48</sup> **Less than 0.5kg of weight loss/day may indicate inadequate diuretic dose or diuretic resistance.**<sup>48</sup>

IIa,B

**9.2.1.3. Vasodilators (Table 14, page 80 for dosing)**

- Vasodilators can confer symptomatic relief and an improvement in hemodynamics but there is, however, a lack of data to draw any firm conclusions concerning their effects on CV outcomes.<sup>270-273</sup>

- In patients with pulmonary congestion who are normotensive, emphasis should be placed on adequate diuresis, with vasodilators reserved for patients who fail to show clinical improvement or have poor perfusion and cool extremities.<sup>274</sup>

- **Nitrates**

- Nitrates are the most widely studied vasodilators.<sup>270,271</sup>

- It should be considered if the BP is adequate (SBP > 100mmHg).

- Early i.v. nitroglycerin administration pre-hospital or in the ER was associated improved post-discharge event rate.<sup>275</sup>

- It is most useful if there is concomitant myocardial ischemia, severe hypertension or aortic or mitral regurgitation.

- It should be administered preferably intravenously for ease of titration.

- Patients should be closely monitored for hypotension. This commonly occurs with concomitant diuretic therapy.

- Extreme caution should be exercised in patients with aortic and mitral stenosis.

- Nitrates are contraindicated in severe valvular stenosis.

I,B

I,C

IIb,C

III,C

**9.2.1.4. Inotropes (Table 14, page 80 for dosing)**

- Inotropes are not routinely administered to patients with an adequate BP.

- They are indicated in the presence of persistent signs of hypoperfusion (hypotension and low cardiac output - "cold patients") despite an adequate filling status.

- These patients are best managed in specialized tertiary centers.

- These agents are best administered via central lines, inserted aseptically with proper infection control.

- **Noradrenaline infusion:**

- ◆ Noradrenaline was as efficacious as dopamine with a trend towards a lower 28-day mortality and safer especially in the subset of patients with cardiogenic shock.<sup>276,277</sup>

- ◆ The combination of noradrenaline-dobutamine appeared to be associated with more favorable hemodynamics and a safer strategy than adrenaline alone.<sup>278</sup>

- ◆ The use of adrenaline in HF complicated with cardiogenic shock should be avoided because of increased mortality.<sup>279</sup>

- **Dopamine infusion:**

- ◆ Dopamine has been shown to improve renal flow and promote natriuresis in patients with HF.<sup>280</sup>

- ◆ The role of low-dose dopamine in helping to improve renal function is still not well proven.<sup>282</sup> Studies seem to indicate that "low-dose" dopamine can worsen renal perfusion in patients with acute renal failure.<sup>281</sup>

IIa,B

IIa,B

IIb,B

**Table 14: Drugs Commonly Used in Acute HF**

|                      | Route of Admin | Dosages                                                                                                                                                |
|----------------------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Diuretics</b>     |                |                                                                                                                                                        |
| Furosemide           | i.v.           | 40-100mg<br><b>Initial dose:</b><br>New onset Acute HF and furosemide-naïve: 20-40mg<br>Known HF and on oral furosemide: 2.5 times the daily oral dose |
|                      | Infusion       | 5-20mg/hour (better than intermittent very high bolus doses)                                                                                           |
| <b>Vasodilators</b>  |                |                                                                                                                                                        |
| Nitroglycerin        | Infusion       | 5-200mcg/min                                                                                                                                           |
| Isosorbide dinitrate | Infusion       | 1-10mg/hr                                                                                                                                              |
| <b>Inotropes</b>     |                |                                                                                                                                                        |
| Noradrenaline        | Infusion       | 0.02-1mcg/kg/min till desired blood pressure is attained                                                                                               |
| Dopamine             | Infusion       | < 2-3mcg/kg/min - renal arterial vasodilation<br>2-5mcg/kg/min - inotropic doses<br>5-15mcg/kg/min - peripheral vasoconstriction                       |
| Dobutamine           | Infusion       | 2-20mcg/kg/min                                                                                                                                         |
| Adrenaline           | Infusion       | 0.05-0.5µg/kg/min                                                                                                                                      |

**➤ Dobutamine infusion:**

- Started at 2-5mcg/kg/minute and titrated by 1-2mcg/kg/minute increments at 30 - minute intervals until the desired clinical and hemodynamic response is attained.
- Dobutamine, when used alone, improved cardiac output but there was a trend towards an increase in mortality.<sup>220,283</sup>

IIb,B

**9.2.1.5 Morphine**

- i.v. 1-3mg bolus (repeated, if necessary, up to a maximum of 10mg) reduces pulmonary venous congestion although its effect on venodilation has been shown to be minimal.<sup>284</sup>
- May reduce anxiety and dyspnea however due to paucity of data, routine use cannot be recommended.<sup>284,285</sup>
- Dose-dependent side effects include nausea, hypotension, bradycardia, and respiratory depression.
- Consider co-administering i.v. antiemetics (metoclopramide 10mg or prochlorperazine 12.5mg).
- In a small study there were no significant difference in mortality between i.v. midazolam and i.v. morphine although the use of morphine was associated with a significantly higher rate of adverse side effects.<sup>286</sup>

IIb,B

**If the blood pressure is low at initial presentation (SBP < 90 mmHg) or drops during treatment:**

IIa,B

- Noradrenaline infusion<sup>276,277</sup> - initial inotrope and if BP is still low, add:

IIa,B

- Dopamine<sup>278</sup>

I,C

- Avoid vasodilators (nitrates) and morphine until the blood pressure has stabilized.
- Over diuresis or hypovolemia - correct accordingly. In Right Ventricular (RV) infarction, the hypotension may respond to volume loading.

**Other measures to be considered**

I,C

- Intubation and mechanical ventilation

IIa,C

- Correction of acidosis

IIb,C

- Invasive hemodynamic monitoring

**If the patient is still unwell with poor perfusion “cool” and/or congested “wet”, Mechanical Circulatory Support may need to be considered:**

**9.2.2 Mechanical Circulatory Support**

This includes:

- **Intra-aortic balloon pump (IABP):**

IIa,B

- This would be useful in patients with ADHF and cardiogenic shock who are not responding optimally to medical therapy and as a bridge to definitive treatment. IABP would be particularly useful in patients with intractable myocardial ischemia or acute moderate to severe mitral regurgitation.<sup>287-289</sup>

IIb,B

- In patients with acute MI complicated by cardiogenic shock and undergoing percutaneous coronary intervention, the use of IABP has not been shown to reduce mortality.<sup>287,290,291</sup> There was however a trend of benefit in patients in Killip class 3 & 4 undergoing reperfusion by fibrinolytic therapy.<sup>292,293</sup>

III,C

- IABP is contraindicated in patients with aortic regurgitation or aortic dissection.

- Others - These include the Impella system, the TandemHeart, and venous-arterial extracorporeal membrane oxygenation (ECMO). These are not widely available in Malaysia.

- Ventricular Assist Devices (VAD) - see also Section 15.2

- These would be useful as a bridge in patients for whom recovery from Acute HF is expected or for whom heart transplant is an option. It may also be used as a destination therapy in selected patients.<sup>294</sup>

**9.2.3 Treat precipitating factors for the Acute HF - “CHAMPION”****C Coronary Artery Disease - Myocardial Ischemia / Infarction:**

- Reversible myocardial ischemia causing Acute HF needs early recognition, rapid stabilization and referral for urgent coronary angiography.
- In STEMI, reperfusion therapy by fibrinolytic or primary Percutaneous Coronary Intervention (PCI) may significantly improve or prevent Acute HF.
- Long term management strategy should include adequate coronary revascularization, antiplatelet therapy, ACE-I and/or ARB,  $\beta$ -blockers and statins.

**H Hypertensive Emergency:**

- Typically presenting as “flash pulmonary edema”. Systolic LV function tends to be normal.
- The blood pressure needs to be reduced relatively quickly.



- This is best achieved with parenteral drugs such as intravenous nitrates.
- No attempt should be made to restore “normal” values of BP as this may cause deterioration of organ perfusion.
- Look for secondary causes of hypertension such as renal artery stenosis and phaeochromocytoma.

**A Arrhythmias**

- Unstable tachy- or bradyarrhythmias need to be identified and treated appropriately e.g., electrical or pharmacological cardioversion or temporary pacemaker.

**M Mechanical causes eg Valvular Heart Disease**

- Acute HF can be caused by valvular conditions such as acute mitral or aortic valve incompetence or stenosis, bacterial endocarditis, aortic dissection and prosthetic valve thrombosis.
- Vasodilator therapy would be beneficial in acute valvular regurgitation, but is contraindicated in severe valvular stenosis.
- Early access to echocardiography is crucial for the diagnosis and management.
- Percutaneous intervention such as mitral valve commissurotomy can be life saving in patients with severe mitral stenosis.

**P Pulmonary Embolism/Pulmonary infections**

- Acute pulmonary embolism can mimic Acute Coronary Syndromes and Acute HF.
- Natriuretic peptides may be raised in acute pulmonary embolism.<sup>295</sup>
- Patients with previously stable HF are also predisposed to deep vein thrombosis which can lead to pulmonary embolism.

**I Infections**

- Severe sepsis can lead to HF by many different mechanisms - inflammatory cytokines, deranged cellular metabolism etc.
- In addition, type 2 MI may also occur as a result of myocardial oxygen supply - demand mismatch as a consequence of hypotension, hypoxia etc and lead to Acute HF.
- Sepsis in a patient with previous stable HF is a management dilemma especially with regards fluid administration and use of HF medications. It tends to be associated with poor clinical outcomes.

**O Other medications**

- These include medications that have negative inotropic effects and/or cause water retention and include agents such as Non Steroidal Anti Inflammatory Agents, Calcium channel blockers, Itraconazole, minoxidil, thiazolidinediones etc.

**N Non compliance to Therapy**

- The reasons for this may be multifactorial and some reasons include poor patient understanding of the disease, financial or transport constraints in getting refills, fears of polypharmacy and the effects of the medications on the kidney etc.
- Patient and family education is important prior to discharge.

**9.2.4 Comorbidities**

Other comorbidities should be identified, and treatment optimized. This includes:

- Diabetes
- Hypertension
- Iron deficiency
- Chronic lung disease



- Chronic Kidney Disease (see section 14.6)
  - Acute HF and renal impairment can co-exist and either may give rise to the other.
  - Renal impairment influences the response to drug therapy.<sup>296</sup> In these patients with refractory fluid retention, continuous ultrafiltration may be considered.

If the patient improves and is now “warm and dry” (no longer congested and with good perfusion), he moves to Phase 3. (Section 8.3)

### 9.3 PHASE 3: Discharge and Post discharge “vulnerable” period

#### 9.3.1. Pre - Discharge

Following “decongestion”, the challenges are to:<sup>297</sup>

- Provide guidance on how to start OMT,
- How to perform up titration at discharge and
- Correctly prioritize or select the most appropriate titration schedule according to the patient profile. (Section 10.2)

##### 9.3.1.1 Conversion to oral therapy

Following adequate response to intravenous therapy, the patient should be converted to optimal oral medications. If the LVEF < 40%, they should be initiated on Foundational HF medications (RAS blockers,  $\beta$ -blockers, MRA and SGLT2-i) as outlined below.

- **Diuretics**<sup>48</sup>

- **Oral diuretics may be commenced following resolution of symptoms of congestion and the patient achieving his “dry weight”.**

I, C

- The initial dose of oral diuretics required is generally higher than the intravenous dose.<sup>298</sup> A reasonable guide is to use a mg-to-mg i.v. to oral conversion.
- In patients who are suspected to have diuretic resistance, bumetanide may be considered as it is better absorbed.<sup>254</sup>

I, C

- The dose of diuretics at discharge needs to be individualized.
- Many patients are discharged with residual clinical congestion, and this is a strong predictor of readmission.<sup>299,300</sup>
- Residual clinical congestion is often difficult to assess. Absence of dyspnea, peripheral edema, the amount of fluid lost during i.v. diuretic therapy and the patient achieving his “dry weight” are inadequate signs to assess decongestion.<sup>48,299,301</sup> This is further compounded by the inter-individual variation in the amount of fluid retention that can precipitate symptoms.<sup>299</sup>

I, C

- Determining the most appropriate outpatient dose of diuretic can be difficult and requires careful follow-up, particularly early in the post-discharge period.<sup>48</sup>
- **It would be prudent to discharge the patient on the same dose of oral diuretics as the individual was on as in - patient and then only cautiously reduce the dose in the post discharge period after careful assessment of the patient’s congestion status.**

I, A

- **MRA**<sup>302-304</sup>

- **These can be commenced within 24-48 hours of admission.**<sup>48</sup>
- Renal function and potassium levels need to be monitored.

I, A

- **Renin Angiotensin System Blockers (RAS blockers)-Angiotensin Converting Enzyme Inhibitor (ACE-I) or Angiotensin Receptor blockers (ARB)**<sup>305,306</sup>

- **Renin Angiotensin System (RAS) Blockers may be commenced at admission if the initial BP is adequate. (systolic BP  $\geq$  100mmHg)**



- In all other cases, it is best to defer for at least 24 hours till the BP is stable. (systolic BP  $\geq 100$ mmHg)
- If the patient is already on a RAS blocker, it is advisable to stop it for at least 24 hours if the BP is low. It can be recommenced at a lower dose once the BP is stable. If the BP is adequate (systolic BP  $\geq 100$ mmHg), it can be continued at the same dose.
- **Start at a low dose depending on the BP and renal function and up titrate as tolerated.**

- **$\beta$ -blockers**<sup>307-313</sup>

I, A

- **It is advisable to commence oral  $\beta$ -blockers if the BP is adequate (systolic BP  $\geq 100$ mmHg) and the patient is no longer congested i.e. his lungs are clear and there is no more edema.** If  $\beta$ -blockers are initiated in a patient who is still congested, it can worsen the congestion.
- If already on a  $\beta$ -blocker, this can be continued depending on the patient's symptoms and hemodynamics.
- **Whenever possible,  $\beta$ -blockers should be continued.** A meta-analysis showed that discontinuation of  $\beta$ -blockers in patients admitted with ADHF was associated with significantly increased in-hospital mortality, short-term mortality, and the combined endpoint of short-term rehospitalization or mortality.<sup>314</sup>

- **RAS blockers- Angiotensin Receptor Neprilysin Inhibitor (ARNI)**<sup>315,316</sup>

IIa,B

- **ARNI can be commenced as first line therapy in Acute HF in lieu of RAS Blockers.** In the clinical trial, it resulted in a greater reduction in NT-Pro BNP levels than an ACE-I (enalapril) without improvement in clinical end points.<sup>315</sup>
- It can be commenced if the systolic BP  $\geq 100$ mmHg.
- Registry data indicates that among patients hospitalized for HF/EF, sacubitril/valsartan at discharge was independently associated with reduced post discharge mortality.<sup>316</sup>
- In-hospital initiation of ARNI in patients with Acute HF in lieu of ACE-I is safe.
- Substituting ARNI for ACE-I/ARB during the in-hospital stay can be considered if the patient is already on the latter drugs and is admitted with decompensation.
- ACE-I needs to be stopped at least 36 hours prior to the initiation of ARNI. ARB can be switched to ARNI at the next dosing interval.

- **SGLT2- Inhibitor (SGLT2-i)**<sup>317,318</sup>

I, B

- **These can be initiated after stabilization of the patient during the in-hospital stay.**
- In most of the clinical trials, SGLT2-i was initiated when patients were:
  - ◆ No longer on inotropes for the last 24 hours *and*
  - ◆ SBP  $> 100$ mmHg *and*
  - ◆ No increase in the diuretic dose in the last 6 hours *and*
  - ◆ Vasodilators have been stopped for at least 6 hours.
- The diuretic dose may need adjustment taking into consideration the degree of diuresis induced by this drug.

### 9.3.1.2 Worsening renal function.

Worsening Renal Function may occur during treatment.<sup>319</sup> See Section 14.6

Following conversion to oral therapy, the patient should be observed for at least 24 hours for the stability of symptoms, weight and hemodynamics prior to discharge. The follow-up plans must be tailored according to the availability of facilities and expertise to manage the patient on an outpatient basis.

**9.3.1.3. Deep vein thrombosis (DVT) prophylaxis****IIa,B**

HF patients especially if they are bed-bound for protracted periods are at risk for DVT. Prophylactic measures include:<sup>320,321</sup>

- TED stockings
- Direct oral anticoagulants (DOAC)
- Unfractionated or low molecular weight heparin.

**9.3.1.4 Comorbidities**

Other comorbidities should be identified, and treatment optimized according to guidelines. This includes:

- Diabetes<sup>322</sup>
- Hypertension<sup>147</sup>
- Iron deficiency -
  - In patients with HF, iron deficiency (ID) is defined as:<sup>155,323</sup>
    - ♦ Serum ferritin <100 ng/mL or,
    - ♦ Serum ferritin:100-299 ng/mL and a transferrin saturation (TSAT) <20%.
  - In patients hospitalized with HF and having ID, treatment with i.v. iron improved symptoms and reduced the risk of HF hospitalizations but did not reduce the risk of CV death.<sup>324,325</sup>
- Chronic lung disease

**IIa,B**

Discharge planning is a very important process after an episode of hospitalization for HF to reduce HF related readmissions. The patient has transitioned into a vulnerable period during which he may develop complications such as hypotension, worsening renal function, electrolyte imbalance or decompensate into HF due to inadequate fluid restriction. If this phase is not optimally treated, it will lead to re-admission. The patient should be given an early follow-up, within 2 weeks of discharge, preferably in the HF clinic. The following steps are recommended:

**I,C**

- A care plan needs to be agreed upon by the patient, family, and main caregiver with the treating physician. (Appendix III & IV, page 164-165)

**I,C**

- A summary of the care plan must be made available to the doctor in the out-patient clinic. (Appendix III & IV, page 164-1654)

Hospitalization is a key opportunity to initiate Foundational HF medications.<sup>326</sup> A large trial showed that starting and optimizing Foundational HF medications as in- patient, followed by regular check-ups and monitoring after hospital discharge, cuts the rate of all-cause death and HF readmissions within 6 months.<sup>327</sup>

**9.3.2 Post-discharge phase (vulnerable period)**

At the time of hospital discharge, the patient should be:

- On all the foundational HF medications as tolerated.
- Given a discharge summary and discharge care plan. (Appendix III & IV, page 164-165)
- Informed of the need and, if indicated, given appointments for further cardiac work up e.g., coronary angiography and cardiac rehabilitation.



At the Follow up visit:

- The instructions in the discharge care plan should be followed and identify/determine if otherwise.
- The Foundational HF medications should be up titrated to maximally tolerated or target doses as soon as possible, preferably within 12 weeks post-discharge. (Section 12.2 - Patient profiling and titration of Foundational HF Medications)
- After each up titration, there should be a safety check of hemodynamics and renal function within one to two weeks.
- Where available, NT-Pro BNP measurements may be used to risk-stratify and guide management.
- The presence of any of the following features has been shown to predict re-congestion and re-hospitalization:<sup>328</sup>
  - Orthopnea,
  - Edema,
  - A raised JVP,
  - Weight gain **and**
  - The need to increase daily diuretic dose because of fluid overload.
- Patients who had none of these at 1-month post discharge had good 2-year survival.<sup>328</sup>
- The presence of any of these clinical features is an indication of the need for optimization of Foundational HF medications.

#### **Key Message # 7: Acute HF**

- Acute HF may present as:
  - De Novo HF - first occurrence of HF *or*,
  - Acute decompensated HF (ADHF) - this is a more common presentation occurring in a previously stable patient with HF who has now deteriorated.
- Three important phases should be considered in the management of these patients.
  - **Phase 1** - Urgent treatment and stabilization usually in the emergency department.
  - **Phase 2** - In-hospital management
  - **Phase 3** - Discharge and Post discharge

**Key Recommendation #4: Acute HF**

- Phase 1 - Urgent treatment and stabilization
  - Assessment and management must be prompt and done concurrently.
    - ◆ Rapid recognition and making the diagnosis of HF based on symptoms and signs and quickly assessing its severity.
    - ◆ Maintaining adequate oxygenation and perfusion of vital organs.
    - ◆ Identification and stabilization of life-threatening hemodynamics (heart rate and rhythm, blood pressure and organ/tissue perfusion).
    - ◆ Relieving clinical symptoms, signs and preventing end organ damage.
  - Identification and treatment of the underlying cause and precipitating/ aggravating factors. The more important precipitants are as in the acronym:  
“**C H A M P I O N**”
    - ◆ **C** - coronary artery disease - Acute Coronary Syndrome (Myocardial infarction/Ischemia)
    - ◆ **H** - Severe and uncontrolled hypertension
    - ◆ **A** - Arrhythmias
    - ◆ **M** - Mechanical e.g., Acute valvular dysfunction (e.g., acute mitral regurgitation from chordal rupture); cardiac tamponade
    - ◆ **P** - Pulmonary embolism, pulmonary infections
    - ◆ **I** - Infections e.g., urinary tract, Covid
    - ◆ **O** - Other medications (medications that increase fluid retention and/or have negative inotropic effects)
    - ◆ **N** - Non-compliance to treatment especially oral diuretics and or dietary/fluid restriction
  - Look for non-cardiovascular comorbidities- Diabetes, chronic kidney disease, thyroid disease (both hyper and hypothyroidism), chronic lung disease, Anemia.

**Key Recommendation #5: Acute HF**

- Phase 2 - In-hospital management
  - After initial clinical assessment, management should be instituted as in Flow Chart II, Page 35.
  - For grading of recommendations and levels of evidence, see Table 2, Page 36.
- Phase 3 - Discharge and Post discharge
  - Discharge planning is a very important process after an episode of hospitalization for HF to reduce HF related readmissions.
  - The patient should be given a discharge summary and care plan. (Appendix III & IV, page 164-165)
  - If the LVEF < 40%, they should be initiated on the Foundational HF medications (RAS blockers,  $\beta$ -blockers. MRA and SGLT2-i) as in-patient.

**10. CHRONIC HEART FAILURE – HF DUE TO REDUCED LV FUNCTION, LVEF<40% (HFrEF)****10.1 PHARMACOLOGICAL MANAGEMENT**

There are five essential classes of medications (Optimal Medical Therapy) in the management of patients with HFrEF. These are:

I, B

- Diuretics

I, A

- Renin-angiotensin system (RAS) blockers:
  - Angiotensin converting enzyme inhibitors<sup>305,306,329,330</sup> **or**
  - Angiotensin receptor blockers<sup>331-334</sup> **or**
  - Angiotensin receptor neprilysin inhibitor<sup>335</sup>

I, A

- $\beta$ -blockers<sup>307-313</sup>

I, A

- Mineralocorticoid Antagonists (MRA)<sup>302-304</sup>

I, A

- Sodium glucose cotransporter 2 inhibitors (SGLT2-i)<sup>127,336-343</sup>

- Diuretics, by relieving congestion, **are the initial step** in the management of patients with symptomatic HF.
- The other four medications (RAS blockers,  $\beta$ -blockers, MRA and SGLT2-i) have been shown to improve survival, reduce hospitalizations for HF and improve quality of life in patients with HFrEF. SGLT2-i has been shown to be effective irrespective of diabetic status.<sup>340-342</sup> For these reasons, these 4 medications are referred to as **“Foundational HF” Medications**.
- The **initiation and up titration** of these 4 Foundational drug classes should be guided by the patient's:
  - Clinical status,
  - Heart rate,
  - Blood pressure,
  - Fluid overload status,
  - Renal function,
  - Electrolytes and
  - Tolerance.
- **Comorbidities** such as atrial fibrillation, diabetes, chronic kidney, and pulmonary disease are also important conditions to be considered.
- These 4 Foundational HF Medications **should be initiated at about the same time** and up titrated if needed to their target or maximally tolerated doses.
  - In general, **starting low doses of the 4 different classes** would be preferred over up titration of each of the individual drugs to the maximally tolerated dose before initiating the next drug.
  - These drugs are **preferably all initiated, albeit at low doses, when the patient is admitted with HF** so that at the time of discharge, the patient is on all the “Foundational HF” Medications.
- These drugs should be **“up titrated” as outpatient to the maximally tolerated dose** according to the patient's profile - hemodynamic and renal function. (Section 10.2 and Flowcharts III, IV & V, page 37-39)



- Drugs with less proven survival benefits (e.g., diuretics) should be re-evaluated for reduction in dosage **when Foundation HF medications are not well tolerated due to a low BP**. If the **patient is no longer congested and appears clinically to be euvolemic, the doses of diuretics can be reduced**. Alternatively, **administering drugs at different timing** may be considered for those with symptomatic hypotension.
- Wherever possible, Foundational HF medications should be continued during an acute illness. If discontinued, they should be restarted as soon as the condition has stabilized.

### 10.1.1 Diuretics (Table 15, page 90)

I,B

- Diuretics are **indicated in all patients in whom there are signs and/or symptoms of congestion** to alleviate symptoms, improve exercise capacity, and decrease hospitalizations for HF.<sup>344,345</sup>
- The goal is **to achieve and maintain euvolemia with the lowest dose of a diuretic**. Euvolemia is, however, often difficult to assess clinically. Important **features to look for clinically** are:<sup>345</sup>
  - Resolution of symptoms and signs of congestion - subclinical congestion is sometimes, however, difficult to detect.
  - Improvement in effort tolerance and walking distance.
- Other useful **investigative tools which may be helpful to assess euvolemia**, if available, include:
  - **Chest X-ray** - this is a specific but not a very sensitive indicator of volume status.<sup>346,347</sup>
  - **Echocardiographic parameters** - such as:
    - ◆ Decrease in chamber sizes.
    - ◆ Decrease in the diameter of the IVC and a 50% decrease with inspiration.
    - ◆ Normalization or at least a decrease in doppler indices of LV and RV filling pressures.
  - **Natriuretic peptides** - These may, however, be persistently elevated in patients with Chronic HF and in addition, there is biological variability. A decrease in levels compared to admission would be a helpful guide although relying on this alone can lead to a false assurance that decongestion has been achieved.<sup>48</sup>
  - **Lung ultrasound (LUS)** - In a small study, tailored LUS-guided diuretic treatment of pulmonary congestion reduced the number of decompensations and improved walking capacity in patients with HF.<sup>348</sup>
- Most of these investigative tools-such as echocardiographic parameters and natriuretic peptides - measure filling pressures and not volume status. These tests, however, may sometimes be insufficient to detect subclinical volume excess without accompanying increases in filling pressures.<sup>349</sup>
- The dose of diuretic used is variable and dependent on individual requirements. In the presence of:
  - **Severe congestive HF and ADHF**, oral diuretic therapy may be ineffective. **Intravenous therapy may be preferred**. The diuretic of choice in these patients with fluid overload is a loop diuretic i.e., furosemide. The **goal is a reduction of body weight of about 1kg/day**.
  - **Mild fluid retention, thiazide diuretics may be preferred** especially in the presence of co-existing hypertension. Thiazides, however, are weak diuretics.
- For most patients however, **a loop diuretic is often required**. Other **Foundational HF Medications** - ARNI, MRAs, and SGLT2-i - **also have diuretic characteristics and this may help modify the dose of the diuretic**.<sup>350-353</sup>



- Patients on diuretics should be monitored closely as over diuresis can cause intravascular volume depletion leading to hypotension and deterioration of renal function. Hypokalemia is a common problem with diuretic use and oral potassium supplementation is usually necessary.
- Patients should be **educated in ‘dry weight’ management** and advised to record their daily weight. If there is a **consistent increase in weight of more than 2kg in 3 days, patients should be educated to self-adjust their diuretic (furosemide) dose together with restriction of their fluid intake until their “dry weight” is regained.** However, if the weight increase is associated with worsening symptoms or the patient fails to respond to these measures, the patient should seek medical help immediately.
- Diuretic therapy may be reduced as the clinical condition of the patient improves and they are assessed to have attained euvolemia using the indicators listed earlier. In one study, however, these indicators had limited predictive capability to determine down titration success/failure.<sup>349</sup>
- In selected patients who have stable symptoms and have been taught to manage their volume status and modify their diuretic dose accordingly, diuretic therapy may be discontinued. This is more likely to succeed if their LVEF improves to above 40-45%. In a study of a small number of patients with stable CHF on OMT, diuretic withdrawal did not result in dyspnea or adverse CV events up to 90 days.<sup>354</sup>

**Table 15: Diuretics Used In Heart Failure**

|                                                       | Route of Administration    | Starting Dose          | Usual Daily Dose          |
|-------------------------------------------------------|----------------------------|------------------------|---------------------------|
| <b>LOOP DIURETICS</b><br>Furosemide<br>Bumetanide     | i.v. / Oral<br>i.v. / Oral | 20 - 40mg<br>0.5 - 1mg | 20 - 80mg<br>0.5 - 2mg    |
| <b>THIAZIDES</b><br>Hydrochlorothiazide<br>Metolazone | Oral<br>Oral               | 25mg<br>2.5mg          | 12.5 - 50mg<br>2.5 - 10mg |

**10.1.2 Renin-angiotensin system (RAS) blockers (Table 3, page 93)**

These include:

- Angiotensin Converting Enzyme Inhibitors (ACE-I)
- Angiotensin Receptor Blockers (ARBs)
- Angiotensin Receptor Neprilysin Inhibitor (ARNI)

**10.1.2.1 ACE-I and ARB**

I,A

- ACE-I improve survival and reduce HF hospitalizations in patients with mild, moderate, and severe symptoms of HF<sub>rEF</sub>.<sup>305,306,329,330</sup> **They are recommended in all HF<sub>rEF</sub> patients.** There are no differences among available ACE-Is in their effects on symptoms or survival.<sup>305</sup>
- In head-to-head analysis, there was no difference between ACE-I and ARBs in terms of CV outcomes such as mortality and HF hospitalization, in patients with HF<sub>rEF</sub>.<sup>355-357</sup>
- **ACE-I are, however, the preferred agents because the data for reduction in mortality and HF hospitalizations are more robust.**<sup>356,357</sup>

I,A



- ARBs are however better tolerated because of their better side effect profile.<sup>356,357</sup>
- I,A ● **ARBs are indicated in HFrEF in ACE-I intolerant patients** and can be considered in patients who develop angioedema with ACE-I or ARNI.<sup>332-334</sup>
- Ila,B ● In patients post MI with impaired LV function, the ARB, Valsartan, was found to be as effective as captopril.<sup>331</sup>

**In the initiation of ACE-I/ARB, the following should be considered:**

- Patients with underlying low systolic BP < 100mmHg and/or elevated serum creatinine or potassium (>5.0 mmol/l) should be initiated with a low-dose of ACE-I/ARB cautiously.
- Avoid excessive diuresis before treatment. If patients are on large doses of diuretics, the BP and renal function should be monitored.
- Start with a low dose. The dose should be increased gradually to the target dose (Flowchart III & IV, pages 37-38, Table 3, page 93) or the maximum tolerated dose.
- Orthostatic hypotension should be avoided. Wherever possible, the BP should be measured in the lying and standing position. The consensus definition of orthostatic hypotension is a reduction of systolic BP of at least 20 mm Hg or a reduction of diastolic BP of at least 10 mm Hg within 3 minutes of erect standing.<sup>358</sup> In the elderly, the presence of orthostatic hypotension should be interpreted cautiously due to both its high prevalence in this population and its low association with underlying pathology directly related to volume status.<sup>345</sup>
- Renal profile should be checked periodically. Serum creatinine may increase up to 30% (or estimated Glomerular Filtrate Rate (eGFR) may decrease by 25%) from baseline at 7-14 days, after introduction of either an ACE-I or an ARB.<sup>359</sup> Dose adjustments is not required if the increase in serum creatinine stabilizes at < 30% (or eGFR reduces < 25%) and serum potassium < 5.5 mmol/l. The renal function should however be monitored periodically on a regular basis. Other causes of deteriorating renal function such as infection, drug therapies or worsening HF should be excluded.
- There is no significant difference in rates of hypotension, hyperkalemia, or renal dysfunction between ACE-Is and ARBs.

- I,A ● ACE-I intolerance denotes the presence of a bothersome cough (most commonly, 5% - 20% although it can be as high as 64.9%) or the experience of angioedema (uncommon, < 1%) with ACE-I therapy.<sup>360,361</sup> Patients with these conditions may be switched to an ARB, although, some may still develop angioedema.<sup>362,363</sup>

- III,A ● **Routine combined use of both ACE-I and an ARB should be avoided**, as this combination causes more adverse effects (hypotension, hyperkalemia, and renal dysfunction).<sup>364,365</sup>

**10.1.2.2. ARNI**

- ARNI is a **combination of angiotensin receptor blocker (ARB) and a neprilysin inhibitor** (an enzyme that degrades natriuretic peptides-bradykinin, adrenomedullin, and other vasoactive peptides). The only drug in this class is sacubitril/valsartan.

- I,B ● ARNI has been **shown to improve symptoms, reduce HF hospitalizations and increase survival in patients with HFrEF** who have previously tolerated ACE-I.<sup>335,366</sup>
- ARNI has also been shown to be safe in patients who are ACEI-naïve and in hospitalized patients who have just recovered from Acute HF.<sup>315</sup>

- I,B ● **ARNI is recommended as a replacement to ACE-I in patients with HFrEF who remain symptomatic despite OMT.** It has been shown to decrease CV death, HF hospitalizations, and symptoms.<sup>335</sup> The benefit of ARNI over an ACE-I, was consistent



regardless of background therapy and irrespective of previous coronary revascularization or  $\beta$ -blocker dose.<sup>366</sup> It has also been shown to have favorable effects on cardiac remodeling.<sup>367</sup>

**Ila,B**

- ARNI may also be considered as a first-line RAS blocker in ACE-I-naïve patients.<sup>315</sup>
- ARNI in Post MI patients with LVEF <40% was not superior to the ACE-I ramipril.<sup>368</sup>
- The tolerability of ARNI and its side effect profile is like that of ACE-I or ARB. Laboratory monitoring is also similar.
- The use of ARNI is, however, associated with hypotension but a low incidence of angioedema.
- When initiating an ARNI, the initial dosing and rate of titration:
  - Is dependent on pre-existing treatment, hemodynamic profile, and associated comorbidities. It should be individualized.
  - Should not be initiated if systolic BP < 100mmHg and used with caution if eGFR < 30 mL/min/1.73m<sup>2</sup>.
  - When switching from ACEI to ARNI, a washout period of at least 36 hours is required to decrease the risk of angioedema.
  - No washout period is required for conversion between ARB and ARNI.
- The drug should be up titrated to the target or maximally tolerated dose shown to improve important HF outcomes.
- Among patients with HFrEF, similar improvement in prognostic biomarkers, health status, and cardiac remodeling were observed across various ARNI doses - average daily dose: 112 mg in Tertile 1 (low dose), 342 mg in Tertile 2 (moderate dose), and 379 mg in Tertile 3 (high dose).<sup>369</sup>
- In patients taking ARNI, NT-Pro BNP is a more reliable biomarker than BNP. BNP levels may be spuriously elevated as the drug prevents its breakdown.

### 10.1.3 $\beta$ -blockers (Table 3, page 94)

**I, A**

- In major clinical trials,  $\beta$ -blockers, on top of standard therapy, have been shown to improve symptoms, reduce hospitalizations and improve survival in patients with HFrEF, regardless of their NYHA Functional class.<sup>307-313,370-372</sup>

**I, A**

- **All patients with HFrEF should be on  $\beta$ -blockers unless there is a contraindication.**<sup>307-313,370-372</sup>

- Objective improvement in cardiac function might not, however, be apparent for 6-12 months after  $\beta$ -blocker initiation. Hence, early initiation of  $\beta$ -blockers for all stable patients with current or prior symptoms of HF upon diagnosis is warranted, unless contraindicated.
- **When initiating a  $\beta$ -blocker, the patient should be in a non-congested state** ie. out of Acute HF and preferably euvolemic with clear lung fields. Caution should be exercised when initiating  $\beta$ -blockers in a patient in NYHA class III-IV.

**Ila,B**

- **Patients who decompensate and are admitted in Acute HF should be maintained on the same dose of  $\beta$ -blockers unless the clinical condition (hypotension or significant bradycardia) warrants a temporary stopping or reduction in the dose.**<sup>314</sup>
- After the patient has been stabilized, an attempt should be made to titrate to the **target or maximum tolerated dose**.
- The contraindications for  $\beta$ -blocker use include:
  - Second- or third-degree atrioventricular block (without a pacemaker).
  - Bronchial asthma (relative contraindication - use with caution).
  - Symptomatic bradycardia or hypotension.
  - Critical limb ischemia.



Ila,B

- Initiating therapy with a  $\beta$ -blocker first is non-inferior to the standard approach of starting with an ACE-I.<sup>310</sup> The benefits seen with both these drugs are additive.

**Table 3: Doses of Foundational HF Medications**

| Drug                 | Initial Daily Dose(s) | Target Dose(s)      | 1/8 Target Dose | 1/4 Target Dose | 1/2 Target Dose | Tablet Strength Available | Renal Dose For CrCl < 30ml/min                                     |
|----------------------|-----------------------|---------------------|-----------------|-----------------|-----------------|---------------------------|--------------------------------------------------------------------|
| <b>ACE-I</b>         |                       |                     |                 |                 |                 |                           |                                                                    |
| Captopril            | 6.25mg BD             | 50mg TDS            | 6.25mg          | 12.5mg          | 25mg            | 12.5mg, 25mg              | Yes<br>6.25mg-12.5mg bd/tds                                        |
| Enalapril            | 2.5mg OD              | 10-20mg BD          |                 | 2.5-5mg         | 5-10mg          | 5mg, 10mg, 20mg           | Yes<br>2.5mg OD                                                    |
| Lisinopril           | 2.5-5mg OD            | 20-40mg OD          | 2.5mg           | 5-10mg          | 10-20mg         | 5mg, 10mg, 20mg           | Yes<br>2.5mg OD<br>CrCl 10-30: 5mg OD<br>CrCl <10: 2.5mg OD        |
| Perindopril          | 2mg OD                | 8-16mg OD           |                 | 2-4mg           | 4-8mg           | 2mg, 4mg, 8mg             | Yes<br>CrCl 15-30: 2mg eod<br>CrCl <15 & HD: 2mg on dialysis day** |
| Ramipril             | 2.5mg OD              | 10mg OD             | 1.25mg          | 2.5mg           | 5mg             | 2.5mg, 5mg, 10mg          | Yes<br>1.25mg OD<br>CrCl <40: 25% normal dose                      |
| <b>ARBs</b>          |                       |                     |                 |                 |                 |                           |                                                                    |
| Candesartan          | 4-8mg OD              | 32mg                | 4mg OD          | 8mg             | 16mg            | 8mg, 16mg                 | Yes<br>4mg OD                                                      |
| Losartan             | 25-50mg OD            | 50-150mg OD         |                 | 12.5-37.5mg     | 25-75mg         | 25mg, 50mg, 100mg         | Yes<br>No adjustment                                               |
| Valsartan            | 40mg OD               | 160mg BD            |                 | 40mg            | 80mg            | 40mg, 80mg, 160mg         | Yes<br>Use with caution                                            |
| <b>ARNI</b>          |                       |                     |                 |                 |                 |                           |                                                                    |
| Sacubitril/valsartan | 100mg (49/51mg) BD*   | 200mg (97/103mg) BD | 25mg            | 50mg            | 100mg           | 50mg, 100mg, 200mg        | Yes<br>50mg BD                                                     |



| Drug                 | Starting Dose | Target Dose | 1/8 Target Dose | 1/4 Target Dose | 1/2 Target Dose | Tablet Strength Available | Renal Dose For CrCl < 30ml/min |                                                  |
|----------------------|---------------|-------------|-----------------|-----------------|-----------------|---------------------------|--------------------------------|--------------------------------------------------|
| β-Blocker            |               |             |                 |                 |                 |                           |                                |                                                  |
| Bisoprolol           | 1.25mg OD     | 10 mg OD    | 1.25mg          | 2.5mg           | 5mg             | 2.5mg, 5mg, 10mg          | Yes                            | No adjustment                                    |
| Carvedilol           | 3.125mg BD    | 25 mg BD    | 3.125mg         | 6.25mg          | 12.5mg          | 6.25mg, 12.5mg, 25mg      | Yes                            | No adjustment                                    |
| Metoprolol Tartrate* | 12.5-25mg BD  | 100 mg BD   | 12.5mg          | 25mg            | 50mg            | 50mg, 100mg               | Yes                            | No adjustment                                    |
| Nebivolol+           | 1.25mg OD     | 10 mg OD    | 1.25mg          | 2.5mg           | 5mg             | 5mg                       | Yes                            | No adjustment                                    |
| MRA                  |               |             |                 |                 |                 |                           |                                |                                                  |
| Spironolactone       | 12.5-25mg OD  | 50 mg OD    |                 | 12.5mg          | 25mg            | 25mg                      | Yes                            | Contraindicated                                  |
| Eplerenone           | 25 mg OD      | 50 mg BD    |                 | 12.5mg          | 25mg            | 25mg, 50mg                | Yes                            | Contraindicated                                  |
| Finerenone           | 20mg OD       | 20mg OD     |                 |                 |                 | 10mg, 20mg                | Yes                            | eGFR 25-60: 10mg OD<br>eGFR <25: contraindicated |
| SGLT2-i              |               |             |                 |                 |                 |                           |                                |                                                  |
| Dapagliflozin        | 10mg OD       | 10mg OD     |                 |                 |                 |                           |                                | CrCl <25: limited experience                     |
| Empagliflozin        | 10mg OD       | 10mg OD     |                 |                 |                 |                           |                                | CrCl <20: not recommended                        |

\* The evidence-based form of metoprolol in HF is metoprolol succinate (CR/XL), which is not available in Malaysia. The only form of metoprolol available in Malaysia is metoprolol tartrate

+ Shown to reduce the composite endpoint of all-cause mortality and CV hospitalization, but not all-cause mortality.

IBM Micromedex Drug Ref (2023) (cited 14<sup>th</sup> January 2023) & product leaflet available from Quest3 (accessed date: 13<sup>th</sup> January 2023)

\*\* At the discretion of the attending physician

CrCl unit: ml/min; eGFR unit: ml/min/1.73m<sup>2</sup>

**10.1.4. MRA (Table 3, page 94)****I,A**

- MRAs improve symptoms, reduces HF hospitalizations and mortality in patients with HFrEF.<sup>303-305</sup> **It is indicated in all symptomatic HFrEF patients (NYHA II-IV) in addition to other Foundational HF Medications.**
- Caution should be exercised in the following conditions :
  - Hyperkalemia
  - Renal dysfunction
  - Concomitant use of:
    - ◆ Potassium-sparing diuretics
    - ◆ Potassium supplements
    - ◆ ACEI/ARB/ARNI
    - ◆ NSAIDs

**IIa,B**

- **MRAs have also been shown to be beneficial in patients with HF and comorbid conditions such as diabetes and chronic kidney disease despite a greater risk of hyperkalemia and acute renal insufficiency.**<sup>373</sup>
- It has also been shown to improve CV outcomes in elderly HF patients > 75 years.<sup>374</sup>
- **Potassium supplements may need to be reduced or discontinued.** If despite these measures, hyperkalemia persists (serum potassium >5.5mmol/l), then the dose of MRA should be reduced or stopped. Alternatively potassium binders may be used. (Section 14.6)
- Spironolactone may uncommonly cause gynaecomastia in men. The incidence of gynaecomastia is lower with eplerenone as it is more selective for aldosterone blockade.
- Finerenone is a non-steroidal MRA which has less risk of hyperkalemia and BP-lowering effects. It has been shown to reduce first and recurrent HF hospitalizations in patients with Type 2 diabetes and CKD.<sup>375-378</sup>

**10.1.5. SGLT2-i (Table 3, page 94)****I,A**

- **SGLT2-i, particularly dapagliflozin and empagliflozin, lower the risk of HF hospitalizations, cardiovascular mortality, and improve quality of life in patients with HFrEF, irrespective of diabetes status.**<sup>340-342,379-381</sup>
- The diuretic/natriuretic characteristics of SGLT2-i may provide further benefits in decreasing congestion and may allow for a reduction in loop diuretic dosage.<sup>351-353</sup>
- In the initiation of SGLT2-i, the following should be considered:
  - Renal profile should be checked periodically and monitored regularly. Although renal function is known to decrease modestly following initiation, SGLT2-i tends to be renoprotective in the long term.<sup>382</sup> A slight decrease in eGFR following initiation is expected, but this should not lead to discontinuation of treatment.
  - Monitor glycemia regularly, especially in diabetic patients. Consider modifying other diabetic medications.
  - Identify and eliminate risk factors for ketoacidosis e.g., fasting, inadequate fluid intake.
  - Monitor fluid balance regularly, especially when a patient is on diuretics, is elderly, and/or frail. Adjust diuretic therapy and fluid intake as necessary.
- Uro-genital infections may occur during treatment with SGLT2-i. Patients should be educated on the signs and symptoms of uro-genital infections as well as the appropriate preventive measures.

**10.1.6. I<sub>f</sub> channel inhibitor- Ivabradine (Table 16, page 96)**

- Ivabradine selectively inhibits the I<sub>f</sub> current in the sinoatrial node, resulting in a reduction in heart rate.
- **It is effective in patients in sinus rhythm.** Ivabradine may, however, be associated with an increased incidence of AF.<sup>383,384</sup>
- Ivabradine resulted in a reduction in the combined endpoints of CV mortality and HF hospitalizations in patients who were still symptomatic while on OMT with an ACE-I (or ARB), a  $\beta$ -blocker and an MRA and:<sup>385</sup>
  - With an LVEF  $\leq 35\%$  and
  - An episode of HF hospitalization in the past 12 months and
  - Who were in sinus rhythm with a heart rate of  $\geq 70$  beats per minute (bpm).
- The magnitude of heart rate reduction by  $\beta$ -blocker plus ivabradine, rather than the background dose of  $\beta$ -blockers, appeared to determine the effect on CV outcomes.<sup>386</sup>
- In patients with a heart rate of  $\geq 75$  bpm, ivabradine has been shown to improve CV outcomes, the best risk reduction being seen in those achieving a heart rate  $< 60$  bpm or a heart rate reduction  $> 10$  bpm.<sup>387</sup>
- **In patients with stable CAD and LVEF  $< 40\%$ , ivabradine did not reduce CV outcomes.** In a subgroup of patients with a heart rate  $> 70$  bpm, there was a reduction in the secondary end point of the incidence of CAD (fatal and non-fatal MI and coronary revascularization).<sup>388</sup>
- **Every effort should be made to achieve target or maximally tolerated doses of  $\beta$ -blockers before initiation of ivabradine.** It would be useful in patients who have contraindications to  $\beta$ -blockers or who are not able to tolerate higher dose of  $\beta$ -blockers due to its side effects. Ivabradine has no effect on BP or myocardial contractility.
- Patients should be educated to measure and record their pulse regularly, to monitor for bradycardia. Side effects of ivabradine include symptomatic bradycardia (dizziness, fatigue, or hypotension) and visual disturbances.

**Table 16: Other Drugs Recommended for HF Management and Their Dose Regime**

| Drug                                   | Starting dose       | Target dose |
|----------------------------------------|---------------------|-------------|
| <b>I<sub>f</sub> channel inhibitor</b> |                     |             |
| Ivabradine                             | 5mg bd <sup>†</sup> | 7.5mg bd    |
| <b>Others</b>                          |                     |             |
| Isosorbide dinitrate <sup>‡</sup>      | 10mg tds            | 40mg tds    |
| Digoxin                                | 0.0625mg od         | 0.25mg od   |
| Vericiguat                             | 2.5mg od            | 10mg od     |

<sup>†</sup>In patients  $>75$  years old, lower starting dose of 2.5mg bd can be used.

<sup>‡</sup>(PO) Hydralazine-nitrate combination is not available in Malaysia.

**10.1.7. Soluble guanylate cyclase stimulator- Vericiguat**

- Vericiguat may be considered in patients in NYHA classes II to IV, who despite being on OMT with a RAS blocker, a  $\beta$ -blocker and an MRA, have evidence of worsening HF. It has been shown to reduce CV mortality or HF hospitalizations.<sup>389</sup>

**10.1.8. Nitrates (Table 16, page 96)**

IIa,C

- Symptoms of HF such as orthopnea, paroxysmal nocturnal dyspnea, exercise-induced dyspnea and/or angina may be relieved with the use of nitrates alone, in the form of tablets, sprays, or transdermal patches.
- Continuous (i.e., around the clock) use should generally be avoided to prevent nitrate tolerance and pseudotolerance.<sup>390</sup>
- Nitrates are mainly used in Acute HF.

I,A

- In chronic HF, the trials on nitrates have been in combination with hydralazine. This combination has been shown to improve survival in the African American population with HF.<sup>391,392</sup>

**10.1.9. Digoxin**

IIa,B

- In patients with HF, digoxin may be considered in the following situations:
  - To relieve symptoms and lower the risk of hospitalization in patients with symptomatic HFrEF and already on OMT with an ACE-I (or ARB), a  $\beta$ -blocker and an MRA.<sup>393-395</sup> It has no survival benefit.

IIb,B

- Rate control in patients with AF.<sup>396-398</sup>

IIa,B

- Where rate control is the preferred strategy,  $\beta$ -blocker alone or in combination with digoxin was associated with a similar decrease in the risk of mortality.<sup>396-398</sup>

- Digoxin alone was associated with an increase in mortality.<sup>399,400</sup>

IIa,B

- **Digoxin may be considered in patients with HF and AF** in the following situations:<sup>397,398</sup>

- Rate control is inadequate on  $\beta$ -blockers alone.
- $\beta$ -blockers are contraindicated.
- Rapid control of the ventricular rate with parenteral drugs is required.
- **Digoxin has a narrow therapeutic range** and thus close monitoring of renal function and serum electrolytes (particularly potassium and magnesium levels) is required, prior to initiation of digoxin and periodically during its use.
- No loading dose is required for the management of chronic HF. Lower doses of digoxin and lower levels of serum digoxin (0.5-0.8ng/ml or 0.65 to 1nmol/L) are efficacious and appear adequate in most patients with compensated HF.<sup>401-403</sup> **The maintenance dose of digoxin may range between 0.0625mg to 0.25mg daily**, which may be lower in elderly patients, women, and those with renal impairment.
- **Regular monitoring of digoxin levels is not required** other than to assess for toxicity as the levels should not be used to guide dose adjustment in chronic therapy. Digoxin levels may be elevated in the presence of worsening renal function, electrolyte imbalance (hypokalemia, hypomagnesemia, or hypocalcemia) or interacting drugs (e.g., amiodarone), which may lead to atrial and ventricular arrhythmias particularly in the presence of hypokalemia.

**10.1.10. Antiplatelet and Anticoagulation Therapy**

IIb,B

- **There is no role for routine antiplatelet or anticoagulant therapy in patients with HF.**<sup>404-412</sup>

- The decision to treat patients with HF with antiplatelet therapy is **largely influenced by the presence or absence of concomitant arterial disease**.
- In patients with HF and sinus rhythm, anticoagulation, when compared to antiplatelet therapy, did not significantly influence the primary outcome of death and stroke, but reduced ischemic stroke rates.<sup>409-411</sup> Treatment with warfarin however, was associated with a higher risk of major bleeding complications.<sup>409</sup>



- HF patients with the following risk factors for thromboembolism should be given an appropriate anticoagulant, unless contraindicated:

- **Atrial fibrillation (AF)**

- **All patients with AF, HF and CHA<sub>2</sub>DS<sub>2</sub>-VASc score > 2 in men or > 3 in women should be given an anticoagulant indefinitely, unless contraindicated.**<sup>412-416</sup>
- **Direct-acting oral anticoagulants (DOACs) are recommended over warfarin** in patients with HF who are eligible, with the exception of those with moderate or severe mitral stenosis and those with mechanical prosthetic heart valves.<sup>412,416</sup>

- **Left ventricular thrombus (LVT).**

- The 1-year risk of stroke with LVT is high (10%) even with anticoagulation.<sup>417</sup> There was no statistically significant difference in thrombus resolution, risks of bleeding complications, stroke, or systemic embolization and mortality in LVT patients treated with warfarin compared to those treated with DOACs.<sup>418,419</sup>
- If an LVT is detected, **the duration of anticoagulation will depend on clinical judgment, including the following factors:**

- **If following an acute MI, then anticoagulation (especially for a duration > 3 months)** was associated with an overall lower risk of MACE or all-cause mortality.<sup>417</sup> A recent review has suggested that if the LV thrombus has resolved, anticoagulation can be discontinued and dual antiplatelet therapy continued per management of MI.<sup>420</sup> If the LV thrombus is persistent, anticoagulation should continue with repeat imaging every 3 months. Once anticoagulation has been discontinued, repeat imaging 3 months later is advised.<sup>420</sup>
- **If there is no obvious acute event and the thrombus was detected during a routine examination**, then the duration of anticoagulation becomes more difficult to decide since the risk of embolisation decreases over time although still present. In these cases, clinical judgment taking into account the bleeding risk and clinical profile of the patient can help in decision making.

### 10.1.11. Antiarrhythmic Drug Therapy

- Arrhythmias are common in HF. The more common ones are:
  - Atrial fibrillation
  - Ventricular tachyarrhythmias
  - Bradyarrhythmias

#### 10.1.11.1. Atrial fibrillation (AF)

- **New-onset AF in a patient with established HF is associated with a poor prognosis irrespective of the LVEF.**<sup>421,422</sup> On the other hand, patients who develop AF first, followed by HF usually have a more benign course.<sup>423</sup>
- Combination of HF and AF may increase the risk of stroke, dementia, HF hospitalization and all-cause mortality.<sup>416,424</sup>
- Patients with AF and HF can be managed **by identifying and treating potential causes of AF and optimizing HF treatment - i.e., Optimal Medical Therapy.**
- **The AF can be treated by either rate control or rhythm control although in patients with recent onset of AF < 1 year duration, rhythm control is preferred to rate control.**<sup>155,398,416,425</sup>

**10.1.11.1. A Rate control**

- **The optimal resting ventricular rate in patients with AF and HF is unknown.** There was no significant difference in clinical events between a strict rate control of < 80bpm vs a more lenient < 100/bpm.<sup>416,426,427</sup>
- **A reasonable approach is to aim for a HR < 110/bpm** unless there are symptoms or a deterioration in LV function, in which case, one should try to target a HR < 80 bpm.<sup>416,427</sup>
- **The optimal ventricular rate during exercise is also uncertain, but may be < 110 bpm during light exercise.**<sup>416,426</sup>
- **In the rate control of AF:**
  - β-blockers should be considered for both short - and long-term rate control.<sup>396,428-430</sup> β-blockers are preferred over digoxin as it provides better rate control during exercise and improves morbidity and mortality in patients with HF although the latter effect is attenuated in patients with AF.<sup>396,426,429</sup>
  - Digoxin should be considered if the ventricular rate remains elevated despite treatment with β-blockers, or if β-blockers are contraindicated or intolerable.<sup>397,426,429,431,432</sup>
  - Intravenous amiodarone may be used for rate control, but only in the acute setting.<sup>416,433</sup>
  - The initial target heart rate for rate control therapy should be < 110 bpm.<sup>416</sup>
  - Combination therapy may be considered if a single rate control therapy does not achieve the target heart rate.<sup>416</sup> Rate control is better when digoxin and β-blockers are used in combination rather than with each drug individually.<sup>396-398,432</sup>
- In patients with marked congestion who cannot tolerate β-blockers, suggest:
  - Oral or IV digoxin
  - Oral or IV amiodarone<sup>433,434</sup>

IIa,B

IIa,B

IIa,B

**10.1.11.1. B Rhythm control**

- In patients with permanent AF and HF and LVEF < 35% (HFrEF), there was no difference in all-cause mortality or secondary outcomes (death from any cause, worsening HF or stroke) comparing a strategy of rate or rhythm control.<sup>435,436</sup>
- In patients with early onset AF of < 1 year duration and/ or paroxysmal AF, and/or cardiovascular problems who received early rhythm-control, there was a significantly decreased primary end point of a composite of CV death, stroke or hospitalization due to HF or ACS than those who received standard care.<sup>416,425</sup> The study included 28% stable HF patients.<sup>425</sup> At 2 years, however, symptoms and LV function did not differ significantly between the groups.<sup>399,437</sup>
- The primary indication for rhythm control is reduction in AF-related symptoms and improvement of quality of life.<sup>399</sup>
- Rhythm control can be achieved by:
  - Pharmacotherapy:
    - ◆ Amiodarone is preferable compared to other antiarrhythmic drugs as others are associated with worse outcomes in HFrEF.<sup>433,434,438</sup>
  - Electrical therapy:
    - ◆ Electrical cardioversion and/or
    - ◆ Radiofrequency ablation.<sup>416,425</sup>

IIa,A

IIa,B

IIa,C

IIa,A

**10.1.11.2 Ventricular Arrhythmias**

- The exact prevalence of sudden cardiac death (SCD) in patients with HF in the contemporary era is not known. It varies depending on the etiology of the HF and the LVEF.<sup>439</sup> Patients with HF and reduced (< 30 or 35%) LVEF (HFrEF) account for < 20% of all SCDs.<sup>440</sup>



- SCD is most often due to either sustained ventricular tachycardia (VT) or ventricular fibrillation (VF) although sometimes it may be due to a bradyarrhythmia or electromechanical dissociation.<sup>440,441</sup> Occasionally rapid supraventricular tachycardias may deteriorate to malignant ventricular tachyarrhythmias.<sup>442</sup>
- Patients with HF with ventricular arrhythmias must be treated aggressively with OMT.
- **The following medications have been shown to reduce the risk of SCD:**
  - **β-blockers:** have been shown to reduce SCD in the clinical trials done on patients post MI as well as in the HF trials.<sup>311,313,443</sup>
  - **ACE-I or ARB:** clinical trials done following MI and heart failure with EF < 40% showed that ACE-I or ARB reduced SCD.<sup>444-449</sup>
  - **ARNI:** reduced both SCD and deaths due to worsening HF.<sup>450,451</sup>
  - **MRA:** have been shown to reduce the incidence of SCD.<sup>303,304,452,453</sup>
  - **Statins:** do not prevent SCD or overall mortality, however statins may minimize hospitalization for worsening heart failure in patients with HF.<sup>454,455</sup>
- In addition to the above, in patients with ventricular tachyarrhythmias, the following are important:
  - Identify contributing factors such as electrolyte imbalances, ischemia and drugs.
  - Implantable cardioverter defibrillator (ICD).<sup>456-460</sup> (Section 8.2.3.2)
  - Antiarrhythmic drug therapy with amiodarone can be considered as adjunctive therapy in patients with ICD to reduce the number of shocks and in patients who are not candidates for ICD.<sup>461,462</sup>
  - Radiofrequency ablation may be considered in the event of VT storms.<sup>463</sup>

#### 10.1.11.3 Bradyarrhythmias- see section 10.3.3

#### 10.1.12. Calcium Channel Blockers (CCBs)

- Routine use of CCBs is not recommended in patients with HF/rEF as they do not confer any morbidity or mortality benefit but worsen HF outcomes.<sup>464-469</sup>
- Diltiazem, verapamil and nifedipine should be avoided.<sup>462-467</sup>
- However, amlodipine and felodipine may be considered for other indications such as persistent hypertension despite OMT.<sup>468,470</sup>

### 10.2 PATIENT PROFILING AND TITRATION OF “FOUNDATIONAL HF MEDICATIONS” (Table 5, page 43)

The “Foundational HF” drug classes should be initiated in all patients, preferably prior to discharge. These “Foundational HF” Medications are:

- Renin-angiotensin system (RAS) blockers:
  - Angiotensin converting enzyme inhibitor (ACE-I) or
  - Angiotensin receptor blocker or
  - Angiotensin receptor neprilysin inhibitor (ARNI)
- β-blockers
- MRA
- SGLT2-i

These drugs should be up titrated cautiously depending on the patient’s BP, heart rate, renal function, and potassium levels.<sup>297</sup> The following is a guide to titration and modifying the doses of these medications depending on the patient’s profile.

**10.2.1 Blood Pressure:****If systolic BP < 90 mmHG**

- Detect and treat causes of hypotension, such as hypovolemia from over diuresis, bleeding, infection accordingly.
- If the patient is euvolemic using the criteria in section 10.1.1, an attempt may be made to reduce the dose of diuretics to the lowest maintenance level e.g. furosemide 20 mg EOD or use a mild thiazide diuretic such as hydrochlorothiazide 12.5-25 mg daily. Occasionally it may be possible to remove the diuretic completely especially if the LVEF > 40-45%.
- Remove all non “Foundational HF” Medications that may also cause hypotension eg - nitrates, calcium channel blockers, alpha blockers.
- The dose of the “Foundational HF” Medications only needs to be reduced or temporarily stopped if there is symptomatic hypotension.
- SGLT2-i and low-dose MRA have minimal effects on BP and may be continued. Spironolactone at a mean dose of 26 mg daily had no significant hemodynamic effects in the RALES trial.<sup>303</sup> However when used as a fourth line drug in patients on a mean of 2.9 other anti-hypertensive drugs, at a mean dose of 25 mg daily, it can drop the SBP/DBP by 21.9/9.5 mm Hg.<sup>471</sup>

**If systolic BP > 110 mmHG**

- The dose of RAS blockers and  $\beta$ -blockers should be up titrated in turn till the target or maximally tolerated dose.
- Whether the RAS blocker or the  $\beta$ -blocker should be up titrated first will depend on the patient's heart rate, renal function, and potassium levels.

**10.2.2 Heart Rate (HR)****10.2.2.1 Sinus Rhythm**

- The optimal HR for patients with HF and in sinus rhythm is 50-60/bpm.<sup>416</sup>

**If the HR > 70/bpm**

- The dose of  $\beta$ -blocker should first be up titrated to the target or maximally tolerated dose.
- Ivabradine may be added to the  $\beta$ -blocker to achieve the target HR.<sup>386,387</sup>

**If the HR < 50/bpm**

- Discontinue non-dihydropyridine calcium channel blockers e.g., diltiazem and verapamil, digoxin, or other antiarrhythmic drugs such as amiodarone.
- If the HR is still < 50/bpm, the dose of ivabradine should first be reduced.
- If the HR is still < 50/bpm or the patient has symptomatic bradycardia, then the dose of  $\beta$ -blocker should be reduced or temporarily discontinued.

**10.2.2.2 Atrial Fibrillation****Heart rate (HR)**

- The optimal resting ventricular rate in patients with AF and HF is unknown.
- It is not unreasonable to aim for a HR < 110bpm.<sup>416</sup>



- If the patient is troubled by palpitations or there is deterioration of LV function, then a more strict rate control of 80bpm may be targeted.<sup>416</sup> The ventricular rate should however, be maintained > 70 bpm.<sup>297</sup> A lower HR has been associated with worse outcomes.<sup>472</sup>
- Excessive rate control, which may be associated with an increase in pauses, carries a risk.<sup>416</sup>
- The optimal ventricular rate during exercise is also uncertain, but may be < 110bpm during light exercise.<sup>416,426</sup>
- In patients with persistent AF and HFrEF, heart rate is not a predictor of mortality unlike in patients in sinus rhythm.<sup>416,426,429,431,472</sup>
- There is no clear evidence for a prognostic benefit of  $\beta$ -blockers in HF patients with AF.<sup>155,416,426</sup>

**Systolic BP < 90mmHg**

- $\beta$ -blockers may be stopped and replaced with digoxin for rate control.
- This may allow for the up titration of RAS blockers.

**10.2.3. Chronic Kidney Disease - see section 14.6****Key Message # 8: Chronic HF due to LV reduced Function (HFrEF)**

- Optimal HF medications are:
  - Diuretics - to be titrated according to presence of congestion.
  - Foundational HF medications:
    - ◆ Renin-angiotensin system (RAS) blockers:
      - Angiotensin converting enzyme inhibitor (ACE-I) or
      - Angiotensin receptor blocker or
      - Angiotensin receptor neprilysin inhibitor (ARNI)
    - ◆  $\beta$ -blockers
    - ◆ Mineralocorticoid Antagonists (MRA)
    - ◆ Sodium glucose cotransporter 2 inhibitors (SGLT2-i)
  - Other drugs (when necessary) include:
    - ◆ Ivabradine
    - ◆ Nitrates
  - For doses of these Foundational HF medications, Grades of recommendation and Levels of Evidence of these medications, see Tables 3 & 4, page 40-42.

**Key Message # 9: Arrhythmias**

- Arrhythmias are common in HF. These include:
  - Atrial Fibrillation
  - Ventricular arrhythmias
  - Bradycardias

**Key Recommendations #6: Chronic HF due to HFrEF: Pharmacotherapy**

- For Grades of recommendation and Levels of Evidence, see Table 4, page 42
- **Foundational HF Medications**
  - The Foundational HF Medications should be initiated at about the same time and up titrated if needed to their target or maximally tolerated doses.
  - Once the patient is stable and no longer volume overloaded, the dose of diuretics may be down titrated to the lowest maintenance dose.
  - In general, starting low doses of the 4 different classes would be preferred over up titration of each of the individual drugs to the maximally tolerated dose before initiating the next drug.
  - These drugs are preferably all initiated, albeit at low doses, when the patient is admitted with HF so that at the time of discharge, the patient is on all of them.
  - For the initiation and up titration of these HF drugs see Flowchart III, IV & V, page 37-39.

**Key Recommendation #7: Arrhythmias**

- **Atrial Fibrillation**
  - To be treated by rate or rhythm control (if onset < 1 year).
  - Anticoagulation with DOAC or Vitamin K antagonists.
- **Ventricular arrhythmias**
  - Identify contributing factors such as electrolyte imbalances, ischemia and drugs.
  - Implantable cardioverter defibrillator (ICD)
- **Bradyarrhythmias**
  - Device Therapy as indicated.

**10.3. DEVICE THERAPY IN HEART FAILURE****10.3.1. Cardiac Resynchronisation Therapy (CRT)**

Patients who remain symptomatic (NYHA class II-III) despite OMT should be considered for CRT.

CRT has been shown to improve symptoms, hospitalizations, and mortality. though up to 30% of patients may be non-responders.<sup>473-477</sup>

Patients **with all the following criteria** can be considered for CRT:<sup>475-484</sup>

- Sinus rhythm
- LVEF ≤ 35%
- LBBB
- QRS duration on resting 12 - lead ECG:
  - ≥ 150ms
  - ≥ 120-149ms

I,A

IIa,B

Mechanical ventricular dyssynchrony is no longer a criterion in selecting patients for CRT.<sup>485-487</sup>



Patients with AF are less likely to respond to CRT. They may be considered for CRT along with atrio-ventricular node ablation.<sup>488-490</sup>

Conduction system pacing may be an alternative to biventricular pacing in patients with pacing indications (expected RV pacing of > 40%) and HF not fulfilling criteria for CRT or with unsuccessful coronary sinus lead implantation.<sup>491-494</sup>

Regular monitoring of the patients after device implantation is mandatory to adjust medical therapies and reprogram the device as necessary.

### 10.3.2. Implantable Cardioverter Defibrillator (ICD)

Sudden cardiac death (SCD) in patients with HF is often due to ventricular fibrillation or ventricular tachycardia. This risk can be reduced with the implantation of an ICD.

An ICD is recommended for secondary prevention in patients with previous sudden cardiac arrest or documented sustained ventricular arrhythmias.<sup>456-460</sup>

It should be considered in patients who fulfil the eligibility criteria, who otherwise have good clinical function and prognosis to improve their survival.

#### Secondary prevention:

The following should be considered for implantation of ICD:<sup>456-460,495-498</sup>

I,A

- Patients resuscitated from SCD due to ventricular fibrillation or hemodynamically unstable sustained ventricular tachycardia. These cardiac arrest survivors have a high risk of recurrent events and implantation of an ICD has been shown to reduce mortality.

IIa,B

- Patients with chronic HF and LVEF  $\leq 35\%$  who experience syncope of unclear origin have a high risk of subsequent SCD should also be considered for placement of an ICD.

I,A

- Prior MI and LVEF  $\leq 40\%$  with non-sustained VT **AND** inducible sustained VT or VF during an electrophysiological (EP) study.

#### Primary prevention (prophylactic ICD implantation)

Prophylactic ICD implantation to reduce the risk of SCD may be considered in patients with:

I,A

- Prior MI and at least 40 days after an MI and 3 months after revascularization by PCI or CABG and:

- LVEF  $\leq 35\%$  with mild to moderate HF symptoms (NYHA class II-III).<sup>496</sup>

IIa,B

- LVEF  $\leq 30\%$  regardless of NYHA class, to reduce mortality.<sup>495</sup>

I,B

- Non ischemic cardiomyopathy LVEF  $< 35\%$ , on guideline directed medical therapy with no reversible causes, and

- Mild to moderate HF symptoms (NYHA class II-III)<sup>497,498</sup>

IIa,B

- No HF symptoms (NYHA class I)<sup>497,498</sup>

- Additional risk factors shown to further increase the risk of sudden cardiac death and ventricular arrhythmias that may be factored to further support prophylactic implantation of a defibrillator device (ICD/ CRTD) are:

- Syncope/ presyncope

- LVEF  $< 25\%$

- Non sustained Ventricular Tachycardia

- PVCs  $> 10/\text{hr}$  on Holter monitoring

**10.3.3. Pacemakers****I,A**

- Patients with significant bradyarrhythmias, trifascicular bundle branch blocks and permanent or paroxysmal third- or high-degree atrioventricular (AV) blocks should be considered for pacemaker therapy.<sup>479</sup> Prior to implanting a conventional pacemaker, the need for an ICD or Cardiac Resynchronisation Therapy (CRT) device should be considered.
- Conduction system pacing (which includes His bundle and left bundle branch area pacing) is a new pacing modality. The early results appear promising but evidence on its safety and efficacy is still lacking.<sup>479</sup>

**Key Recommendation #8: Arrhythmias****• Device Therapy For HF**

- The grade of recommendations and levels of evidence for device therapy (ICD, CRT, pacemakers) are as in Table 4, page 42.

**10.4. SURGERY FOR HEART FAILURE**

Patients with HF should undergo surgery if the pathology causing the HF is amenable to surgical treatment. The decision to subject a patient to surgery should however consider the functional status, prognosis, and comorbid conditions of the patient.

Surgical procedures include the following:

**10.4.1. Revascularization Procedures**

Patients with CAD and HF may benefit from revascularization particularly if they have angina and anatomy that is suitable for revascularization (left main stem or triple vessel disease). The benefit of revascularization is likely to be more in younger patients with more severe left ventricular dysfunction, severe CAD with angina, viable myocardium, and reversible ischemia.

The STICH trial did not find any difference in survival between optimal medical therapy and CABG in patients with CAD amenable to surgery and LVEF < 35% at the end of the trial at a median follow up of 56 months.<sup>499</sup> An extended follow up of the trial, however, after a median follow-up of 9.8 years, found lower total mortality, CV death and the combined outcome of all cause death and CV hospitalizations in patients with severe HF (LVEF < 35%) who underwent CABG.<sup>500</sup> There was an early risk of mortality following CABG, but the benefits of CABG were seen after 2 years, and these were seen whether myocardial viability or angina was present or absent.

**Ila,B**

Coronary revascularization by CABG should be considered in patients with HF and suitable coronary anatomy.

A recently completed trial comparing PCI to OMT in patients with demonstrable myocardial viability and LVEF < 35% did not result in an improvement in CV outcomes.<sup>501</sup>

These high-risk patients undergoing CABG have a high surgical morbidity with longer intensive care unit stays, a median in-hospital stay of 9.0 days and higher 30-day mortality.<sup>499</sup> The treatment decision should be individualized after a discussion with the patient and family and considering the recommendations of the Heart Team.

**10.4.2 LV Reduction Surgery**

In the STICH trial, surgical ventricular restoration (SVR) resulted in a significant reduction of the end-systolic volume index but despite this, there were no differences in symptoms, exercise tolerance, rate of death or hospitalization compared with patients who underwent CABG alone.<sup>502</sup>

**III,B**

As such, SVR is not routinely recommended in patients with HF who have areas of LV dyskinesia or akinesia and undergoing CABG.<sup>502</sup>

**IIb,B**

LV aneurysmectomy may be considered in patients with a large discrete LV aneurysm who develop HF, angina pectoris, thromboembolism, and tachyarrhythmias due to the aneurysm.<sup>503</sup>

**10.4.3. Valve Surgery**

Please see section Section 14.2.5.1

**10.4.4. LV Assist Devices**

Left ventricular assist devices have been used to:

- Bridge patients with HF to heart transplant.<sup>504</sup>
- Support patients as a bridge to recovery.<sup>505</sup>
- Provide long term hemodynamic support in patients ineligible for heart transplantation (destination therapy).<sup>506</sup> In this group of patients, permanent treatment with LVADs improves survival and quality of life compared with optimal medical therapy alone. It is however expensive and is associated with significant adverse events.<sup>507-509</sup>

**IIa,B**

Patients awaiting heart transplant who have become refractory to medical therapy and requiring inotropic support should be considered for a mechanical support device as a bridge to transplant.<sup>507-509</sup>

**Key Messages # 10: Surgery for HF**

- Patients with HF should undergo surgery if the pathology causing the HF is amenable to surgical treatment.
- This is particularly so if they have ischemia demonstrated as angina or by non-invasive testing and an anatomy that is suitable for revascularization (left main stem or triple vessel disease).
- The decision to subject a patient to surgery should however consider the functional status, prognosis, and comorbid conditions of the patient.

**Key Recommendations #8: Chronic HF due to HFrEF**

## ● Surgery For HF

## ➤ CAD

- ◆ Patients with HF should undergo surgery if they have angina and anatomy that is suitable for revascularization (left main stem or triple vessel disease).

## ➤ Valve Disease

- ◆ All patients with VHD should be assessed periodically on the need for early intervention before they begin to develop symptoms of reduced effort tolerance and decompensate.
- ◆ Patients who are assessed to require intervention should be seen by a heart team to help decide the timing and type of intervention.
- ◆ The indications for valve intervention are as in the Appendix V, page 166.

**11. HEART FAILURE WITH MILDLY REDUCED LV EJECTION FRACTION (HFmrEF)**

Symptomatic HF patients with LVEF between 41-49% are now classified under this category (previously known as heart failure with mid-range ejection fraction).<sup>35</sup> HFmrEF, makes up about 10-20% of patients with HF.<sup>510</sup>

There are limited number of randomized control trials focusing on this category of patients specifically. Hence, the evidence for treatment of this group of patients are derived from post-hoc or sub-group analysis of previous HF trials conducted in patients with HFrEF and HFpEF.

HFmrEF is a clinical entity between HFrEF and HFpEF. The LVEF can:

- Remain 40 - 49%.
- Improved to > 50% - HFimpEF.
- Worsen < 40% HFpEF.

It may be helpful to re-evaluate the LVEF periodically to determine the trajectory of the disease process. It is important to appreciate that these trajectories may not be linear and uni-directional.

In the management of these patients:

I,A

- SGLT2-i have been shown to reduce the combined end point of worsening HF or CV death.<sup>511-513</sup>

I,B

- Diuretics are recommended in patients who remain symptomatic and show signs of congestion despite a SGLT2-i.<sup>514,515</sup>

IIa,B

- Sub-group analysis of large trials indicates that  $\beta$ -blockers, RAS blockers (ARNI, ACEI or ARB) and MRAs can be considered to reduce the risk of HF hospitalization and CV death.<sup>516-519</sup>
- CV risk factors must be optimized.

**Key Messages # 11: Chronic HF due to HFmrEF**

- HFmrEF includes symptomatic HF patients with LVEF between 41-49% and is a clinical entity between HFrEF and HFpEF.
- There are limited number of randomized control trials focusing on this category of patients - most of the evidence is derived from post-hoc or sub-group analysis of previous HF trials.

**Key Recommendations #9: Chronic HF due to HFmrEF**

- The management of these patients include:
  - Optimal treatment of CV risk factors.
  - SGLT2- i
  - Diuretics for patients who remain symptomatic and show signs of congestion.
  - $\beta$ -blockers, RAS blockers (ARNI, ACEI or ARB) and MRAs can be considered to reduce the risk of HF hospitalization and CV death.

**12. HEART FAILURE WITH IMPROVED LV EJECTION FRACTION (HFimpEF)**

This is a new category which includes patients with a:<sup>520</sup>

- Baseline LVEF of < 40% **and**
- > 10-point increase from baseline LVEF **and**
- A second measure of LVEF > 40%.

Patients with a baseline LVEF of 41-49% who have improved to a new baseline of LVEF > 50% may be categorized in this group.<sup>35</sup>

LV function and structural abnormalities do not fully normalize despite improvement in symptoms, functional capacity, and near normalization of biomarkers. Symptoms and signs may however relapse and cardiac biomarkers may increase if HF treatment is withdrawn.<sup>41</sup> This "remission" status requires continued treatment for HF.

Ila,B

Hence in patients with HFimpEF, Foundational HF medications should be continued to prevent relapse of symptoms and subsequent deterioration of LV function.<sup>41</sup>

**Key Messages # 12: Chronic HF due to HFimpEF**

- HFimpEF includes patients with a:
  - Baseline LVEF of < 40% **and**
  - >10-point increase from baseline LVEF **and**
  - A second measure of LVEF > 40%.
- LV function and structural abnormalities do not fully normalize despite improvement in symptoms, functional capacity, and near normalization of biomarkers.

**Key Recommendations #10: Chronic HF due to HF<sub>imp</sub>EF**

- Symptoms and signs may relapse, and cardiac biomarkers may increase if HF treatment is withdrawn.
- Foundational HF medications should be continued to prevent relapse of symptoms and subsequent deterioration of LV function.

**13. CHRONIC HEART FAILURE - HF DUE TO PRESERVED LV SYSTOLIC FUNCTION, LVEF > 50% - (HF<sub>p</sub>EF)**

HF<sub>p</sub>EF is a heterogenous disease with various phenotypes and comorbidities.<sup>521-524</sup> It is highly prevalent, accounting for up to 50% of all patients with HF and is becoming the dominant form of HF in aging populations worldwide.<sup>525,526</sup>

In this guideline HF<sub>p</sub>EF refer to LVEF  $\geq$  50%. Importantly, clinicians should be aware that LVEF is a continuous variable and the EF cut-offs used in definitions are therefore arbitrary.<sup>155,527,528</sup>

HF<sub>p</sub>EF differs from HF<sub>r</sub>EF in that HF<sub>p</sub>EF patients are older and more often female. AF, CKD, and non-CV comorbidities are more common in patients with HF<sub>p</sub>EF than in those with HF<sub>r</sub>EF.<sup>529</sup>

In the ASIAN-HF registry, patients with HF<sub>p</sub>EF were younger and leaner. They, however, carried a high comorbidity burden, with 70% of patients having a least two comorbidities -most commonly hypertension, followed by anemia, CKD, diabetes, ischemic heart disease (IHD), AF, and obesity.<sup>525</sup> Similarly in the MyHF registry, patients with HF<sub>p</sub>EF had a high comorbidity burden, commonly hypertension, diabetes, IHD and CKD.<sup>530</sup>

**13.1. Diagnosis**

The diagnosis of HF<sub>p</sub>EF is challenging as symptoms and signs can be attributable to other co-existing conditions and LVEF is normal. Chronic obstructive pulmonary disease and obesity per se can also contribute to dyspnea in these patients.

The main hemodynamic finding in HF<sub>p</sub>EF is an elevation in LV filling pressures i.e. end diastolic LV pressure (LVEDP). In the early stages of the disease, these become elevated only during exercise or exertion giving rise to exertional shortness of breath.<sup>531</sup> As the disease progresses, it becomes elevated even at rest.

As outlined previously, the definition of HF includes the presence of:

- Clinical symptoms and signs of HF **and**
- Raised NP - In general, the level of elevation of these peptides are less than that seen in patients with HF<sub>r</sub>EF.<sup>532</sup> NP levels may be normal in patients with HF<sub>p</sub>EF who are obese and falsely raised in patients without HF<sub>p</sub>EF but with AF.<sup>533-535</sup> Thus the diagnosis of HF<sub>p</sub>EF cannot be relied on solely on elevated NP levels.



If the patient has HF as per above definition, then basic investigations should be performed as outlined previously to look for possible etiology and the presence of comorbidities:

- ECG
- Chest X-ray
- Lab investigations
- Echocardiogram - Some features supporting the diagnosis of HFpEF include:<sup>536</sup>
  - LVEF  $\geq 50\%$  within 72 hours of the clinical event.
  - Left ventricular hypertrophy (increased LV wall thickness or LV mass index  $> 115\text{g/m}^2$  for men and  $> 95\text{g/m}^2$  for women).
  - Left atrial enlargement (LA volume index  $> 34\text{mL/m}^2$  or  $> 40\text{mL/m}^2$  in the presence of AF).
  - Diastolic dysfunction if  $E/e' \geq 15$ .<sup>536</sup> A mean  $E/e'$  index  $> 15$  at rest has good diagnostic value for identifying a high mean pulmonary capillary wedge pressure (mPCWP), supporting the likelihood of HFpEF but an  $E/e'$  ratio within the intermediate range (9-14) is less sensitive. The  $E/e'$  index cannot be recommended as a single diagnostic index above all other non-invasive measures of filling pressures (such as retrograde pulmonary venous flow).<sup>536</sup>
  - Tricuspid valve regurgitation velocity  $> 2.8\text{m/s}$ .

HFpEF may be suspected in patients who have dyspnea and in whom non-cardiac causes of breathlessness have been excluded. In addition, the:

- The LVEF  $> 50\%$  **and**
- There is no significant heart valve disease or cardiac ischemia.

Many patients who are elderly, obese and who have hypertension, diabetes and/or AF presenting with dyspnea would fall into this category.

The diagnosis of HFpEF is often not clear cut. It does not depend on a single parameter but on a combination of parameters derived from clinical presentation, laboratory results and imaging tests that together will give a probability for the diagnosis.<sup>536,537</sup>

In difficult diagnostic situations, 2 scores have been proposed to aid in the initial assessment of patients suspected of having HFpEF (Appendix VI & VII, page 167-168)

- H2FPEF score<sup>538</sup>
- HFA-PEFF score<sup>536</sup>

In cases of intermediate probability of the disease based on the scores, confirmatory tests such as exercise echocardiography, invasive measurement of intracardiac pressures i.e. the mPCWP at rest and sometimes with exercise is necessary to make the diagnosis. (Appendix VI, page 167)

These tests are best performed in specialized cardiology centers and most doctors may not have access to them. This limits the broad clinical applicability of the scores and demonstrates the ongoing difficulty in making the diagnosis of HFpEF.<sup>539</sup>

The following patients who continue to have dyspnea may be referred to these centers for further evaluation and management:

- Cases where the diagnosis is unclear.
- Non responders to treatment especially those with multiple comorbidities.

**13.2. Etiology and Associated Comorbidities**

- HFpEF may also share similar clinical characteristics with valvular heart disease, pericardial disease, and high-output HF. In all these conditions, patients present with HF and have normal LVEF and even normal LV dimensions. In HFpEF however, the HF is due to myocardial disease resulting in abnormal myocardial relaxation, decreased compliance, and increased filling pressure.<sup>539</sup> For this reason, a basic echocardiogram is essential in the initial work up of these patients.
- The etiological factors affecting HFpEF and HFrEF seem to be different. When compared to patients with HFrEF, patients with HFpEF are more likely to have hypertension, valvular heart disease (e.g., aortic stenosis) and AF and are less likely to have a MI or left bundle branch block.<sup>540</sup>
- Comorbidities play an important role in the pathophysiology of HFpEF. These contribute to systemic and endomyocardial inflammation with fibrosis which are important in the pathogenesis of the disease.<sup>524</sup> It is these comorbidities that primarily correlate with outcome in patients with HFpEF rather than NP levels or echocardiographic parameters.<sup>541</sup>
  - Hypertension remains the most prevalent comorbidity of HFpEF, with a prevalence of 60% to 89%.<sup>542-544</sup>
  - Other comorbidities include :<sup>542-544</sup>
    - ◆ Overweight or obesity- Obesity is not only a comorbidity, but it is also an important risk factor for HFpEF development. The mechanisms involved are complex and include neurohormonal mechanisms and oxidative stress.<sup>545</sup> Morbid obesity also has an adverse effect on cardiac remodeling.<sup>545</sup>
    - ◆ Diabetes mellitus (DM)
    - ◆ Chronic obstructive pulmonary disease (COPD)
    - ◆ Obstructive sleep apnea (OSA)
    - ◆ Anemia
    - ◆ Coronary artery disease (CAD)
    - ◆ Chronic kidney disease (CKD)
  - The presence of diabetes, a lower systolic BP, lower hemoglobin and a lower eGFR were associated with a poorer outcome.<sup>541</sup>
  - AF is common in HFpEF and increases the risk of adverse outcomes.<sup>546</sup>

**13.3 Management (Table 6, page 44)**

Compared with HFrEF patients, hospitalizations, and deaths in patients with HFpEF are more likely to be due to non-cardiovascular causes. As LVEF increases, the proportional contribution of non-cardiac and non-HF events to death or hospitalization increases, highlighting the importance of managing comorbidities.

The important aim of therapy is to alleviate symptoms, improve well-being and reduce hospitalizations. Screening for comorbidities and treating these appropriately is important.

**13.3.1 Lifestyle measures**

- **Overweight or obesity** are important comorbidities in HFpEF.
  - A caloric restriction diet is feasible and safe in older, obese patients with HFpEF. It has been shown to significantly improves patient's dyspnea, peak oxygen consumption, and quality of life.<sup>547</sup>
  - Caloric restriction should ideally be combined with exercise, such as walking exercise for one hour three or more times per week.<sup>169,547,548</sup>
  - Bariatric Surgery in patients with HFpEF and obesity was associated with improved symptoms, reduction in HF hospitalizations, reverse cardiac remodelling, and improved LV distensibility.<sup>549-551</sup>

**● Exercise training:**

- This is safe and improves exercise capacity and quality of life.<sup>169,547,551,552</sup>
- Combined endurance/resistance training appears safe for patients with HFpEF and improves exercise capacity (as reflected by an increase in peak oxygen consumption), physical functioning score and diastolic function.<sup>169,548,549</sup> It should consist of dynamic isotonic (e.g., walking or cycling) and not static exercise.
- High-intensity interval training or moderate continuous training has not proved to be better than guideline-based physical activity for patients with HFpEF.<sup>173</sup>

**13.3.2 Identifying and treating the underlying cause(s) and co-morbidities.**

- **Hypertension** should be treated to target goals.<sup>147</sup> Improved BP control has been shown to reduce morbidity and hospitalizations for HF.<sup>119,554-555</sup> More intensive BP control in patients with high CV risk including those > 75 years of age, significantly reduces HF and other cardiovascular outcomes.<sup>553,556,557</sup>

- **CAD** is common in patients with HFpEF and this should be treated appropriately.

- **Diabetes** - See section 14.1

- **Chronic Kidney Disease** - See Section 14.6

- **Tachyarrhythmias** should be treated, and sinus rhythm restored whenever possible.

- If the patient remains in persistent AF,  $\beta$ -blockers or non-dihydropyridine calcium channel blockers (verapamil, diltiazem) alone or in combination are the usual first line agents used for rate control.

- In patients with recent onset AF < 1 year duration or paroxysmal AF who received early rhythm-control, there was a significantly decreased primary end point of a composite of CV death, stroke, or hospitalization due to HF than those who received rate control.<sup>416,425</sup>

- **Anticoagulation** - Patients with paroxysmal or persistent AF should be anticoagulated to reduce the risk of thromboembolic events.<sup>412,416</sup>

**13.3.3 Pharmacological options**

Pharmacotherapy has not been shown to reduce total or CV mortality. These include:

- **Diuretics:** These are necessary to control pulmonary congestion and peripheral edema but should be used cautiously so as not to lower preload excessively and thereby reduce stroke volume and cardiac output. One should aim to use the lowest dose of a diuretic necessary to maintain euvolemia.

- **$\beta$ -blockers:**  $\beta$ -blockers are often prescribed for the management of comorbidities such as coronary artery disease or atrial fibrillation. At present, however, there is no good data to show that  $\beta$ -blockers are beneficial in the treatment of HFpEF.<sup>558</sup> Furthermore, the type of  $\beta$ -blockers (non-selective and vasodilating, such as nebivolol and carvedilol, vs. rate controlling only) may have differential effects in different HFpEF phenotypes.<sup>559</sup>  $\beta$ -blocker withdrawal was shown to improve maximal functional capacity and chronotropic incompetence in patients with HFpEF.<sup>560</sup> Further studies are warranted on the use of  $\beta$ -blockers in these patients.

**● RAS blockers**

- Current evidence does not support their routine use in HFpEF in the absence of an alternative indication.<sup>561,562</sup> Trial data show a reduction in HF hospitalizations, but no reduction in all-cause or CV mortality in HFpEF.<sup>559</sup>

- **ARB** - the use of ARB may be considered to decrease hospitalizations, particularly among patients with LVEF on the lower end of this spectrum.<sup>563</sup>

- **ACE-I** - may be considered to reduce hospitalization in patients with HFpEF.<sup>564</sup>



IIb,B

➤ **ARNI** - did not reduce the composite endpoint of CV death and total hospitalizations for HF in patients with a LVEF > 45% when compared to an ARB.<sup>565</sup> In a combined analysis of 2 large ARNI trials, there was a suggestion of benefit in patients with LVEF < 57% and in women (benefits of sacubitril-valsartan in women were sustained up to LVEF 60%, while for men the benefit was restricted to LVEF < 45%).<sup>566</sup>

IIb,B

- **MRA:** MRAs improve diastolic function in patients with HFpEF.<sup>567</sup> It may be considered to decrease HF hospitalizations, particularly among patients with LVEF on the lower end of this spectrum.<sup>562,568,569</sup> Benefits were seen in patients with LVEF > 45%, elevated BNP level or HF admission within 1 year, eGFR > 30 mL/min/1.73 m<sup>2</sup>, creatinine < 2.5 mg/dL, and potassium < 5.0 mEq/L).<sup>323</sup> MRAs showed a greater benefit in patients who demonstrated more functional impairment, obesity, diabetes, CKD, concentric LV hypertrophy, high renin, and biomarkers of tumor necrosis factor-alpha-mediated inflammation, liver fibrosis, and tissue remodeling.<sup>570</sup> Hyperkalemia was more common in those on MRA and close monitoring of potassium is important.

I,A

- **SGLT2-i** - These have been shown to decrease HF hospitalizations and CV mortality.<sup>511-513</sup> There was also a decrease in the slope of the eGFR decline, and a modest improvement in quality of life at 52 weeks.<sup>511-513</sup> The benefit was similar irrespective of the presence or absence of diabetes at baseline. In a subgroup analysis by LVEF, there was a signal for lower benefit on the primary composite endpoint, first and recurrent HF hospitalizations, at higher LVEF > 62.5%.<sup>511</sup>
- **Non-dihydropyridine Calcium Channel Blockers (verapamil and diltiazem):**
  - Clinical data regarding use of calcium-channel blockers in HFpEF is sparse. Its use is mainly to treat hypertension or for rate control in AF.

**Key Message #13: Chronic HF due to HFpEF**

- HFpEF is a heterogeneous disease that is highly prevalent, accounting for up to 50% of all patients with HF and is becoming the dominant form of HF in aging populations worldwide.
- In this guideline HFpEF refer to LVEF ≥ 50%. LVEF is a continuous variable and the EF cut-offs used in definitions are therefore arbitrary.
- HFpEF differs from HFrEF in that HFpEF patients are older, more often female with AF, CKD and non-CV comorbidities being more common.
- The main hemodynamic finding in HFpEF is an elevation in LV filling pressures i.e. end diastolic LV pressure (LVEDP), initially only on exertion and later even at rest.

**Key Recommendation # 11: Chronic HF due to HFpEF**

- Timely and early diagnosis of HFpEF leads to a better outcome.
- The management of patients with HFpEF includes:
  - Lifestyle measures
    - ◆ Weight reduction
    - ◆ Exercise training
  - Managing comorbidities such as hypertension, CAD, CKD, atrial fibrillation, obesity according to guidelines.
  - Pharmacotherapy:
    - ◆ Diuretics - for volume overload.
    - ◆ SGLT2-i - decreases HF hospitalizations and CV mortality.
    - ◆ RAS blockers (ARB, ACE-I, ARNI) - As a group, RAS blockers have not been shown to reduce total or CV mortality. There was a suggestion of benefit of ARNI in women up to LVEF 60%, while for men the benefit was restricted to LVEF < 45%.
    - ◆ MRA - it may be considered to decrease HF hospitalizations, particularly among patients with LVEF on the lower end of this spectrum.
    - ◆  $\beta$ -blockers - No good data that they are beneficial in the treatment of HFpEF although they are often prescribed for the management of comorbidities such as CAD or AF.

**14. SPECIAL GROUPS****14.1. DIABETES AND HEART FAILURE**

- HF and diabetes mellitus (DM) often co-exist, each increasing the likelihood of developing the other.<sup>571</sup> Diabetes is one of the risk factors for the development of HF and similarly among patients with HF there is a high prevalence of often unrecognized dysglycemia.<sup>572-574</sup>
- In Malaysia, about 60-65% of patients admitted with HF, have diabetes as a comorbidity.<sup>13-15</sup>
- Almost all the epidemiological and clinical trial data on DM and HF have been based on trials done on patients with type 2 DM.

**14.1.1 Risk of developing dysglycemia in Patients with HF**

- Patients with HF irrespective of ejection fraction phenotype, were shown to be at an increased risk of developing dysglycemia- both prediabetes and DM.<sup>573,574</sup>
- The prevalence of DM varies<sup>571</sup>:
  - 6-25% in patients with left ventricular systolic dysfunction (LVSD).
  - 12-30% in patients with symptomatic HF.
  - Up to 40% in patients hospitalized with HF.
- In the clinical trials, however, the incidence of new onset diabetes varied from 6-7%.<sup>571,575,576</sup>
- Predictors for developing new onset DM among patients with HF include:<sup>573,574,577-581</sup>
  - Severity (as reflected by a higher NYHA functional class) and duration of HF.
  - Elevated body mass index and waist circumference.
  - History of smoking.
  - Higher systolic blood pressure.
  - Certain medications such as loop diuretics, metoprolol.
- The presence of DM has been shown to be linked to adverse outcomes in HF of both ischemic and non-ischemic etiologies, irrespective of LVEF.



- All patients at Risk (Stage A) and who already have HF (Stage B-D) should be screened for dysglycemia and this should be treated appropriately according to guidelines.<sup>322</sup>

#### **14.1.2 Risk of developing HF among patients with dysglycemia**

- Diabetic patients have between 2-5x risk of developing HF.<sup>572</sup> Whether this indicates a causal effect or just a comorbidity is still unclear. It is an important CV complication and the prognosis of diabetic patients admitted to hospital for HF were worse than those admitted with MI.<sup>582,583</sup>
- The prevalence of HF ranges between 4-30% in various diabetic drug trials, with almost a third of patients being undiagnosed.<sup>584</sup>
- Even pre-diabetics (depending on the definition used) have a 9-58% risk of developing HF.<sup>585</sup> It is often the first manifestation of a CV event.<sup>571</sup>
- Predictors for the development of HF in patients with DM include:<sup>571,586-589</sup>
  - Female sex
  - Longer duration of DM
  - Older age
  - Increased body mass index
  - Poorer glycemic control
  - Insulin use
  - Renal impairment

#### **14.1.3. Prognosis of patients with Dysglycemia and HF**

- HF per se is associated with a high morbidity and mortality and concomitant DM compounds this risk.<sup>590-593</sup>
- This is particularly so in patients with HFpEF who have co-existing microvascular complications.<sup>594</sup> Both poor glycemic control (HbA1c >9.5%) and tight glycemic control (HbA1c <5.5%) are associated with increased all-cause mortality in this group of patients.<sup>595</sup>
- In Acute HF, incidence of hospitalization (adjusted for age and sex) is 2x higher in patients with DM compared with those without.<sup>596,597</sup> It is an independent predictor for overall mortality in acute HF.<sup>598</sup>
- Every 1% HbA1c reduction is associated with a 15% relative risk reduction in hospitalization for HF.<sup>599</sup>
- Inadequate glycemic control (HbA1c ≥ 7.0%) within one year after hospital discharge for HF, was significantly associated with a higher risk for all-cause mortality compared with an adequately controlled diabetes (HbA1c < 7.0%).<sup>598</sup>
- Once HF develops, the clinical course is marked by frequent hospitalizations and eventually death.<sup>600-602</sup> Most sudden deaths are due to LV dysfunction rather than a new ischemic event.
- Advanced age, duration of the disease, insulin use, the presence of CAD and an elevated serum creatinine are all independent risk factors for the development of HF.<sup>603</sup>
- However, despite having five risk factor variables within the target range i.e. glycated hemoglobin level, low-density lipoprotein cholesterol level, blood pressure, absence of albuminuria and smoking, the risk for hospitalizations for HF was increased by almost 45% over a 5.7-year period.<sup>604</sup>

**14.1.4. Pathophysiology of HF in DM**

- The development of HF in DM involves a complex interconnected process between systemic, myocardial, and cellular mechanisms that lead to coronary atherosclerosis, myocardial ischemia and infarction, with co-existing hypertension and dyslipidemia.<sup>605</sup>
- Diabetic cardiomyopathy is defined as the presence of diastolic or systolic dysfunction in a patient with DM without other obvious causes for cardiomyopathy, such as CAD, hypertension, or valvular heart disease.<sup>605-607</sup> However, whether diabetes directly causes a cardiomyopathy is, however, uncertain.<sup>582,584</sup>

**14.1.5. Diagnosis of HF in diabetes**

- The diagnosis of HF in patients with diabetes is like those without diabetes (see section 6).
- In those patients suspected of having HF, appropriate investigations as outlined in section 6 should be performed.

**14.1.6. Management****14.1.6.1. Prevention of HF**

- Diabetes is a CVD defining disease, and patients should have their other CV risk factors treated aggressively and closely monitored.<sup>322</sup>
- Lifestyle measures are important and have been shown to lower risk for HF, particularly in patients with HFpEF.<sup>608</sup> These include:<sup>322</sup>
  - Weight reduction,
  - Increased physical activity; and
  - Smoking cessation.
- The BP and LDL-C should be treated to target:<sup>322</sup>
  - BP 130-139/70-79mmHg, SBP < 130 in those without pre-existing CAD (but not < 120 mmHg) and who are at a higher risk of CKD or stroke.
  - LDL-C according to CV risk category.
- Ambulatory or home BP monitoring should be encouraged as undetected hypertension and masked hypertension is highly prevalent among patients with DM.<sup>609</sup>
- In asymptomatic patients, routine screening for CAD is not recommended. Routine screening does not improve outcomes if CV risk factors are not treated to target.<sup>322</sup>

I,A

I,B

I,B

I,A

I,A

**14.1.6.2. Treatment of HF**

- The treatment of both Acute and Chronic HF in patients with DM is similar to those without DM with the use of the Foundational HF medications.

**14.1.6.3. Treatment of Diabetes -Glucose Lowering Agents**

- Intensive glucose lowering is not associated with any significant reduction in CV risk.<sup>610</sup> In fact, tight control of DM especially with the occurrence of hypoglycemia is associated with increased mortality.<sup>611-613</sup>
- In addition, HbA1c variability, has been shown to be an independent risk factor for CVD in T2DM patients, even when the HbA1c is within the target range.<sup>614</sup>
- Glycemic targets for patients with DM and HF should be individualized to include considerations of comorbidities, severity of HF, and to balance benefits versus potential risks.<sup>322,605</sup>

IIb,B

I,A



I,A

- However, due to limited HF-specific data in DM patients, current recommendations suggest a target range of HbA1c 7%- 8%, while minimising adverse effects of treatments particularly hypoglycaemia.<sup>322,584,605</sup> This target however should be individualized based on patient characteristics and comorbidities.

**A) Sodium-glucose cotransporter-2 inhibitors (SGLT2-i)**

I,A

- SGLT2 -i showed a reduced risk of major adverse CV events, with the greatest benefit being seen for reduction in hospitalizations for HF.<sup>615,616</sup> This is the most consistent observation in CV outcome trials across this class.<sup>615,616</sup>
- In patients with DM and established CVD or at high CV risk, SGLT2-i should be given to prevent hospitalizations for HF.<sup>155</sup>
- SGLT2-i with the strongest with the strongest proven benefits for HF hospitalizations should be used.
- However, SGLT2-i should be used with caution, taking into consideration recognized side effects including genital candidiasis, euglycemic diabetic ketoacidosis, as well as lower-limb amputation and fractures with Canagliflozin.<sup>605</sup>

**B) Glucagon-Like Peptide 1 (GLP-1) Receptor Agonists**

IIa,A

- In patients with DM and established CVD or multiple risk factors for CVD, GLP-1 receptor agonists with proven CV benefits is recommended to reduce the risk of major adverse cardiovascular events.<sup>322,584,617</sup>
- GLP-1 receptor agonists show a 9%-12% reduction of the risk of HF hospitalization.<sup>618,619</sup> However, to date, there is insufficient data to include GLP-1 receptor agonists in the recommended treatment of patients with diabetes for HF risk reduction.<sup>620,621</sup>

**C) DPP4 inhibitors**

IIb,B

- DPP4 inhibitors confer no added benefit for therapy of DM in HF patients compared to placebo.<sup>617</sup>

III,A

- Saxagliptin and Alogliptin are not recommended in patients with DM and HF.<sup>155,156,622</sup>

**D) Biguanides (Metformin)**

IIa,B

- Observational studies indicate that metformin can be safely used in HF with no increased mortality even in those with NYHA class III and IV.<sup>623</sup>
- It also reduced the risk of hospitalization due to HF and all-cause hospitalization.<sup>623,624</sup>
- Lactic acidosis during hospitalization was uncommon.<sup>625</sup>
- Metformin should be discontinued when the eGFR is <30ml/min/1.73m<sup>2</sup>.
- In older patients with DM who were hospitalized for HF, metformin initiation was independently associated with a significant reduction in HF hospitalization particularly in patients with LVEF > 40%.<sup>626</sup> Conversely, sulfonylurea initiation was associated with excess risk of death and HF hospitalization, regardless of LVEF.<sup>626</sup>
- Metformin is recommended for the treatment of DM in patients with HF if the eGFR is stable and ≥ 30 mL/min/1.73m<sup>2</sup>.
- It should be used cautiously in unstable or hospitalized patients with HF.<sup>584,617,627</sup>

**E) Sulfonylureas****IIb,B**

- Current evidence based on observational cohort studies, suggests that sulfonylureas (including 2<sup>nd</sup> generation drugs) increase the risk of HF and therefore should be avoided if possible.<sup>605,628-631</sup>
- Sulfonylureas have been shown to be as safe as the DPP4 inhibitors as an add-on therapy to metformin or other diabetic medications for atherosclerotic CVD end points i.e. CVD death, nonfatal MI or nonfatal stroke. The occurrence of HF, however, was either not a key end point or not reported in these trials.<sup>632-634</sup> A large cohort study however, found the use of the second-generation sulfonylurea, glimepiride to be safe in patients with diabetes and chronic HF.<sup>635</sup>

**F) Insulin****IIb,C**

- Insulin therapy may be considered in patients with advanced systolic HF<sub>r</sub>EF for better glycemic control.<sup>584</sup>
- However, insulin has been associated with increased all-cause mortality and hospitalization for HF.<sup>636,637</sup> This is especially in patients with low HbA1c < 7%.<sup>636</sup>

**G) Thiazolidinedione****III,A**

- TZD use including pioglitazone and rosiglitazone should be avoided in patients with T2DM and symptomatic NYHA class III & IV heart failure.<sup>584,617,638,639</sup>

**Key Messages #14: HF and Diabetes**

- HF and diabetes mellitus (DM) often co-exist, each increasing the likelihood of developing the other. Whether this indicates a causal effect or just a comorbidity is still unclear.
  - Patients with HF irrespective of ejection fraction phenotype, have been shown to be at an increased risk of developing dysglycemia.
  - Also, diabetic patients have between 2-5x risk of developing HF.
- HF per se is associated with a high morbidity and mortality and concomitant DM compounds this risk. This is particularly so in patients with HF<sub>p</sub>EF.

**Key Recommendations #12: HF and Diabetes****• Prevention of HF:**

- Lifestyle measures are important. They have been shown to lower the risk for HF, particularly HFpEF. These measures include:
  - ◆ Attainment of appropriate weight,
  - ◆ Increased physical activity and
  - ◆ Smoking cessation.
- The BP and LDL-C should be treated to target:
  - ◆ BP 130-139/70-79mmHg, SBP < 130 in those without pre-existing CAD (but not < 120 mmHg) and who are at a higher risk of CKD or stroke
  - ◆ LDL-C according to CV risk category.

**• Treatment of HF**

- The treatment of both Acute and Chronic HF in patients with DM is similar to those without DM with the use of the Foundational HF medications.

**• Treatment of Diabetes - Glucose lowering Agents**

- Glycemic targets for patients with DM and HF should be individualized.
- Current recommendations suggest a target range of HbA1c 7%- 8%, while minimizing adverse effects of treatments particularly hypoglycemia.
- Glucose lowering drugs:
  - ◆ With proven CV benefits:
    - SGLT2-i
    - GLP-1 receptor agonists
    - Metformin
  - ◆ With no proven CV benefits but which will help with glucose control:
    - DPP4 inhibitors
  - ◆ With no proven CV benefits and need to be used with caution:
    - Insulin -This has been associated with increased all-cause mortality and hospitalization for HF especially in patients with low HbA1c < 7%.
    - Sulfonylurea
  - ◆ That need to be avoided:
    - Thiazolidinediones

**14.2. VALVULAR HEART DISEASE (VHD) AND HF****14.2.1. Etiology**

Common etiological factors are:

- Rheumatic Carditis and Rheumatic Heart disease (RHD)
  - In a patient with acute rheumatic carditis, the early detection and appropriate treatment of beta-hemolytic streptococcal infections of the pharynx (positive throat swab and raised ASOT) during the infection and with subsequent appropriate penicillin prophylaxis against recurrent infection will help prevent the development of RHD in later life.
  - RHD is a sequela of acute rheumatic carditis. It occurs because of inflammation and scarring of the heart valves leading to fusion of the commissures and/or fusion and shortening of the chordae and deformation of the valve cusps. The incidence of Rheumatic Fever and RHD is still high among the rural and urban poor.<sup>640</sup>
- Infective Endocarditis
  - This can result in acute valvular damage or may exacerbate an underlying valve pathology.
  - A common source is poor oral hygiene.
  - An often-unrecognized source is i.v. lines.



- Congenital
  - This includes congenital bicuspid aortic valves, congenital mitral valve prolapse (Barlow's disease) and pulmonary stenosis.
- Degenerative disease
  - Commonest cause in the elderly.
  - This includes calcific aortic stenosis, aortic regurgitation secondary to aortic root dilatation from any cause, and degenerative mitral regurgitation.

#### 14.2.2. Valve pathology

- **Mitral Valve Disease**
  - Mitral stenosis (MS)
    - ◆ This is almost always due to RHD.
  - Mitral Regurgitation
    - ◆ Primary mitral regurgitation is a disease of the mitral valve leaflets or chordae. It is the most common primary VHD seen locally.
      - In young adults and middle-aged persons, it is commonly due to RHD. It may also be secondary to collagen disorders such as Marfan's.
      - In the older age group (>70 years of age), degenerative causes leading to fibroelastic deficiency, myxomatous degeneration, chordal elongation and rupture are more common.
    - ◆ Secondary mitral regurgitation also known as functional MR, is a consequence of LV dysfunction with normal mitral valve leaflets and chordae. The LV dilatation may be secondary to CAD or any other non-ischemic etiology.
- **Aortic Valve Disease**
  - Aortic stenosis (AS)
    - ◆ May occur due to congenital bicuspid aortic valve.
    - ◆ More commonly develops in the elderly due to progressive degeneration, scarring and calcification of the valve.
  - Aortic regurgitation (AR)
    - ◆ Locally it is commonly due to RHD.
    - ◆ Bicuspid aortic valve is the commonest congenital cause.
    - ◆ Other causes include infective endocarditis or secondary to dilatation of the ascending aorta due to Marfan's or idiopathic aortic dilatation etc.

#### 14.2.3. History and Physical Examination

- Common presenting symptoms are dyspnea, reduced effort tolerance and palpitations.
  - The presence of symptoms of HF in a stenotic lesion is a strong indication for intervention.
  - In regurgitant lesions, particularly with aortic regurgitation, it is usually a late presentation.
- The presence of valvular AF indicates significant valve pathology - usually due to mitral valve disease.

#### 14.2.4. Investigations

Essential investigations include:

- ECG
- Chest Radiograph - Chest X-rays
- Echocardiography -
  - Transthoracic echocardiography is usually adequate to make the diagnosis and plan management.



- Transesophageal echocardiography may be required if the transthoracic echocardiography is of suboptimal quality or occasionally for prosthetic valve dysfunction, prosthetic valve endocarditis or left atrial thrombi. It is also necessary preoperatively when detailed valve anatomy is required to assess valve repairability.

#### 14.2.5. Management

**I,C**

All patients with VHD should be assessed periodically on the need for early intervention before they begin to develop symptoms of reduced effort tolerance and decompensate. This can be achieved by:

- Serial Chest X-rays to look for progressive cardiac enlargement.
- Serial echocardiograms are more sensitive to look for progressive deterioration. It can assess valve severity, chamber dilatation, pulmonary artery pressures and cardiac (both LV and RV) function.
- Objective assessment of the patient's effort tolerance and symptoms in doubtful cases.

Patients with VHD who are assessed to require intervention should be seen by a heart team to help decide the timing and type of intervention.<sup>641</sup> Valve intervention include:

- Surgical repair or valve replacement.
- Percutaneous intervention.

The initial medical management of patients with VHD and HF (especially those with secondary mitral regurgitation) is as outlined earlier with the Foundational HF medications.

**III,C**

In addition,

- Vasodilators should be avoided in severe valve stenosis.

**IIa,B**

- Atrial Fibrillation should be managed with:

**IIb,B**

- Rate control using:

- ♦  $\beta$ -blockers and/or
- ♦ Digoxin for rate control

**I,A**

- Anticoagulation with DOAC or vitamin K antagonists to prevent thrombo-embolism.

##### 14.2.5.1. Valve Intervention in patients with HF

The indications for valve intervention are listed in Appendix V, page 166.

- Mitral valve intervention
  - Mitral stenosis
    - ♦ If the anatomy is suitable, percutaneous balloon mitral valvuloplasty is the procedure of choice.
    - ♦ If not suitable, then open mitral commissurotomy or mitral valve replacement may be indicated.
  - Mitral Regurgitation (MR)
    - ♦ Patients with HF of non-ischemic origin and severe functional mitral regurgitation may have symptomatic improvement after mitral valve surgery.<sup>641</sup> If the LVEF < 30%, mitral valve repair is preferred as mitral valve replacement is associated with poorer outcomes.<sup>642</sup>
    - ♦ Patients with LV systolic dysfunction undergoing surgical coronary revascularization who also have severe mitral regurgitation secondary to ventricular dilatation (functional mitral regurgitation) may be considered for concomitant mitral valve repair or replacement.<sup>643,644</sup>



- ♦ In patients with functional moderate to severe MR and who are not surgical candidates, the use of a MitraClip on a background of Foundational HF medications is superior to medical therapy alone.<sup>645-647</sup> At 2 years, there was a reduction in HF hospitalization and mortality in symptomatic HF patients with grade 3-4+ MR.<sup>648</sup>
- ♦ Compared to surgery, MitraClip demonstrated a similar safety profile and shorter length of stay in high-risk patients, at the expense of increased residual mitral regurgitation and higher reoperation rate.<sup>649</sup>
- ♦ The other landmark transcatheter mitral valve intervention study, however, did not meet its primary end point although there was a trend towards benefit in the cumulative rate of HF hospitalization between 12 and 24 months among the treated patients.<sup>650,651</sup>
- ♦ In view of the conflicting results, additional studies are needed to identify patients who will benefit the most from transcatheter mitral valve interventions.<sup>641</sup>
- ♦ At present, it may be considered in patients with moderate to severe MR and who are not surgical candidates and in high volume centers where this is done routinely.<sup>645-651</sup>

**Ila,B****• Aortic valve intervention****➤ Aortic stenosis (AS)**

- ♦ Severe AS has been defined as a:<sup>641</sup>
  - Valve area  $< 1.0 \text{ cm}^2$ ,
  - A mean transvalvular gradient  $> 40 \text{ mm Hg}$ , and
  - Peak flow velocity  $> 4.0 \text{ m/s}$ .
- ♦ In the absence of symptoms and adverse prognostic features, watchful waiting has generally been recommended with prompt intervention at symptom onset.<sup>641</sup>
- ♦ Variants include:
  - Severe AS with high gradient.
  - Severe AS with low gradient and poor LV function.
  - Severe AS with low/ normal gradient and preserved LV function.
- ♦ Low dose dobutamine stress echocardiography can help in the diagnosis when the LVEF  $< 50\%$  and the stroke volume index  $< 35 \text{ mL/m}^2$ .<sup>641,652</sup>
  - In patients with true AS, the increased flow across a fixed valve orifice results in increased transvalvular flow velocity and gradients, without a change in calculated valve area.
  - In pseudo-AS, the augmented flow results in only a mild increase in transvalvular gradient and an increase in valve area to  $\geq 1.0 \text{ cm}^2$ .
- ♦ In the presence of preserved LV function, LVEF  $> 50\%$ , the diagnosis is more challenging.<sup>641</sup>
- ♦ In elderly patients  $> 70$  years of age at moderate to high surgical risk, transaortic valve replacement is a reasonable alternative to surgical valve replacement.<sup>641,653-658</sup>

**Ila,B****➤ Aortic Regurgitation**

- ♦ In patients not reaching the thresholds for surgery, close follow-up is advised. They should be considered for surgery if:<sup>641</sup>
  - Exercise testing indicates borderline symptomatic patients or
  - There is progressive enlargement of the left ventricular end-diastolic diameter (LVEDD)  $> 65 \text{ mm}$ , or
  - Progressive decrease in LVEF.

**Key Messages # 15: HF and Valvular Heart Disease (VHD)**

- VHD is an important cause of HF.
- In the young, RHD is an important cause. The incidence of RHD in rural and urban poor is high.
- In the older population, degenerative valve disease is more common.

**Key Recommendations #8: HF and Valvular Heart Disease**

- All patients with VHD should be assessed periodically on the need for early intervention before they begin to develop symptoms of reduced effort tolerance and decompensate.
- Patients who are assessed to require intervention should be seen by a heart team to help decide the timing and type of intervention.
- The indications for valve intervention are as in the Appendix V, page 166.

**14.3. CARDIOMYOPATHY AND HEART FAILURE**

Cardiomyopathies are a heterogeneous group of myocardial disorders which frequently present as HF.<sup>659</sup> They can be inherited (familial/genetic) or acquired and may be accelerated by disease modifiers such as infection, immunological diseases, drugs, and comorbidities such as hypertension and diabetes mellitus.<sup>659-661</sup> In clinical practice they are usually classified according to anatomic and physiologic features into:

- Dilated cardiomyopathy (DCM)
- Hypertrophic cardiomyopathy (HCM)
- Restrictive cardiomyopathy (RCM)
- Arrhythmogenic cardiomyopathy
- Unclassified cardiomyopathy

As the presentation of arrhythmogenic cardiomyopathy is typically ventricular arrhythmias rather than HF, it will not be discussed further in this document.

It is worthwhile noting that specific etiologies of cardiomyopathy can have overlapping clinical phenotypes or transform from one clinical phenotype into another. For instance, Anderson-Fabry disease can have overlapping features of HCM and RCM, while RCM due to amyloidosis may transform into DCM as the disease progresses.

Where applicable, genetic testing and family screening should be considered.

**14.3.1 Dilated Cardiomyopathy (DCM)**

- DCM is one of the leading causes of HF/rEF globally.<sup>662</sup>
- It is characterized by systolic dysfunction and ventricular dilatation in the absence of abnormal loading conditions (e.g., hypertension, valvular stenosis) or CAD<sup>659</sup>. Other morphological changes include atrial dilatation, reduction in ventricular wall thickness, and functional mitral and tricuspid regurgitation.
- The etiologies are myriad but can generally be classified into genetic or non-genetic causes (Appendix VIII, page 169). In some circumstances these causes can overlap and interact with one another e.g., cardiotoxicity from chemotherapeutic agents in a patient predisposed genetically to DCM.



- A family history can be detected in 30-50% of cases<sup>663</sup> and a genetic determinant in up to 40% of DCM patients.<sup>659,664</sup> More than 60 genes coding for various myocardial proteins and molecules have been implicated in the pathogenesis of DCM.<sup>664,665</sup> The most common inheritance pattern is autosomal dominant.
- The clinical course of DCM is variable, but it generally follows of one these pathways<sup>666</sup>:
  - Recovery following incident HF.
  - Remission with improvement or stabilization of LV systolic function.
  - Progression to advanced HF and death.
- Reverse remodeling is usually observed in DCM resulting from potentially reversible causes such as alcohol-related cardiomyopathy, peripartum cardiomyopathy and tachycardia-induced cardiomyopathy.

#### **14.3.1.1. Management**

- Management of dilated cardiomyopathy is as outlined in the earlier sections with the Foundational HF Medications.

#### **14.3.2. Hypertrophic Cardiomyopathy**

- HCM accounts for 2-3% of HF<sup>667</sup>. Etiologies of HCM are shown in (Appendix VIII, page 179-180)
- It is characterized by an increase in LV wall thickness (commonly defined as >15 mm, or >13 mm in adult first degree relatives of HCM patients) in one or more myocardial segments which cannot be explained by abnormal loading conditions.<sup>668,669</sup>
- The predominant form of HF in HCM is HFpEF. A small number of patients develop HFrEF later in the disease.
- HCM can be obstructive or non-obstructive. HF is prevalent in the majority of obstructive HCM and in 10% of non-obstructive HCM.<sup>670</sup>
- The LV outflow tract obstruction may be due to:<sup>671</sup>
  - Systolic anterior motion (SAM) of the mitral valve with prolonged septal contact.
  - Midcavity muscular apposition, usually caused by anomalous papillary muscle insertion directly into anterior mitral leaflet.
- Dynamic Left Ventricular outflow tract obstruction (gradient,  $\geq 30$  mm Hg) may be associated with mitral regurgitation. It is a strong independent determinant of HF.<sup>671</sup> Surgical myomectomy has been shown to reduce mortality in these patients.<sup>672</sup>
- About 3.5-17% of HCM patients progress to advanced HF. The pathophysiologic processes leading to progression of HF include severe LV obstruction, hypertrophy and adverse remodeling.<sup>673-676</sup>

##### **14.3.2.1 Management** <sup>668,670,677</sup>

- Referral to multidisciplinary HCM centers can help optimize care for patients with HCM.
- Genetic counseling of the patient and family is important.
- In patients with:
  - Arrhythmias:
    - ◆ Assessment of the individual's risk for sudden cardiac death (SCD) and consideration for ICD implantation.<sup>678</sup>
    - ◆ To treat AF by rate or rhythm control and appropriate anticoagulation.
  - Significant Left Ventricular Outflow Tract Obstruction:
    - ◆ Septal reduction procedures by either:
      - Surgical myomectomy - procedure of choice in experienced centers.<sup>672</sup>
      - Trans-catheter alcohol ablation in patients with advanced age or significant comorbidities.



- ♦ Pharmacotherapy - drugs that have negative inotropic effects such as  $\beta$ -blockers, verapamil or disopyramide. Emerging novel agents for obstructive HCM include agents such as mavacamten - a first-in-class cardiac myosin inhibitor. This has been shown to improve symptoms, physical and social function, and quality of life in a phase 3 study.<sup>679,680</sup> It has been approved for the treatment of symptomatic New York Heart Association (NYHA) class II-III obstructive HCM.

➤ HF - to treat accordingly as per clinical presentation as HFpEF or HFrEF.

#### **14.3.3. Restrictive Cardiomyopathy**

- RCM is characterized by restrictive physiology with normal or reduced diastolic volumes, and normal or reduced systolic volumes (in LV, RV or both).<sup>669</sup>
- There is typically no LVH, but some forms of infiltrative or storage diseases such as amyloidosis or Fabry-Anderson disease can cause an increase in LV wall thickness.<sup>681</sup>
- The etiology of RCM is heterogenous. It comprises infiltrative or non-infiltrative myocardial disorders, storage diseases and endomyocardial disorders that are either idiopathic, hereditary, or acquired (Appendix VIII, page 169).
- The prevalence of HF in RCM has been reported to be as high as 83%.<sup>682</sup> The predominant form of HF in RCM is HFpEF. HFrEF may manifest late in the disease and is more prevalent in amyloidosis and hemochromatosis.<sup>683,684</sup>
- The pathophysiology of HF in RCM is increased myocardial wall stiffness, resulting in diastolic dysfunction with subsequent predisposition to AF<sup>681</sup>. The LV systolic function is usually preserved but may be reduced at late stages of the disease, such as in cardiac amyloidosis.<sup>685</sup>
- The prognosis of RCM is generally poor, regardless of the underlying etiology.<sup>686</sup>

##### **14.3.3.1 Management**

- Management includes:
  - The Foundational HF medications are generally used but caution should be exercised as these patients tolerate over diuresis and hypotension poorly.
  - Consider disease-specific therapy if available e.g., amyloidosis and Fabry's disease.

#### **Key Messages # 16: HF and Cardiomyopathies**

- Cardiomyopathies are a heterogenous group of myocardial disorders which frequently present as HF.
- They can be inherited (familial/genetic) or acquired.
- They are usually classified according to anatomic and physiologic features into:
  - Dilated cardiomyopathy (DCM)
  - Hypertrophic cardiomyopathy (HCM)
  - Restrictive cardiomyopathy (RCM)
  - Arrhythmogenic cardiomyopathy
  - Unclassified cardiomyopathy
- In general, treatment of patients with cardiomyopathy encompasses the Foundational HF Medications as well as etiology-specific therapy (if available).

**14.4. CARDIOMYOPATHY DUE TO ARRHYTHMIAS OR CONDUCTION ABNORMALITIES****14.4.1 Arrhythmia Induced Cardiomyopathy**

Arrhythmia-induced cardiomyopathy (AiCM) includes:<sup>687,688</sup>

- Tachycardia-induced cardiomyopathy.
- Atrial Fibrillation (AF) induced cardiomyopathy.
- Premature Ventricular Complexes (PVC) induced cardiomyopathy.

Arrhythmia-induced Cardiomyopathy is a reversible cause of HF characterized by LV dysfunction resulting from an increased ventricular rate. The degree of LV dysfunction correlates with the duration as well as rate of the tachyarrhythmia. The cardiomyopathy may present weeks, months, or years after the onset of the tachycardia.<sup>687,689</sup> It can occur in the setting of either an incessant or paroxysmal tachycardia and it should be suspected if no other cause of LV dysfunction is identified.<sup>687</sup>

Recognition of this entity is important clinically, as treatment of the underlying arrhythmia can result in either partial or complete recovery of LV function which, in turn, would result in an improvement in morbidity and mortality.<sup>687</sup>

In managing arrhythmias in patients with HF:

**I,A**

- Initiate the “Foundational HF” Medications to improve LV function and optimize reverse LV remodeling.

**I,C**

- Identify and treat arrhythmias accordingly. Arrhythmia recognition and suppression should be considered as part of the holistic evaluation and management of HF.

**I,C**

- Treat non cardiovascular comorbidities particularly, lung disease, obstructive sleep apnea and obesity.

**14.4.1.1. Detection of Arrhythmias**

Arrhythmias are sometimes difficult to detect, especially if they are paroxysmal and infrequent. In any person who presents with a deterioration in cardiac function, paroxysmal palpitations, near faints or syncope and the clinical suspicion that these could possibly be due to arrhythmias are high, the following modalities may be considered to aid in the diagnosis:

- Prolonged ECG rhythm monitoring using 3-day, 7-day or even 1- month rhythm monitors.
- Using a watch that can monitor ECG heart rhythm or
- An implantable loop recorder.

**14.4.1.2. Supraventricular Arrhythmias**

Any supraventricular tachycardia (SVT) with a rapid ventricular response may induce HF. Commonly encountered SVTs in clinical practice include:

- Incessant atrial tachycardia (AT),
- Very frequent episodes of atrioventricular nodal re-entrant tachycardia (AVNRT), and
- Atrioventricular re-entrant tachycardia (AVRT).

Arrhythmia-induced Cardiomyopathy has been reported in 10% of patients with AT, and as high as 37% of patients with incessant AT.<sup>688,690</sup>

**I,A**

- Successful treatment, usually via electrophysiological study and radiofrequency ablation, could potentially restore LV function.<sup>416</sup>

**14.4.1.3. Atrial Fibrillation**

AF is the commonest cardiac arrhythmia encountered in clinical practice.<sup>691</sup> AF and atrial flutter with rapid ventricular response is the most common cause of tachycardia induced cardiomyopathy.<sup>687</sup> AF induced cardiomyopathy may be suspected if LV function improves after rhythm control has been achieved and no other underlying cause has been identified.<sup>687</sup>

Management of AF in patients with HF would include the use of:<sup>690</sup>

**IIa,B**

- Anti-arrhythmic drugs for either rate control or rhythm control. Most anti-arrhythmic drugs are however contraindicated in the presence of HF.

**IIa,A**

- Catheter ablation - This has a higher chance of achieving rhythm control in patients with AF than anti-arrhythmic drugs.<sup>437</sup> Catheter ablation has been shown to reduce the risk of death and HF hospitalization in patients with AF and HF when compared to standard therapy with anti-arrhythmic drugs.<sup>692,693</sup> The success of the procedure and benefits seen were more likely in patients who underwent the procedure within 1 year of diagnosing AF.<sup>692</sup>

**14.4.1.4. Ventricular Arrhythmias**

Ventricular arrhythmias, including frequent PVCs or VT, may also induce HF.<sup>689</sup> It is however, sometimes difficult to ascertain whether the PVCs caused LV dysfunction or whether progressive LV dysfunction caused the frequent PVCs.<sup>694,695</sup> Even if the PVCs are the result of LV dysfunction, if frequent, may contribute to and increase the risk of HF and mortality even after adjusting for age and other ECG abnormalities.<sup>689</sup>

**I,B**

Thus, maintenance of sinus rhythm and/ or control of ventricular rate is important in treating patients with HF.<sup>689,694,695</sup>

PVC induced cardiomyopathy is a diagnosis of exclusion, to be suspected in patients with frequent PVCs > 10%, especially in the nonischemic setting. Generally, LV dysfunction has been associated with greater PVC burden (>10% and usually > 20%).<sup>694</sup>

Curative or suppressive therapies with either radiofrequency ablation or anti-arrhythmic drugs may be considered in patients with PVC burden of:

**IIb,C**

- > 10% over 24 hours (high) - The prevalence of PVC induced cardiomyopathy in this group of patients has been reported to be about 7% but this could be an underestimate.<sup>687,696</sup>

**IIb,B**

- > 20% over 24 hours (very high) - The exact prevalence in this group is not known.<sup>694,696,697</sup>

**I,C**

In patients with cardiomyopathy suspected to be caused by frequent and predominately monomorphic PVCs, catheter ablation is recommended.

**IIa,B**

Anti-arrhythmic drug therapy that may be considered include  $\beta$ -blockers and class 3 anti-arrhythmic drugs such as amiodarone and sotalol.<sup>696,698</sup>

**I,B**

Radiofrequency ablation is preferred, since most anti-arrhythmic drug therapy are contraindicated and may be pro-arrhythmic in the presence of HF.<sup>694,699,700</sup>

**14.4.2. Cardiomyopathy due to Conduction Abnormalities**

This will include cardiomyopathy due to conduction abnormalities/dyssynchrony, such as chronic RV pacing and left bundle branch block.

**14.4.2.1. Pacing induced Cardiomyopathy**

Pacing induced cardiomyopathy is defined as a reduction in LVEF of > 10% after pacemaker placement.<sup>701</sup> Paced QRS durations  $\geq 150$  milliseconds was associated with an increased risk.<sup>701</sup>

It is a complication of single- and dual-chambered pacemakers and may be present in up to 9% of patients. It is most prevalent within the first year after implantation and is associated with more pronounced intraventricular dyssynchrony.<sup>702</sup> The prevalence of this condition varies, depending on its definition, from 9% at 1 year, 7 up to 15% at 25 years.<sup>703</sup>

A high RV pacing percentage of > 40% and long-term pacing may exacerbate HF symptoms, increase hospitalization for HF, and increase mortality.<sup>704-709</sup>

**I,A**

In patients with HF who have bradyarrhythmias and where pacing is indicated, biventricular pacing (Cardiac Resynchronisation Therapy - CRT) is the pacing mode of choice.<sup>708-710</sup>

**14.4.2.2. Others:**

This would include left bundle branch block (LBBB) - induced cardiomyopathy and other dyssynchronopathies.

**Key Messages # 17: HF due to Arrhythmias and Conduction Abnormalities**

- Cardiomyopathy due to arrhythmias and conduction abnormalities are potentially reversible causes of HF.
- Successful treatment of the arrhythmia by drug therapy or catheter ablation can result in normalization of LV function.

**Key Recommendations #13: HF Due to Arrhythmias and Conduction Abnormalities**

- In managing cardiomyopathy due to arrhythmias, radiofrequency ablation is the preferred therapy, since most antiarrhythmic drugs are contraindicated in the presence of HF.
- In patients with HF who have bradyarrhythmias and where pacing is indicated, biventricular pacing (Cardiac Resynchronisation Therapy) is the pacing mode of choice.

**14.5 CARDIO-ONCOLOGY AND HEART FAILURE**

Heart disease and cancer are often linked due to:

- Common risk factors (e.g., increasing age and cigarette smoking)
- Treatment strategies
  - Chemotherapy drugs has been associated with HF, arrhythmias, vasculitis and thromboembolic disease.
  - Radiotherapy of the mediastinum and left chest can lead to CAD, myopericardial fibrosis and valvular dysfunction.
  - New oncological therapeutic agents like targeted therapies affecting signalling pathway and immune checkpoint inhibitors.



Dyspnea in cancer patients could be due to:

- Fluid overload.
- Cardiomyopathy due to chemotherapeutic agents, stress (Takotsubo), underlying CAD, coronary vasospasm or pericardial diseases.
- The primary cancer causing anemia, lung and pericardial involvement.

Chemotherapy-induced cardiomyopathy is not common, clinical HF occurs in 1-5% and an asymptomatic decrease in LV function in the range of 5% to 20%.<sup>711</sup>

Cardiotoxicity can develop in a subacute, acute, or chronic manner. Acute or subacute cardiotoxicity may occur at any time from the commencement of therapy up to 2 weeks after termination of treatment.<sup>709</sup> Chronic toxicity can occur early (within 1 year) or late.<sup>711,712</sup> Prognosis in the patients who develop HF, especially late onset HF, is poor.<sup>712,713</sup>

The anthracycline class of chemotherapeutic agents remain the major cause of chemotherapy-induced cardiomyopathy.<sup>712,713</sup> In the current era, newer agents have also been implicated in a reversible form of cardiomyopathy.<sup>714-716</sup>

At present, there is no consensus definition of cardiotoxicity.<sup>717</sup> The definition that is currently most often used is by the Cardiac Review and Evaluation Committee of trastuzumab-associated cardiotoxicity.<sup>718</sup> By this definition, chemotherapy-induced cardiomyopathy is said to occur in the presence of at least one or more of the following: <sup>714,718</sup>

1. Cardiomyopathy characterised by a decrease in Ejection Fraction (EF) globally or due to regional changes in interventricular septum contraction.
2. Signs and symptoms of HF.
3. Signs associated with HF including but not limited to S3 gallop, tachycardia, or both.
4. Decline in initial EF of at least 5% to < 55% with signs and symptoms of HF or asymptomatic decrease in EF of at least 10% to < 55%.

Chemotherapy drugs that have been associated with HF are as listed in the (Appendix IX & X, pages 170-171)

One must consider both drug efficacy and toxicity in choosing chemotherapeutic agents. Many of the newer targeted agents cause a reversible form of HF and symptoms usually resolve after the initiation of anti-failure medications.<sup>715,718-720</sup>

Risk factors for anthracycline toxicity include:<sup>713,721-726</sup>

- The total cumulative dose- there is however, no safe dose for doxorubicin.<sup>713</sup>
- Intravenous bolus administration versus infusion.
- Higher single doses.
- History of prior irradiation.
- Use of concomitant agents known to have cardiotoxicity.
- Female gender.
- Underlying CV disease.
- Age (children and elderly > 65 years).

An increase in cardiac biomarkers such as troponins during and after administration is an indication of toxicity.<sup>726,727</sup>

**14.5.1 Management**

I,C

- Patients undergoing chemotherapy should have a careful clinical evaluation and assessment and treatment of CV risk factors.<sup>714-716</sup>

I,C

- Blood pressure control is important in all patients especially in those being considered for Vascular endothelial growth factor signaling pathway (VSP) inhibitors.

I,C

- All patients with potentially cardiotoxic chemotherapy should have an echocardiogram prior to treatment. An important parameter is the LVEF determined using the biplane method of discs (Simpson's method) or three-dimensional echocardiography (preferred) where available. Newer techniques to detect and quantitate regional and global myocardial dysfunction (strain assessment with global longitudinal strain) can be used to detect pre-clinical and subtle changes in function.<sup>724,727,728</sup> A 15% change from the baseline global longitudinal strain measure indicates an abnormality.<sup>724</sup>

IIb,C

- Biomarkers such as troponin and natriuretic peptides can help identify patients at higher risk.<sup>729</sup>

Close collaboration between the oncologist and the cardiologist is important.

For the oncologists, the strategy (prior to commencement) includes:<sup>724</sup>

I,C

- Identifying high risk patients (pre-existing heart disease, presence of CV risk factors, age-both the very young and old- female gender, use of high dose anthracycline regimens).

I,C

- High risk patients should:

I,C

- Have a pre-treatment cardiac function evaluation. If the LVEF is < 50%, refer to the cardiologist.

I,C

- Be considered for non-cardiotoxic alternatives.

I,C

- Have their therapy protocols adjusted where necessary (e.g., reduction in doses, continuous infusions rather than bolus injections, liposomal doxorubicin, dexrazoxane etc).

I,C

- Have regular echocardiography evaluation to detect deteriorating myocardial function.

For the cardiologists/general physicians, the strategy includes:

I,C

- Treating CV risk factors.

I,C

- Assessing, repeating (if necessary) imaging studies. (e.g., using high quality LVEF measurement, strain evaluation etc).

I,C

- Assessing cardiac biomarkers (troponin and/or Natriuretic Peptides).

IIa,B

- Considering cardio-protection prior to/or during treatment using  $\beta$ -blockers, MRA and/or ACE-I/ARB if:<sup>726,730-733</sup>

- EF < 50%,

- EF drops by > 10%

- Abnormal global longitudinal strain (GLS) (> 15% drop).<sup>724</sup>

- Monitoring LVEF during therapy is important with repeat echocardiography at 3-monthly intervals and/or according to symptoms. If cardioprotective medications are given, monitoring may be necessary at closer intervals of time depending on the clinical condition of the patient e.g., at monthly intervals.

- Withholding cardiotoxic therapy is a last resort. (for anthracycline LVEF < 45%, for anti-HERS2 therapy LVEF < 40%).

- Monitoring after completion of therapy:

- Obtain post therapy LVEF.

- Repeat echocardiography in 6 months or 1 year. Most cases of treatment-associated cardiac dysfunction develop within the first year after completion of therapy.<sup>733</sup>

- If EF remains abnormal, follow guidelines for management of HF.<sup>725</sup>

**Key Message # 18: Cardio-Oncology**

- Heart disease and cancer are often linked due to common etiologic factors and chemotherapeutic treatment strategies.
- Chemotherapy-induced cardiotoxicity is not common:
  - Clinical HF occurs in 1-5% and
  - An asymptomatic decrease in LV function in the range of 5% to 20%.
- It can develop in a subacute, acute (within 2 weeks of termination of drug administration) or a chronic manner.
- One must consider both drug efficacy and toxicity in choosing chemotherapeutic agents.

**Key Recommendations #17: Cardio-Oncology**

- Close collaboration between the oncologist and the cardiologist is important.
- Patients undergoing chemotherapy should have a careful clinical evaluation and assessment. Specifically:
  - All CV risk factors should be treated adequately.
  - High risk patients should be identified and in these patients:
    - ◆ A pre-treatment cardiac echocardiogram is advisable. If the LVEF < 50%, they should be referred to the cardiologist.
    - ◆ Reassessing and repeating (if necessary) imaging studies during and after treatment.
    - ◆ Assessing cardiac biomarkers when indicated - troponins and/or Natriuretic Peptides.
  - Considering cardio-protection prior to/or during treatment using  $\beta$ -blockers, MRA and/or ACE-I/ARB if:
    - ◆ EF < 50%,
    - ◆ EF drops by > 10%
    - ◆ Abnormal global longitudinal strain (GLS) (> 15% drop).

**14.6. HEART FAILURE AND CHRONIC KIDNEY DISEASE (CKD)****14.6.1. Epidemiology, definitions, and classifications**

HF and CKD frequently co-exist. The presence of both at the time of admission for HF, varies from 45.4% in patients with chronic HF to > 60% in those with Acute HF.<sup>734,735</sup>

In addition, during treatment of acute HF, a significant proportion of patients will develop varying degrees of worsening renal function.

The definition of worsening renal function is:<sup>219</sup>

- An increase in serum creatinine by  $\geq 26.5 \mu\text{mol/L}$  (0.3mg/dl) **and/or**
- A  $\geq 25\%$  increase in serum creatinine or a  $\geq 20\%$  drop in eGFR.

The rise in serum creatinine usually occurs in the first three to five days of hospitalization.

The incidence of worsening renal function is estimated to be between 19 and 45%.<sup>736</sup> This large observed range is due to variations in the definitions of worsening renal function, the observed time-at-risk, and the study population.<sup>736</sup>



Risk factors for worsening renal function during admission for HF include:<sup>737</sup>

- A prior history of HF or diabetes **and/or**
- An admission serum creatinine of  $> 133\mu\text{mol/L}$  **and/or**
- Systolic blood pressure  $> 160\text{mmHg}$ .

Worsening renal function may fulfill criteria for type 1 or type 2 cardiorenal syndrome (CRS). The term “cardiorenal syndrome” refers to disorders of the heart and kidneys whereby acute or chronic dysfunction in one organ may induce acute or chronic impairment of the other.<sup>736</sup>

This syndrome has been classified by the Acute Dialysis Quality Initiative working group into 5 subtypes as shown in Table 17, page 133.<sup>736</sup> Many patients, however, may belong to more than one subtype and may move between subtypes during the course of their disease.<sup>736</sup>

#### **14.6.2. Pathogenesis of Cardio-Renal Syndrome (CRS)**

Multiple mechanisms are involved in the pathogenesis of CRS. These include:

- Increased renal venous pressure - venous congestion is probably the most important factor.<sup>738</sup>
- Right ventricular dysfunction.
- Reduced renal perfusion due to reduced cardiac output.
- Volume overload.
- Neurohumoral adaptations (e.g., activation of the sympathetic nervous system and the renin-angiotensin-aldosterone system, increases in vasopressin and endothelin-1).
- Anemia.
- Metabolic abnormalities e.g., disorders of calcium, magnesium, and potassium metabolism.

#### **14.6.3 Clinical significance /Impact of kidney dysfunction in HF**

The combination of cardiac and kidney disease increases the complexity and costs of care and may interact to worsen prognosis.

#### **A. Pharmacologic considerations**

- Safety:
  - Dosing of renally-excreted cardiac drugs need adjustment in the presence of renal impairment (e.g., digoxin, insulin, low molecular weight heparin, direct oral anticoagulants).
  - Patients with HF are at increased risk of contrast-induced acute kidney injury.
  - Safety of newer drugs:
    - ♦ ARNI - safety has been demonstrated for  $\text{eGFR} > 30 \text{ mls/min/1.73m}^2$ . A small study found that it was as safe as irbesartan in patients with  $\text{eGFR} > 20 \text{ mls/min/1.73m}^2$ .<sup>739</sup>
    - ♦ SGLT2-i - safety has been shown for  $\text{eGFR} > 20 \text{ mls/min/1.73m}^2$ .<sup>740,741</sup>
    - ♦ MRA - only finerenone has been shown to be safe in patients with  $\text{eGFR}$  down to  $25 \text{ mls/min/1.73m}^2$ .<sup>373-376</sup>
  - There is a lack of published data on newer drugs such as ARNI and SGLT2-i at  $\text{eGFR} < 20 \text{ mls/min/1.73m}^2$ . If the patients are already on this drug and the  $\text{eGFR}$  drops to  $< 20 \text{ mls/min/1.73m}^2$ , or if there is a compelling need for initiation despite  $\text{eGFR} < 20 \text{ mls/min/1.73m}^2$ , shared decision-making with patients for off-label use is necessary.
- Efficacy:
  - Impaired renal function affects drug choices and dosing. If  $\text{eGFR} < 30 \text{ mls/min/1.73m}^2$ , most thiazide diuretics are no longer effective and loop diuretics are preferred.
  - Higher doses of loop diuretics may be required with increasing renal impairment.

**B. Prognostic implications**

- CKD is a risk multiplier in patients with HF - as the disease severity worsens, the risks of CV events, hospitalizations, and mortality increase.<sup>742</sup> In patients with HF, worsening renal function may not always indicate a poor outcome.<sup>743</sup> The prognostic value of worsening renal function is mainly determined by:
  - The presence of persistent congestion.
  - Baseline renal function and magnitude of renal changes.
  - Duration - a persistently worsening renal function is usually associated with hemodynamic derangements and poor prognosis as compared with a transient worsening of renal function as a result of aggressive decongestive therapy.<sup>744-746</sup>

**14.6.4 Management****A) General Considerations:**

I,C

- A multi-disciplinary approach with an early referral to a nephrologist is recommended.

I,C

- Exclude potentially reversible causes for increasing renal dysfunction such as hyper- or hypovolemia, concomitant medications such as aminoglycosides and NSAIDs, and renal artery stenosis.

**Table 17: Classification of Cardiorenal Syndrome (CRS)**

| Cardiorenal (CRS) Subtypes                | Description                                                                                                                                 |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| CRS Type 1 (Acute CRS)                    | Rapid worsening of cardiac function leading to acute kidney injury e.g., MI with cardiogenic shock.                                         |
| CRS Type 2 (Chronic CRS)                  | Chronic abnormalities in cardiac function leading to progressive Chronic Kidney Disease (CKD) e.g., chronic HFrEF or HFpEF.                 |
| CRS Type 3 (Acute Renocardiac Syndrome)   | Worsening of renal function leading to acute cardiac dysfunction e.g., in Acute Kidney Injury.                                              |
| CRS Type 4 (Chronic Renocardiac Syndrome) | Primary chronic kidney disease contributing to decreased cardiac function and an increased risk of CV events e.g., CKD leading to LVH, CAD. |
| CRS Type 5 (Secondary CRS)                | Presence of comorbid cardiac and renal dysfunction due to either acute or chronic systemic disease e.g., diabetes, sepsis, amyloidosis.     |

Adapted from Ronco C, McCullough P, Anker SD, et al. Cardio-renal syndromes: report from the consensus conference of the Acute Dialysis Quality Initiative. *Eur Heart J* 2010;31:703-11.



I,C

- Closely monitor electrolytes and kidney function, especially during acute illnesses, dehydration and when increasing doses of cardiac drugs including diuretics. The baseline renal function will determine how frequently this should be done.

- The recommended range for serum potassium is 4 - 5.5mmol/L.<sup>747</sup>

I,C

- Wherever possible, avoid nephrotoxins, e.g. contrast media for angiography.

**B) Choice of Pharmacotherapy**

I,B

- Intravenous diuretics

- Loop diuretics are the first choice.

IIa,B

- Continuous infusion may not have greater benefits compared with bolus dosing.<sup>247, 249-252</sup>

- Start initially with 2.5 x the usual oral dose.

IIa,B

- Combination therapy (loop diuretic and thiazide/thiazide-like diuretic/mineralocorticoid) may be required to enhance diuresis.<sup>255, 256, 748</sup> However, care is required to avoid electrolyte disturbances, hypovolemia and worsening renal function.

- Foundational HF Medications in patients with concomitant CKD:

I,A

- Foundational HF Medications have been proven to be equally efficacious in patients with CKD.

- Many of these agents can still be initiated in patients with lower GFR (Table 19, page 136).

- However, most of the treatments used for HF can cause worsening renal function and/or hyperkalemia. Therefore, close monitoring of renal function is essential:

- ◆ In the landmark MRA trials, potassium was checked after 7 days (and again after 72 hours if dose reduction was needed).

- ◆ In the RAS blocker (ACE-I/ARB/ARNI) and SGLT2-i trials, laboratory assessment of creatinine, urea, eGFR and electrolytes was typically done after 14 days during drug titration and every 4 months thereafter.

- ◆ The interval for monitoring renal function tests needs to be individualized depending on baseline renal function, concomitant medications and comorbidities. With more advanced CKD at baseline, renal function tests should be repeated within 3-7 days after drug initiation and every time doses are titrated.

- ACE-i/ARB/ARNI and SGLT2-i can lead to an initial drop in eGFR but this should not be a reason to automatically stop or down titrate these agents. For HF<sub>r</sub>EF, both ARNI and SGLT2-i have renoprotective effects.<sup>749</sup>

- ACEi/ARB should be up titrated to the maximum recommended dose to achieve optimal dosing provided:<sup>750</sup>

- ◆ SCr levels remain < 30% from the baseline (or eGFR reduces < 25%) **and**
- ◆ Serum potassium < 5.5 mmol/L

- Consider reducing or discontinuing ACEi/ARB within two months from commencement (after excluding other precipitating factors) when:<sup>750</sup>

- ◆ SCr levels remain ≥ 30% from the baseline (or eGFR reduces ≥ 25%) **or**
- ◆ Serum potassium ≥ 5.5 mmol/L

- A decrease in eGFR over time (beyond 2 months from commencement of medication) may reflect the underlying disease process and is not necessarily a signal to automatically discontinue or down titrate the RAS blocker or SGLT2-i.<sup>749</sup>



I, C

➤ In patients with advanced renal disease, the decision to start or stop RAS blockers should be individualized.<sup>751</sup> The discontinuation of RAS blockers in patients with advanced and progressive CKD did not make any significant difference in kidney disease progression and the long-term rate of decrease in the eGFR.<sup>752</sup> Hence the decision to continue or discontinue RAS blockers should be made according to the individual patient's level of proteinuria, blood pressure control, tolerability and cardiovascular risk profile.

IIa,B

- Novel potassium binders such as sodium zirconium cyclosilicate and patiromer have been proven to be safe and efficacious even with long term use of up to one year. These new agents may be considered to allow continuation or dosing optimization of Renin Angiotensin Aldosterone System (RAAS) inhibitors in patients with hyperkalemia.<sup>155</sup>
- A guide to adjustment of dose of RAS blockers with changes in eGFR is as in Table 18, below.

**Table18: Management of RAS Blockers in response to changes in renal function**

| Compared to baseline, serum creatinine increases by: | HFpEF                                                 | HFrEF                                                                         |
|------------------------------------------------------|-------------------------------------------------------|-------------------------------------------------------------------------------|
| < 30%                                                | Consider stopping RAS blocker<br>Review Dosage of MRA | Continue unless there is symptomatic hypotension                              |
| 30 - 50%                                             | Stop RAS Blocker                                      | Consider reducing the dose of RAS blocker and/or temporary withdrawal         |
| < 50%                                                | Stop RAS Blocker                                      | Temporarily stop RAS blocker. Exclude other causes and consider re-challenge. |

*Adapted from Clark AL, Kalra PR, Petrie MC, Mark PB, Tomlinson LA, Tomson CR. Change in renal function associated with drug treatment in heart failure: national guidance. Heart. 2019 Jun;105(12):904-910.*

**C) Ultrafiltration**

- This involves the removal of plasma water across a semi-permeable membrane in response to a transmembrane pressure gradient.
- It may be considered for congestive symptoms refractory to diuresis but should be used in consultation with a nephrologist.<sup>753</sup>
- Evidence for its efficacy in fluid removal is, however, mixed at present.<sup>753</sup>

IIa,B

**D) Renal Replacement Therapy (Hemodialysis or Peritoneal Dialysis)**

- Criteria for possible need to initiate renal replacement therapy include:
- ♦ Oliguria (urine output < 200mls/12 hours) or anuria unresponsive to judicious fluid resuscitation.
  - ♦ Intractable pulmonary edema.
  - ♦ Hyperkalemia refractory to medical therapy or where urgent correction is required (serum potassium > 6.5mmol/L).



- ♦ Severe acidemia (pH < 7.2).
- ♦ Serum urea > 25mmol/L.
- ♦ Serum creatinine > 300μmol/L.
- ♦ Uremic symptoms or signs e.g., uremic pericarditis

**Table 19: Initiation of HF drugs in relation to baseline CKD status\***

| Medications | EVIDENCE ACROSS eGFR STRATA                                     |                                          |                                                                            |                                                           |
|-------------|-----------------------------------------------------------------|------------------------------------------|----------------------------------------------------------------------------|-----------------------------------------------------------|
|             | eGFR > 60<br>mls/min/1.73m <sup>2</sup>                         | eGFR 30-60<br>mls/min/1.73m <sup>2</sup> | eGFR 15-30<br>mls/min/1.73m <sup>2</sup>                                   | ESRD                                                      |
| ACE-i/ARB   | Strong evidence                                                 |                                          | Moderate evidence                                                          | Moderate evidence if on dialysis, weak if not on dialysis |
| ARNI        | Strong evidence                                                 |                                          | Weak evidence<br>Small study only for eGFR > 25 mls/min/1.73m <sup>2</sup> | Not recommended                                           |
| β-blockers  | Strong evidence                                                 |                                          | Moderate evidence                                                          |                                                           |
| MRA         | Strong evidence                                                 |                                          | Only finerenone at eGFR > 25 mls/min/1.73m <sup>2</sup>                    | Not recommended                                           |
| SGLT2-i     | Strong evidence<br>(From eGFR > 20 mls/min/1.73m <sup>2</sup> ) |                                          |                                                                            | No Evidence                                               |
| Ivabradine  | Strong evidence                                                 |                                          | No Evidence                                                                |                                                           |
| Vericiguat  | Strong evidence                                                 |                                          |                                                                            | No Evidence                                               |

\*Adapted from Mullens W, Martens P, Testani JM, et al. Renal effects of guideline-directed medical therapies in heart failure: a consensus document from the Heart Failure Association of the European Society of Cardiology. *Eur J of Heart Fail* 2022;24: 603-619

### Key Message # 19: HF and CKD

- Cardiac and kidney disease frequently co-exist, and this increases the complexity and costs of care, and may interact to worsen prognosis.
- During treatment of Acute HF, a significant proportion of patients will develop varying degrees of worsening renal function usually in the first three to five days of hospitalization.
- In patients with HF, worsening renal function may not always indicate a poor outcome—especially if it is due to overdiuresis and hypotension or drug therapy.

**Key Recommendations #15: HF and CKD**

- A multi-disciplinary approach with early referral to a nephrologist is recommended.
- Almost all Foundational HF Medications can be used in patients with eGFR > 30mls/min/1.73m<sup>2</sup>
- In patients with eGFR < 30mls/min/1.73m<sup>2</sup>, the following drugs can be used:
  - Diuretics- usually higher maintenance doses
  - Careful use of RAS blockers
  - $\beta$ -blockers
  - SGLT2-i (eGFR > 20mls/min/1.73m<sup>2</sup>)
- Closely monitor electrolytes and kidney function. The baseline renal function will determine how frequently this should be done.
- RAS blockers and SGLT2-i can lead to an initial drop in eGFR, but this should not be a reason to automatically stop or down titrate these agents. See table 18, page 135 for the management of RAS blockers in response to changes in renal function.
- Occasionally ultrafiltration and renal replacement therapy (hemodialysis or peritoneal dialysis) may be necessary.

**14.7 HEART FAILURE IN PREGNANCY**

About 0.5-4% of pregnant women have cardiac disease.<sup>754</sup> HF remains the most common complication among all pregnant women with heart disease regardless of the cause (valvular, congenital, cardiomyopathy or pulmonary hypertension).<sup>755</sup>

In Malaysia, in 2020, the maternal mortality ratio was at 24.9 maternal deaths per one hundred thousand live births.<sup>756</sup> This refers to the annual number of female deaths caused by or related to pregnancy per one hundred thousand live births.

In general, pregnancy is well-tolerated in women with cardiomyopathy and NYHA class I pre- pregnancy. Factors increasing maternal and fetal risks include:<sup>757-761</sup>

- Previous cardiac events
- NYHA class III to IV **or**
- LVEF < 40%.

In the ROPAC Registry, women with pre-pregnancy or peripartum cardiomyopathy had the highest mortality rate (2.4%)<sup>762,763</sup>

Normal hemodynamic changes that occur in pregnancy are:

- An increase in cardiac output by 30-50% during normal pregnancy.
- An increase in cardiac output to 80% above baseline during labour and delivery.
- Hemodynamic changes return to baseline 2-4 weeks after vaginal delivery and up to 6 weeks after Caesarean delivery.

In women with heart disease, these changes may have a deleterious effect on their cardiovascular system and may precipitate HF.

The periods of greatest risk for cardiac events during pregnancy are early third trimester, at delivery and in the immediate post partum period.



HF may develop in pregnancy:<sup>192,764</sup>

- For the first time in a patient with pre-existing heart disease (congenital and/or valvular) due to decompensation from the stress.
- May occur in a patient who had HF previously and still has a depressed myocardial function. (LVEF < 40%).
- In a patient with a previously unrecognized genetic cardiomyopathy or a latent cardiac viral infection which has been unmasked or activated by the stress of pregnancy.<sup>765-768</sup>
- In a patient with a previously normal heart due to:<sup>769-771</sup>
  - Hypertensive complications of pregnancy i.e. gestational hypertension and the more severe forms preeclampsia, the HELLP syndrome.  
(H: hemolysis, EL: elevated liver enzymes, LP: low platelet count).
  - Peripartum cardiomyopathy.
  - Takotsubo syndrome.<sup>755</sup>

#### 14.7.1. Diagnosis

Most forms of cardiac disease can be detected by physical examination, ECG and echocardiography. In cases of new HF or if there is diagnostic uncertainty, non-contrast CMR may be considered.

#### 14.7.2. Risk stratification

- Pregnant women with cardiac disease are at risk of adverse maternal and fetal outcomes.
- Their risk should be assessed before conception or early in the pregnancy to optimize the outcome of the pregnancy.<sup>192,764</sup>
- Maternal cardiovascular risk can be assessed using the modified World Health Organization (WHO) or NYHA classification.<sup>192</sup> (Appendix XI & XII, page 172-173)
- Level of Care will depend on the maternal CV risk:<sup>192</sup>
  - **Low risk:** can be managed at their local center after review by a family medicine specialist/physician or cardiologist.
  - **Moderate risk:** should be managed at a tertiary center by a multidisciplinary team with cardiac expertise.
  - **High risk:** should be referred early to the tertiary center for assessment. If termination of pregnancy is considered, it can be performed up to 22 weeks.

#### 14.7.3. Management

The management of HF in pregnancy is more difficult than in the non-pregnant state and should be managed by a multidisciplinary team consisting of physicians, obstetricians, and pediatricians.<sup>789,754,755,764</sup>

In the management of HF in pregnancy, the following issues need to be considered:<sup>192,755,764</sup>

- Gestational age at presentation.
- Clinical presentation, either as Acute HF or Chronic HF.
- Response to medical therapy.
- Potential maternal and fetal risks.
- Review and replace all fetotoxic drugs
- Timing and mode of delivery.

Predictors of maternal cardiac complications are as in Table 20, Page 139.

**14.7.3.1. Preconceptual counselling**

I,C

- Patient-centered counselling regarding contraception and the risks of cardiovascular deterioration during pregnancy should be provided to all women in the childbearing age with known or suspected heart disease.<sup>192-194,764</sup>
- Patients with LVEF < 30% and those in NYHA Class III and IV should be strongly advised not to get pregnant.<sup>192,772</sup> If pregnant, termination should be considered.

**14.7.3.2. Management of HF during pregnancy**

This includes:

- **Non-pharmacological measures**

I,B

The management of patients with mild symptoms consists mainly of non-pharmacological measures such as:<sup>192</sup>

- Limiting strenuous exercise.
- Adequate rest - maintaining a low salt diet.
- Treating anemia and infections early.
- Frequent antenatal examinations.

- **Pharmacological measures**

The following drugs may be used in the pregnant patient with HF:<sup>192</sup>

I,C

- Diuretics are the first line therapy in patients who are fluid overloaded.

IIa,C

- Nitrates and/or hydralazine are used for preload and afterload reduction.

IIa,C

- $\beta$ -blockers can be used cautiously, most commonly metoprolol.<sup>773,774</sup>

IIa,C

- Digoxin is safe in pregnancy and during breast feeding.

III,C

- ACE-I, ARB, ARNI, MRA, SGLT2-i and ivabradine are contraindicated in pregnancy.<sup>755,775,776</sup>

IIa,C

- ACE-I (enalapril and captopril) can be used in the post partum period.<sup>775,776</sup>

- In the postpartum period, Foundational HF medications may be given. The patient should however be advised not to breast feed.

**Table 20: Predictors of Maternal Risk for Cardiac Complications**<sup>192,754,768,769</sup>

|                                                                                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------|
| • Cyanosis (oxygen saturation < 90%)                                                                                              |
| • Repaired or unrepaired cyanotic heart disease                                                                                   |
| • History of HF before pregnancy                                                                                                  |
| • Prior cardiac event (HF, transient ischemic attack, stroke, arrhythmia)                                                         |
| • Prior arrhythmia (symptomatic sustained tachyarrhythmia or bradyarrhythmia requiring treatment)                                 |
| • NYHA class > II                                                                                                                 |
| • Valvular stenosis (aortic or mitral valve area < 1.5cm <sup>2</sup> ) and LV outflow tract obstruction (peak gradient > 30mmHg) |
| • Reduced systemic ventricular dysfunction (LVEF < 40%)                                                                           |
| • Mechanical valve                                                                                                                |
| • High risk aortopathy                                                                                                            |
| • Reduced subpulmonary ventricular function (TAPSE < 16 mm)                                                                       |
| • Pulmonary arterial hypertension                                                                                                 |
| • Systemic and pulmonary atrio-ventricular valve regurgitation ( moderate to severe)                                              |
| • Natriuretic peptide levels (NT-Pro BNP > 128pg/mL at 20 weeks predictive of event later in pregnancy)                           |
| • Cardiac medication before pregnancy                                                                                             |
| • Maternal history of smoking                                                                                                     |
| • No prior cardiac intervention                                                                                                   |

**● Other treatment considerations in the pregnant patient.**

- Patients with AF who are hemodynamically unstable should be promptly electrically cardioverted. This is safe in pregnancy.
- Anticoagulation is indicated in the presence of AF, dilated left atrium or mechanical prosthetic heart valve.
- Patients with valvular lesions who remain symptomatic despite optimal medical treatment may be considered for percutaneous valve intervention or surgery.
- Commonly recommended antihypertensive drugs include methyldopa, labetalol, calcium channel blockers and hydralazine.<sup>192,775</sup>
- Echocardiographic reexamination may be considered:
  - ◆ In the third trimester for reassessment of myocardial structure and function before labor.
  - ◆ When there are significant changes in HF symptoms or signs during pregnancy.
  - ◆ When HF medications are reduced or discontinued.
- In selected patients, Natriuretic peptide (BNP or NT-Pro BNP) monitoring may have some value for prediction of cardiovascular events.<sup>777,778</sup>

I,C

**A. Antenatal care**

The principles of management of HF in pregnancy are like that in the non-pregnant state. If the patient is in decompensated HF requiring inotropes, she should be transferred to a cardiac center.<sup>192,754</sup>

**B. Labour and delivery**

Timing and mode of delivery should be carefully planned by a multidisciplinary team. In the majority of patients, vaginal delivery with epidural anesthesia is the preferred mode of delivery.

- Cesarean section is indicated:<sup>192</sup>
  - For obstetric reasons.
  - In patients on warfarin or who have discontinued their warfarin for < 2 weeks and who now are in imminent labour.
  - In patients with severe pulmonary hypertension.
- It is beneficial to shorten the second stage of labour by forceps or vacuum assisted delivery.<sup>192</sup>
  - Left lateral decubitus position is preferred to attenuate the hemodynamic effects in the supine position.
  - A slow i.v. infusion of oxytocin immediately after birth, (2 U of oxytocin given over 10 min followed by 12 mU/min for 4 h) reduces the risk of post-partum hemorrhage and has a minimal impact on cardiovascular parameters.<sup>192</sup>
- Routine antibiotic prophylaxis is not recommended in patients with valvular heart disease undergoing uncomplicated vaginal delivery or Cesarean section.

**C. Post partum care**

- After delivery, careful monitoring of hemodynamic status should be done for at least 24 hours, or longer in high-risk patients. In patients with severe cardiac lesions, hemodynamics may be abnormal up to 10 days after delivery.<sup>192</sup>
- These patients should be evaluated post-partum to assess the need for corrective surgery.



- The risk of recurrence of HF in subsequent pregnancies should also be made known to the patient.
- Follow-up visit at 6 weeks post-partum should be attended by the multidisciplinary team, a full cardiac assessment should be done, and appropriate contraception should be advised.
- Postpartum women who breast feed can start ACE-I (enalapril or captopril preferred). Metoprolol remains the preferred  $\beta$ -blocker.<sup>773-776</sup>

#### **14.7.4 Peripartum Cardiomyopathy (PPCM)**

- PPCM presents as HF secondary to LV systolic dysfunction, usually shown by an LVEF < 45% occurring during the third trimester or in the months following delivery without any other identifiable cause.
- The majority of PPCM cases are diagnosed in the post-partum period. PPCM frequently presents with Acute HF but may also present with ventricular arrhythmias and/or cardiac arrest.
- Cardiac recovery may occur in the first 3-6 months though it may be delayed up to 2 years. Recovery rates vary among regions, from 75% to less than 50%.<sup>779-781</sup>
- Subsequent pregnancies for women with previous peripartum cardiomyopathy have been associated with further decreases in LV function, maternal death, and adverse fetal outcomes.<sup>779</sup> The strongest prognostic determinant is LVEF <50% before a subsequent pregnancy.<sup>780-782</sup>
- The incidence of intracardiac thrombi during Acute HF caused by peripartum cardiomyopathy has been reported to be around 16%-17% with a 9% thromboembolic event rate.<sup>782</sup>
- Predictors of adverse pregnancy outcomes include:<sup>780,781</sup>
  - LVEF < 30%
  - RV involvement.
- Prognosis is related to the initial LVEF, presence of LV thrombus, RV involvement, preeclampsia, geographical region, and race.<sup>779,780</sup>

##### **14.7.4.1. Management:**

- This is as listed in section 14.7.3.2
- In addition:
  - In those patients presenting with severe HF and cardiogenic shock requiring inotropic or vasopressor support, it is advisable to transfer them to an advanced HF center. Urgent delivery by cesarean section (irrespective of gestation) should be considered.
  - Bromocriptine has been proposed for patients with acute PPCM to reduce the production of a cleaved 16 kDa prolactin fragment, which may contribute to the pathophysiology of PPCM. The efficacy and safety of bromocriptine for acute PPCM treatment, however, remains uncertain currently, particularly in the setting of contemporary HF and cardiogenic shock management.<sup>780,783</sup> The use of bromocriptine increases thromboembolic risk and thus, adequate anti-coagulation is important.
  - Some patients may present with HF late post-partum and thus, they should be followed up closely long-term. They should also be counselled about subsequent pregnancies and the risk of relapse.

**Key Message #20: HF and Pregnancy**

- About 0.5–4% of pregnant women have cardiac disease. HF remains the most common complication among all women with heart disease regardless of the cause.
- Women with cardiac disease should be assessed:
  - Before conception to assess their risk and to be advised accordingly and
  - Early in the pregnancy to optimize the outcome of the pregnancy.
- Maternal cardiovascular risk can be assessed using the modified World Health Organization (WHO) or NYHA classification. (Appendix XI & XII, page 172–173)
- Level of Care will depend on the maternal CV risk.
  - **Low risk:** can be managed at their local center after review by a family medicine specialist/physician or cardiologist.
  - **Moderate risk:** should be managed at a tertiary center by a multidisciplinary team with cardiac expertise.
  - **High risk:** should be referred early to the tertiary center for assessment.
- Patients with LVEF < 30% and those in NYHA Class III and IV should be strongly advised not to get pregnant. If pregnant, termination should be considered.

**Key Recommendations #16: HF and Pregnancy**

- HF in pregnancy should be managed by a multidisciplinary team consisting of physicians, obstetricians and pediatricians.
- HF that develops during pregnancy can be managed with the judicious use of diuretics, digoxin, nitrates,  $\beta$ -blockers (most commonly metoprolol) and/or hydralazine.
- For post-partum women with severe acute HF caused by Peripartum cardiomyopathy and LVEF < 35%, Foundational HF medications to improve LVEF recovery and prophylactic anticoagulation are recommended.

**14.8. CORONAVIRUS 2019 (COVID 19) +/- VACCINE AND HEART FAILURE**

COVID-19 is an infectious disease caused by severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2).

In hospitalized patients with COVID-19, the prevalence of HF varied between 4% and 21% and about 8% - 33% required critical care admissions.<sup>784</sup> Patients with HF who develop COVID-19 had an overall mortality rate between 20% and 40%.<sup>784,785</sup>

HF is an independent predictor of mortality in hospitalized COVID-19 patients and these patients were more likely to be sicker and develop complications. Patients with HFrEF did worse than those with HFmrEF or HFpEF.<sup>784</sup>

Patients hospitalized for COVID-19 may develop both an acute decompensation of chronic HF as well as new onset HF. New onset HF was observed in as much as a quarter of hospitalized COVID-19 patients and about one-third of those admitted to the intensive care unit (ICU).<sup>786</sup>



New onset HF may be due to multiple mechanisms. These include myocardial injury due to the direct effects of the virus, myocardial ischemia (Type 1 or 2 Myocardial infarction), multisystem inflammatory syndromes, cytokine storm, stress cardiomyopathy, acute cor pulmonale from micro and macro pulmonary emboli.<sup>787,788</sup> Myocardial injury occurred in at least 10% of unselected COVID-19 cases and up to 41% in critically ill patients or in those with concomitant CV comorbidities.<sup>789</sup>

#### **14.8.1. Myocarditis and Myocardial Involvement**

This is a rare but serious complication of SARS-CoV-2 infection as well as COVID-19 mRNA vaccination.<sup>788</sup>

Myocarditis is defined by the presence of all of the following:<sup>788</sup>

- Cardiac symptoms (e.g., chest pain, dyspnea, palpitations, syncope),
- Elevated cardiac troponin (cTn) **and**
- Abnormal electrocardiographic, and /or echocardiographic, cardiac magnetic resonance imaging (CMR), in the absence of flow-limiting coronary artery disease.

There may also be myocardial involvement only without myocarditis and this is defined by:<sup>788</sup>

- An abnormal myocardium manifest by electrocardiographic, echocardiographic, CMR and /or histopathologic findings, with or without symptoms and with or without an elevated cTn.

#### **Prevalence**

The Center for Disease Control (CDC) reported the rate of myocarditis or pericarditis after SARS-CoV-2 infection ranged from 12.6 - 114 per 100,000 for males and 5.4 - 61.7 per 100,000 for females across different age groups.<sup>790</sup>

Myocarditis following COVID-19 mRNA vaccination is rare. The rate of myocarditis or pericarditis after mRNA COVID-19 vaccination was 0 - 35.9 per 100,000 for males and 0 - 10.9 per 100,000 for females across different age groups and vaccine cohorts.<sup>790</sup> The highest observed rates have been in young male individuals (aged 12-17 years) after the second vaccine dose.<sup>791</sup> Vaccination-linked myocarditis is milder than that associated with the virus itself.<sup>791</sup> Myocarditis following administration of the COVID-19 vaccine have a significantly lower rate of mortality compared with myocarditis associated with non-COVID-19 viral infection.<sup>792</sup>

Despite this, COVID-19 vaccination is associated with a very favorable benefit-to-risk ratio for all age and sex groups evaluated thus far.

#### **Management<sup>788</sup>**

- Hospitalization is recommended for patients with definite myocarditis, ideally at an advanced HF center.
- Patients with myocarditis and COVID-19 pneumonia (with an ongoing need for supplemental oxygen) should be treated with corticosteroids.
- For patients with suspected pericardial involvement, treatment with nonsteroidal anti-inflammatory drugs, colchicine, and/or prednisone is reasonable.
- Intravenous corticosteroids may be considered in those with suspected or confirmed COVID-19 myocarditis with hemodynamic compromise or multisystem inflammatory syndrome in adults.



- As appropriate, guideline-directed medical therapy for HF should be initiated and continued after hospital discharge.
- In general, vaccine-associated myocarditis should be diagnosed, categorized, and treated in a manner similar analogous to myocarditis following Covid-19 infection.

**Long-term cardiovascular outcomes of COVID-19**

- The risk and 1-year burden of cardiovascular disease in survivors of acute COVID-19 are substantial.<sup>793</sup>
- Individuals with COVID-19 are at increased risk of other CV diseases such as cerebrovascular disorders, arrhythmias, ischemic and non-ischemic heart disease, pericarditis, myocarditis, HF, and thromboembolic disease.
- These risks and burdens were evident even among individuals who were not hospitalized during the acute phase of the infection.
- Patients with more severe initial illness were more likely to develop long-term CV complications and this increased in a graded fashion according to the care setting during the acute phase (non-hospitalized, hospitalized and admitted to intensive care).
- Care pathways of those surviving the acute episode of COVID-19 should include attention to CV health and disease.

**Key Message # 21: HF and Covid 19 Infection and Vaccine**

- In hospitalized patients with COVID-19, the prevalence of HF varied between 4% and 21% and about 8%-33% required critical care admissions.
- Patients with HF who develop COVID-19 had an overall mortality rate between 20% and 40%.
- Myocarditis is a rare but serious complication of SARS-CoV-2 infection as well as COVID-19 mRNA vaccination.
- The rate of myocarditis or pericarditis across different age groups:
  - After SARS-CoV-2 infection:
    - ♦ 12.6-114 per 100,000 for males and 5.4-61.7 per 100,000 for females.
  - After mRNA COVID-19:
    - ♦ 0-35.9 per 100,000 for males and 0-10.9 per 100,000 for females.

**14.9. HEART FAILURE IN ADULT CONGENITAL HEART DISEASE (ACHD)**

Majority of pediatric patients with congenital heart disease are now surviving into adulthood due to advances in medical and surgical treatment.<sup>794,795</sup> However many of these adults with congenital heart disease (ACHD) are at risk of long term sequelae with increased morbidity and morbidity.

**14.9.1. Definition**

Definition of HF in the ACHD in patients with biventricular circulation is like that of those with acquired heart disease.<sup>796</sup>

However, defining HF in patients with Fontan circulation, may be more complicated.<sup>797</sup>

**14.9.2. Burden of disease**

- Prevalence of HF in ACHD ranges from 3.3% (median follow up of 27.5 years)<sup>798</sup> to 6.4% at a mean age of 35 years.<sup>799</sup>
- HF is the most common cause of death in ACHD patients, accounting for 17- 42% of all deaths.<sup>800-803</sup>
- The median age of death from HF was 48.4 years (range 20.2- 91.2 years)<sup>798</sup> and varied based on the complexity of the underlying lesions and interventions.<sup>799,800,804,805</sup>
- Those patients with more complex CHD including systemic RV, single ventricle palliated with Fontan circulation, and unrepaired cyanotic CHD with Eisenmenger physiology comprise the majority of HF- related deaths.<sup>799,803,806</sup>
- Hospitalization related to ACHD associated HF has increased significantly over the recent years accounting for 11.8%-20% of all ACHD admissions.<sup>806,807</sup>
- ACHD patients with HF had longer length of stay, higher burden of arrhythmias and were more likely to die.<sup>807</sup> One-year mortality after HF admission was 24% and 3-year mortality was 35%.<sup>808</sup> Male gender, pacemaker implantation, admission duration, non-cardiac medication use, and high serum creatinine were identified as independent risk factors for 3-year mortality.<sup>808</sup>

**14.9.3. Predisposing risk factors**

- The risk factors for developing HF and related mortality in ACHD patients includes advancing age, worsening NYHA functional class, complexity of the underlying congenital heart defect(s), pulmonary hypertension/ Eisenmenger syndrome, endocarditis, atrial arrhythmias, ventricular dysfunction, end organ dysfunction and previous interventions e.g., Fontan surgery.<sup>799,802,803</sup>

**14.9.4. Pathophysiology of HF in ACHD**

- Unlike acquired heart disease, the underlying mechanism for HF is more heterogenous and complex.
- It is highly dependent on the underlying anatomy, presence of pulmonary arterial hypertension, timing and type of intervention, myocardial protection during surgery, presence of residual hemodynamic lesions and acquired comorbidities.
- The most common reasons for ACHD-HF include presence of native, new, or residual hemodynamic lesions, myocardial failure and arrhythmias.
- For the different causes of ACHD-HF based on the underlying pathophysiology, related congenital heart disease and etiology tailored management see Appendix XIII, pages 174-175. <sup>186-187,796,809,810</sup>

**14.9.5. Diagnosis of HF in ACHD**

- Diagnosing HF is challenging as typical clinical findings may not be present and most patients do not report symptoms of decreased functional capacity due to chronic adaptation since childhood.<sup>811</sup>
- Interpreting the routine investigations also poses a challenge as many are abnormal even at baseline.
- Assessment of HF in these ACHD patients should include a comprehensive review of clinical history, underlying congenital heart defect, previous interventions, signs, symptoms, and relevant investigations.
- Due to the multifactorial etiology of HF in these patients, a patient-centered approach is crucial.



- Signs and symptoms of HF are non-specific. It can vary depending on whether it is due to biventricular failure, systemic or sub pulmonary ventricle failure.<sup>812,813</sup> See Section 6.
- Chronic lymphatic dysfunction in failing Fontan circulation causes plastic bronchitis and protein losing enteropathy (PLE) characterized by enteric protein loss.<sup>814</sup>
- Right sided HF is more prevalent in ACHD compared to other acquired heart disease.
- Patients with palliated or unrepaired cyanotic congenital heart disease, Eisenmenger syndrome or pulmonary to systemic shunts may have worsening cyanosis contributing to impaired functional capacity with or without HF. These patients are also at risk of thromboembolic events.<sup>815</sup>
- Arrhythmias may be the cause or the first manifestation of HF in these patients.
- Identifying the cause of HF is important as it may be reversible (e.g., new or residual hemodynamic lesions, acute arrhythmias, compounding medical conditions like iron deficiency anemia or thyrotoxicosis.)

#### **14.9.6. Investigations**

- Investigations are like that of acquired heart disease as outlined earlier in Section 6. In addition, the following investigations are important to assess both structural and hemodynamic anomalies: (Appendix XIV, page 176)
  - Transthoracic and transesophageal echocardiogram
  - Cardiac magnetic resonance (CMR)
  - Computer tomography scan
  - Cardiac catheterization for invasive hemodynamic assessment.
  - Rhythm monitoring and analysis.
- Patients with ACHD are usually well adapted to their functional limitations. Thus, objective assessment of their functional capacity and response to therapy is important. This can be done by exercise tests (Cardiopulmonary exercise test (CPET) or a six-minute walk test). The CPET predictive values of poor prognosis may, however, be different in ACHD compared to acquired heart disease especially in single ventricle physiology<sup>816,817</sup> and in patients with cyanotic heart disease.<sup>818,819</sup>

#### **14.9.7. Management (Appendix XV: Flowchart of the management of ACHD-HF, page 177)**

- The aim of treatment is to alleviate symptoms, improve quality of life and to improve survival. These patients are preferably managed by a multidisciplinary team that includes both the ACHD and HF cardiologist.
- The principles of managing HF in ACHD HF are:
  - First to access and address all reversible causes.
  - To initiate pharmacotherapy if HF control is still not optimal.
  - To consider cardiac resynchronization and more advanced therapies if these patients continue to have worsening HF despite optimal medical therapy.
- Potentially reversible causes in ACHD-HF are:
  - Native, new, or residual structural anomalies that are amenable to surgical or catheter interventions. These include:
    - ◆ Shunts (VSD/ASD/PDA)
    - ◆ Valvular lesions (free flow Pulmonary regurgitation post TOF repair/aortic regurgitation e.g., post arterial switch)
    - ◆ Outflow tract obstruction (conduit stenosis/ LV outflow tract obstruction).<sup>820</sup>
  - Arrhythmias- atrial<sup>821</sup> or ventricular tachyarrhythmias
  - Other medical problems such as-sepsis, thyrotoxicosis, anemia, uncontrolled hypertension.



- **Pharmacotherapy**<sup>809,810,812</sup>

In general, drug therapy for HF in ACHD lacks evidence unlike that in acquired HF. The medical treatment is based on the varying subgroups of ACHD-HF.

**A) Impaired systolic function**

- **Systemic left ventricle failure (LVEF<40%)**

- The treatment is similar as acquired HF.
- Patients with HF/rEF may be treated with RAS blockers (ACE-I, ARB, ARNI),  $\beta$ -blockers, MRA and SGLT2-i. Loop diuretics are given for symptom relief.

- **Systolic failure of subpulmonic RV (RVEF<40%)**<sup>822</sup>

- No medical treatment is indicated for asymptomatic patients.
- There is limited evidence for the use of RAS blockers or  $\beta$ -blockers.
- Diuretics are the mainstay treatment for symptomatic patients.
- It is important to identify hemodynamically significant lesions and to address these.<sup>820</sup>
- Patients with Right Heart Failure (RHF) due to Pulmonary Arterial Hypertension (PAH)/Eisenmenger would benefit from PAH targeted therapy.<sup>823</sup>

- **Systolic failure of systemic RV**<sup>824,825</sup>

- There are no standard guidelines for medical therapy for asymptomatic patients.
- In symptomatic patients, standard heart failure medications may be given but evidence for their use in this setting is lacking.
- An increase in biomarkers and worsening VO2 max may be useful parameters to consider for initiating medical therapy.

- **Systolic failure of single ventricle**<sup>826-828</sup>

- There are no guidelines for the use of standard drugs in asymptomatic patients.
- Loop diuretics may be used judiciously in symptomatic patients.
- PAH targeted therapy (Phosphodiesterase-5 Inhibitors and Endothelin receptor antagonists) may improve pulmonary blood flow and improve cardiac output.<sup>819</sup>

**B) Arrhythmias**<sup>829</sup>

- Diagnostic and interventional catheter and device-based electrophysiologic procedures in adults with moderate or complex CHD or complex arrhythmias should be performed in a regional ACHD center by a cardiac electrophysiologist with expertise in CHD, and in a laboratory with appropriate personnel and equipment.<sup>821,830</sup>

I,C

- These should be managed promptly and aggressively with appropriate antiarrhythmic drug therapy, catheter, or surgical ablation.

I,B

- Permanent pacing is recommended in:<sup>829-831</sup>

I,B

- Advanced second- or third-degree AV block associated with symptomatic bradycardia, ventricular dysfunction, or low cardiac output.

I,B

- Post operative advanced second- or third-degree AV block that persists more than 7days after cardiac surgery.

I,B

- Congenital 3<sup>rd</sup> degree AV block.<sup>831</sup>

Ila,C

- Cardiac implantable cardioverter defibrillator (ICD) may be considered for the secondary prevention of sudden death in:

- High risk ACHD patients resuscitated from cardiac arrest due to VT/VF *and*

- In those with spontaneous sustained ventricular tachycardia with no clear reversible cause.<sup>832</sup>

- CRT may be helpful in managing HF in patients with subpulmonary and systemic RV.<sup>833,834</sup>

**C) Cyanosis<sup>815</sup>**

- Iron deficiency anemia is common in those with cyanotic heart disease and may be missed due to secondary erythrocytosis.
- Routine phlebotomy based on a predetermined hematocrit should be avoided.
- These patients are also at risk of thromboembolic events and bleeding. Anticoagulants should be given with caution.

**D) Advanced therapies for ACHD- HF<sup>835,836</sup>**

- The use of mechanical circulatory support (MCS), total artificial heart, and heart transplantation in ACHD patients are challenging due to:
  - Complex anatomy with abnormal pulmonary arteries, anomalous pulmonary or systemic venous drainage, aortopulmonary collaterals.
  - Prior surgical intervention/repeated sternotomies.
  - Allosensitization due to blood transfusion, use of homografts.
  - Pulmonary hypertension - thus a heart-lung transplant may be required.
  - End organ involvement e.g. Fontan associated liver disease (may need heart-liver transplant).

**E) General measures**

- Infective endocarditis<sup>837</sup>
  - Optimal oral hygiene is recommended to prevent infective endocarditis which has been identified as a predisposing risk for ACHD-HF.<sup>837</sup>
  - Antibiotic prophylaxis should be considered in selected high-risk patients to prevent endocarditis during invasive dental procedures.
- Pregnancy (refer to section 14.7).
  - It is important that in all women in the childbearing age with ACHD, preconception counselling should be initiated early, and all pregnancies should be pre-planned.
  - Exercise rehabilitation<sup>838</sup> (refer to section 16)
  - Even though the evidence is less extensive in ACHD-HF, individualized exercise programs have been shown to be beneficial.<sup>839</sup>
  - Cardiopulmonary exercise testing is a useful tool performed both at baseline and post exercise training.

**F) Palliative care and advance care planning (refer to section 15.3)<sup>840</sup>**

- Advanced care planning is preferably done in an ACHD center with support from a multidisciplinary team involving both the ACHD cardiologist and palliative care specialist.
- It should be patient-tailored and age appropriate due to the heterogenous nature and chronicity of ACHD and counselling should be given based on the projected course of the respective disease and prognosis.<sup>841</sup>

**Key Message # 22: HF and ACHD**

- HF is the most common cause of death in ACHD patients, accounting for 17- 42% of all deaths.
- Patients with isolated simple defects generally do well with mortality rates like those in the general population.
- Patients with complex heart defects such as systemic RV, single ventricle palliated with Fontan circulation, and unrepaired cyanotic CHD with Eisenmenger physiology comprise the majority of HF-related deaths.

**Key Recommendations #17: HF and ACHD**

- The principals of managing HF in ACHD HF are: (see Appendix XV, page 177)
  - First to access and address all reversible causes.
  - To initiate pharmacotherapy if HF control is still not optimal.
  - To consider cardiac resynchronization and more advanced therapies if these patients continue to have worsening HF despite optimal medical therapy.

**15. ADVANCED HEART FAILURE****15.1. Heart Transplant****I,A**

Heart transplantation is well-established for refractory end stage HF. It is the gold standard for the treatment of advanced HF in the absence of contraindications. World-wide the 1-year survival rate is 91% and the median survival post-transplant is 12 to 13 years.<sup>842</sup>

This definitive therapy for HF is however limited by the lack of donor organs.<sup>843</sup> To address this, worldwide, efforts are being made to increase this organ donor shortage by increasing the donor age limit criteria, improving donor recovery strategies and being more stringent in the selection criteria.<sup>155,843</sup>

All patients with severe symptomatic HF despite OMT and no other alternative therapeutic options should be considered for heart transplant. They need to be referred to a HF specialist hospital for further evaluation.

Assessment for heart transplant is done by a multispecialty, multidisciplinary team and appropriate work up will be performed for eligibility.

Eligibility criteria to be considered for heart transplant include:

- Poor LVEF (< 25%).
- Recurrent HF hospitalizations.
- Major limitation of the patient's daily activities.
- Poor effort tolerance i.e. peak VO<sub>2</sub> (maximal oxygen consumption or peak oxygen uptake) < 10ml/kg/min (or < 50% predicted). VO<sub>2</sub> max is the maximum rate of oxygen consumption measured during incremental exercise i.e. exercise of increasing intensity. This is widely used as an indicator of cardiorespiratory fitness.
- Intravenous inotropic dependence for symptomatic relief or to maintain end organ function.
- Motivated, psychologically stable, and compliant with therapy.

Contraindications to heart transplant:

- Active infection.
- Severe peripheral arterial or cerebrovascular disease.
- Malignancy within 5 years.
- Diabetes mellitus with widespread microvascular complications.
- Systemic disease with multi-organ dysfunction.
- Irreversible chronic kidney, liver, or lung disease.
- Pharmacologically irreversible pulmonary hypertension.
- Other medical or psychosocial issues that would impact survival.

**15.2. Mechanical Circulatory Support (MCS)-see Section 9.2.2.**

The use of a mechanical circulatory support (e.g., Left Ventricular Assist Device -LVAD) may be considered as a:<sup>844-848</sup>

- Bridge to recovery in patients with potentially reversible or treatable conditions **or**
- Bridge to heart transplant in suitable candidates **or**
- Destination therapy in candidates who are not suitable for transplant.

Patients have improvement in their symptoms when compared to OMT.<sup>844,845</sup>

However, the rate of rehospitalizations due to complications of bleeding, thrombosis and infections are high.<sup>844-846,848</sup> Many patients also go into major depression. Thus, extensive discussion with the patient and family is necessary prior to LVAD implantation.

**15.3. Palliative and End of Life Care**

Despite recent advances in therapy, for some patients, HF remains a progressive disease and carries a poor prognosis. Patients with refractory symptoms despite guideline-directed medical therapy, should be considered for cardiac transplantation or LVAD implantation if deemed eligible. This includes:

- Those with < 50% survival at 1 year (using established scoring tools)
- Persistently elevated natriuretic peptide levels and/or
- A low V<sub>O2</sub> max measurement (if available) and
- other eligibility criteria as stated in section 15.1.

In patients who are unable to receive advanced therapy, palliative care consultation should be considered.<sup>219,849</sup> The aim of palliative care in HF is to prevent and relieve suffering and to promote the best quality of life for patients and their families.<sup>850</sup>

Although majority of guidelines do not specifically address when best to refer end-stage HF patients for palliative care consideration, due to the often-unpredictable trajectory in decline over time, palliative care is particularly important in a subgroup of patients. A recent international consensus referral criterion for specialist palliative care for patients with advanced HF includes 6 major categories:<sup>851</sup>

- Patients suffering from advanced HF alongside complications including cardiac cachexia or multiorgan failure, having concurrent non-cardiac life-threatening diseases, or being intolerant to guideline-directed therapies.
- Patients likely eligible for advanced therapies but not receiving them due to various factors.
- Patients with repeated hospitalizations or emergency department visits within the past 3 months.
- Patients with poor prognostic estimate, with estimated life expectancy of 6 months or less.
- Patients suffering from refractory symptoms requiring palliative sedation, severe emotional or physical symptoms, and severe spiritual or existential distress.
- Patients, their families, or care teams requesting palliative care, including assistance in discussing goals of care and other planning surrounding withdrawal or de-escalation of life-prolonging interventions.

End-of-life issues surrounding advanced HF remains complex to this day, with minimal evidence in the literature to guide management.<sup>852</sup> Nonetheless, there exists several



position papers and expert consensus documents highlighting key components in providing palliative care for HF patients, including the following:<sup>853,854</sup>

- **Advanced care planning**

- Clearly define goals of care and preference for future and pre-existing treatment, especially those which are life-sustaining. (e.g., intubation and mechanical ventilation, ICD implantation, etc)
- Address disease-specific aspects of care with clear limits to escalation.
- Record and clearly document these plans in the medical records.
- When requested and possible, clearly nominate a personal representative for medical decision-making.
- When available, clearly indicate where palliative care is to be provided (home versus healthcare facility).<sup>855</sup>

- **Medical management**

- Medical management with clear goals of care, prioritizing symptom control over disease control, and maximizing quality of life.
- Common symptoms and signs to be addressed include dyspnea, fatigue, pain, irregular bowel habits including constipation, urinary retention, or incontinence, and most importantly depression and anxiety.
- Validity of previous medications used in disease control should be continuously re-evaluated for benefit and harm, with appropriate adjustment.
- Religious and spiritual needs, values and existential quest need to be continually addressed during each consultation or visit.
- Professional help offered by specialists in spiritual care should be engaged when requested for.
- Care for the dying.
- Understanding that dying is a medical diagnosis that should neither be neglected nor postponed.
- Understanding that dying is a complex and dynamic process.
- Counselling, support, and reassurance should be provided to patients and their families.
- The aim should be to provide the highest level of comfort to the patient and their families, and therapy that contradict this should be discontinued (e.g., ICD deactivation)

- **Managing ethical dilemma**

- An ethical dilemma arises when at least two of the four ethical principles conflict with one another.
- Patient autonomy remains central in the decision making for palliative care, although the 3 other ethical principles remain paramount (beneficence, non-maleficence, and justice).
- In the event of an ethical dilemma, ethical consultation through palliative care services or a professional ethics committee should be sought.

### **Key Messages #23: Advanced HF**

- All patients with severe symptomatic HF despite OMT and no other alternative therapeutic options should be considered for heart transplant.
- If they are not eligible or a donor heart is not available, they may be considered for LVAD as a destination therapy or a bridge to heart transplant.
- Patients who are unable to receive either should be considered for palliative care.

**Key Recommendations #18: Advanced HF**

- Heart transplantation is well-established for refractory end stage HF.
- Patients with severe symptomatic HF despite OMT and no other alternative therapeutic options should be considered for palliative care.

**16. HEART FAILURE REHABILITATION**

Fatigue and breathlessness lead to individuals restricting their physical activities. This in turn leads to deconditioning - a phenomenon of reduced capacity to perform physical, mental, and cognitive tasks. This adds to the humanistic burden of HF, increasing morbidity and recurrent hospitalizations.

- Exercise training in HF addresses 3 main impairments:
  - Exercise intolerance due to muscle wasting leading to a loss of strength and endurance.<sup>856</sup>
  - Cardiopulmonary and musculoskeletal deconditioning.<sup>857</sup>
  - Exertional dyspnea that occurs from:<sup>857</sup>
    - ◆ Increased respiratory muscle work because of excessive ventilation to compensate for increased muscle lactate release.
    - ◆ Increased lung dead space.
    - ◆ Decreased lung compliance caused by chronic pulmonary congestion and fibrosis.
- Exercise training in HF is safe and effective.<sup>858</sup> Long-term exercise training leads to improvement in quality of life, functional capacity and survival rates.<sup>155,164,165,859-865</sup>

**16.1 Cardiac rehabilitation (CR) in HF**

- HF is now endorsed as an indication for CR.<sup>866</sup> CR spans throughout the continuum of HF care.<sup>862</sup>
- A formal cardiac rehabilitation program usually includes:
  - A medical and functional evaluation.
  - Education of the patient and family on the importance of:
    - ◆ Medical adherence
    - ◆ Lifestyle changes
    - ◆ Fluid management
    - ◆ Dietary recommendations *and*
    - ◆ Addressing psychosocial concerns
  - Exercise prescription, training, counselling on physical and daily activities.
  - Functional training and work hardening.
  - Stress management and adaptation.
- CR should be recommended to all stable HF patients, in NYHA II-III. This includes patients with no recent ( $\leq 6$  weeks) or planned ( $\leq 6$  months) major CV hospitalizations or procedures.<sup>858</sup>
- There is insufficient data at present to recommend outpatient CR for patients in NYHA IV.
- CR involves:
  - An **inpatient phase** which aims towards safe home discharge. Compared to routine care, early CR significantly improves the effect of physical capacity, physiological outcomes, and clinical outcomes in acute HF patients.<sup>867</sup>

IIa,B



- An **outpatient phase** to improve functional gains. Three exercise training modalities in HF include:<sup>862</sup>
  - ◆ Endurance aerobic training.
  - ◆ Resistance strengthening - this addresses skeletal muscle alterations and sarcopenia.
  - ◆ Respiratory muscle training.
- A **maintenance phase** that aims at improving long term survival by focusing on identifying and treating CV risk factors and optimizing HF treatment.<sup>868</sup>
- Hydrotherapy or aquatic exercise training can improve exercise capacity, muscle strength and quality of life like land-based training programs. It is a safe and effective alternative for those unable to participate in traditional exercise programs.<sup>869</sup>
- Active participation in any kind of exercise training program is sufficient to improve the prognosis, quality of life and functional capacity.<sup>870</sup> Active participation is a more important factor in achieving improvement than how the exercise is performed.<sup>871</sup>
- The details of the Cardiac Rehabilitation program in HF can be obtained from several resources.<sup>868,872</sup>

### 16.2. Settings for CR in HF

- A community-based CR program provides patients with a structured exercise training intervention alongside educational support and psychological counselling.<sup>873</sup>
- Settings for CR in HF include:<sup>874</sup>
  - Traditional center-based CR programs are safe and effective but are resource intensive and of limited availability. It is recommended during the initial sessions following discharge particularly in patients with severe HF symptoms.
  - Home-based programs are available for stable and well-treated patients. It is a cost-effective treatment option even in HFrEF.<sup>165</sup>
  - Emerging CR models include:
    - ◆ Hybrid CR that combines short-term center-based CR with home-based CR.<sup>875</sup>
    - ◆ Mobile phone based CR<sup>876</sup>
    - ◆ Telerehabilitation services.<sup>877</sup> There is still inadequate evidence at present on the impact of this modality on hospitalization and CV death reduction.<sup>878</sup>

### 16.3. Heart failure CR in special populations

These include:

- Individuals with implantable devices
  - CR in HF patients with ICD and CRT is safe and beneficial.<sup>862,879-881</sup>
  - Activities such as hydrotherapy or pronounced arm - shoulder movements should be avoided as this may lead to an ICD discharge and loss of consciousness.
- Elderly patients with HF
  - Frailty is common among the elderly and is associated with increased morbidity and mortality. It is partially reversible and potentially preventable in HF with early rehabilitation referral.<sup>882</sup>
  - Rehabilitation non-enrolment results in worsening frailty status.<sup>883</sup>
  - Cognitive impairment and HF frequently coexist - either as acute delirium with decompensated HF during hospital admission or dementia.
  - Patients with a high frailty score and cognitive impairment will benefit from closer contact with the HF specialist team, more frequent follow-up and monitoring and individualized self-care support.

**16.4 Barriers to HF rehabilitation**

- Despite benefits, cost-effectiveness, and strong practice guideline recommendations, CR remains underused.<sup>172</sup>
- Barriers include<sup>884</sup>
  - Physician Factors:
    - ◆ Fewer referrals from cardiologists.
    - ◆ Fewer well trained CR staff.
    - ◆ Overworked doctors.
  - Patient Factors:
    - ◆ Lack of motivation
  - Reluctance to change in lifestyle.
    - ◆ Transport difficulties.
    - ◆ Lack of family support.
    - ◆ Time constraints.
  - Service Factors:
    - ◆ Lack of insurance coverage.
    - ◆ Accessibility of programs.

**Key Messages # 24: HF Rehabilitation**

- Fatigue and breathlessness lead to individuals restricting their physical activities and this in turn, leads to deconditioning.
- Exercise training in patients with HF is safe and leads to an improvement in functional capacity, exercise duration, and health related quality of life.
- HF is now endorsed as an indication for Cardiac Rehabilitation, and it spans throughout the continuum of HF care.

**Key Recommendations #19: HF Rehabilitation**

- Cardiac Rehabilitation should be recommended to all stable HF patients, in NYHA II-III.

**17 ORGANISATION OF CARE****17.1 LEVEL OF CARE AND SHARED MANAGEMENT**

The care of patients with HF should ideally take place in a multidisciplinary system, allowing for shared care between the hospital (secondary or tertiary settings) and community (primary setting). A multidisciplinary approach encompasses patient education, cardiac rehabilitation, psychosocial support and palliative care, and has been proven to reduce HF hospitalizations and mortality in discharged patients.<sup>884-886</sup>

The multidisciplinary team usually consists of cardiologists and/or general physicians, HF nurses, pharmacists, dietitians, physiotherapists, primary care providers, social workers as well as geriatricians, psychologists, cardiac rehabilitation physicians, occupational therapists and when necessary, palliative care specialists.



Care can be done in two different settings:

- In the patient's own home - home-based interventions are associated with significantly lower healthcare costs, reduced hospital readmissions and an improvement in the patient's quality of life.<sup>855</sup> This may now be possible with telemedicine.
- Specialist outpatient clinic - the HF clinic.

HF clinics can either be:

- Nurse-directed - these are run by nurses with special training in HF.
- Physician-directed - run by general physicians and/or cardiologists.

HF clinics can be established in the tertiary hospitals or in the primary care setting. The minimum human resource requirements are:

- A cardiologist or general physician with an interest in cardiology and heart failure,
- A dedicated nurse and
- A medical technologist for blood taking, doing echocardiography and 6 minute-walk tests.

In bigger clinic settings, the involvement of a cardiac rehabilitation team and physiotherapists to encourage physical activities, pharmacists and counsellors for end-of-life care would be advisable. The needs of these patients vary, and these may need to be addressed by a multidisciplinary team.

These clinics will be the intermediary between in-patient hospital care and community primary care. Patients who can be seen in these clinics include those:<sup>887</sup>

- Recently discharged after an admission for decompensated HF (a waiting time of 7-12 days post discharge has to be the maximum wait-time).
- Who are in the early decompensation phase and need treatment modification.
- Who are stable but need up titration of HF medications.
- With ICD or CRTs.
- With comorbidities, such as renal dysfunction, diabetes and COPD.
- With advanced HF who may benefit from:
  - Heart transplant.
  - Left Ventricular Assist Device (LVAD).
  - Palliative care.

The objectives of these HF clinics may vary based on local settings. They include:<sup>887,888</sup>

- Optimization of Foundational HF medications particularly the up titration of  $\beta$ -blockers, ACE-I and MRA.
- Initiating referral for Cardiac Rehabilitation
- Education of the patient and caregiver on the nature of the disease and its progression.
- Promotion of self-care such as :
  - Compliance to medical therapy and fluid restriction.
  - Regular weighing and adjusting diuretic doses according to symptoms and body weight.

Patient with optimized HF medications/treatment plans can be discharged to the community with appropriate care plans to primary care. Close partnership between these HF clinics and primary care helps to reduce unnecessary admissions to hospital.

Moving forward, it may be timely, to suggest establishing community medical care centers for chronic illnesses such as diabetes, HF, CKD etc. This could serve as a resource and



rehabilitation center for individuals with chronic illnesses. It could be run by both volunteers - doctors, nurses, physiotherapists, rehabilitation physicians - with funding from government and Non Government Organizations and offering tax incentives to all those participating in the program.

In 2018, the Heart Failure Medication Therapy Adherence Clinic (MTAC) was introduced. These clinics are conducted by pharmacists in collaboration with doctors and other healthcare providers to improve heart failure management. The objectives of the Heart Failure MTAC are:<sup>889</sup>

- Enhance patient's adherence to HF pharmacotherapy and non-pharmacological interventions.
- To reduce unscheduled emergency department visits or hospitalizations of HF patient due to acute decompensation.
- To provide consultative service to doctors on evidence-based HF pharmacotherapy and related issues.
- To collaborate with doctors and other health care professionals in HF management program.

## **17.2. MONITORING AND FOLLOW-UP**

Patients with HF require regular follow-up and monitoring. Serial evaluations serve to assess a patient's status, response to therapy, development of complications and disease progression.

### **17.2.1. Assessment during Follow Up Visits**

Key components of assessment include:

- **History:**
  - Functional capacity - NYHA functional class or 6-minute walk test
  - Cognitive status and nutritional status
  - Diet and sodium intake
  - Consumption of alcohol or illicit drugs
  - Smoking history
- **Physical examination:**
  - Fluid status and body weight
  - Blood pressure, heart rate and rhythm
  - Examination of the cardiovascular and respiratory systems
- **Investigations:**
  - Blood tests - serum urea, electrolytes, creatinine and eGFR as necessary.
  - Serial brain natriuretic peptide measurements to guide and tailor HF therapy cannot be recommended at the present time due to a lack of consistent evidence.<sup>77,81,891,892</sup>
  - Echocardiogram:
    - Routine serial echocardiogram is not recommended. However, if there has been a recent change in clinical status (e.g., episode of acute coronary syndrome) or if the patient has received treatment (e.g., revascularization or device therapy) that might significantly change certain echocardiographic parameters, a follow-up echocardiogram is reasonable to re-assess the LVEF, degree of valvular function and structural remodeling.
- **Others:**
  - Review of pharmacotherapy - compliance, side effects and titration of dosages as necessary.
  - Control of disease specific management - hypothyroidism, hyperthyroidism, acromegaly, connective tissue disease, HIV, sarcoidosis, amyloidosis, hemochromatosis.
  - Review use of potentially detrimental medication to HF control - e.g., NSAIDs.



- Education:
  - Another key component during follow up is education regarding the name, dose and function of each medication the patient is on. This is not only important in terms of enhancing compliance, but this is also to avoid polypharmacy from a different facility the patient may also be seeing.

### 17.2.2. Frequency of follow-up

This will depend on:

- The patient's clinical stability **and**
- Need for pharmacotherapy optimization.

A patient with a recent episode of decompensation or clinical instability, for instance, should ideally be seen again soon, usually within 2 weeks. Ultimately, the intensity and type of follow-up would be determined by the local organization of care and resources.

### 17.3. CARDIOLOGY REFERRALS

HF patients with stable symptoms may be managed at the primary care level.

Referral to the cardiologist should be considered in the following situations:

- De novo HF for a comprehensive workup to confirm the diagnosis and determine the etiology, and to devise a management plan.
- Episodes of acute decompensation
- Worsening HF symptoms despite appropriate therapy
- HF that is complicated with symptomatic hypotension or excessive bradycardia, limiting up titration of pharmacotherapy.
- Symptomatic stable CAD and/or acute coronary syndrome for consideration of revascularization (PCI or CABG)
- Resuscitated cardiac arrest.
- Documented or suspected significant arrhythmias e.g., AF, VT
- ECG demonstrating a LBBB morphology with a QRS duration of > 150ms for consideration of device therapy.
- Significant valvular disease not previously assessed or worsening valvular dysfunction.
- Preconception assessment and counselling of women with significant structural heart disease and a history of HF or LV dysfunction.
- Complex congenital cardiac lesions and/or Eisenmenger's syndrome.

### 17.4. TELEMEDICINE/TELEHEALTH

Telemedicine services may improve healthcare accessibility and geographical limitations. In the unforeseen circumstances of a future pandemic, telemedicine may provide uninterrupted patient consultation when community lockdowns or social distancing is required.<sup>893</sup>

Early escalation of HF care with a multidisciplinary team is possible without the need for face-to-face consultation.

**17.4.1. Telemedicine Strategies****Requirements**

- Home Monitoring Equipment -The following are required:
  - Computer, tablet, or smart phone
  - BP machines
  - Weighing scales
  - Optional:
    - ◆ ECG recording device on smart phone.
    - ◆ Mobile monitor (e.g., Cardio Mobile)

Patients, and/or their younger family members can record and transmit the information to the physician.

**Virtual visits**

- This will help maintain the doctor-patient relationship.
- These virtual visits can be performed via audio-visual telecommunication system or through an online portal to review the following:
  - Assessment of symptoms.
  - Review information from the Home Monitoring devices.
  - Where available, laboratory results from a laboratory close to the patient's home.
  - Optimization of Foundational HF Medications.

**Telerehabilitation**

Telemedicine can allow:

- Initiation of dietary and physical cardiac rehabilitation.
- Identification of cardiac decompensation and advise hospitalization, when necessary.

**Key Messages # 25: Level of Care, Shared Care, Monitoring and Follow - Up & Telemedicine/Telehealth**

- The care of patients with HF should ideally take place in a multidisciplinary system, allowing for shared care between the hospital (secondary or tertiary settings) and community (primary setting).
- Heart Failure clinics will serve as an intermediary between in-patient hospital care and community primary care.
- HF clinics can either be:
  - Nurse-directed - these are run by nurses with special training in HF.
  - Physician-directed
- At these clinics, patients with HF undergo close follow-up and monitoring to evaluate their status, response to therapy, development of complications and disease progression.
- When stable with optimized HF medications/treatment plans, patients can be discharged to the community with appropriate care plans to primary care.
- Where necessary, (decompensation, development of significant arrhythmias) they can be referred to the cardiologists/specialists.
- Telemedicine services may improve healthcare accessibility and geographical limitations. It can complement existing HF clinic services and provide venues for early escalation of HF therapy before decompensation.

**18. OTHER THERAPIES FOR HEART FAILURE**

Despite taking conventional HF therapy, patients may seek alternative therapy and healing approaches that are not considered as allopathic medicine. The National Center for Complementary and Alternative Medicine (NCCAM) defines complementary and alternative medicine (CAM) as a group of diverse medical and healthcare interventions, practices, products, or disciplines that are not generally considered part of conventional medicine.

**18.1. Enhanced External Counter Pulsation (EECP)**

There is inadequate evidence of clinical effectiveness of EECP in HF.<sup>894-898</sup> Data seem to indicate that there is improved functional status, walk distance, and symptoms and NYHA Functional class.<sup>896-897</sup> There is however, concern that it could precipitate or exacerbate symptoms in those with a history of HF.<sup>898</sup>

**18.2. Stem cell therapy**

Stem cells, which are derived from bone marrow or umbilical cord blood, have been infused into the coronary arteries or injected directly into the myocardium. The original hypothesis was that stem cells would engraft in the myocardium and replace damaged cardiomyocytes. Subsequent research has suggested that stem cells probably do not differentiate into functional cardiomyocytes, but as an alternative hypothesis, it was hoped that they might have paracrine effects that would benefit myocardial function.

There is still limited knowledge on the optimal cell type, dosing, route of administration, patient parameters, patient safety, and other important variables that contribute to successful stem cell therapy. There is, however, still a lot of research left to do into the effectiveness and safety of these interventions.<sup>899-901</sup>

**18.3. Omega 3 fatty acids**

This may be considered as an adjunctive therapy to standard HF therapy, based on observations and clinical trials which showed a small benefit in CV death and/or hospitalizations especially among diabetics.<sup>902-907</sup>

**18.4. Coenzyme Q10**

There is some evidence that coenzyme Q10 probably reduces all-cause mortality and hospitalization for HF. However there still needs to be further research to determine the role of Coenzyme Q10 in patients with HF, the optimal dose, and the duration of Coenzyme Q10 supplementation.<sup>908-910</sup>

**18.5. Tai Chi**

Tai Chi may improve 6 minute walk test distance, quality of life and LVEF in patients with HF.<sup>911-914</sup> Its effect on hard CV outcomes such as rehospitalization, MI and mortality is not known.

**18.6. Yoga**

Yoga improves peak VO<sub>2</sub> (exercise capacity) and quality of life in chronic HF patients.<sup>915</sup> In addition, it has been reported to reduce cardiovascular events, morbidity, and mortality although the evidence supporting these conclusions is somewhat limited.<sup>916</sup>

**19. PERFORMANCE MEASURES**

Performance measures should be used with the goal of improving quality of care for HF and conversion of scientific evidence into clinical practice.<sup>155,917</sup> It has been shown that higher hospital performance measures were inversely associated with HF readmissions.<sup>918,919</sup>

**Process performance measures** focus on the aspects of care that are delivered to a patient, while outcome measures focus on the endpoints such as mortality or hospitalization.

Process performance indicators for in-patients with HF includes:<sup>155,917-919</sup>

- % of patients who had documentation of NYHA Functional Class.
- % of patients who had LVEF measurement.
- % of patients with current or prior LVEF < 40% and without contraindications discharged with ACE-I/ARB/ARNI.
- % of patients with current or prior LVEF < 40% and without contraindications discharged on  $\beta$ -blockers.
- % of patients with current or prior LVEF < 40% and without contraindications discharged on MRA.
- % of patients with chronic or paroxysmal AF/Atrial Flutter without contraindications on anticoagulant therapy at discharge.
- % of patients given a post discharge appointment within 14 days.
- % of patients who had up titration of their Foundational HF medications to target or maximally tolerated doses at 3 months.

The accepted performance measure is 60%.

**Outcome Measures** indicators include:

- In-hospital mortality
- 30-day readmission for heart failure

Refer to Appendix XVI, page 178 for calculation of these measures.

**Key Recommendations #20: Performance Measures**

- Performance measures are used with the goal of improving quality of care for HF.
- This includes:
  - Process performance which measures the aspects of care that are delivered to a patient and
  - Outcome measures which focus on hard endpoints such as mortality or hospitalization.
- For the Quality indicators in HF, see section 19 & Appendix XVI, page 178.

**Appendix I: Causes of Elevated Natriuretic Peptide Levels\*****Cardiac**

- Heart Failure, including RV syndromes
- Acute coronary syndromes
- Heart muscle disease, including Left Ventricular Hypertrophy
- Valvular heart disease
- Pericardial disease
- Atrial fibrillation
- Myocarditis
- Infective Endocarditis
- Cardiac surgery
- Cardioversion
- Toxic-metabolic myocardial insults, including cancer chemotherapy

**Non-cardiac**

- Advancing age (Table 12, page 60 for optimal cut off values according to age)
- Anemia
- Renal failure:
  - NProBNP loses its prognostic value in patients with  $GFR < 30\text{ml/min/1.73 m}^2$ .
  - BNP levels are relatively independent of GFR.
- Right ventricular overload from:
  - Pulmonary causes: obstructive sleep apnea, severe pneumonia
  - Pulmonary hypertension whatever the cause
- Shock
- Critical illness
- Bacterial sepsis
- Severe burns
- Liver cirrhosis
- Intracranial pathologies
- Thyrotoxicosis

\*Adapted from

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## Appendix II: Salt Content of Common Malaysian Foods Daily Reference Intake (DRI) for salt: 5 g per day (1 teaspoon)

| SALT CONTENT IN FOOD               |                       |                                  |                    |                                                               |                     |
|------------------------------------|-----------------------|----------------------------------|--------------------|---------------------------------------------------------------|---------------------|
| LOW<br>(< 5% of DRI)               |                       | MODERATE<br>(5% - 20% of DRI)    |                    | HIGH<br>(> 20% of DRI)                                        |                     |
| Tosai without Gravy (1 piece)      | 0.5g                  | Preserved Bean Curd (16g)        | 0.8g (16% of DRI)  | Fried Chicken (3 pieces), Coleslaw, Mashed potato, Soft Drink | 10.4g (208% of DRI) |
| Cereal (Cornflakes) (1 cup, 30g)   | 0.4g                  | Luncheon Meat (1 piece)          | 0.9g (18% of DRI)  | Curry Mee (1 large bowl)                                      | 8.2g (164% of DRI)  |
| Fresh Fruit (1 medium)             | <0.15 (3% of DRI)     | Beef Burger Patty (1 piece, 74g) | 0.8g (16% of DRI)  | Wonton Noodle Soup (1 large bowl)                             | 4.5g (90% of DRI)   |
| Fresh Vegetables (1 cup, cooked)   | <0.3g (5% of DRI)     | Nuggets (2 pieces, 40g)          | 0.7g (14% of DRI)  | Bihun Soup (1 large bowl)                                     | 4.0g (80% of DRI)   |
| Tuna In Water, no salt (1 can)     | 0.2g (4% of DRI)      | Chili Sauce (1 tbsp, 15g)        | 0.56g (11% of DRI) | Char Kway Teow (1 plate)                                      | 3.6g (72% of DRI)   |
| Low Fat Milk (1 packet)            | 0.2g (4% of DRI)      |                                  |                    | Fried Hokkien Noodles with prawn (1 plate)                    | 3.6g (72% of DRI)   |
| Putu Mayam (1 piece, 50g)          | 0.15g (3% of DRI)     |                                  |                    | Pizza with Tomato Sauce, Chicken, and Mushroom                | 2.75g (55% of DRI)  |
| Baked Potato (1 whole)             | <0.15g (<3% of DRI)   |                                  |                    | Chicken Burger + Fries and Soft Drink (1 regular set)         | 2.6g (52% of DRI)   |
| Fresh Fish (1 whole)               | <0.13g (2.6% of DRI)  |                                  |                    | Budu (1 tbsp)                                                 | 2.2g (44% of DRI)   |
| Oat Crackers (3 pieces)            | 0.11g (2.2% of DRI)   |                                  |                    | Salted Egg (1/2 piece)                                        | 2.0g (40% of DRI)   |
| Low Salt Biscuit (3 pieces)        | 0.09g (2% of DRI)     |                                  |                    | Beef Noodles (789g)                                           | 9.9g (198% of DRI)  |
| Chicken Breast Part (1 piece, 38g) | 0.06g (1% of DRI)     |                                  |                    | Hot Dog Set                                                   | 3.9g (78% of DRI)   |
| Margarine Reduced Salt (1 tsp)     | 0.05g (1% of DRI)     |                                  |                    | Nasi Lemak (210g)                                             | 2.1g (42% of DRI)   |
| Unsalted Nuts (1 cup)              | 0.04g (0.9% of DRI)   |                                  |                    | Yong Tau Fu (324g)                                            | 4.3g (86% of DRI)   |
| Chapati with Gravy (1 pc)          | 0.04g (0.8% of DRI)   |                                  |                    | Instant Noodles (85g)                                         | 3.8g (76% of DRI)   |
| White Rice (1 scoop)               | 0.04g (0.8% of DRI)   |                                  |                    | Chinese Sausage (1 piece, 68g)                                | 2.6g (52% of DRI)   |
| Soy Milk (1 glass, 250ml)          | 0.01g (0.2% of DRI)   |                                  |                    | Instant Porridge (1 cup, 35g)                                 | 2.4g (48% of DRI)   |
| Tau Kua (1 pc)                     | 0.005g (0.1% of DRI)  |                                  |                    | Instant Soup (1 packet, 63g)                                  | 2.0g (40% of DRI)   |
| Oat (6 tbsp, 35g)                  | 0.001g (0.06% of DRI) |                                  |                    | Cube/Stock Powder (110% of DRI)                               | 5.52g (1 cube, 12g) |



## Appendix II: Salt Content of Common Malaysian Foods,cont'd Daily Reference Intake (DRI) for salt: 5g per day (1 teaspoon)

| SALT CONTENT IN FOOD |  |                               |  |                                     |                        |
|----------------------|--|-------------------------------|--|-------------------------------------|------------------------|
| LOW<br>(< 5% of DRI) |  | MODERATE<br>(5% - 20% of DRI) |  | HIGH<br>(> 20% of DRI)              |                        |
|                      |  |                               |  | Light Soya Sauce<br>(1 tbsp,15g)    | 3.14g<br>(63 % of DRI) |
|                      |  |                               |  | Fish Sauce<br>(1 tbsp,15g)          | 2.58g<br>(52% of DRI)  |
|                      |  |                               |  | Salted Fish<br>(1 piece, 25g)       | 1.1g<br>(22% of DRI)   |
|                      |  |                               |  | Dried Anchovies<br>(¼ cup)          | 1.2g<br>(24% of DRI)   |
|                      |  |                               |  | Cheddar Cheese<br>(2 slices)        | 1.2g<br>(24% of DRI)   |
|                      |  |                               |  | Oyster Sauce<br>(1 tbsp)            | 1.6g<br>(32% of DRI)   |
|                      |  |                               |  | Spaghetti Sauce<br>(½ cup)          | 1.3g<br>(26% of DRI)   |
|                      |  |                               |  | Baked Beans<br>(¾ cups)             | 1.3g<br>(26% of DRI)   |
|                      |  |                               |  | Belacan<br>(1 tbsp)                 | 1.7g<br>(34% of DRI)   |
|                      |  |                               |  | Asam Jawa<br>(1 tbsp)               | 1.5g<br>(30% of DRI)   |
|                      |  |                               |  | Taucu<br>(1 tbsp)                   | 1.4g<br>(28% od DRI)   |
|                      |  |                               |  | Dried Prawn<br>(1 tbsp)             | 1.2g<br>(24% of DRI)   |
|                      |  |                               |  | Cheese Sausage<br>(2 pieces,70g)    | 1.4g<br>(28% of DRI)   |
|                      |  |                               |  | Meatballs<br>(5 pieces)             | 1.4g<br>(28% of DRI)   |
|                      |  |                               |  | Keropok Lekur<br>(4 pieces,120g)    | 1.0g<br>(20% of DRI)   |
|                      |  |                               |  | Salted Nuts<br>(1 cup, 155g)        | 1.6g<br>(32% of DRI)   |
|                      |  |                               |  | Canned Chicken Curry<br>(150g)      | 1.4g<br>(28% of DRI)   |
|                      |  |                               |  | Pickled Lettuce<br>(9 pieces,30g)   | 1.3g<br>(26% of DRI)   |
|                      |  |                               |  | Mono Sodium Glutamate<br>(1 tsp,5g) | 1.54g<br>(31% of DRI)  |
|                      |  |                               |  | Dark Soya Sauce<br>(1 tbsp,15g)     | 1.41g<br>(28% of DRI)  |

Source: Malaysian Food Composition Database

**Appendix III: Discharge Care Plan**

The following discharge care plans should be discussed and agreed upon by the patient and treating physician.

| NO | PARAMETER                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | TICK |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 1  | Discussion on: <ul style="list-style-type: none"><li>• The precipitating factor(s) leading to this current exacerbation,</li><li>• How to avoid such circumstances and</li><li>• Other precipitating factors for decompensation.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                       |      |
| 2  | To re-enforce education regarding: <ul style="list-style-type: none"><li>• The current status of the patient and the disease trajectory</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |      |
| 3  | <p>To establish the role of:</p> <ul style="list-style-type: none"><li>• Fluid management and</li><li>• Daily weight measurement at the same time each day (if possible). If weight measurement is not able to be done daily, at least 2-3 times a week is encouraged.</li></ul> <p>A sudden weight gain of more than 2kg in 3 days is a sign of worsening HF. Patients should be educated to increase their diuretic (furosemide) dose together with restriction of their fluid intake until their “dry weight” is regained. However, if the weight increase is associated with worsening symptoms or fails to respond to these measures, the patient should seek medical help immediately.</p> |      |
| 4  | To ensure compliance with medication by explaining the reason and why the medication is integral to the patient's wellbeing.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |      |
| 5  | <p>To encourage:</p> <ul style="list-style-type: none"><li>• Appropriate physical activity and exercise.</li><li>• Lifestyle modification:<ul style="list-style-type: none"><li>➢ Healthy balanced diet with no “added salt and flavoring sauces.”</li><li>➢ Weight management.</li><li>➢ Smoking cessation.</li></ul></li></ul>                                                                                                                                                                                                                                                                                                                                                                 |      |
| 6  | Stressing on the importance of follow up visit to ensure a safe post discharge period.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |      |
| 7  | Referral for cardiac rehabilitation as soon as possible.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |      |

**Appendix IV: Discharge Care Summary**

A summary of the discharge care plan should be made available to the physician at the outpatient clinic upon follow up.

| NO | PARAMETER                                                                                                                                                                                                               | TICK |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 1  | Establish: <ul style="list-style-type: none"><li>• Type of HF (HF<sub>r</sub>EF, HF<sub>p</sub>EF, HF<sub>mr</sub>EF)</li><li>• Underlying cause</li><li>• Precipitating factors (if any)</li></ul>                     |      |
| 2  | Brief synopsis of admission with pertinent findings: <ul style="list-style-type: none"><li>• Blood results</li><li>• ECG abnormalities</li><li>• Echocardiogram findings with LVEF</li></ul>                            |      |
| 3  | Highlight: <ul style="list-style-type: none"><li>• Blood tests that need to be re checked - e.g., electrolytes</li><li>• Tracing blood results e.g., iron studies</li></ul>                                             |      |
| 4  | Medication progress <ul style="list-style-type: none"><li>• Outline plans for resuming medications that had been stopped temporarily.</li><li>• Up titrating Foundational HF Medications and titration goals)</li></ul> |      |
| 5  | Further cardiac work and referrals if indicated: <ul style="list-style-type: none"><li>• Appointment for further cardiac investigations e.g., coronary angiography, cardiac imaging etc.</li></ul>                      |      |
| 6  | Management of non-cardiac diseases (pulmonary disease, substance abuse).                                                                                                                                                |      |
| 7  | Identify problems like poor psychosocial support, impaired health literacy, cognitive impairment, frailty.                                                                                                              |      |
| 8  | Additional services needed like rehabilitation, device therapy, palliative care and destination therapy discussed.                                                                                                      |      |
| 9  | List of patient's and family expectations                                                                                                                                                                               |      |



## Appendix V: Indications for Valve Intervention\*

| Valve  | Pathology     | Recommendations                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mitral | Stenosis      | <p><b>Intervention</b> is indicated in patients with MVA &lt; 1.0 cm<sup>2</sup> and who are:</p> <ul style="list-style-type: none"> <li>• Symptomatic</li> <li>• Asymptomatic but have new onset AF, Pulmonary Hypertension or planning a pregnancy.</li> </ul>                                                                                                                                                                                                                                                                                                                                          |
|        | Regurgitation | <p>Surgery <b>is recommended</b> for:</p> <ul style="list-style-type: none"> <li>• Acute severe mitral regurgitation</li> <li>• Chronic severe mitral regurgitation with: <ul style="list-style-type: none"> <li>➢ Symptoms of decreasing effort tolerance</li> <li>➢ Asymptomatic but: <ul style="list-style-type: none"> <li>♦ LVEF ≤ 60%,</li> <li>♦ LVESD ≥ 40 mm</li> <li>♦ LA volume ≥ 60 mL/m<sup>2</sup> or diameter ≥ 55 mm</li> <li>♦ Systolic pulmonary arterial pressure &gt; 50 mmHg and/or</li> <li>♦ AF</li> </ul> </li> </ul> </li> </ul>                                                 |
| Aortic | Stenosis      | <p>Surgery <b>is recommended</b> for patients who are:</p> <ul style="list-style-type: none"> <li>• Symptomatic</li> <li>• Asymptomatic patients with severe aortic stenosis and: <ul style="list-style-type: none"> <li>➢ Impaired LV function of no other cause</li> <li>➢ Those who are asymptomatic during normal activities but develop symptoms during exercise testing.</li> </ul> </li> </ul>                                                                                                                                                                                                     |
|        | Regurgitation | <p>Aortic valve replacement <b>should be advised</b> for:</p> <ul style="list-style-type: none"> <li>• Symptomatic patients irrespective of the LVEF</li> <li>• Asymptomatic patients with: <ul style="list-style-type: none"> <li>➢ LVEF ≤ 50% or</li> <li>➢ Left ventricular end-systolic diameter (LVESD) &gt; 50 mm</li> </ul> </li> </ul> <p>Aortic valve replacement <b>may be considered</b> in asymptomatic patients and:</p> <ul style="list-style-type: none"> <li>• LVEF &lt; 55% without any other cause</li> <li>• Sustained fall in systolic BP &lt; 20 mmHg on exercise testing</li> </ul> |

\*Adapted from: Vahanian, A., Beyersdorf, F., Praz, F., Milojevic, M., Baldus, S. et al. 2021 ESC/EACTS Guidelines for the management of valvular heart disease. Eur Heart J 2022; 43(7), 561-632.



## Appendix VI: ESC Heart Failure Association algorithm (HFA-PEFF)

**P - Pretest Assessment**

This includes a detailed clinical evaluation, electrocardiogram, laboratory tests, and basic echocardiogram. HFpEF is suggested by:

- Normal LVEF **and**
- Nondilated left ventricle with concentric remodelling, or left ventricular hypertrophy **and**
- Left atrial enlargement.

**E - Diagnostic Workup**

This will include a more detailed echocardiographic study and measurement of natriuretic peptides. They have proposed a scoring system using major and minor diagnostic criteria as outlined below.

**Natriuretic Peptides:****● NT-Pro BNP:**

- *Major Criterion:* > 220 pg/mL - sinus rhythm  
> 660 pg/mL - in AF
- *Minor Criterion:* 125-220 pg/mL - sinus rhythm  
375-660 pg/mL - in AF

**● BNP**

- *Major Criterion:* > 80 pg/mL - sinus rhythm  
> 240 pg/mL - in AF
- *Minor Criterion:* 35-80 pg/mL - sinus rhythm  
105-240 pg/mL - in AF

**Echocardiographic Criteria:**

HFpEF cannot be diagnosed from a single echocardiographic measure. Suggestive criteria include:<sup>16</sup>

**● Septal e':**

- *Major Criterion:* e' < 7 cm/s; or lateral e' < 10 cm/s (subjects aged < 75 years)
- *Major Criterion:* e' < 5 cm/s; or lateral e' < 7 cm/s (subjects aged ≥ 75 years)

**● Average Septal-Lateral E/e' Ratio:** - The E/e' index cannot be recommended as a single diagnostic index above all other non-invasive measures of filling pressures (such as retrograde pulmonary venous flow).<sup>16</sup>

- *Major Criterion:* ≥ 15 - This has good diagnostic value for identifying a high mean pulmonary capillary wedge pressure rest has (mPCWP) supporting the likelihood of HFpEF
- *Minor Criterion:* 9 -14 - This is less sensitive in identifying a high mPCWP.

**● Tricuspid Regurgitant (TR) Peak Velocity**

- *Major Criterion:* > 2.8m/s

**● Pulmonary Artery Systolic Pressure** - This is estimated 4x peak TR velocity plus estimated right atrial pressure.

- *Major Criterion:* > 35mmHg

**● Left Atrial Volume Index:** This is an indirect correlate of LV filling pressures. It is more accurate as a marker of chronic LA remodelling than either LA area or diameter.

- *Major Criterion:* > 34mL/m<sup>2</sup> - in sinus rhythm
- *Major Criterion:* > 40mL/m<sup>2</sup> - in atrial fibrillation
- *Minor Criterion:* 29-34mL/m<sup>2</sup> - in sinus rhythm
- *Minor Criterion:* 34-40mL/m<sup>2</sup> - in atrial fibrillation

**● Left Ventricular Mass Index and Relative Wall Thickness (RWT)**

- *Major Criterion:* LVMI ≥ 149 g/m<sup>2</sup> in men or ≥ 122 g/m<sup>2</sup> in women and RWT > 0.42
- *Minor Criterion:* LVMI ≥ 115 g/m<sup>2</sup> in men or ≥ 95 g/m<sup>2</sup> in women or RWT > 0.42 or LV end-diastolic wall thickness ≥ 12mm

**Scoring System Based on Echocardiographic and Natriuretic Peptide Levels:**

For each major criterion met, 2 points are awarded, and 1 point is awarded for a minor criterion.

- A score of ≥ 5 is diagnostic of HFpEF.
- A score of ≤ 1 makes a diagnosis of HFpEF very unlikely.
- A score of 2-4 makes the diagnosis uncertain and additional work up is recommended.

**F- Advanced Work Up**

This includes functional testing in cases of uncertainty:

- Exercise echocardiogram - To assess diastolic dysfunction during exercise.
  - An average E/e' ratio at peak stress of  $\geq 15$ , with or without a peak TR velocity  $> 3.4$  m/s
- Invasive hemodynamic assessment - If criteria for diastolic dysfunction during an exercise echocardiogram through E/e' ratio and TR velocity are not met, then a right heart catheterization at rest and/or with exercise is the next step.
  - Invasive hemodynamic assessment through a right heart catheterization at rest or at exercise.
 

The following are diagnostic of HFpEF:

    - ◆ At rest PCWP  $\geq 15$  mmHg
    - ◆ At rest left ventricular end-diastolic pressure (LVEDP)  $\geq 16$  mmHg
    - ◆ Exercise PCWP  $\geq 25$  mmHg

**F- Final Etiological Workup**

This consists of establishing the etiology of the HFpEF. This includes assessment of blood pressure control, chronotropic competence, arrhythmias, and ischemia. Cardiac magnetic resonance imaging should be considered where specific etiology such as amyloidosis or hypertrophic cardiomyopathy are suspected.

In general, **steps E, F, F** are best performed in specialized cardiology centers. The following patients who continue to have dyspnea may be referred for further evaluation and management:

- Cases where the diagnosis is unclear.
- Non responders to treatment especially those with multiple comorbidities.

**Appendix VII: H2FPEF score**

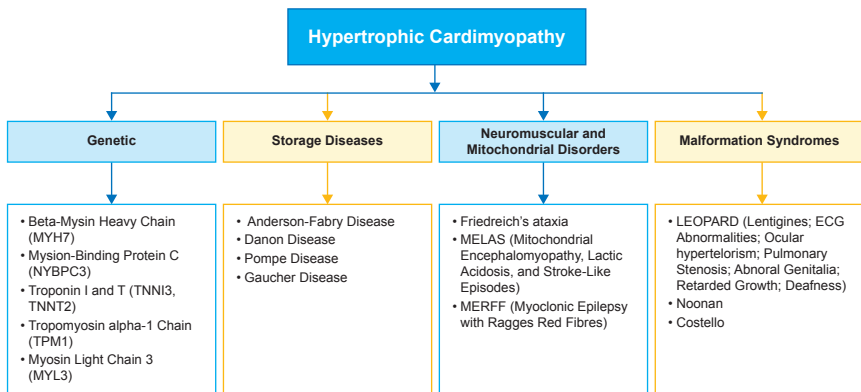
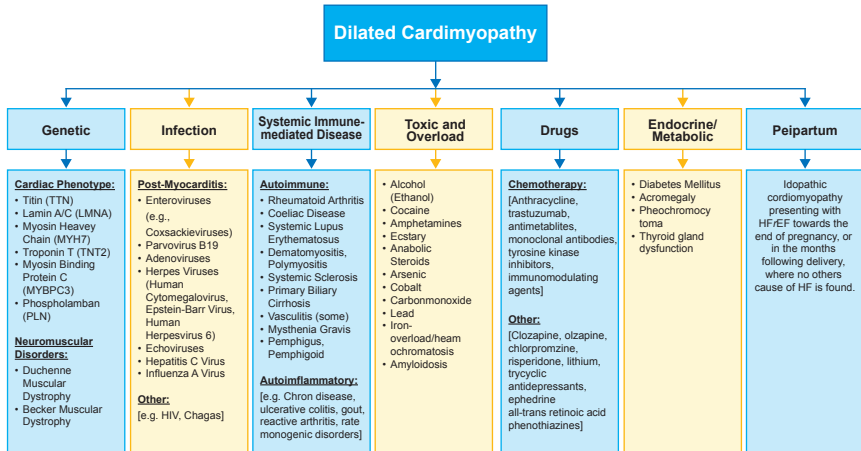
|          | Characteristic                        | Measurement                                    | Score |
|----------|---------------------------------------|------------------------------------------------|-------|
| <b>H</b> | Heavy                                 | Body mass index $> 30$ kg/m <sup>2</sup>       | 2     |
| <b>H</b> | Hypertension                          | Two or more hypertensive medications           | 1     |
| <b>F</b> | Atrial Fibrillation                   | Paroxysmal or persistent                       | 3     |
| <b>P</b> | Pulmonary Artery Pressure             | Pulmonary artery systolic pressure $> 35$ mmHg | 1     |
| <b>E</b> | Elderly                               | Age $> 60$ years                               | 1     |
| <b>F</b> | LV filling pressure by echocardiogram | Echocardiographic E/e' $> 9$                   | 1     |

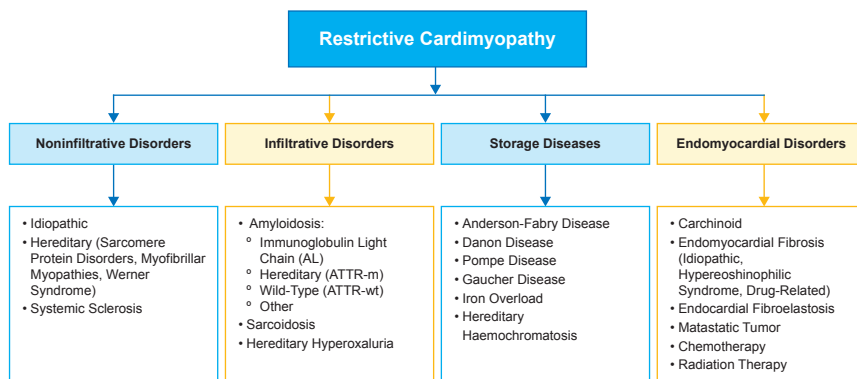
| Score | Probability of HFpEF                                                                       |
|-------|--------------------------------------------------------------------------------------------|
| 6-9   | $>90\%$                                                                                    |
| 0-1   | $\leq 23\%$                                                                                |
| 2-5   | Intermediate probability and require additional testing to determine the cause of dyspnea. |



## Appendix VIII: Etiology of Cardiomyopathies

Adapted from "Heart failure in cardiomyopathies: a position paper from the Heart Failure Association of the European Society of Cardiology")





## Appendix IX: Anticancer Agents Associated with Heart Failure / LV Dysfunction)

| Chemotherapy Agents                                                                                                               | Prevention/Treatment                                                                                                                                                                          |
|-----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Anthracyclines</b><br>Doxorubicin<br>Epirubicin<br>Idarubicin                                                                  | Monitor LVEF, strain assessment with global longitudinal strain.<br>Measure troponins.<br>Consider use of dexrazoxane, continuous infusion, liposomal preparations, $\beta$ -blockers, ACE-I. |
| <b>Alkylating Agents</b><br>Cyclophosphamide<br>Ifosfamide                                                                        |                                                                                                                                                                                               |
| <b>Antimetabolites</b><br>Decitabine<br>Clofarabine                                                                               |                                                                                                                                                                                               |
| <b>Antimicrotubule Agents</b><br>Docetaxel                                                                                        |                                                                                                                                                                                               |
| <b>Monoclonal Antibody-Based Tyrosine Kinase Inhibitors</b><br>Trastuzumab<br>Bevacizumab<br>Adostratuzumab emtacin<br>Pertuzumab | Avoid concomitant use with anthracyclines.                                                                                                                                                    |
| <b>Small Molecule Tyrosine Kinase Inhibitors</b><br>Pazopanib<br>Sorafenib<br>Sunitinib<br>Lapatinib                              | Treat hypertension aggressively                                                                                                                                                               |
| <b>Proteasome Inhibitor</b><br>Carfilzomib<br>Bortezomib                                                                          |                                                                                                                                                                                               |

**Appendix X: Chemotherapy drugs that have been associated with HF**

| Class of Drugs                     | Examples                               |
|------------------------------------|----------------------------------------|
| <b>Anthracyclines</b>              | Doxorubicin, daunorubicin, epirubicin, |
| <b>Antimetabolite</b>              | Capecitabine, 5 Fluorouracil           |
| <b>Alkylating Agents</b>           | Cisplatin, cyclophosphamide            |
| <b>Vinca Alkaloids</b>             | Vincristine, vinblastine               |
| <b>Taxanes</b>                     | Paclitaxel, docetaxel                  |
| <b>Immune Checkpoint Inhibitor</b> | Pembrolizumab, nivolumab               |
| <b>Targeted Therapies</b>          | Herceptin, imatinib, bevacizumab       |

**Appendix XI: Modified World Health Organization Maternal Cardiovascular Risk Assessment<sup>a-f</sup>****WHO CLASS I: No Increase or a Mild Increase in Morbidity**

- Uncomplicated, small, or mild
  - Pulmonary stenosis.
  - Patent ductus arteriosus.
  - Ventricular septal defect.
  - Mitral valve prolapse (with no more than trivial mitral regurgitation).
- Successfully repaired simple lesions (secundum atrial or ventricular septal defect, patent ductus arteriosus, anomalous pulmonary venous drainage).
- Isolated Atrial ectopic beats or ventricular extrasystoles.

**WHO CLASS II: (If otherwise well and uncomplicated)  
Moderate Increase in Maternal Morbidity  
Small Increase in Maternal Mortality (< 1%)**

- Unoperated atrial or ventricular septal defect (moderate size, velocity < 4m/s).
- Repaired tetralogy of Fallot (without significant residual lesions).
- Most asymptomatic arrhythmias without cardiac decompensation - atrial fibrillation, supraventricular tachycardia, Wolff Parkinson White syndrome, long QT syndrome without any cardiac decompensation.

**WHO CLASS II-III: Maternal Mortality, (1-5%)  
Depending on the individual or other co-existing conditions**

- Mild left ventricular dysfunction (LVEF 40-50%).
- Hypertrophic cardiomyopathy.
  - No LV outflow tract obstruction: WHO II
  - LV outflow tract obstruction present: WHO III
- Marfan syndrome without aortic dilation.
- Aorta < 45 mm in aortic disease associated with bicuspid aortic valve.
- Native or tissue valve disease not considered WHO I or IV.
- Mild native or repaired coarctation of aorta (without hypertension or significant obstruction).

**Appendix XI: Modified World Health Organization Maternal Cardiovascular Risk Assessment (cont'd)<sup>a-f</sup>**

**WHO CLASS III: Severe Increase in Maternal Morbidity  
Significant Increase in Maternal Mortality (5-15%)  
Expert counselling required. If pregnancy is decided upon, needs an individualized pregnancy care plan with a multidisciplinary team management.**

- Left ventricular dysfunction (LVEF 35-40%).
- Mechanical valve.
- Systemic right ventricle (cCTGA, post Senning/Mustard).
- Fontan circulation.
- Repaired Tetralogy of Fallot with severe pulmonary regurgitation, right ventricular failure, right ventricular outflow tract obstruction.
- Cyanotic heart disease (unrepaired).
- Other complex congenital heart disease.
- Aortic dilatation 40 - 45 mm in Marfan syndrome.
- Aortic dilatation 45 - 50 mm in aortic disease associated with bicuspid aortic valve.

**WHO CLASS IV: Maternal mortality is 25-50%  
Pregnancy Is Not Recommended or Is Contraindicated.  
If pregnancy occurs, termination should be discussed.  
If pregnancy continues, care as for class III.**

- Severe pulmonary arterial hypertension of any cause (mortality 17-33%).
- Severe systemic ventricular dysfunction (LVEF < 30%, NYHA III-IV).
- Previous peripartum cardiomyopathy with any residual impairment of left ventricular function.
- Severe mitral stenosis (MVA < 1.0 cm<sup>2</sup>), severe symptomatic aortic stenosis (AVA < 1.0 cm<sup>2</sup>).
- Marfan syndrome with aorta dilated > 45 mm .
- Aortic dilatation > 50 mm in aortic disease associated with bicuspid aortic valve.
- Uncorrected severe coarctation.

Modified from:

- Thorne S, MacGregor A, NelsonPiercy C. Risks of contraception and pregnancy in heart disease. *Heart* 2006; 92:15205;
- RegitzZagrosek V, Blomstrom Lundqvist C, Borghi C et al. European Society of Cardiology guidelines on the management of cardiovascular diseases during pregnancy. *Eur Heart J* 2011; 32:314797
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**Appendix XII: New York Heart Association (NYHA) Functional Classification**

| Functional Class | Symptoms                                                                                                                                                                                   | Maternal Cardiovascular Risk |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| <b>CLASS I</b>   | No limitation. Ordinary physical activity does not cause undue fatigue, dyspnea or palpitation.                                                                                            | <b>Low</b>                   |
| <b>CLASS II</b>  | Slight limitation of physical activity. Such patients are comfortable at rest. Ordinary physical activity results in fatigue, palpitation, dyspnea or angina.                              | <b>Low</b>                   |
| <b>CLASS III</b> | Marked limitation of physical activity. Although patients are comfortable at rest, less than ordinary will activity lead to fatigue, palpitation, dyspnea or angina.                       | <b>High</b>                  |
| <b>CLASS IV</b>  | Inability to carry on any physical activity without discomfort. Symptoms of congestive heart failure are present at rest. With any physical activity, increased discomfort is experienced. | <b>High</b>                  |

Modified from:

*The Criteria Committee of the New York Heart Association. Nomenclature and Criteria for Diagnosis of Diseases of the Heart and Great Vessels. 9th ed. Boston, Mass: Little, Brown & Co; 1994:253-256.*



## Appendix XIII: The different causes of ACHD-HF based on the underlying pathophysiology, related CHD and etiology tailored management

| Etiology                                                           | Pathophysiology                                                                                                                                                                                                                                                                                                              | Congenital heart defect                                                                                                                                                                                                                   | Management                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|--------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Systemic left ventricle failure                                    | Pressure overload                                                                                                                                                                                                                                                                                                            | Sub, supra valvular or valvular aortic stenosis, coarctation of the aorta, congenital MS, Shone syndrome                                                                                                                                  | <b>Surgical or catheter intervention</b> to address obstructive or regurgitant valve lesions, persistent shunts, coronary anomalies.<br>For coarctation of aorta - correct residual lesion, treat hypertension<br><b>Medical therapy</b><br>ACEi, ARB ± sacubitril, RAS blockers<br>Beta blockers<br>Mineralocorticoid receptor antagonist<br>Loop diuretics for symptom relief<br><b>CRT/ Biventricular pacing</b> in patients with chronic RV pacing and desynchrony            |
|                                                                    | Volume overload                                                                                                                                                                                                                                                                                                              | Aortic/mitral valve regurgitation, persistent shunts: VSD, ASD, PDA<br>atrioventricular valve regurgitation (post AVSD repair)<br>Aorto-pulmonary collaterals                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                                                    | Myocardial injury /ischemia                                                                                                                                                                                                                                                                                                  | Inadequate myocardial protection during surgery, supply-demand mismatch, ventriculotomy, myocardial hypertrophy, cyanosis, coronary anomalies (ALCAPA)                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                                                                    | Sub-pulmonary Ventricle - systemic ventricle interaction                                                                                                                                                                                                                                                                     | Severe pulmonary regurgitation in TOF, Chronic right ventricular pacing                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Sub-pulmonary morphological right ventricle failure <sup>a,b</sup> | Volume overload                                                                                                                                                                                                                                                                                                              | Severe pulmonary regurgitation in TOF, atrial septal defect with large left-to-right shunt<br>Ebsteins with severe tricuspid regurgitation                                                                                                | <b>Surgical or catheter intervention</b> to address obstructive or regurgitant valve lesions, persistent shunts, conduit stenosis<br><b>Medical therapy</b><br>PAH targeted therapy if RV failure due to PAH<br>Loop diuretics for symptom relieve                                                                                                                                                                                                                                |
|                                                                    | Pressure overload                                                                                                                                                                                                                                                                                                            | Severe RV outflow tract obstruction, RV to PA conduit stenosis, double chambered RV<br>Pulmonary arterial hypertension                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Morphological systemic right ventricle failure <sup>c</sup>        | Pressure Overload<br>Volume Overload<br>Myocardial architecture - RV morphology, Tricuspid systemic atrioventricular valve/ Ebsteinoid                                                                                                                                                                                       | Congenitally corrected transposition of the great arteries, transposition of the great arteries after atrial switch repair (Mustard or Senning) <sup>d,e</sup><br>systemic atrioventricular valve regurgitation - Tricuspid Regurgitation | <b>No standard guidelines for medical therapy<sup>f,g</sup></b><br><b>Medical therapy if symptomatic</b><br><b>Treat arrhythmias, correct residual hemodynamic lesions</b><br>E.g.baffle stenting / systemic TV replacement where appropriate<br>CRT may have a role                                                                                                                                                                                                              |
| Systemic single ventricle failure <sup>h</sup>                     | Complicated multifactor etiology<br>Volume overload (valve regurgitation, collaterals)<br>Decrease in preload - post Fontan<br>Myocardial injury / underlying morphology<br>Elevated systemic venous pressure<br>Diastolic dysfunction (impaired relaxation and ventricular filling) <sup>1</sup> , Tachy & bradyarrhythmias | Single ventricle physiology (unrepaired)<br>Heterotaxy<br>Single ventricle physiology palliated (BTshunt, Glenn, PDA stenting)<br>Post Fontan - PLE, valve regurgitations, arterial desaturation fenestration.                            | <b>Transcatheter Intervention</b><br>eg fenestration closure or creation of fenestration, coil occlusion of aorto - pulmonary collaterals<br>pulmonary artery stenting<br><b>surgical Fontan conversion</b><br><b>Manage arrhythmias</b> - EPS ±RFA, pacemaker implantation<br><b>No standard guidelines for medical therapy</b><br>Judicious use of Loop Diuretics for symptom relief<br><b>PDE5 inhibitors &amp; ERA</b> to decrease PA pressures and to improve cardiac output |



## Appendix XIII: The different causes of ACHD-HF based on the underlying pathophysiology, related CHD and etiology tailored management

| Etiology                                                                                             | Pathophysiology                                                                                                            | Congenital heart defect                                                                                                                                                                       | Management                                                                                                                                                                                                                                                        |
|------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cyanotic systemic and/or sub-pulmonary ventricle failure with or without pulmonary hypertension (PH) | Myocardial injury by chronic hypoxia                                                                                       | Uncorrected or palliated cyanotic congenital heart disease (TOF, TOF with pulmonary atresia, single ventricle)<br>Baffle leak (post atrial switch)<br><br>Veno-veno collateral, pulmonary A/V | <b>Surgical Correction/repair</b> all hemodynamic lesions<br><b>Transcatheter intervention</b> to occlude any veno-veno collateral, occlude baffle leaks<br><b>Treat iron deficiency anemia</b><br><b>Routine phlebotomy not recommended PAH targeted therapy</b> |
|                                                                                                      | Pulmonary to systemic shunts                                                                                               |                                                                                                                                                                                               |                                                                                                                                                                                                                                                                   |
|                                                                                                      | Pressure overload                                                                                                          | Eisenmenger syndrome                                                                                                                                                                          |                                                                                                                                                                                                                                                                   |
| Systolic dysfunction of the systemic ventricle due to arrhythmias <sup>a</sup>                       | Atrial or ventricular tachyarrhythmias<br>Sinus node dysfunction/AV node dysfunction<br>Interatrial re-entrant tachycardia | Post TOF repair, Senning/Mustard (Atrial switch)<br>cCTGA, Ebstein Anomaly, post Fontan <sup>1</sup>                                                                                          | <b>Antiarrhythmic medication</b><br>Electrophysiology study and catheter or surgical ablation<br><b>Devices:</b> Pacemaker, ICD, CRT                                                                                                                              |
| Acquired ischemic heart disease and ventricular dysfunction                                          | Coronary artery disease & Cardiovascular risk factors <sup>m</sup>                                                         | Systemic hypertension, hyperlipidemia, diabetes mellitus, smoking                                                                                                                             | Treat appropriately                                                                                                                                                                                                                                               |
|                                                                                                      | Congenital coronary artery abnormalities                                                                                   | Anomalous origin and/or course, extrinsic compression by a dilated pulmonary artery, coronary kinking after re-implantation of coronary arteries                                              | Treat appropriately                                                                                                                                                                                                                                               |
| Non cardiac cause of ventricular dysfunction                                                         | Anemia, Thyroid diseases, renal failure, Liver failure                                                                     | Can affect any CHD<br>Fontan associated liver disease<br>Iron deficiency anemia more common in cyanotic heart disease                                                                         | Treat appropriately                                                                                                                                                                                                                                               |

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## Appendix XIV: Investigations in ACHD<sup>a</sup>

| Investigations                                                                     | Important                                                                                                                                                                    | Values/Parameter                                                                                                                                                                                                                                      |
|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Electrocardiogram (ECG)</b>                                                     | To identify new onset/change in baseline ECG e.g., right/left ventricular hypertrophy, Conduction abnormalities.                                                             | Baseline ECG is abnormal to begin with in many ACHD.<br>New or recent change in baseline ECG should trigger further review.                                                                                                                           |
| <b>Chest X-ray (CXR)</b>                                                           | Abnormalities in pulmonary artery and cardiac size, lungs (oligemic/plethoric) and thorax/skeletal                                                                           | Baseline CXR may be abnormal.                                                                                                                                                                                                                         |
| <b>Transthoracic echo</b>                                                          | Anatomical diagnosis, identify residual or new lesions and sequelae, assess ventricular function, screening for pulmonary hypertension (PH).                                 | Poor echo window (obesity, multiple scars, skeletal anomalies, lung pathology.)<br>Echo parameters for left ventricular assessment are not suitable for single/ Right Ventricle (RV) function.<br>Extracardiac lesions may not be clearly visualised. |
| <b>Transesophageal echo</b>                                                        | Anatomical diagnosis, conduits, valvular lesions                                                                                                                             | Useful to assess intracardiac shunts (especially ASD), valvular lesion, anomalous venous return.<br>Used to assist in transcatheter interventions                                                                                                     |
| <b>CMR</b>                                                                         | Ventricular volumes and function especially RV<br>Flow and regurgitation fractions.<br>Perfusion defects/fibrosis.<br>Cardiac and extracardiac anatomy.                      | Excellent tool for assessment of RV and to study flow in Fontan and calculate shunts.<br>Artifacts if patients have coils/ pacemakers that are not MRI compatible.<br>Not optimal for coronary anatomy.                                               |
| <b>Cardiac CT</b>                                                                  | Assessment of extracardiac anomalies.<br>pulmonary artery anatomy.<br>stents, conduit and aorto- pulmonary collateral arteries & shunts, aortic anomalies, coronary arteries | 3D reconstruction is invaluable in planning surgical intervention and assessing surrounding non cardiac structures.<br>Risk of radiation and contrast induced nephropathy.<br>Tachycardia impairs resolution.                                         |
| <b>Cardiac catheterisation</b>                                                     | Hemodynamic assessment of shunts.<br>Pressure gradients across stenotic lesions, extracardiac shunts<br>severity of PH and vasoreactivity .                                  | Allows for transcatheter interventions for appropriate lesions.                                                                                                                                                                                       |
| <b>Biomarkers (BNP/NT ProBNP)<sup>b,c,d</sup></b>                                  | Reflects hemodynamic significant lesions, ventricular dysfunction.                                                                                                           | Levels may be raised at baseline<br>- increasing trends/acute rise may suggest HF.<br>Assess response to treatment.                                                                                                                                   |
| <b>Cardiopulmonary exercise test<sup>e</sup></b>                                   | Objective exercise capacity-peak Vo <sub>2</sub> , chronotropic competence, exercise induced arrhythmias, desaturation on exercise.                                          | Heterogenous population <sup>f</sup><br>(Trending may be more important than absolute values)<br>Prognostic values not standardized.<br>Assess response to treatment.                                                                                 |
| <b>Rhythm Analysis (24 hours &amp; up to 7 days ECG monitoring, loop recorders</b> | Identifies baseline and new onset conduction abnormalities (atrial/ventricular premature contractions, tachy/bradyarrhythmias, pauses, heart rate variabilities              | May miss paroxysmal arrhythmias.                                                                                                                                                                                                                      |

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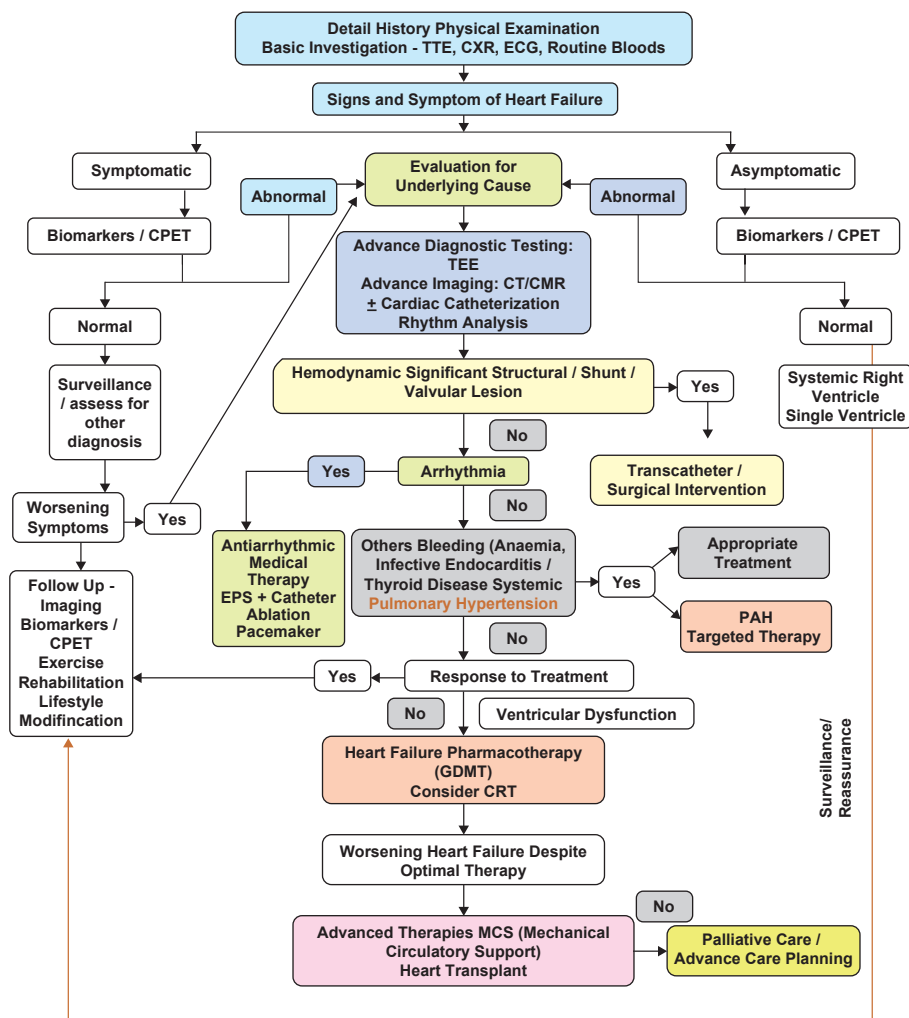
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**Appendix XV: Flow Chart of the Management of ACHD-HF**


**APPENDIX XVI Calculation of Performance and Outcome Measures**

|                                                                                                                                    |   |                                                                                                                                                                                                                            |       |
|------------------------------------------------------------------------------------------------------------------------------------|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>% of patients who had documentation of NYHA Functional Class</b>                                                                | = | $\frac{\text{Number of patients who had documentation of NYHA Functional Class}}{\text{Number of HF patients who were seen during that time period}}$                                                                      | x 100 |
| <b>% of patients who had LVEF measurement</b>                                                                                      | = | $\frac{\text{Number of patients who had LVEF measurement}}{\text{Number of HF patients who were seen during that time period}}$                                                                                            | x 100 |
| <b>% of patients discharged with ACE-I / ARB</b>                                                                                   | = | $\frac{\text{Number of patients who were on ACE-I / ARB at discharge}}{\text{Number of HF patients who were discharged during this time period who had no contraindications to ACE-I/ARB}}$                                | x 100 |
| <b>% of patients discharged on <math>\beta</math>-blockers</b>                                                                     | = | $\frac{\text{Number of patients who were on } \beta\text{-blockers at discharge}}{\text{Number of HF patients who were discharged during that time period who had no contraindications to } \beta\text{-blockers}}$        | x 100 |
| <b>% of patients discharged on MRA</b>                                                                                             | = | $\frac{\text{Number of patients who were on MRA at discharge}}{\text{Number of HF patients who were discharged during that time period who had no contraindications to MRA}}$                                              | x 100 |
| <b>% of patients with chronic or paroxysmal AF/Atrial Flutter on anticoagulant therapy (OAC) at discharge.</b>                     | = | $\frac{\text{Number of patients who had AF/Atrial Flutter who were on OAC at discharge}}{\text{Number of HF patients who had AF/Atrial Flutter during that time period who had no contraindications to OAC at discharge}}$ | x 100 |
| <b>% of patients given a post discharge appointment within 14 days</b>                                                             | = | $\frac{\text{Number of patients who were given a post discharge appointment}}{\text{Number of HF patients who were seen during that time period}}$                                                                         | x 100 |
| <b>% of patients who had their Foundational HF medications up titrated to maximally tolerated doses by 3 months post discharge</b> | = | $\frac{\text{Number of patients who Had the HF medications up titrated within 3 months}}{\text{Number of HF patients who were seen during that time period}}$                                                              | x 100 |



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